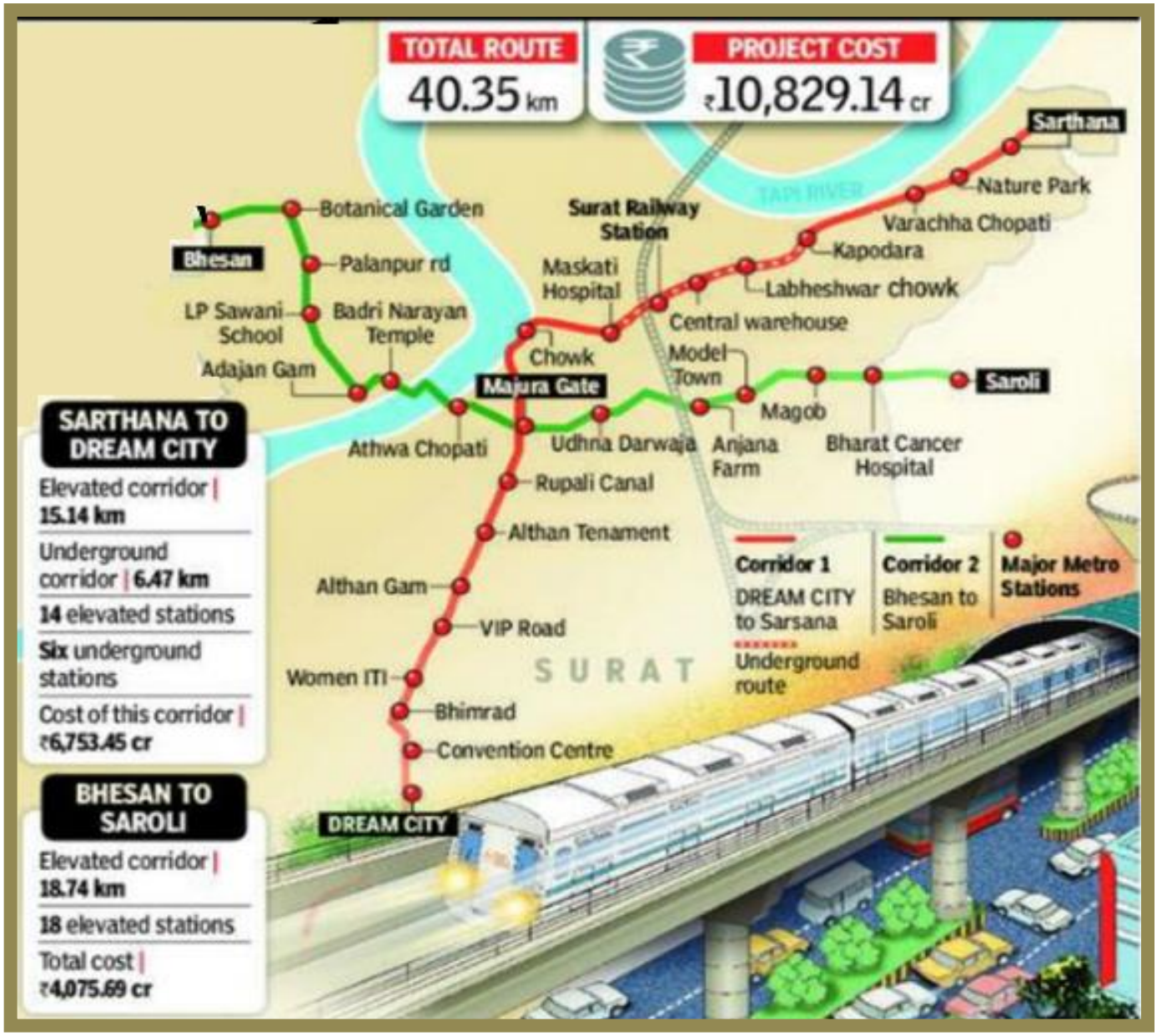
Gujarat Metro Rail Corporation (GMRC) Limited

(SPV of Government of India and Government of Gujarat)

[Formerly known as Metro-Link Express for Gandhinagar and Ahmedabad (MEGA) Company Limited]

ENVIRONMENTAL IMPACT ASSESSMENT STUDY FOR SURAT METRO RAIL PROJECT (PHASE-I)



FINAL REPORT

DECEMBER 2020

RITES LIMITED

(A Government of India Enterprise)

RITES Bhawan,Plot No.1,Sector-29

GURGAON-122001(INDIA)

www.rites.com

Gujarat Metro Rail Corporation (GMRC) Limited

(SPV of Government of India and Government of Gujarat)

[Formerly known as Metro-Link Express for Gandhinagar and Ahmedabad (MEGA) Company Limited]

ENVIRONMENTAL IMPACT ASSESSMENT STUDY FOR SURAT METRO RAIL PROJECT (PHASE-I)

FINAL REPORT

December 2020

RITES LIMITED

(A Government of India Enterprise)

RITES Bhawan,Plot No.1,Sector-29

GURGAON-122001(INDIA)

Ph: 0124-2818760,Fax:0124-2571660

www.rites.com

G	14th December 2020	Final EIA Report	S Harinag	Amita Jain
F	20th July 2020	Draft EIA Report-R3	S Harinag	Amita Jain
E	30th June 2020	Draft EIA Report-R2	S Harinag	Amita Jain
D	06th February 2020	Draft EIA Report-R1	Sanjay Raut	Manish Indurkar
C	21st August 2019	Draft EIA Report	Sanjay Raut	Manish Indurkar
B	15th July 2019	Field Data Report	Sanjay Raut	Manish Indurkar
A	29th April 2019	Inception Report	Sanjay Raut	Manish Indurkar
Rev.	**Date**	**Description**	**Originated**	**Assisted**

Client:

Gujarat Metro Rail Corporation (GMRC) Limited

(SPV of Government of India and Government of Gujarat)

[Formerly known as Metro-Link Express for Gandhinagar and Ahmedabad (MEGA) Company Limited]

Project:

ENVIRONMENTAL IMPACT ASSESSMENT STUDY FOR SURAT METRO RAIL PROJECT (PHASE-I)

Prepared by:

RITES LIMITED
(A Government of India Enterprise)
RITES Bhawan,Plot No.1,Sector-29
GURGAON-122001(INDIA)
www.rites.com

Doc. No.: GMRC/EIA&SIA/SURAT/PH-1/2019/EIA-7

File: RITES/UE/SURAT METRO/RT610-00042/2019

CONTENT

Chapter 0 : Executive Summary 1
Chapter 1 : Introduction 12
1.1. Background 12
1.2. Project Area 13
1.3. Objective of the Study 13
1.4. Scope of EIA Study 14
1.5. Approach and Methodology 15
1.6. Structure of the Report 17
Chapter 2 : Policy, Legal and Institutional Requirements 20
2.1. Environmental Policy Framework 20
2.2. Legal Framework 20
2.2.1. Environment Protection Act 27
2.2.2. Water and Water Pollution 27
2.2.3. Air Quality 28
2.2.4. Noise Quality 28
2.2.5. Solid Waste Management 28
2.2.6. The Ancient Monuments and Archaeological Sites and Remains Act, 2010 29
2.2.7. The Building and Other Construction Workers Act, 1996 30
2.2.8. National Environment Policy, 2006 30
2.3. Institutional Framework 31
2.3.1. Central and State Pollution Control Boards 31
2.3.2. Environmental, Health, and Safety Guidelines for Railways by IFC 32
2.4. World Bank Requirement 32
2.5. KfW Requirement 38
2.6. AFD Requirements 39
2.7. Gap Analysis 40
Chapter 3 : Project Description 49
3.1. Existing Transportation System in Surat City 49
3.2. Study Area 49
3.3. Proposed Metro Rail System in Surat City 50
3.3.1. Stations 55
3.3.2. Permanent Way 57
3.3.3. Rolling Stock, Traction and Signalling 58
3.3.4. Ventilation and Air-Conditioning System 58
3.3.5. Maintenance Depots 58
3.3.6. Power Requirements 59
3.3.7. Sub Station 60
3.3.8. Construction Activities 60
3.4. Cost Estimate 61
3.5. Analysis of Alternatives 61
3.5.1. Alignment Planning 61
3.5.2. Demand Assessment Along The Corridor 62
3.5.3. Corridor Characteristics 64
3.5.4. Capacity of existing transport system (With and without project Assessment) 66
3.5.5. Observations 69
3.5.6. Economic Effects (Social Cost Benefit analysis) 70
3.5.7. Conclusions 71
3.5.8. Environmental and Social Considerations for Alternative Analysis 72

Chapter 4 : Environmental Baseline Data 77
4.1. General 77
4.2. Land Environment 77
4.2.1. Physiography 78
4.2.2. Geology and Hydrogeology 79
4.2.3. Soils 81
4.2.4. Land Use Pattern 84
4.2.5. Seismicity 87
4.3. Water Environment 88
4.3.1. Water Resources of the District 88
4.3.2. Drainage 90
4.3.3. Water Quality of the Project Area 91
4.4. Meteorology and Ambient Air Environment 96
4.4.1. Meteorology 96
4.4.2. Air Quality 100
4.4.3. Noise Environment 104
4.4.4. Vibration 106
4.5. Ecological Environment 118
4.5.1. Forest 118
4.5.2. Flora 118
4.5.3. Fauna 123
4.5.4. Protected Areas in the region 124
4.6. Sensitive Receptors 124
Chapter 5 : Negative Environmental Impacts 131
5.1. General 131
5.2. Environmental Impacts 133
5.3. Impacts due to Project Location 133
4.3.1. Displacement and loss of livelihood of Project Affected People (PAPs) 134
4.3.2. Change of Land Use 134
4.3.3. Impact on Ecological Environment 134
4.3.4. Utility/Drainage Problems 135
4.3.5. Impact on Archaeological Monuments/ Heritage Sites 135
5.4. IMPACTS DUE TO PROJECT DESIGN 135
5.4.1. Right of Way 135
5.4.2. Alignment, Stations, Track design and Architecture 136
5.4.3. Inter Modal Integration 136
5.4.4. Use of Energy and Water at stations and depots 136
5.4.5. Risk Due to Natural Hazards 136
5.5. IMPACT DUE TO PROJECT CONSTRUCTION 136
5.5.1. Air pollution 137
5.5.2. Noise Pollution 142
5.5.3. Vibration Impacts and Risk to Existing Buildings 145
5.5.4. Impact due to Muck Disposal 157
5.5.5. Impact due to construction/demolition waste Disposal 157
5.5.6. Impact due to Hazardous Waste 157
5.5.7. Increased Water Demand 158
5.5.8. Impact due to Pre-casting yards and Material stockpiling 158
5.5.9. Impact on Ground and Surface Water Quality 158
5.5.10. Soil Erosion and land subsidence 158
5.5.11. Traffic Diversions 158
5.5.12. Impacts due to Labour Camps 158
5.5.13. Welfare of Labour on construction site 159

5.5.14. Safety of Labour 159
5.5.15. Impact due to Supply of Construction Material 159
5.5.16. Impacts of Air Pollution on Human Health 159
5.5.17. Impact on Flamingo 160
5.6. IMPACTS DUE TO PROJECT OPERATION 161
5.6.1. Noise Pollution 161
5.6.2. Vibration 167
5.6.3. Energy Consumption at Stations 169
5.6.4. Water Supply, Waste Water and Municipal Solid Waste disposal at Stations 169
5.6.5. Traffic Congestion around Stations 169
5.6.6. Impacts due to Depot 172
5.6.7. Impact on Flamingo 173
Chapter 6 : Positive Environmental Impacts 174
6.1. Employment Opportunities 174
6.2. Benefits to Economy 174
6.3. Direct Benefits to Passengers 174
6.4. Safety 175
6.5. Traffic Noise Reduction 175
6.6. Reduction of Traffic on Road 175
6.7. Reduced Air Pollution 176
6.8. Carbon Credits 176
6.8.1. Carbon Credits for Surat Metro Rail Project 178
Chapter 7 : Environmental Management Plan 180
7.1. Approvals/Clearances 180
7.2. Mitigation Measures 181
7.3. EMP During Project Location and Design 182
7.2.1. Compensatory Afforestation 182
7.2.2. Right of Way, Alignment, Track design and Architecture 186
7.2.3. Spatial Planning of Stations and Inter-Modal Integration 186
7.2.4. Robust Design 186
7.2.5. Green Buildings 187
7.2.6. Solar Power 187
7.2.7. Use of Energy and Water 188
7.2.8. Utility Plan 188
7.2.9. Site Selection Criteria for Construction Yards 189
7.4. EMP During Construction 189
7.3.1. Supply of Construction Material 189
7.3.2. Pre-casting yards and Material Stockpiling 189
7.3.3. Construction Material Management and Housekeeping 190
7.3.4. Hazardous Waste Management 190
7.3.5. Construction and Demolition Waste Management 191
7.3.6. Muck Disposal 191
7.3.7. Energy Management 192
7.3.8. Labour Camp 192
7.3.9. Welfare of Labour on Construction Site 193
7.3.10. Safety of Labour 193
7.3.11. Air Pollution Control Measures 195
7.3.12. Noise Control Measures 197
7.3.13. Vibration Management 199
7.3.14. Water pollution Control 200
7.3.15. Traffic Diversion/Management 200
7.3.16. Soil Erosion and land subsidence control 200

7.3.17. Dewatering of underground works 201
7.3.18. Management Plan for Flamingos 201
7.5. EMP During Operation 202
7.4.1. Noise and Vibration Management 202
7.4.2. Water Supply and Sanitation at Stations 204
7.4.3. Rain Water Harvesting 207
7.4.4. Management Plan for Depots 207
7.4.5. Management Plan for Flamingos 209
7.6. EMP Reporting Arrangement and Institutional Strengthening 209
7.7. Disaster Risk Management 212
7.8. Provisions at Metro Stations/Other Installations 213
7.9. Emergency Measures 215
7.10. Training 218
7.11. Environment Management System (EMS) 218
Chapter 8 : Public Consultations 229
8.1 Introduction 229
8.2 Methods & Approach for Consultation 229
8.3 Public Consultation at Project Level 229
Chapter 9 : Environmental Monitoring Plan 236
9.1. Environment Monitoring Plan 236
9.1.1. Pre-Construction Phase 236
9.1.2. Construction Phase 236
9.1.3. Operation Phase 238
9.2. Establishment of Environmental Division 240
Chapter 10 : Cost Estimate 241
10.1. Summary of Costs 241

LIST OF TABLES

Table 1.1: Structure of EIA Report 17
Table 2.1: Environmental Regulation and Legislation 21
Table 2.2: Gap Analysis 41
Table 3.1: Salient Features of the Corridors 53
Table 3.2: Proposed Stations along Corridor-1 53
Table 3.3: Proposed Stations along W-E Corridor 54
Table 3.4: Power Demand in MVA 59
Table 3.5: Source of Power Supply 60
Table 3.6: Corridor-wise Details of Capital Cost 61
Table 3.7: Model Outputs of Public Transport Assignments (2046) 62
Table 3.8: Mode Selection based on Daily Ridership and PHPDT 63
Table 3.9: Analysis of Vehicular Traffic along the Corridors 67
Table 3.10: Ridership Potential 70
Table 3.11: Ridership Quantifiable and Non Quantifiable Benefits 70
Table 4.1: Environmental Attributes and Frequency of Monitoring 77
Table 4.2: Geological Succession of Surat District 80
Table 4.3: Description of Soil Quality Monitoring Locations 82
Table 4.4: Soil Quality Data 83
Table 4.5: Landuse Class Statistics of Study Area 85

Table 4.6: Details of Water Sampling Locations 91
Table 4.7: Physico-Chemical Analysis of Water Samples 93
Table 4.8: Zooplankton Analysis of Surface Water Samples 95
Table 4.9: Phytoplankton Analysis of Surface Water Samples 95
Table 4.10: Qualitative Benthos Group Wise Analysis 96
Table 4.11: Temperature, Rainfall & Humidity for Year 2014 97
Table 4.12: Temperature, Rainfall & Humidity for Year 2015 97
Table 4.13: Temperature, Rainfall & Humidity for Year 2016 98
Table 4.14: Temperature, Rainfall & Humidity for Year 2017 98
Table 4.15: Temperature, Rainfall & Humidity for Year 2018 99
Table 4.16: Air Quality Monitoring Locations 101
Table 4.17: Ambient Air Quality Monitoring Results 103
Table 4.18: National Amibent Air Quality Standard (MOEFCC) 103
Table 4.19: Noise Level Monitoring Results 104
Table 4.20: Ambient Noise Standards Criteria 105
Table 4.21: Vibration Monitoring Locations 107
Table 4.22: Forest Cover Area of Surat District 118
Table 4.23: Summary of Tree Enumeration 118
Table 4.24: Detailed Tree Inventory For Bhesan To Saroli Corridor 119
Table 4.25: Detailed Tree Inventory for SarthanaTo Dream City Corridor 119
Table 4.26: Predominant Tree Species Along The Corridors 120
Table 4.27: Sensitive Receptors within 100m from The Centre Line 124
Table 5.1: Rating of Environmental Impacts 131
Table 5.2: Number of Trees along the Corridor/Depots 134
Table 5.3: Total Emissions due to Truck Movement during Construction 137
Table 5.4: Emission Source Characteristics 139
Table 5.5: Typical Noise Levels Generated from Construction Equipment 144
Table 5.6: Guideline Vibration Damage Threshold Criteria 145
Table 5.7: Vibration Source levels for Construction Equipment 145
Table 5.8: Vibration due to Vibratory Pile Driver 147
Table 5.9: Vibration due to Impact Pile Driver 151
Table 5.10: Tunneling Methods 155
Table 5.11: Vibration due to TBM 156
Table 5.12: Inputs to Noise Modelling 162
Table 5.13: Predicted Noise Levels due to Metro Operations 162
Table 5.14: Cumulative Noise Levels at Monitoring Locations due to Metro Train Operations 166
Table 5.15: Predicted Vibration Level due to Rapid Transit Rail 168
Table 5.16: Water Demand, Waste Water and SW Generation for Dream City to Sarthana Corridor 170
Table 5.17: Water Demand, Waste Water and SW Generation for Bheshan to Saroli Corridor 171
Table 6.1: Travel Benefits due to Mode Shift 175
Table 6.2: Reduction in Air Pollutants due to Vehicle Reduction 176
Table 6.3: Traction Energy Requirement for Bhesan to Saroli Corridor 179
Table 6.4: Traction Energy Requirement for Sarthana to Dream City Corridor 179
Table 6.5: CO_2 Emissions from Energy Requirement of Metro Operations 179

Table 7.1: Roles and Responsibilities - Securing Approvals/Clearances 180
Table 7.2: Cost of Compensatory Afforestation & Transplantation 183
Table 7.3: Scientific Names Of Tree For Plantation 186
Table 7.4: Solar Power Generation Potential for each Corridor 188
Table 7.5: Organizations Responsible for Utilities and Services 188
Table 7.6: Details of Noise Barriers 202
Table 7.7: Capacity of Bio Digester at Each Station 204
Table 7.8: Wastewater Quality Parameters (Influent & Effluent) 206
Table 7.9: Capacity Estimated Cost of Bio Digester & ETP at Depots 208
Table 7.10: Cost for Training Program 218
Table 7.11: Roles and Responsibilities – Preparation and Implementation EMP 219
Table 8.1: Minutes of Public Consultations 230
Table 9.1: Construction Stage Monitoring Schedule 236
Table 9.2: Environmental Monitoring Cost during Construction 237
Table 9.3: Operation Stage Monitoring Schedule 238
Table 9.4: Environmental Monitoring Cost during Operation 239
Table 9.5: Environmental Division Cost 240
Table 10.1: Cost of Environmental Management Plan 241

LIST OF FIGURES

Figure 1.1: Flow Chart for EIA Study 19
Figure 3.1: Location Map 50
Figure 3.2: Proposed Metro Corridors in Surat City 51
Figure 3.3: Google Map showing Proposed Metro Corridors 52
Figure 3.4: Typical Drawing of Elevated Station 56
Figure 3.5: Typical Drawing of Underground Station 56
Figure 3.6: Typical Drawing of Interchange Station 57
Figure 3.7: Prospective Future Metro And BRT Corridor (2046) 64
Figure 3.8: Entry/Exit of Saroli Station 73
Figure 3.9: Dream City Depot Location 74
Figure 4.1: Physiographic Map of Study Area 78
Figure 4.2: Contour Map of the Study Area 79
Figure 4.3: Geological Map of Surat District 80
Figure 4.4: Soil Map of Gujarat 81
Figure 4.5: Water And Soil Sample Monitoring Locations 82
Figure 4.6: Satellite Imagery of the Study Area 84
Figure 4.7: Land use/Landcover Map of the Study Area 86
Figure 4.8: Land use Classification of the Corridors 87
Figure 4.9: Seismic Zoning Map of Gujarat 88
Figure 4.10: Depth to Water Level Map for Surat District 89
Figure 4.11: Flood Prone Areas of Gujarat State 90
Figure 4.12: Drainage Map of the Study Area 92
Figure 4.13: Wind Rose Diagram for Year 2009-2018 08:30 Hrs 99
Figure 4.14: Wind Rose Diagram for Year 2009-2018 17:30 Hrs 100

Figure 4.15: Air Quality and Noise Level Monitoring Locations 102
Figure 4.16: Vibration Monitoring Locations 106
Figure 4.17: Vibration Monitoring Results at BAPS Pramukh Swami Hospital 108
Figure 4.18: Vibration Monitoring Results at Tristar Hospital, Athwa Gate 109
Figure 4.19: Vibration Monitoring Results Near G D Mehta Mahavir Heart Hospital 110
Figure 4.20: Vibration Monitoring Results Near Nirmal Nagar Society 111
Figure 4.21: Vibration Monitoring Results at Swami Narayan Temple 112
Figure 4.22: Vibration Monitoring Results at Shri Harihar Mahadev Temple 113
Figure 4.23: Vibration Monitoring Results at Lambe Hanuman Temple 114
Figure 4.24: Vibration Monitoring Results at Clock Tower 115
Figure 4.25: Vibration Monitoring Results at Rani Talab Masjid 116
Figure 4.26: Vibration Monitoring Results at Andrew Library 117
Figure 4.27: Giant African Baobab Trees 121
Figure 4.28: Avicennia Marina Mangroves Near Chowpati Garden 122
Figure 4.29: Flock of Lesser Flamingo In Tapi River 123
Figure 4.30: Map of Sensitive Receptors 130
Figure 5.1: Spatial Distribution of PM2.5 (μg/m3) during Construction: Without Control 138
Figure 5.2: Spatial Distribution of PM10 (μg/m3) during Construction: Without Control 139
Figure 5.3: Spatial distribution of 24-hours average PM2.5 conc. (μg/m3):DG Set 140
Figure 5.4: Spatial distribution of 24-hours Average PM10 conc. (μg/m3): DG Set 140
Figure 5.5: Spatial distribution of 24-hours average SO2 conc. (μg/m3): DG Set 141
Figure 5.6: Spatial distribution of 24-hours average NOx conc. (μg/m3): DG SET 141
Figure 5.7: Spatial distribution of 8-hours average CO conc. (μg/m3): DG SET 142
Figure 5.8: Spatial Variation of Construction Equipment Noise Levels dB(A) 144
Figure 5.9: Graphical Representation of the Vibrations due to Vibratory Pile 147
Figure 5.10: BAPS Pramukh Swami Hospital 148
Figure 5.11: Tristar Hospital, Athwa Gate 148
Figure 5.12: D Mehta Mahavir Heart Hospital 149
Figure 5.13: Near Nirmal Nagar Society 149
Figure 5.14: Swami Narayan Temple 150
Figure 5.15: Graphical Representation of the Vibrations due to Impact Pile Driver 151
Figure 5.16: BAPS Pramukh Swami Hospital 152
Figure 5.17: Tristar Hospital, Athwa Gate 153
Figure 5.18: D Mehta Mahavir Heart Hospital 153
Figure 5.19: Near Nirmal Nagar Society 154
Figure 5.20: Swami Narayan Temple 154
Figure 5.21: Proposed alignment Crossing Tapi River 160
Figure 5.22: Noise Contours of Proposed Corridors 163
Figure 7.1: Location of Land Parcel for Compensatory Afforestation 185
Figure 7.2: Silo Top Aero Filter 196
Figure 7.3: Typical Cross Section of Bio Digester 207
Figure 7.4: Institutional Mechanism for EMP Implementation (Construction Phase) 210
Figure 7.5: Institutional Mechanism for EMP Implementation (Operation Phase) 210
Figure 7.6: EMP Monitoring Structure 211
Figure 7.7: EMS Organization 227

Figure 8.1: Photographs of Public Consultations 233

LIST OF ANNEXURE

Annexure 2.1: Drinking Water Quality Standards (IS 10500:2012) 243
Annexure 2.2: Effluent Discharge Standards (Inland Surface Water) 245
Annexure 2.3: Tolerance Limits for Inland Surface Water Quality 246
Annexure 2.4: Ambient Air Quality Standards 247
Annexure 2.5: National Ambient Noise Standards 248
Annexure 5.1: List of Trees along the Corridors/Depots 249
Annexure 5.2: Vibration Levels at Sensitive Receptors 274
Annexure 5.3: Cumulative Noise Levels at Receptors due to Metro Train Operations 282
Annexure 7.1: Muck Dumping Site 287
Annexure 7.2: Details of Batching Plant 289
Annexure 7.3: Details of Muck Dumping Operations 290
Annexure 7.4: Details of Machinery during Construction 291
Annexure 7.5: Safety Check List 292
Annexure 7.6: Accident Report 294
Annexure 7.7: Pollution Monitoring 296
Annexure 7.8: Format for Vibration Monitoring 298
Annexure 7.9: Restoration of Construction Sites 299
Annexure 7.10: Format for Keeping Records of Consent obtained by Contractor 300
Annexure 7.11: Checklist for Environment Inspection 301
Annexure 7.12: Summary Sheet 302
Annexure 8.1: Public Consultations 304

NOMENCLATURE

ANQS	Ambient Noise Quality Standards
ASI	Archaeological Survey of India
BGL	Below Ground Level
BIS	Bureau of Indian Standards
BOQ	Bill of Quantities
C&D	Construction and Demolition
CGWB	Central Ground Water Board
CIRT	Central Institute of Road Transport
CMP	Comprehensive Mobility Plan
CNG	Compressed Natural Gas
CPCB	Central Pollution Control Board
CPHEEO	Central Public Health and Environmental Engineering Organisation
CSC	Construction Supervision Consultant
CTE	Consent to Establish
CTO	Consent to Operate

DG	Diesel Generator
DMRC	Delhi Metro Rail Corporation
DPR	Detailed Project Report
DRDO	Defense Research and Development Organization
E&M	Electrical and Mechanical
EA	Environmental Assessment
EC	Electrical Conductivity
EIA	Environmental Impact Assessment
EMAP	Environmental Management Action Plan
EMoP	Environmental Monitoring Plan
EMP	Environmental Management Plan
EMS	Environmental Management System
EPA	Environment Protection Act
ESA	Environmental and Social Assessment
ESIA	Environmental and Social Impact Assessment
ESMP	Environmental and Social Management Plan
FHWA	Federal Highway Administration
GC	General Consultancy
GHG	Green House Gases
GIS	Geographic Information System
GoI	Government of India
GW	Ground Water
HT	High Tension
HW	Hazardous Waste
IMD	Indian Meteorological Department
IS	Indian Standard
ISO	International Organization for Standardization
KLD	Kilo Litres per Day
LISS	Linear Imaging Self Scanning Sensor
MLD	Million Litres per Day
MoEFCC	Ministry of Environment, Forests and Climate Change
NABL	National Accreditation Board for Testing and Calibration Laboratories
NGO	Non-Governmental Organisation
NOC	No Objection Certificate
OHSAS	Occupational Health and Safety Assessment Series
PAP	Project Affected People
PIU	Project Implementation Unit
PM	Particulate Matter
PPE	Personal Protective Equipment
PPV	Peak Particle Velocity
PWD	Public Works Department
RDSO	Research Designs & Standards Organisation

RMC	Ready Mix Concrete
RTFCT-LARR	Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement
SEA	Strategic Environmental Assessment
SHE	Safety, Health and Environment
SPCB	State Pollution Control Board
STP	Sewage Treatment Plant
SW	Surface Water
TBM	Tunnel Boring Machine
UG	Under Ground
UN	United Nations
UNECE	United Nations Economic Commission for Europe
UNFCCC	United Nation Framework Convention on Climate Change
UNISDR	United Nations International Strategy for Disaster Reduction
USEPA	United States Environmental Protection Agency
VEC	Valued Environmental Component
VOC	Vehicle Operating Cost
WB	World Bank
WGS	World Geodetic System

Chapter 0 : Executive Summary

0.1 INTRODUCTION

Gujarat Metro Rail Corporation Ltd. intends to develop the Surat Metro Rail Phase-I from Sarthana to Dream City (21.61 Km) and Bhesan to Saroli (18.74 km).

0.1.1 Objective of the Study

The objective of the present study is to carry out Environmental Impact Assessment (EIA) and preparation of Environmental Management Plan (EMP) for both the corridors.

0.1.2 Scope of EIA Study

The scope of work for the EIA study was issued vides LOA No. GMRC/EIA & SIA/Surat/PH-1/2019/LOA dated 20-03-2019. The scope of Environmental Impact Assessment (EIA) comprises environmental baseline, impacts resulting from pre-construction, construction and operation phases and measures to mitigate adverse impacts due to the project on the environment.

0.2 LEGAL, POLICY AND INSTITUTIONAL REQUIREMENTS

The Indian Acts, Rules and Notifications applicable to the proposed project have been considered while preparing the EIA report. In addition to these, requirement as per AFD, KfW and World Bank's Environment and Social Framework 2017 have also been reffered. As per the MoEFCC, GoI Notification metro rail project does not require prior environmental clearance. Gap Analysis between worldbank guidelines, IFC and GOI regulations are briefed in Chapter 2.

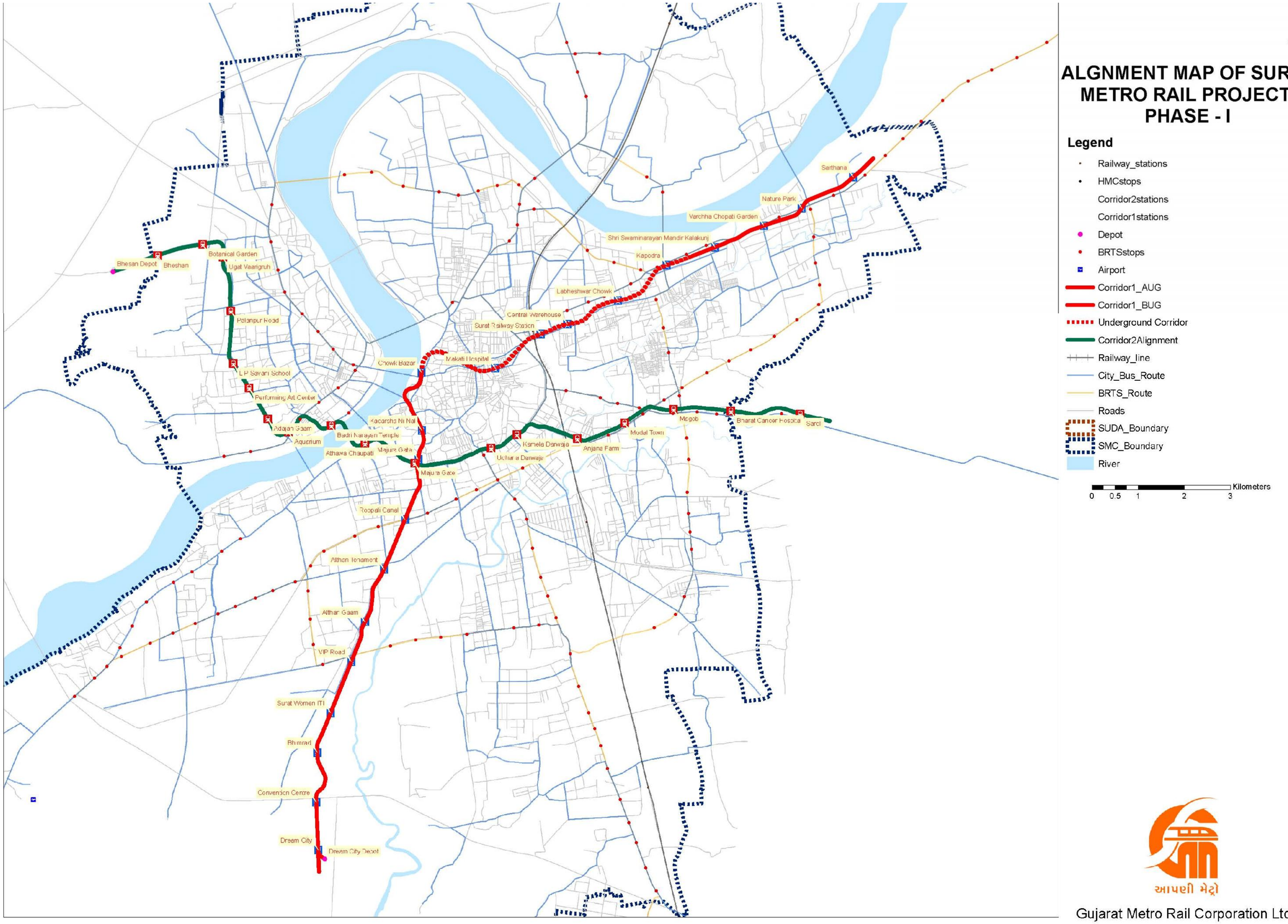
0.3 PROJECT DESCRIPTION

Surat metro Phase I project having two corridors has a total length of 40.35 km with 38 stations. Sartthana as elevated section and heads towards Dream City as UG section of 6.47km. Total 20 stations have been proposed along the corridor 1 of which 14 stations are elevated and 6 stations are UG. Second corridor starts from Bhesan and heads towards Saroli, this section is fully elevated. Total 18 stations have been proposed along this corridor. Alignments of proposed Surat Metro Rail Corridors are showin in Figure 0.1.

0.3.1 Analysis of Alternatives

A Comprehensive Mobility Plan (CMP) aims to retain the compact structure of the city of Surat and proposes intensification of development along the rapid transit corridors. The report recommends the 3 corridors as potential metro network as 1. Dream city – Kamrej (28.9 km), 2. Bhesan – Umbhel (26.3 km) and 3. Majuragate to Karamala (15.8 km). Study was done on the basis of demand analysis and network characteristics, in order to assess suitability of a rapid transit mode for these three corridors.

Figure 0.1: Surat Metro Rail Corridors



Economic analysis have been undertaken to assess economic effects of alternative transit system options for identified corridors. Alignments selected on the considerations of direct and indirect benefits with metro like time save, pollution reduction, reductions in road accidents etc. During the DPR preparation the section finalized based on various design considerations and alternative construction techniques to reduce the impact on environmental and social aspects.

0.4 ENVIRONMENTAL BASELINE DATA

The Environmental baseline data collected for the study includes secondary data and primary data on quality of air, water and soil; noise; vibration; flora and fauna and sensitive receptors along the corridors.

0.4.1 Land Environment

Physiography: Surat is a port city situated on the banks of the Tapti river which was formally known as Tapi river. The city is located at 21°10'N and 72°50'E. It has an average elevation of 13 metres. The project area is with elevation range of 0m to 30m. However as per the current ground levels, the proposed metro alignment is passing through 6m to 18m elevations.

Geology and Hydrogeology: Major geological formations exposed in the district are Quaternary alluvium, Tertiary limestone and sandstones and Deccan Trap basalt. Surat comes under cenozoic era of tertiary period of Palaeocene series/epoch. The major aquifers in the district are formed by alluvium and Deccan Trap basalt with Tertiary formations occupying a small patch. The alluvium occurs in the western part of the district and along the streams. In Alluvial areas the depth of the tube wells range from 36m to 169m with piezometric levels varying between 2m to 10m.

Soils: In coastal region the soil is deep to very deep dark grey to black colour. The texture is clay loam to silty loam. Surat having costal alluvium and costal alluvium (saline) soils. To get the actual charectaristics of soil in the project area, soil samples from various locations were collected and analyzed to understand the soil characteristics of the area with respect to its productivity potential. Ten representative soil samples have been collected along the proposed metro alignment during 29/5/2019 to 7/6/2019.

Land Use Pattern: The land use of 15 km belt on either side of the project corridors has been identified: built-up area is 18.14%, agriculture land 49.45%, Water Bodies 9.67%, Scrub Land 2.94%, River 9.24%, open forest 0.24% and prawn ponds 9.88%.

Seismicity: The proposed project corridors fall in Zone III as per revised seismic zoning map of India.

0.4.2 Water Environment

Water Resources of the District: The western and north western parts of the district is covered by Alluvium. This aquifer can broadly be demarcated into two zones namely newer alluvium and older alluvium. As per the CGWB report of Surat district, the depths of the

wells in alluvium generally range from 3 to 30 m with some of the wells having bores down to 50m below bottom of the wells. The depth to water level in Surat city varies from 10m to 20m but at some places it goes up up to 5m.

Flood Prone Areas: Surat lies in the flood plain and also lies in an estuarial region. There are number of creeks in the south- western part of the city. The city has been historically susceptible to flooding. Development of Surat in the past has been affected due to the high flood in Tapi however appropriate measures have been taken like construction of embankment, river training and river front development etc.

Drainage: The Tapi is the major river which passes through the central parts of the district and flows towards the west. The walled city area has no good drainage system. There are few pockets of low lying area, where water gets logged in the rainy season. It is observed that the area of Navasari Bazar, Khapatia Chakla, Panini Bhit, Vada Chauhta area get flooded in rainy season.

Water Quality: The water sampling was done on the basis of availability of ground water and surface water source along the proposed metro alignments. There were no water in the canal during the field study (May 19). Hence 10 surface water samples from small water ponds were collected and other 4 samples were collected from ground water source. Parameters like pH, calcium, floride and ntrate are found within permissible limits in both surface and ground water samples. Hard water found in 3 locatons, TDS is very hign i.e. 18716 mg/l in Tapi river due to salt water intrusion in high tide from creek. Also It is found high in 2 other locations. High total cound to phytoplanktones found in creek water and one of canal water due to coastal water pushing in more phytoplankton population towards mouth of the estuary. Zooplankton standing stock was studied in terms of biomass, population. Highest count was found in creek water sample.

0.4.3 Ambient Environment

Climate: Meteorological data like monthly total rainfall, maximum & minimum temperature and relative humidity of the Surat for the period of Jan 2014 to Dec 2018 have been collected from IMD. Monthly wind rose diagram for last 10 years i.e. 2009 to 2018 at 8:30 hrs and 17:30 hrs have also been collected from IMD. The predominant direction is from NW and SW for 08:30 hrs and N, NW and SW direction for 17:30 hrs.

Ambient Air Quality: 20 ambient air monitoring stations were selected strategically to generate the representative samples for ambient air quality covering residential, institutional and industrial area along the corridors. Two days continuous air quality monitoring at 20 locations along the metro corridors has been done in the month of May – June 2019. Five major air pollutants viz. particulate matter (PM10 & PM2.5), Sulphur Dioxide (SO2), Nitrogen Dioxide (NO2) and Carbon Monoxide (CO) representing the air quality were monitored. The air quality monitoring results indicate that all parameters are within the permissible limits of National Standards**.** When these results compared with IFC standards, PM_{10} is exceeding the permissible limits at all locations and $PM_{2.5}$ exceeding at few locations.

Noise Environment: Noise level survey was carried out at 20 locations along the metro corridor. The noise levels were collected over 24 hours in the month of May - June 2019 at hourly interval during morning, afternoon and evening such that peak and off peak hours are covered. Noise Monitoring results shows that noise level at all locations is exceeding the noise level standards prescribed by CPCB either day or night or both for Residential Zone, Commercial Zone; and when compared with IFC standards within the permissible limits at few locations and at remaning locations exceeding the permissible limits. The main source of noise in the project area is the traffic movement on the road.

Vibration: Vibration monitoring was carried out at 10 locations for 24 hrs continuous along the corridor in the month of May & June 2019. Vibration monitoring was carried out using the Nomis Seismograph equipment which can measure the radial, transverse and vertical vibration of ground-borne vibrations.

0.4.4 Ecological Environment

Trees survey was carried out in the month of May - June 2019 along the proposed metro alignments and at station locations. About 2093 trees observed along both the corridors and depot. The height and girth of the trees ranges from 3m to 14m and 30cm to 400cm respectively. Two giant African Baobab (*Adansonia digitata)* trees are observed within ROW opposite Croma Shopping Complex & Jodiyar Mata Mandir near proposed Adajan Gam-station along Bhesan to Saroli corridor. Out of these two Giant biobab tree, one tree is collapsed due to weather disturbance on 02-05-2020. Surat Municipal Corporation proposed to preserve the felled tree at the same place.

Mangrove: The *Avicennia marina* is the predominant mangrove species found at the left bank of Tapi river. Four mangrove shrubs of about 3m height and 30cm girth are found at the left bank of river Tapi near Chowpati Garden (Jawaharlal Nehru Park).

0.4.5 Sensitive Receptors

Sensitive receptors such as educational institutes (schools, colleges, universities etc), hospitals, and religious places within 100 m from the central line of the proposed metro rail corridor has been demarcated. In the field study about 188 sensitive receptors found within 100 m from the centre line of proposed metro corridors.

0.5 Negative Environmental Impacts

0.5.1 Impacts due to Project Location

Change of Land use: Land will be required permanently for stations, and running sections. Both government and private land will be acquired for the project and the details are given in civil engineering in the DPR. No forest land is required for the proposed project corridors.

Impact on Ecological Environment: Total about 2093 number of trees are likely to be affected due to the proposed project, out of which about 932 trees to be cut and 1161 trees are to be transplant. No rare endangered endemic and threatened (REET) species are observed during the field study.

Impact on Archaeological Monuments/Heritage : No Archaeological and Heritage are directly affected due to the proposed metro rail project.

0.5.2 Impacts due to Project Design

Right of Way, Alignment and Architecture: In case of underground metro the space at ground level and above can continue to be used for roads and light structures and visual intrusion will be minimize.

Inter Model Integraion: Physical and operational integration of metro with other modes especially walk, public transport and intermediate public transport (hired modes) is found to increase ridership and decrease congestion inside and outside the stations.

Use of Energy and Water at stations and depots: Consumption of energy is significantly reduced by proper design of open spaces at stations and multimodal integration facilities outside stations. Water required at stations and Depot for non-drinking purposes like horticulture and toilets can be sourced from recycled water.

0.5.3 Impacts due to Project Construction

Air pollution: Particulate pollution occurs due to excavation, loading and unloading of construction material, vehicular movement, construction equipment and DG sets etc. It also occurs in sites of muck disposal, debris disposal and pre-casting yards. Resulting pollution is short term.

Noise Pollution: Noise generated from construction equipment (Lmax) in the range of 94-124 dB(A), which decreases with increase in distance. The noise levels due to different activities during construction phase will be reduced to ambient noise levels at a distance of 125 m. These estimated noise levels are typical and will vary with specifications of the equipment, usage pattern etc.

Vibration Impacts and Risk to Existing Buildings: Threshold criteria for various structures are given in the report. Vibration source levels for typical construction equipment are also listed in the report. Vibration threshold criteria is predicted for all the sensitive receptors.

Impact due to Muck Disposal: Construction activities will result in excavation of about 6.89 lakh cum and it will be used in backfilling of the piles and rest will be disposed off properly. t. Muck disposal if not properly done can result in air and water pollution, noise, diversion of green parks and temporary displacement. Site selection criteria for muck is proposed in the Chapter 7.

Impact due to Hazardous Waste: Hazardous waste would mainly arise from the maintenance of equipment. Unsafe disposal can result in water and soil pollution.

Increased Water Demand: The water demand is estimated to be 670 kilolitres per day (KLD) for construction activities.

Impact on Water Quality: Ground water contamination can take place if sewage at labour camps or chemical substances from construction site or dumped muck or construction/demolition waste or used water from the RMC plant get leached by precipitation of water and percolate to the ground water table.

Soil Erosion and land subsidence: Run off from unprotected excavated areas can result in excessive soil erosion.

Traffic Diversions: Complete/partial traffic diversions on road will be required, as most of the construction activities are on the road.

Impacts due to Labour Camps: Improper disposal of solid waste generated by labour camps can pollute surface water bodies and groundwater. The water requirement at labour camp is the responsibility of the contractor. Construction workers are more prone to infectious diseases due to unsafe sexual activity and lack of sanitation facilities and insect vectors. Problems could arise due to cultural differences between workers from outside and local residents.

Impact due to Supply of Construction Material: Poor choice of source and quarrying operations cause dust pollution and wastage of natural resources.

0.5.4 Impacts due to Project Operation

Noise Pollution: Effect of predicted day-time noise level has low impact with respect to the existing ambient noise environment. The predicted noise levels at all sensitive receptors along the corridor were predicted. Noise mitigation measures proposed to mitigate operational noise of metro.

Vibration: Passing of trains on elevated section causes vibrations. The dominant component of vibration due to passing on elevated section is horizontal. The resultant magnitudes of PPV at all sensitive receptors are predicted and are within permissible limits.

Energy Consumption at Stations: Energy is required at stations for facilities like lighting, passenger information, access, security, climate control, escalators/elevators etc.

Water Supply, Waste Water and municipal solid waste disposal at Stations: The total water demand, waste water gnereated and solid waste generation has been estimated for the horizon years 2021, 2026, 2036 and 2046. It is assumed that 15% of Alighting passengers at each station will use the toilets. The water demand during operation phase will be supplied by Municipal Corporation.

0.5.5 Impacts due to Depot

Water supply: It will be required for different purposes in the depot. The water demand at both the depots would be about 259 KLD for train washing and 46 KLD for domestic purpose for official staffs and staff quarters. Water requirement for other purposes like horticulture, flushing urinals/closet will be met from recycled water.

Effluent/Wastewater Generation: About 210 KLD of waste water will be generated from washing of trains and 41 KLD sewage will be generated at both the depots.

Oil Pollution: Oil spillage during change of lubricants, cleaning and repair processes, in the maintenance Depot cum workshop for maintenance of rolling stock, is very common. The spilled oil should be trapped in oil and grease trap.

Noise Pollution: The main source of noise from depot is the operation of workshop.

Loss of Trees: About 319 number of trees are observed at Bhesan depot site and 176 trees are to be removed at Dream City depot. Out of this 453 trees will be cut and 42 trees will be transplanted.

0.6 Positive Environmental Impacts

Employment Opportunities: In operation phase, about 2856 people will be employed for operation and maintenance of the system. Thus, the project would provide substantial direct employment and consequent indirect employment.

Benefits to Economy: These corridors will yield benefits in terms of growth in economic activity due to better accessibility, savings in fuel consumption, corresponding reduction in cost of road construction and maintenance, reduction in vehicle operating costs, savings in travel time, improvement in quality of life and reduction in loss of productivity due to health disorders resulting from pollution and accidents.

Direct Benefits to Passengers: The project will result in direct benefits to users of Metro like reduction in vehicle operating costs, savings in travel time, improvement in quality of life, reduction in loss of productivity due to health disorders resulting from pollution and reduction in road accidents.

Traffic Noise Reduction: Reduction in traffic volume of 10% & 50% reduces noise at the tune of 0.5 dB & 3.0 dB respectively.

Reduction of Traffic on Road: The basis of reduction of vehicle is shift of ridership from road vehicle to the metro railway. The reduction in number of vehicles gives benefits to economy by reduction in Vehicle Operating Cost (VOC), Fuel Consumption, Pollution Load, Accidents and Travel Time etc.

Reduced Air Pollution: The operation of metro will reduce generation of CO, PM, HC & NOx and CO_2; which will reduce the health impacts along with reduction in cost of carbon capture.

Carbon Credits: CO_2 emissions due Energy consumption (Traction and Auxiliary) and reduction due to regeneration energy for year 2021 are estimated based on the CO_2 Intensity of Electricity from grid as given in CMP Tool Kit 2014.

0.7 Environment Management Plan

Compensatory Afforestation: It is proposed to plant three saplings for each tree to be cut and one sapling for each tree to be transplant; hence 3957 trees need to be planted. The native plant species and miscellaneous indigenous tree species are recommended for afforestation. Forest Department has identified 15 ha of Forest land near village Anaval for Afforestation.

Right of Way, Alignment and Architecture: Viaduct and elevated stations shall be shaped to minimize visual intrusion.

Spatial Planning of Stations and Inter-Modal Integration: Adequate and well-laid out space shall be designed for concourses and platforms, escalators, elevators and staircases, lighting, turnstiles for normal and abnormal operating conditions; optimal height / depth of the stations, forced ventilation shall be provided. Physical and operational integration of metro with other modes shall be planned.

Green Buildings: Green building (also known as sustainable building) refers to both a structure and the using of processes that are environmentally responsible and resource-efficient throughout a building's life-cycle: from siting to design, construction, operation, maintenance, renovation, and demolition. Green buildings help in better preservation of environment as in such structures there are provisions for better saving of energy, water and CO2. Such buildings also have better waste management arrangements. All stations and Depot buildings can be designed as green buildings.

Solar Power: For the utilization of renewable energy, wherever feasible, installations for solar power can be implemented on the roof of elevated stations and in Depots. It is assumed that 75% of roof area of each station is considered for solar power generation. Solar energy generation per year is estimated to be 3.45 Giga-watt-hr for Sarthana to Dream City corridor and 4.44 Giga-watt-hr for Bhesan to Saroli Corridor.

Use of Energy and Water: Requirement of electrical energy for climate control, lighting and other facilities at stations shall be optimized by proper use of day light and design of passenger flow inside stations and on streets outside stations. To augment the water requirement Rain Water Harvesting is proposed at elevated station and alignment.

Utility Plan: Prior to the execution of work at site, detailed investigation of all utilities will be undertaken and plans for their retention in situ with precautions or temporary/permanent diversions prepared and got approved by respective agencies

Construction and Demoliation Waste Management: The procurement source of the construction materials will be decided by the Contractor, but it will be from the licensed supplier. Sites for casting of structural concrete elements and material stockpiling will be decided before start of construction.

Muck Disposal: The excavated soil generated from the construction of proposed metro will be utilized for backfilling of piles and remaining will be disposed off environmental friendly manner.

Energy Management: The contractor shall use and maintain equipment and employ recommended practices so as to conserve energy.

Labour Camp, welfare and Safety: The facilities to be provided by the Contractor are water supply, waste water and sewage treatment; Solid Waste Management; Health care awareness and clinics; Shelter at Workplace, Canteen Facilities, and First aid facilities. Construction works shall be executed as laid down in the Safety Health and Environment (SH&E) Manual prepared by the Contractor and approved by the Implementing Agency.

Hazardous Waste Management: It shall be stored and disposed of by the Contractor as per Hazardous and Other Wastes (Management, Handling & Trans-boundary movement) Rules 2019.

Air Pollution and Noise Pollution Control: Mitigation measures as per SH&E manual shall be adopted during the construction period.

Vibration Control: Construction activities shall be scheduled such that demolition, earthmoving and ground-impacting operations do not occur in the same time period. At locations where the alignment is close to sensitive structures, the contractor shall prepare a monitoring scheme prior to construction at such locations. In case of sensitive structures, vibration mitigation measures will be implemented.

Water pollution Control: Wash water shall be led through separate drains into precipitation chambers before discharging into the sewage drain to the standards prescribed for disposal. Traffic Diversion/Management: Measures such as road widening, traffic segregation, one-way movements, traffic diversions, acquisition of service lanes, etc. will be employed.

Soil Erosion and Land Subsidnece Control: Excavation shall be limited; temporary berms and use of temporary mulches, fabrics, mats, seeding, or other control devices or methods shall be implemented.

Noise Management: Noise barriers are proposed wherever required to reduce the noise impact.

Water Supply and Sanitation at Stations: Water supply for drinking, washing of stations, air conditioning and other uses will be procured from municipal authorities. This will be supplemented by re-use of treated waste water generated by staff and passengers. The waste water will be treated by installing bio digesters at each station. Non-hazardous solid waste generated at stations will be collected, segregated and disposed as per Solid Waste Management Rules 2016.

Rain Water Harvesting: To conserve and augment groundwater, it has been proposed to construct rainwater harvesting structure of suitable capacity along the viaduct and at depot sites.

Disaster Risk Management: The recommended approach (UNISDR) is to manage disaster risk rather than managing disasters. Disaster risk is the combination of the severity and

frequency of a hazard, the numbers of people and assets exposed to the hazard, and their vulnerability to damage.

Training: The training for engineers and managers will be imparted by GMRCL on regular basis to implement the environmental protection clauses of the tender document and to implement the best environmental practices during the construction phase and ensure preparedness for disaster prevention.

By implementing the mitigation measures and management plans detailed in Environmental Managmenet Plan, the project is expected to comply with the Indian Standards and WB ESF guidelines.

0.8 Public Consultations

Project level public consultations consultations were conducted at various locations along the alignments during first week of November 2019. Public consultations were conducted at 10 locations where people from different communities were participated. During public consultation, issues or suggestion raised by the stakeholders related to tree removal, air and noise pollution, comfort and safety, traffic congestion, employment generations etc.

0.9 Environmental Monitoring Plan

Environmental monitoring will be carried out for air quality, noise levels, vibration, water quality and ecology during construction and operation phases of the project. The monitoring report of environmental parameters (Air, Noise, Water and soil) will be prepared by the environmental engineer and submitted to the Project Management Consultant. Project diaries would record environmental problems (soil erosion, air quality, water quality, noise level etc), as well as safety incidents and will be retained.

0.9.1 Establishment of Environment Division

It is recommended that establishment of an Environment Division at the initial stage of the project itself with an Environmental Officer and an Environment Engineer. The task of the division would be to supervise and coordinate studies, environmental monitoring and implementation of environmental mitigation measures, and it should report directly to Chief Engineer of the Project Authority.

0.10 Cost Estimates

The total estimated environmental management and monitoring cost for the proposed project is Rs 3656.44 lakh.

Chapter 1 : Introduction

1.1. Background

Surat, a city in the Indian state of Gujarat is the eighth largest city and ninth largest urban agglomeration in India. The city with its centre on the banks of river Tapti, is located 284 km south of the state capital, Gandhinagar; and 289 km north of Mumbai. The city Surat was the world's 4th fastest growing city in 2016 according to a study conducted by the City Mayors Foundation and has been selected as one of twenty Indian cities to be developed as a smart city under PM Narendra Modi's flagship Smart Cities Mission. Famous for its diamond cutting and polishing is known as the Diamond City of India. Traditionally, the major industries in the city were textile manufacturing and shipbuilding. The British established their first Indian factory (trading post) at Surat in 1612. The city gradually declined throughout the 18th century. However, by the mid-19th century Surat was the fastest growing city of the Gujarat state with a population of more than 5.5 million with an area of 326.515 sq km.

Recognizing the need to cater to ever increasing population of the city and provide commuters an environment friendly, fast and reliable travel option, present Government of Gujarat decided to take steps to implement metro rail transit systems (MRTS) in cities having population more than 2 Million. As a part of this decision, Metro Rail Project for Surat is being initiated as the city population is more than 5.5 million. Surat Municipal Corporation in association with Gujarat Urban Development Company (GUDC) Ltd., Gandhinagar is exploring the feasibility of the project. The detailed project report (DPR) for Surat Metro Rail was prepared by Delhi Metro Rail Corporation (DMRC) in June 2018.

In August 2012, Surat Municipal Corporation (SMC) proposed four routes for the Metro project. The Government of Gujarat approved two corridors in January 2017. In March 2019, Union Government approved the Surat Metro Rail Project with two metro rail corridors and with a combined length of 40.35 km. The first corridor is proposed from Sarthana to Dream City Line covering a distance of 21.61 km. The second corridor is proposed from Bhesan to Saroli covering a distance of 18.74 km. These two corridors will intersect at Majura Gate.

Gujarat Metro Rail Corporation (GMRC) Ltd. has further awarded the consultancy work to RITES Limited for preparation of Environmental Impact Assessment and Social Impact Assessment study reports vide LOA No. GMRC/EIA & SIA/SURAT/PH-1/ 2019 Dated 20th March, 2019. The assignment will cover the Surat Metro Rail project's two corridors, i.e. from Bhesan to Saroli and Sarthana to Dream city along with two depot sites.

1.2. Project Area

The proposed metro rail system will pass through Sarthana to Dream City Line and from Bhesan to Saroli of Surat, Gujarat. The coordinates for Surat are 21°15:21°8' N and 72°44':72°53' E with a total geographical area of about 327 sq. km. The city is situated in the southern part of the state and is bounded by the Arabian Sea on the west, Bharuch in the North, the Valsad district in the south, the Dangs district in the south-east, and the Tapi district in the east. The Surat district has been divided into nine talukas namely Bardoli, Choryasi, Kamrej, Mahuva, Mandvi, Mangrol, Olpad, Palsana and Umarpada. Geographically, the district belongs to the western coastlands of the Deccan peninsula. Major geological formations exposed in the district are Quaternary alluvium, Tertiary limestone and sandstones and Deccan Trap basalt. The Tapi is the major river which passes through the central parts of the district also throughout Surat city and flows towards the west. In the last 50 years, maximum temperature recorded in Surat ranged from 30.6 0C to 37.7 0C however the average temperature of Surat is 27.2 0C. Surat's climate is classified as tropical. The summers here have a good deal of rainfall, while the winters have very little. April and May are the hottest months. With the onset of monsoons, temperature decreases appreciably in June but remains steady thereafter till September. The average annual rainfall is 1192 mm.

As per provisional reports of Census of India, population of Surat in 2011 was 4,467,797; of which male and female are 2,543,623 and 1,924,174 respectively. Population of Surat metropolitan was 4,591,246 of which 2,619,373 are males and 1,971,873 are females. The literacy rate of Surat is higher than the national average at 88%. Male literacy has been recorded at 91.22%, while it is 83.44% for females. There are more men than women in the city. There is also a high rate of young people, with 12% of the total population being under the age of 6. The official language of the city is Gujarati. The majority of people in Surat associate themselves with Hinduism, minor religions including Islam, Jainism and Christianity.

Industrial development in Surat district is attributed to the presence of a large number of diamond processing, textiles and chemical & petrochemical industries.

1.3. Objective of the Study

The objective of the study is to facilitate the Gujarat Metro Rail Corporation (GMRC) Ltd. in preparation of EIA report as per the requirement of regulatory and funding agency. The objective of the study as per the Terms of Reference is as follows:

- To collect all possible primary and secondary data on environmental baseline information, that may affect environmental components of the aligned area;
- To assess and calculate potential impacts on natural environment and impacts caused by the proposed alignment.

- To prepare Environmental Management and Monitoring Plan (EMMP) for the necessary actions to minimise potential environmental impacts as well as to propose proper mitigation measures.
- To disseminate the information of Environmental Impact Assessment (EIA) studies.

As per LOA, EIA study shall cover the entire requirement under the environmental related legislation in India, JICA Guidelines 2010 for environmental and social considerations and World Bank Operational Policy (OP) 4.12. As informed by GMRCL, the project is funded by AFD & KfW, accordingly the report has been prepared as per requirement of world bank ESF 2017 and performance standards of IFC.

1.4. Scope of EIA Study

The scope of work for the EIA study as given in the Letter of Award vide letter GMRC/EIA & SIA/SURAT/PH-1/2019 dated March 20, 2019 is as follows:

The EIA study will include the environmental management plan (EMP) and environmental monitoring plan. The proposed EIA study shall cover following in detail.

- Description of the Proposed Project (Provide a brief description of the relevant parts of the project, using maps (at appropriate scale) where necessary, and including the following information: location; general layout; size, capacity, etc.; pre-construction activities; construction activities; schedule; facilities and services; operation and maintenance activies; required offsite investments; and life span)
- Legislative and Regulatory Considerations (Describe the pertinent regulations and standards governing environmental quality, health and safety, protection of sensitive areas, protection of endangered species, sitting, land use control, etc., at international, national, regional and local levels)
- Description of the Environment.
  (a) Physical environment: geology; topography; soils; climate and meteorology; ambient air quality; surface and ground- water hydrology; coastal and oceanic parameters; existing sources of air emissions; existing water pollution discharges; and receiving water quality.
  (b) Biological environment: flora; fauna; rare or endangered species; sensitive habitats, including parks or preserves, significant natural sites, etc.; species of commercial importance; and species with potential to become nuisances, vectors or dangerous.
- Identification of the boundary of recorded forests / mangroves;
- Collection of information from / and interview with forest department, academicians and local NGOs;
- Estimate the number of trees to be cut (Height, diameter, species etc.);
- Field reconnaissance: Vegetation distribution, important fauna and endangered fauna;

- Inventory Survey of Sensitive Receptor;
- Noise & Vibration measurement;
- Identification of major rivers/ creeks likely to be affected, measurement of water quality in the identified rivers/creeks and collection of existing and published data on water quality;
- Existing data on air quality, water quality, soil, status on solid waste, etc. will be collected, interviews with the experts on the relevant field will be conducted, if necessary;
- Description of the alternatives extends to siting, design, technology selection, construction techniques and phasing, and operating and maintenance procedures. Compare alternatives in terms of potential environmental impacts, capital and operating costs, suitability under local conditions, and institutional, training, and monitoring requirements. This alternative analysis will be done subject to the availability of alternative alignments from GMRCL.
- Determination of the potential impacts due to proposed project (negative, positive, direct and indirect) during the project cycle.
- Preparation of Environment Management Plan (EMP) and Environment Monitoring Plan (EMoP) based on analysis of collected data, mitigation strategy, EMoP will be finalised incorporating feedback from local residents participated in Public Consultation Meetings and ensuring good environmental quality through monitoring;
- Identification of Institutional needs to Implement Environmental Assessment Recommendations (Review the authority and capability of institutions and recommend steps to strengthen or expand them so that the management and monitoring plans in the environmental assessment can be implemented);
- Assist in coordinating the environmental assessment with other government agencies, in obtaining the views of local NGO's and affected groups,
- Explore opportunities for claiming carbon credits against this project including methodologies and estimated value of credits.

1.5. Approach and Methodology

The Environmental Impact Assessment (EIA) report will be conducted as per the ToR issued by the GMRC Limited and EIA & EMP guidelines of World bank. The impact zone for the trees for this report comprises area under Right of Way (RoW) and working space of the proposed Metro line, Stations, Depots and permanent ancillary facilities if any. However the study area comprises green area, river and environmental sensitive areas around the project site. So that, secondary data has been collected within 15km radius of proposed alignment and inventory of environmental sensitive features has been carried out within 100m on either side of centre line of alignment. The methodology is highlighted in the following paragraphs.

The study should ascertain the existing baseline conditions and assess the impacts as a result of construction and operation of the project. The changes likely to occur in different components of the environment viz. physical, biological and socio-economic etc. shall be studied, analyzed and quantified, wherever possible. The accurate analysis of assessment depends upon the reliable data generated/available on environment. The consultant has documented the baseline data for various parameters of ecology (Flora & Fauna), environmental pollution (air, water, noise, vibration and solid waste) and socio-economic (public health, education and economics).

The standard methodology for the data collection, impact assessment and formulation of management plans is adopted. The National Acts, Legislation and Laws along with World bank are consulted with a view to ensure compliance with various requirements.

Baseline Data Collection: The existing land-use pattern of the area has been identified mainly as urban human settlements, roads, Trees and water bodies. Physical and chemical parameters of soils along the project corridor are studied from the samples collected and analysed. Based on this data, impact on soil has been predicted.

The water samples are collected from surface water sources and ground water source (bore wells) for analysis as per IS: 10500-2012. The secondary data, if any, has been studied and compiled. The impact on water quality due to the proposed project activities has been evaluated.

Air and Noise quality is an important consideration during construction and operation phases. Ambient air quality and noise levels are monitored in and around project area to develop present baseline levels in the area. Location of Air & Noise monitoring have been identified on the basis of landuse of the area.

Vibration levels are measured along the project corridor along the sensitive identified locations with the objective to establish the baseline data and assess the impacts of vibration at important receptors. There is no sensitive receptors i.e. heritage structures etc.

Terrestrial ecology was studied along the proposed metro corridors. The vegetation types documented through the visual inspection, past research and field investigations. A survey was carried out in the project area to find the existing flora and fauna. The list of birds, animals, aquatic ecology etc. of the area is compiled along with the existence of any rare and endangered species.

Meteorological data: The monthly temperature, relative humidity, rainfall, Wind rose and cloud cover of meteorological stations at Surat was collected from the Indian Meteorological Department, Ahmedabad.

Based on project particulars and the existing environmental conditions, potential impacts are identified that are expected to be affected as a result of the proposed project and

wherever possible, these are quantified. Both positive and negative impacts are evaluated to have an idea about resultant impacts. These impacts are assessed for various phases of project cycle namely, location & design, construction and operation. The standard methodology is adopted for impact prediction and assessment.

An environmental management strategy is developed to mitigate the adverse impacts during construction and operation phases of the project. The strategy includes evaluation of alternative methods to reduce or eliminate adverse impacts of the most critical areas likely to contribute to the most significant environmental burdens. The Environmental Management Plan (EMP) would highlight the proposed mitigation measures to be implemented during project construction phase like compensatory afforestation plan, infrastructure facilities like sanitation, labour camps, and refuse disposal etc. Cost estimates for each of the proposed mitigation measures are given.

Consultation and communication with stakeholders during the project preparation is an integral part of the process of gathering relevant data for impact assessment, and facilitates the development of appropriate options for the affected population.

It is necessary to monitor the environmental attributes during construction and operation. Monitoring indicates any environmental problems. This will facilitate to assess the effectiveness of management / mitigation measures. The post project environmental monitoring program is designed for implementation along with the cost. The Budget estimates and equipments necessary for the effective implementation of the Environmental Management Plan and environmental monitoring are worked out. The approach for the study is presented in **Figure 1.1.**

For the preparation of this EIA, the project team and environmental experts have liaised with the GMRC Limited in order to discuss the proposed scope of the EIA, available data in the specific area on environmental attributes and general comments / observations that these authorities may have on the project and its environs.

1.6. Structure of the Report

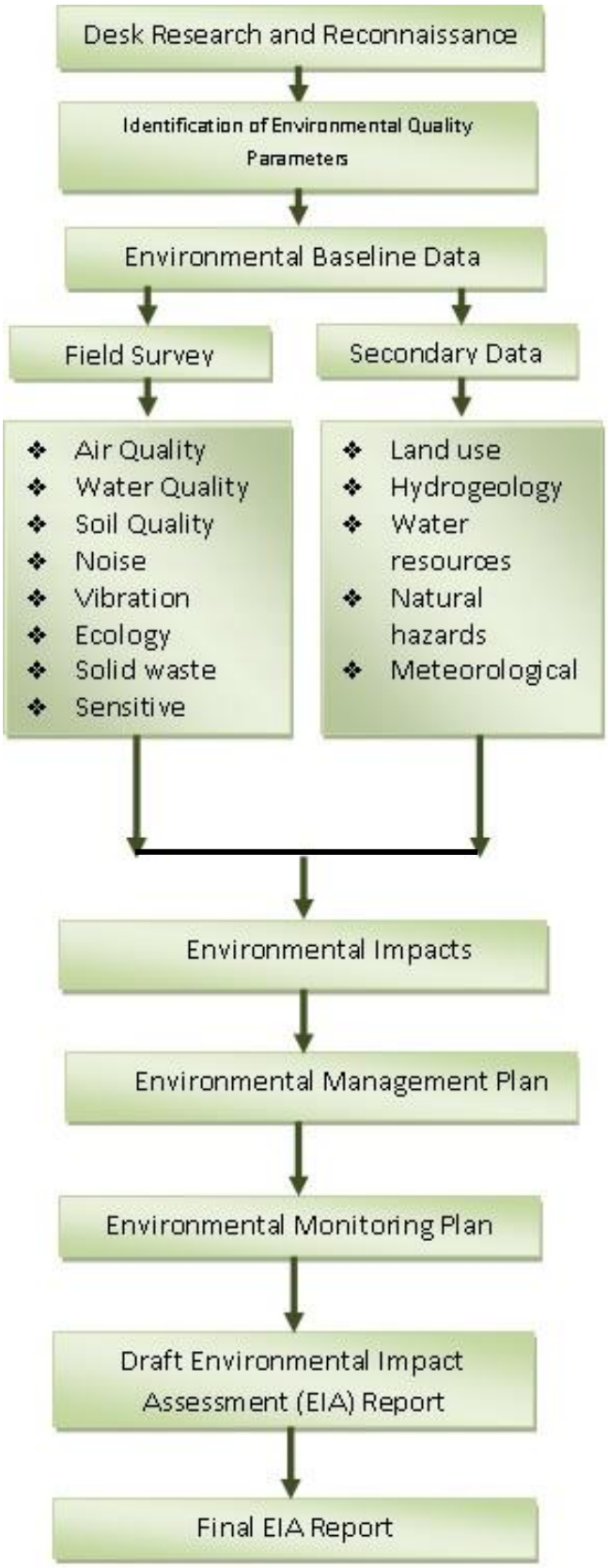
The structure of the Environmental Impact Assessment Report is as per **Table 1.1**.

Table 1.1: Structure of EIA Report

Chapter 1	Introduction	Introduction about the project, objectives and scope of work.
Chapter 2	Policies, Legal and Institutional Requirement	Provides over all legal frame work in relation to required regulatory compliance and institutional arrangement.

Chapter 3	Project Description	This chapter describes the details of the proposed metro corridors.
Chapter 4	Environmental Baseline Data	This chapter provides description on the present environmental setting of the project area.
Chapter 5	Negative Environmental Impacts	This chapter describes the negative environmental impacts associated with the proposed project during construction and operation phases.
Chapter 6	Positive Environmental Impacts	Project benefits are described under this chapter
Chapter 7	Environmental Management Plan	Environmental strategy to offset/mitigate the probable adverse impacts has been outlined
Chapter 8	Public Consultation	Project level public consultations
Chapter 9	Environmental Monitoring Plan and Environmental Management System	Environmental monitoring and management systems are described
Chapter 10	Cost Estimate	Summary of Environmental Costs are detailed in this chapter

Figure 1.1: Flow Chart for EIA Study



Chapter 2 : Policy, Legal and Institutional Requirements

2.1. Environmental Policy Framework

The environmental policy framework include existing institutions and legislations relevant to the project at the International, National and State levels. The various statutory clearances/ permissions from state and central government authorities and institutional framework are discussed in the subsequent section.

2.2. Legal Framework

The proposed project would be governed by various Acts, Rules and regulations set by the Ministry of Environment, Forests and Climate Change (MoEFCC) at the Central level and other regulatory agencies at the State and local level. Various environmental standards, specifications and guidelines of Central Pollution Control Board (CPCB) and state level agencies will also be applicable.

It is important to mention over here that the Central government framed an 'umbrella law', called the Environment (Protection) Act, 1986 to broadly encompass and regulate an array of environmental issues. The overall purpose of EPA was to establish an overall coherent policy and provide a basis for the coordinated work of various government agencies with operational responsibility for the environment and natural resources. The legislation also invests authorities with regulatory powers to address specific issues affecting the environment. The Act also does not allow any person to carry on an industry, operation or process that discharge or emit any environmental pollutants in excess of standards prescribed under specific rules and notifications.

The Acts, Rules and Notifications applicable to environmental aspects of the constructional and operational phases of the proposed project are summarized and briefly described in the Table 2.1 below.

Table 2.1: Environmental Regulation and Legislation

S. No	Legislation	Purpose	Relevant to Development of Metro	Reason for Applicability/ Non applicability	Applicable Yes/ No	Authority
1.	Environment Protection Act, 1986	To protect and improve overall environment	Section 7: Not to allow emission or discharge of environmental pollutants in excess of the standards. Section 8: Handling Hazardous substances to comply with procedural safeguards Section 24: Effect of other laws	As all environmental notifications, rules and schedules are issued under this act.	Yes	MoEFCC. Gol; DoE, State Govt. CPCB; SPCB
2.	Environmental Impact Assessment Notification 14th Sep-2006 amended in 2009 and 2013	To provide environmental clearance to new development activities following environmental impact assessment.	EC will be required in case of commercial development is proposed.	No commercial development will take place.	No	MoEFCC/ SEIAA
3.	Coastal Regulation Zone Rules 2019	The purpose of this act to declare the coastal stretches and imposing restriction of industries operation and process in the CRZ.	Land area between HTL and LTL. Developed land areas up to or close to the shoreline, within the existing municipal limits or other existing legally designated urban areas. Area shall include the water area and the bed area between LTL at the bank of the tidal influenced water body to the LTL on the opposite side of the bank. Land area from High Tide line to 500m on the landward side. Land area between HTL to 50 m or width of the creek.	Alignment crossing at Tapi river	Yes	SCZMA/MoEFCC/ Town planning
4.	Forest (Conservation) Act 1980 amended in 1988;	To protect and manage forests, to check deforestation by restricting conversion of forest	Section 2, Restriction on the use of forest land for non forest purpose. Chapter 2, Clause 2.1, Sub clause (iv),	There is no protected forest area along the alignment but there are 4 mangroves along	Yes	Nodal officer of state

S. No	Legislation	Purpose	Relevant to Development of Metro	Reason for Applicability/ Non applicability	Applicable Yes/ No	Authority
	Forest (Conservation) Rules 2003, 2004 Gujarat Government Gazette August 9, 1979	areas into non-forest areas Road side area of Road under P.W.D. to be protected forest.	Re diversion of forest land	the alignment at the bay of tapi river		
5	Indian Wildlife Protection Act, 1972, amended in 2002	Protection of animals and specified plants.	Section 27, Restriction on entry in sanctuary Section 28, Grant of permit Section 33, Control of sanctuaries	No wildlife sanctuary exists	No	Chief conservator wildlife, wildlife wing, Forest Department,
6.	Wetlands (Conservation and Management) Rules 2017	To conserve and management of the wetland	Section 3, Applicability of Rules. Section 4, Restriction of activities in wetlands.	No Wetland of international importance, wetland as notified by the central government lies along the alignment.	No	State wetland Authority
7.	Saurashtra Felling of trees (infliction of Punishment) Act, 1951 & amendment Rule 2008.	An act to provide for heavier punishment for the unauthorized felling of trees.	Section 3, clause 1, sub clause (a), no person can fell tree without the written permission of the collector. Section 3, Clause 2, felling of babul trees. Section 1, Clause E, of Amendment Section 4 , and section 5 for transit pass	2093 trees need to be felled/ transplant.	Yes	Collector/ Forest Department/Gram Panchyat
8.	Bombay Forest Rules 1942, Bombay Forest (Gujarat Amendment) Rules, 2018,	To protect the un legal transport of timber.	Rule 66 of 1942, for transit pass.	Permission for transit of timber of trees of Nilgiri, Subabul, Babool, Ashoka and drumstick under Indian Forest Act, 1927 and Bombay Forest	Yes	DFO/CCF

S. No	Legislation	Purpose	Relevant to Development of Metro	Reason for Applicability/ Non applicability	Applicable Yes/ No	Authority
				Rules, 1942 shall be required in case of trees cut in the areas within 200 m from the edge of the National Highways, State Highways and District Road. As project site lies within 200 m from the edge of the National Highways, State Highways and District Road.		
9.	The Ancient Monuments and Archaeological sites and Remains Act,1958 amended in 2010 with Ancient Monuments and Archaeological Sites and Remains (Amendment) Act, 2017 which is yet to be passed by Rajya Sabha.	Preservation of ancient and historical monuments and archaeological sites and remains of national importance. To regulate the archaeological excavations and protection of sculptures, carvings etc.	Amendment of Section 20A: Construction of public works in prohibited area of protected monuments is permitted which is yet to be passed by Rajya Sabha.	No ancient monument and archaeological site lies near the alignment.	No	National Monument Authority (NMA)
10.	Aircraft Act 1934 with Aircraft (Amendment) Act, 2007 (44 of 2007)& Ministry of Civil Aviation (Height Restrictions for Safeguarding of	Height Restrictions	**Rule 4,** Restriction on Constructions, erection, trees **Sub Rule 3**, No structure higher than the height specified within the 20 km of Aerodrome Reference Point of the civil and defence aerodromes	Surat International Airport is about 4.5 km far from the project site and all proposed structures height are lower than the permissible height. So, Sub rule 3 is not applicable.	No	Airport Authority of India (AAI)

S. No	Legislation	Purpose	Relevant to Development of Metro	Reason for Applicability/ Non applicability	Applicable Yes/ No	Authority
	Aircraft Operations) Rules, 2015 that are further amended in 2018.					
11.	Air (Prevention and Control of Pollution) Act, 1981 amended in 1987; Air (Prevention and Control of Pollution) Rules, 1982	The purpose of this regulation is to determine and build the air quality targets, to assess the air quality, to protect the current condition where the air quality is good and to improve it where needed.	Section 21: Restriction on use of certain industrial plant requires consent to establish or operation of any plant from the board. Section 22: Persons carrying on industry not to allow emission of air pollutants in excess of the standards laid down by the board under section 17(1)(19)	Operation of batching plants, material yards depot.	Yes	CPCB
12.	CPCB Order Dated 7th March, 2016	The purpose of this order is to revise the list of category of industrial sector.	Ready mix concrete plant covers under Green Category of industrial sector.	RMC plant covers under the green category of Industrial Sector as per revised list of CPCB.	Yes	CPCB
13.	CPCB order Dated 18th November, 2009	The purpose of this order is to revise the NAAQS.	The regulation sets limits for air quality parameters for Residential, Commercial, ecologically sensitive area.	Project site lies near to the residential, commercial, ecologically sensitive area.	Yes	SPCB
14.	Water (Prevention and Control of Pollution) Act, 1974 amended in 1988; Water (Prevention and Control of Pollution) Rules, 1975 amended in 2011	The purpose of this regulation is to set the legal and technical principles to be followed in the control of water pollution, in order to protect the ground and surface waters, and to prevent water pollution.	Section 24: Prohibition on use of stream or well for disposal of polluting matter. Section 25 & 26: Require consent to establish or operation of any industry from the board.	During construction for establishment of hot mix plant, workers' camp, construction camp, etc. A permit must be obtained from the SPCB for the direct discharge of any domestic and/or industrial wastewaters and site run off where this is discharged from a point	Yes	SPCB

S. No	Legislation	Purpose	Relevant to Development of Metro	Reason for Applicability/ Non applicability	Applicable Yes/ No	Authority
				source into receiving waters.		
15.	Noise Pollution (Regulation and Control) Rules, 2000 amendment in 2010	Compliance with Ambient Noise Standards in accordance to land use of the area.	Rule 3 & 4: The ambient Air quality Standards in respect of noise for different areas. Rule 5 sub rule 5 A: Restriction on the use of horns, sound emitting construction equipments during night time in the residential area.	Project site lies in the different areas/Zones. During construction and operation for the vehicular movement and use of machinery in construction.	Yes	SPCB
16.	Solid Waste Management Rules, 2016 & its ammendments	Management (Collection, Handling, Storage and disposal) of solid waste	Section 4: Segregate and store the waste in suitable bins and handover segregated waste to authorized waste picker or collector.	During construction for establishment of hot mix plant, workers' camp, construction camp, etc	Yes	SPCB
17.	Hazardous and Other Wastes (Management and Trans boundary Movement) Rules and its amendments , 2016.	Obtaining Authorization from SPCB for handling and storing of hazardous material (fuel)/ waste	Schedule-I , activity Drill cuttings excluding those from water based mud , Used or spent oil	During construction for establishment of hot mix plant, workers' camp, construction camp, etc	Yes	SPCB
18.	Construction and Demolition Waste Management Rules, 2016 and its amendments.	Management of waste resulting from waste resulting from construction, re-modeling, repair and demolition of any civil structure.	Section 4, Clause 3 and 5: Duties of the waste generator. Waste generators who generate more than 20 tons or more in one day or 300 tons per project in a month shall have to segregate the waste, submit waste management plan.	During Construction of metro station and alignment.	Yes	SPCB
19.	Plastic Waste Management (amendment) Rules 2018 and its amendments.	Management of Plastic waste	Rule 8, Responsibility of the waste generators. Storage of waste at source and handover the segregated waste to urban local body.	During construction for establishment of hot mix plant, workers' camp, construction camp, etc	Yes	SPCB

S. No	Legislation	Purpose	Relevant to Development of Metro	Reason for Applicability/ Non applicability	Applicable Yes/ No	Authority
20.	E waste (Management) Rules 2016 and its amendments.	Management of E waste	Rule 9, Responsibility of Consumer or Bulk Consumer.	Metro station, offices, maintenance Depot	Yes	SPCB
21.	Notification for use of fly ash	Reuse large quantity of fly ash discharged from thermal power plant to minimize land use for disposal	Paragraph 1, Subparagraph 1A: use of flyash products for constructions.	Thermal power plant (Nani Naroli village) located about 30 km of the proposed project site.	Yes	SPCB
22.	Mineral and concession Rules 1960 as amended 2012.	For opening new quarry.	Chapter IV, Chapter V , Grant of license	Non metallic industrial mineral will be procured during construction.	Yes	District Collector
23.	Central Motor Vehicle Act, 1988 and Central Motor Vehicle Rules, 1989	To check vehicular air and noise pollution.	Chapter V: Construction Equipment and maintenance of Motor Vehicle	Construction Equipment handled in construction sites.	Yes	Motor Vehicle Department
24.	Metro Rail Transit System Guideline for Vibration RDSO, Ministry of Railways September 2015	Norms for Noise and vibration for metro railways in India.	Chapter1 to 7: This document lays guidelines for noise reference levels, vibration impact criteria for detailed analysis, noise and vibration mitigation measures, and ways of reducing excessive noise and vibration caused by metro railway projects.	During construction and operation of metro.	Yes	
25.	The Building and Other construction Workers (Regulation of Employment and conditions of Services) Act, 1996	To Regulate the employment and conditions of service of building and other construction workers and to provide for their safety, health and welfare measures.	Chapter VI: Hours of Work, Welfare Measures and other condition of Service of Building Workers. Chapter VII: Safety and Health Measures.	Workers camp, construction camp etc	Yes	State Welfare Board

2.2.1. Environment Protection Act

The Act is for the purpose of protecting and improving the quality of the environment and preventing, controlling and abating environmental pollution. Protect and improve environment under this Act by

- Planning and execution of a nation-wide programme for the prevention, control and abatement of environmental pollution;
- Laying down standards for the quality of environment in its various aspects;
- Laying down standards for emission or discharge of environmental pollutants from various sources whatsoever;
- Restriction of areas in which any industries, operations or processes or class of industries, operations or processes shall not be carried out or shall be carried out subject to certain safeguards;
- Laying down procedures and safeguards for the prevention of accidents which may cause environmental pollution and remedial measures for such accidents;
- Laying down procedures and safeguards for the handling of hazardous substances.

No person carrying on any industry, operation or process shall discharge or emit or permit to be discharged or emitted any environmental pollutant in excess of such standards as may be prescribed.

2.2.2. Water and Water Pollution

The use of water resources and also the discharge of polluted water (sewerage) are primarily regulated by the Water (Prevention and Control of Pollution) Act, 1974 amended in 1988. The Water Cess Act, 1977 amended in 1992 and 2003, including Rules 1978 and 1991 provides for levy and collection of Cess on water consumed with a view to generate resources for prevention and control of water pollution. The Act assigns functions and powers to the Central Pollution Control Board (CPCB) and State Pollution Control Board (SPCBs) for prevention and control of water pollution.

The Environment (Protection) Act 1986 amended in 1991 and Rules also lays down specific standards for quality of water effluents to be discharged into different type of water bodies (sewers, surface water bodies like lakes and rivers, marine discharge). Additionally, the water supplied to users for drinking shall also conform to the National Drinking Water Standard; IS-10500 are given at Annexure 2.1. The general standards for discharge effluent in Inland Surface Water Bodies are given at Annexure 2.2. Tolerance limits for Inland Surface Water Quality are given at Annexure 2.3.

The Central Ground Water Board (CGWB) the statutory authority set up by the Central Government has also restricted the drilling of tube wells and bore wells in certain water

scarce areas in the country. Surat City does not figure in the list of Notified areas where permission to abstract ground water through any energized means will not be accorded for any purpose other than drinking water, it will be advisable to optimize extraction of ground water for drinking purpose during construction.

2.2.3. Air Quality

The Air (Prevention and Control of Pollution) Act, 1981 and amended in 1987 including Rules 1982 and 1983 was enacted to prevent, control and reduce air pollution. According to Section 21 of the Act, no person shall establish or operate any activity, which can cause air pollution without obtaining Consent to Establish (CTE) as per the Air Act. The Act also lays down national ambient air quality standards for pollutants like PM_{10}, $PM_{2.5}$ Sulphur dioxide, Nitrogen dioxide, Carbon monoxide, Lead, Ozone, Ammonia, Benzene and Benzo pyrene, Arsenic and Nickel with the intent of managing air quality for different category of areas (Industrial, Residential, Rural and Ecological sensitive areas). Ambient Air Quality Standards have been notified by the CPCB vide Gazette Notification dated 16th November 2009, are given at Annexure 2.4.

2.2.4. Noise Quality

With the objective of regulating ambient noise quality in the environment, the Central Government has notified the Noise Pollution (Regulation and Control) Rules, 2000 amended in 2010 under the EPA. The noise standards for different category of areas are based on the weighted equivalent noise level (Leq). The notified ambient noise standards are presented in Annexure 2.5.

2.2.5. Solid Waste Management

Construction and Demolition Waste Management Rules, 2016 and its amenments identify roles of waste generator, service provider, local authorities, SPCB, State Government, CPCB, BIS and Central Government. The Rules specify procedure for reporting accidents during waste processing or treatment or disposal, roles and criteria for site selection for storage and processing or recycling facilities, applications of waste made from waste materials.

Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 and its amendments specify the following:

- Occupier`s responsibility for safe and environmentally sound management of hazardous and other wastes in terms of sending or selling to an authorised actual user or disposal in an authorised disposal facility.
- Responsibilities of State Government,
- Rules for grant of authorization to manage wastes and for utilization of wastes.

- Roles of waste processor and State Government in treatment, storage and disposal facility for hazardous and other wastes.
- Procedures for packaging, labelling, and transport of hazardous and other wastes.

Solid Waste Management Rules, 2016 and its amendments are applicable to every domestic, institutional, commercial and any other non residential solid waste generator except industrial waste, hazardous waste, hazardous chemicals, bio medical wastes, e-waste, lead acid batteries and radio-active waste. Duties of waste generators, manufacturers, local authorities, various Officers and ministries of Government, Pollution Control agencies are stipulated in these Rules.

2.2.6. The Ancient Monuments and Archaeological Sites and Remains Act, 2010

The Ancient Monuments and Archaeological Sites and Remains (Amendment and Validation) Act, 2010 has been enacted to amend the Ancient Monuments and Archaeological Sites and Remains Act, 1958 and to make provision for validation of certain actions taken by the Central Government under the said Act. The act has come into force on January 23, 2010.

The Act states that the limits of prohibited area and regulated area around the monuments, archaeological sites and remains declared by the Central Government as protected have been specified in the principle Act as 100 m and 200 m respectively. The limits so fixed may be further extended on the basis of gradation and classification of the monuments, archaeological sites and remains to be done by the National Monuments Authority, which is to be constituted by the Central Government by virtue of the Amendment in the principle Act. The Act defines regulated area and prohibited area as follows:

Prohibited Area: It is the area beginning at the limit of the protected area or the protected monument declared as of national importance, and extending to a distance of 100 m in all directions. There is also a provision in the Act to further extend the prohibited area beyond 100 m having regard to the classification of any protected monument or protected area on the recommendation of National Monument Authority by the Central Government.

Regulated Area: It is the area beginning at the limit of the prohibited area in respect of every protected archaeological monument/site and remains and extending to a distance of 200 m in all directions. This 200 m regulated area could further be extended having regard to the classification of any protected monument or protected area on the recommendation of National Monument Authority by the Central Government. The regulated area has extent not only horizontally but also vertically and covers even below the surface.

The Act provides that in exceptional cases where Central Government or DG/ASI is satisfied that the works/project is in public interest and does not have significant adverse impact on

the monument/site, permission can be granted for such work in prohibited area. The Act provides that none other than an archaeological officer can carry out any construction in any prohibited area. The Act provides that no permission, including carrying out any public work or project essential to the public or other constructions, shall be granted in any prohibited area on and after the date on which the Ancient Monuments and Archaeological Sites and Remains (Amendment and Validation) Act, 2010 comes in to force.

Amendment to this Act vide The Ancient Monuments And Archaeological Sites And Remains (Amendment) Bill, 2017 defines "public works" to mean construction works related to infrastructure financed and carried out by any department or offices of the Central Government for public purposes which is necessary for the safety or security of the public at large and emergent necessity is based on specific instance of danger to the safety or security of the public at large and there is no reasonable possibility of any other viable alternative to such construction beyond the limits of the prohibited area. The Bill 2017 states that the clause in Act 2010 which *barred permission to construct in prohibited area* shall not apply to the public works.

2.2.7. The Building and Other Construction Workers Act, 1996

The Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996 aims to provide for regulation of employment and conditions of service of the building and other construction workers as also their safety, health and welfare measures in every establishment which employs or employed ten or more workers. The provisions in the Act for health and safety measures for the construction workers are in conformity with International Labour Organisation Convention.

2.2.8. National Environment Policy, 2006

National Environment Policy 2006 is a response to India's national commitment to a clean environment, mandated in the Constitution in Articles 48 A and 51 A (g), (DPSP) strengthened by judicial interpretation of Article 21.

The existing policies have recognized the need for sustainable development in their specific contexts and formulated necessary strategies to give effect to such recognition. The National Environment Policy seeks to extend the coverage, and fill in gaps that still exist, in light of present knowledge and accumulated experience. It does not displace, but builds on the earlier policies. The objectives of the National Environmental Policy are:

- Conservation of Critical Environmental Resources
- Intra-generational Equity: Livelihood Security for the Poor
- Inter-generational Equity
- Integration of Environmental Concerns in Economic and Social Development:

- Efficiency in Environmental Resource Use
- Environmental Governance
- Enhancement of Resources for Environmental Conservation

The policy focuses on encouraging the regulatory authorities, Central and State, to institutionalize regional and cumulative environmental impact assessments to ensure that environmental concerns are identified and addressed at the planning stage itself. The policy adopts the civil liability for environmental damage that would deter environmentally harmful actions, and compensate the victims of environmental damage.

2.3. Institutional Framework

The Ministry of Environment Forest and Climate Change (MoEFCC) is the nodal agency in the administrative structure of the central government for planning, promotions, co-ordination and overseeing the implementation of India's environmental and forestry policies and programs. The major responsibilities of MoEFCC include:

- Environmental resource conservation and protection, including environmental impact assessment, clearance of developmental projects;
- Co-ordination with the other ministries and agencies, voluntary organizations and professional bodies for environmental action plans;
- Promotion of research and development, manpower planning and training and creation of environmental awareness;
- Liaison and coordination with international agencies involved in environmental matters.

2.3.1. Central and State Pollution Control Boards

The Central Pollution Control Board is responsible for pollution control throughout the country. In addition to the control of air, noise and water pollution it is also responsible to ensure effective control of disposal of hazardous wastes and storage and handling of hazardous chemicals and substances. With the enactment of air and water pollution laws, states have set-up their own State Pollution Control Boards (SPCBs) to monitor industrial emissions and effluents and to approve the operation of new industries after careful scrutiny. The functions of the SPCBs include:

- The planning of comprehensive state programs for the prevention and control of air and water pollution and to ensure the implementation thereof;
- Inspection of pollution control equipment/ plants for monitoring of their efficiency.

The SPCB in consultation with the Central Pollution Control Board may establish norms for air quality, gaseous emission and noise level etc.

2.3.2. Environmental, Health, and Safety Guidelines for Railways by IFC

The Environmental, Health, and Safety (EHS) Guidelines are technical reference documents with general and industryspecific examples of Good International Industry Practice (GIIP). When one or more members of the World Bank Group are involved in a project, these EHS Guidelines are applied as required by their respective policies and standards. These industry sector EHS guidelines are designed to be used together with the General EHS Guidelines document, which provides guidance to users on common EHS issues potentially applicable to all industry sectors. Ambient Air Quaity and Noise standards published by IFC are given in Annexure 2.2, 2.4 and 2.5.

2.4. World Bank Requirement

The World Bank Environmental and Social Framework sets out the World Bank's commitment to sustainable development, through a Bank Policy and a set of Environmental and Social Standards that are designed to support Borrowers' projects. The Environmental and Social Standards set out the requirements for Borrowers relating to the identification and assessment of environmental and social risks and impacts associated with projects supported by the Bank through Investment Project Financing. The Bank believes that the application of these standards, by focusing on the identification and management of environmental and social risks, will support Borrowers in their goal to reduce poverty and increase prosperity in a sustainable manner for the benefit of the environment and their citizens. The standards will: (a) support Borrowers in achieving good international practice relating to environmental and social sustainability; (b) assist Borrowers in fulfilling their national and international environmental and social obligations; (c) enhance nondiscrimination, transparency, participation, accountability and governance; and (d) enhance the sustainable development outcomes of projects through ongoing stakeholder engagement.

The ten Environmental and Social Standards (ESS) establish the standards that the Borrower and the project will meet through the project life cycle, as follows:

ESS1: Assessment and Management of Environmental and Social Risks and Impacts

ESS1 sets out the Borrower's responsibilities for assessing, managing and monitoring environmental and social risks and impacts associated with each stage of a project supported by the Bank through Investment Project Financing, in order to achieve environmental and social outcomes consistent with the Environmental and Social Standards (ESSs). The term 'environmental and social assessment' is a generic term that describes the process of analysis and planning used by the Borrower to ensure the environmental and social impacts and risks of a project are identified, avoided, minimized, reduced or mitigated.

A. Use of borrower's environmental and social framework

When a project is proposed for Bank support, the Borrower and the Bank will consider whether to use all, or part, of the Borrower's ES Framework in the assessment, development and implementation of a project. Such use may be proposed provided this is likely to address the risks and impacts of the project, and enable the project to achieve objectives materially consistent with the ESSs.

If the assessment identifies gaps in the Borrower's ES Framework, the Borrower will work with the Bank to identify measures and actions to address such gaps. Such measures and actions may be implemented during project preparation or project implementation and will include, where necessary, measures and actions to address any capacity development issues pertaining to the Borrower, any relevant national, subnational or sectoral implementing institution, and any implementing agency. The agreed measures and actions, together with the timeframes for their completion, will form part of the ESCP.

The Borrower will take all actions necessary to maintain the Borrower's ES Framework, as well as acceptable implementation practices, track record, and capacity, in accordance with the measures and actions identified in the ESCP, throughout the project life cycle. The Borrower will notify and discuss with the Bank any significant changes in the Borrower's ES Framework that may affect the project. If the Borrower's ES Framework is changed in a manner inconsistent with the requirement of above paragraph and the ESCP, the Borrower will carry out, as appropriate, additional assessment and stakeholder engagement in accordance with the ESSs, and propose changes, for approval by the Bank, to the ESCP.

B. Environmental and social assessment

The Borrower will carry out an environmental and social assessment of the project to assess the environmental and social risks and impacts of the project throughout the project life cycle. The assessment will be proportionate to the potential risks and impacts of the project, and will assess, in an integrated way, all relevant direct, indirect and cumulative environmental and social risks and impacts throughout the project life cycle, including those specifically identified in ESSs2–10.

The environmental and social assessment will be based on current information, including an accurate description and delineation of the project and any associated aspects, and environmental and social baseline data at an appropriate level of detail sufficient to inform characterization and identification of risks and impacts and mitigation measures. The assessment will evaluate the project's potential environmental and social risks and impacts; examine project alternatives; identify ways of improving project selection, siting, planning, design and implementation in order to apply the mitigation hierarchy for adverse environmental and social impacts and seek opportunities to enhance the positive impacts of the project. The environmental and social assessment will include stakeholder engagement as an integral part of the assessment, in accordance with ESS10.

C. Environmental and Social Commitment Plan (ESCP)

The Borrower will develop and implement an ESCP, which will set out measures and actions required for the project to achieve compliance with the ESSs over a specified timeframe. The ESCP will be agreed with the Bank and will form part of the legal agreement. The draft ESCP will be disclosed as early as possible, and before project appraisal.

The ESCP will take into account the findings of the environmental and social assessment, the Bank's environmental and social due diligence, and the results of engagement with stakeholders. It will be an accurate summary of the material measures and actions required to avoid, minimize, reduce or otherwise mitigate the potential environmental and social risks and impacts of the project. A completion date for each action will be specified in the ESCP.

The ESCP will describe the different management tools that the Borrower will use to develop and implement the agreed measures and actions. These management tools will include, as appropriate, environmental and social management plans, environmental and social management frameworks, operational policies, operational manuals, management systems, procedures, practices and capital investments. All management tools will apply the mitigation hierarchy, and incorporate measures so that the project will meet the requirements of applicable laws and regulations and the ESSs in accordance with the ESCP throughout the project life cycle.

D. Project monitoring and reporting

The Borrower will monitor the environmental and social performance of the project in accordance with the legal agreement (including the ESCP). The extent and mode of monitoring will be agreed upon with the Bank, and will be proportionate to the nature of the project, the project's environmental and social risks and impacts, and compliance requirements. The Borrower will ensure that adequate institutional arrangements, systems, resources and personnel are in place to carry out monitoring. Where appropriate and as set out in the ESCP, the Borrower will engage stakeholders and third parties, such as independent experts, local communities or NGOs, to complement or verify its own monitoring activities. Where other agencies or third parties are responsible for managing specific risks and impacts and implementing mitigation measures, the Borrower will collaborate with such agencies and third parties to establish and monitor such mitigation measures.

Monitoring will normally include recording information to track performance, and establishing relevant operational controls to verify and compare compliance and progress. Monitoring will be adjusted according to performance experience, as well as actions requested by relevant regulatory authorities and feedback from stakeholders such as community members. The Borrower will document monitoring results.

E. Stakeholder engagement and information disclosure

As set out in ESS10, the Borrower will continue to engage with, and provide sufficient information to stakeholders throughout the life cycle of the project, in a manner appropriate to the nature of their interests and the potential environmental and social risks and impacts of the project. For High Risk and Substantial Risk projects, the Borrower will provide to the Bank and disclose documentation, as agreed with the Bank, relating to the environmental and social risks and impacts of the project prior to project appraisal. The documentation will address, in an adequate manner, the key risks and impacts of the project, and will provide sufficient detail to inform stakeholder engagement and Bank decision making. The Borrower will provide to the Bank and disclose final or updated documentation as specified in the ESCP.

ESS2: Labor and Working Conditions

ESS2 recognizes the importance of employment creation and income generation in the pursuit of poverty reduction and inclusive economic growth. Borrowers can promote sound worker-management relationships and enhance the development benefits of a project by treating workers in the project fairly and providing safe and healthy working conditions. The objectives of the ESS2 are;

- To promote safety and health at work.
- To promote the fair treatment, non-discrimination and equal opportunity of project workers.
- To protect project workers, including vulnerable workers such as women, persons with disabilities, children (of working age, in accordance with this ESS) and migrant workers, contracted workers, community workers and primary supply workers, as appropriate.
- To prevent the use of all forms of forced labor and child labor.
- To support the principles of freedom of association and collective bargaining of project workers in a manner consistent with national law.
- To provide project workers with accessible means to raise workplace concerns.

ESS3: Resource Efficiency and Pollution Prevention and Management

ESS3 recognizes that economic activity and urbanization often generate pollution to air, water, and land, and consume finite resources that may threaten people, ecosystem services and the environment at the local, regional, and global levels. The current and projected atmospheric concentration of greenhouse gases (GHG) threatens the welfare of current and future generations. At the same time, more efficient and effective resource use, pollution prevention and GHG emission avoidance, and mitigation technologies and practices have become more accessible and achievable.

This ESS sets out the requirements to address resource efficiency and pollution prevention and management throughout the project life cycle consistent with GIIP. The objectives of this ESS 3 are;

- To promote the sustainable use of resources, including energy, water and raw materials.
- To avoid or minimize adverse impacts on human health and the environment by avoiding or minimizing pollution from project activities.
- To avoid or minimize project-related emissions of short and long-lived climate pollutants.
- To avoid or minimize generation of hazardous and non-hazardous waste.

ESS4: Community Health and Safety

ESS4 recognizes that project activities, equipment, and infrastructure can increase community exposure to risks and impacts. In addition, communities that are already subjected to impacts from climate change may also experience an acceleration or intensification of impacts due to project activities. ESS4 addresses the health, safety, and security risks and impacts on project-affected communities and the corresponding responsibility of Borrowers to avoid or minimize such risks and impacts, with particular attention to people who, because of their particular circumstances, may be vulnerable. The objectives of ESS4 are;

- To anticipate and avoid adverse impacts on the health and safety of project-affected communities during the project life cycle from both routine and nonroutine circumstances.
- To promote quality and safety, and considerations relating to climate change, in the design and construction of infrastructure.
- To avoid or minimize community exposure to project-related traffic and road safety risks, diseases and hazardous materials.
- To have in place effective measures to address emergency events.
- To ensure that the safeguarding of personnel and property is carried out in a manner that avoids or minimizes risks to the project-affected communities.

ESS5: Land Acquisition, Restrictions on Land Use and Involuntary Resettlement

ESS5 recognizes that project-related land acquisition and restrictions on land use can have adverse impacts on communities and persons. Project-related land acquisition or restrictions on land use may cause physical displacement (relocation, loss of residential land or loss of shelter), economic displacement (loss of land, assets or access to assets, leading to loss of income sources or other means of livelihood), or both. The term “involuntary resettlement”refers to these impacts. Resettlement is considered involuntary when affected persons or communities do not have the right to refuse land acquisition or restrictions on

land use that result in displacement. Experience and research indicate that physical and economic displacement, if unmitigated, may give rise to severe economic, social and environmental risks: production systems may be dismantled; people face impoverishment if their productive resources or other income sources are lost; people may be relocated to environments where their productive skills are less applicable and the competition for resources greater; community institutions and social networks may be weakened; kin groups may be dispersed; and cultural identity, traditional authority, and the potential for mutual help may be diminished or lost. For these reasons, involuntary resettlement should be avoided. Where involuntary resettlement is unavoidable, it will be minimized and appropriate measures to mitigate adverse impacts on displaced persons (and on host communities receiving displaced persons) will be carefully planned and implemented. The objectives of ESS5 are;

- To avoid involuntary resettlement or, when unavoidable, minimize involuntary resettlement by exploring project design alternatives.
- To avoid forced eviction.
- To mitigate unavoidable adverse social and economic impacts from land acquisition or restrictions on land use by: (a) providing timely compensation for loss of assets at replacement

ESS6: Biodiversity Conservation and Sustainable Management of Living Natural Resources

ESS6 recognizes that protecting and conserving biodiversity and sustainably managing living natural resources are fundamental to sustainable development. Biodiversity is defined as the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are a part; this includes diversity within species, between species, and of ecosystems. Biodiversity often underpins ecosystem services valued by humans. Impacts on biodiversity can therefore often adversely affect the delivery of ecosystem services.

ESS6 recognizes the importance of maintaining core ecological functions of habitats, including forests, and the biodiversity they support. All habitats support complexities of living organisms and vary in terms of species diversity, abundance and importance. This ESS also addresses sustainable management of primary production and harvesting of living natural resources.

ESS6 recognizes the need to consider the livelihood of project-affected parties, including Indigenous Peoples, whose access to, or use of, biodiversity or living natural resources may be affected by a project. The potential, positive role of projectaffected parties, including Indigenous Peoples, in biodiversity conservation and sustainable management of living natural resources is also considered.

ESS8: Cultural Heritage

ESS8 recognizes that cultural heritage provides continuity in tangible and intangible forms between the past, present and future. People identify with cultural heritage as a reflection and expression of their constantly evolving values, beliefs, knowledge and traditions. Cultural heritage, in its many manifestations, is important as a source of valuable scientific and historical information, as an economic and social asset for development, and as an integral part of people's cultural identity and practice. ESS8 sets out measures designed to protect cultural heritage throughout the project life cycle. The objectives of this ESS are;

- To protect cultural heritage from the adverse impacts of project activities and support its preservation.
- To address cultural heritage as an integral aspect of sustainable development.
- To promote meaningful consultation with stakeholders regarding cultural heritage.
- To promote the equitable sharing of benefits from the use of cultural heritage.

ESS10: Stakeholder Engagement and Information Disclosure

This ESS recognizes the importance of open and transparent engagement between the Borrower and project stakeholders as an essential element of good international practice. Effective stakeholder engagement can improve the environmental and social sustainability of projects, enhance project acceptance, and make a significant contribution to successful project design and implementation. Stakeholder engagement is an inclusive process conducted throughout the project life cycle. Where properly designed and implemented, it supports the development of strong, constructive and responsive relationships that are important for successful management of a project's environmental and social risks. Stakeholder engagement is most effective when initiated at an early stage of the project development process, and is an integral part of early project decisions and the assessment, management and monitoring of the project's environmental and social risks and impacts.

2.5. KfW Requirement

According to the Sustanability Guidelines Assessment and management of Environmental, Social, and Climate Aspects: Principles and Procedures, October 01, 2019 by KfW Development Bank, their assessment standards are the Environmental and Social Standards of the World Bank Group (i.e. for public agencies the Environmental and Social Standards (ESS) as well as relevant Operational Polices of the World Bank and the IFC Performance Standards (PS) for cooperation with the private sector) and their General and sector-specific Environmental, Health and Safety (EHS) Guidelines as well as the Core Labour Standards of the International Labour Organization (ILO).

ESIAs are an integral component of project and programme appraisals conducted by KfW Group. The aim of an ESIA is to assess the risks or negative impacts an investment, project

or programme may have on the physical, biological and social environment and to determine measures together with the project partner or business client for avoiding, mitigating or offsetting such undesired effects. KfW applies internationally recognised environmental and social standards, including those of the EU, the World Bank Group and the International Labour Organisation (ILO). Concerns about the protection of human rights form an integral part of the ESIA.

The categorisation into A, B and C projects follows the international standards of the World Bank/IFC (International Finance Corporation) and is subject to an internal verification by a non-market unit. The categorisation determines the scope and depth of the assessment.

Category A projects imply potentially diverse significant adverse or irreversible impacts or risks to human health, the environment or the climate. Examples of projects that may cause irreversible impacts – such as resettlement – include infrastruc-ture projects (for example the construction of new highways), basic industry projects (including the construction of refineries) or energy projects (such as the construction of large hydropower projects). Significant adverse environmental or social impacts or risks may also arise from category B projects but these are less severe and can usually be mitigated with state-of-the-art measures or standard solutions. Examples include metalworking or pharmaceutical industry projects in existing industrial zones. No or only minor adverse effects are expected from category C projects.

According to categerisation of the project, this Surat Metro Rail project falls into the Category A.

2.6. AFD Requirements

According to Environmenal and Social Framework of AFD, in terms of E&S risk management, AFD Group does not have specific E&S Standards and applies World Bank Group Standards including the World Bank E&S Framework and IFC Performance Standards and the related Guidance and Interpretation Notes.

AFD approach to ESRM

The aim of AFD E&S risk management approach is to:

- Assess the environmental and social risks and impacts of each project submitted to the decision-making bodies;
- Propose appropriate measures to avoid, minimize, and/or offset these risks and their impacts;
- Monitor the implementation of these measures during the implementation phase for the operation;
- Manage unforeseen events and;

- Improve the quality of projects and the environmental and social performance of counterparts.

AFD's environmental and social due diligence is adapted to the nature and scale of the project and is proportional to the level of E&S risks and impacts. For this purpose, depending on the potential E&S risks, AFD classifies projects into the following four categories: High risks (A), Substantial risks (B+), Moderate risks (B) or Low risks (C).

According to categerisation of the project, this Surat Metro Rail project falls into the High Risk (A).

2.7. Gap Analysis

The significant difference between the Indian laws and regulations relating to Metro project and World Bank standards are briefly given in Table 2.2. As per the Indian Laws, Environmental Impact Assessment studies are not applicable for Railway/metro projects regardless of potential impacts, whereas World Bank standards requires an assessment depending on the classification of activity based on risk. Bank and country's own systems will be followed, and the content of documents will be harmonized wherever possible.

Table 2.2: Gap Analysis

Aspect	World Bank Standards	IFC Performance Standards	Government of India Regulations	Gaps and Action
Assessment and Management of Environmental and Social Risks and Impacts	**ESS1:** Identify, assess, manage and monitor the environmental and social risk and impacts of the project. Mitigation measures are identified.	**PS1:** Identify and evaluate environmental and social risk and impacts of the project.	**EIA notification 2006 and Its Amendment:** EIA notification states that the project or activities listed in the Schedule of above said notification shall be undertaken in any part of India only after environmental clearance. As, Metro projects are not listed in the any schedule of the above said notification, hence, EIA/EMP are not required for the metro projects.	Gaps exist between the National and International regulations. However, no gap has been identified between ESS1 and PS1. Therefore, EIA/EMP will be carried out in line with the PS1 and ESS1.
Project Description (Area of Influence)	**ESS1**: Describes the project and its geographic, environmental, social, and temporal Context, maps of the project site and the area that may be affected by the projects.	**PS1:** Identifies the physical elements, aspects and facilities that are likely to generate environmental and social impacts. Impact is identified in the context of the project area of influence.	As project is not covered under the EIA notification 2006 and its amendments, hence, Area of influence is not applicable.	Gaps exist between the National and International regulations. Therefore, Project Description will meet the requirements of PS1 and ESS1.
Baseline Data	**ESS1:** Conduct monitoring and reporting on the environmental and social performance of the Project.	**PS1:** The risks and impacts identification process will be based on recent environmental and social baseline data at an appropriate level of details.	As project is not covered under the EIA notification 2006 and its amendments, hence, Baseline data collection is not applicable.	Gaps exist between the Gaps exist between the National and International regulations. Therefore, Baseline data collection will be carried out in line with the PS1 and ESS1.
Environmental risks and impacts	**ESS1:** Takes into account all relevant environmental and social risks and impacts of the project.	**PS1:** Identification of Environmental and social risk of the project.	As project is not covered under the EIA notification 2006 and its amendments, hence, risks assessment and impacts identification are not applicable.	Gaps exist between the National and International regulations. Therefore, Environmental risk identification and impacts will be carried out in line with the PS1 and ESS1.
Mitigation Measures	**ESS1**: Identifies mitigation measures and significant	**PS1:** Mitigation and performance improvement	As project is not covered under the EIA notification 2006 and its amendments, hence,	Gaps exist between the National and International regulations.

Aspect	World Bank Standards	IFC Performance Standards	Government of India Regulations	Gaps and Action
	residual negative impacts that cannot be mitigated and, to the extent possible,	measures and actions that address and the identified environmental and social risks and impacts of the project.	mitigation measures adopted for the impacts are not applicable.	Therefore, Mitigation measures will be provided to meet the requirements of PS1 and ESS1.
Capacity Development and Training	**ESS1:** specific description of institutional arrangements, identifying which party is responsible for carrying out the mitigation and monitoring measures, operation.	**PS1:** organizational structure that defines roles, responsibilities, and authority to implement the ESMS	As project is not covered under the EIA notification 2006 and its amendments, hence, this requirement is not applicable.	Gaps exist between the National and International regulations. Therefore, Capacity development and training will be undertaken to meet the requirements of PS1 and ESS1.
Environment commitment plan	**ESS1**: Summarizes key measures and actions and the time frame required for the project to meet the requirements of the ESSs.	**PS1:** Will establish, maintain, and strengthen as necessary an organizational structure that defines roles, responsibilities, and authority to implement the ESMS.	As project is not covered under the EIA notification 2006 and its amendments, hence, this requirement is not applicable.	Gaps exist between the National and International regulations. Therefore, Environment commitment plan will be undertaken to meet the requirements of PS1 and ESS1.
Resource efficiency	**ESS3:** Implementation of technically and financially feasible measures for improving efficient consumption of energy, water, and raw materials, as well as other resources.	**PS3:** Implementation of technically and financially feasible measures for improving efficient consumption of energy, water, and raw materials, as well as other resources.	**National Environment Policy, 2006:** efficient use of environmental resources in the sense of reduction in their use per unit of economic output. **National Water Policy, 2012:** States that the utilisation of water uses for diverse purpose should be optimized and an awareness of water as a scarce resource should be fostered. **G20 Agenda:** Resource efficiency has also been one of the important discussion points in G20 Agenda, whereby G20 countries are integrating circular economy as part of their	Gaps exist between the National and International regulations. Water legislation at national level is aimed only at preventing pollution of water and its various resources. Current legislations don't cover the areas of conservation of water, energy, raw material as a resource and its usage in an efficient and sustainable manner. Presently, no

Aspect	World Bank Standards	IFC Performance Standards	Government of India Regulations	Gaps and Action
			implementation strategies for SDGs. India is a signatory to the Global Development Agenda - 2030 of the United Nations (UN). In 2015, the Government of India set up the India Resource Panel (InRP) to mainstream resource efficiency in policy formulation and foster its spread. Draft National Resource Efficiency Policy released for public comments.	comprehensive or framework law at national level exists that deals with the issue of use efficiency. Therefore, Resource efficiency will be undertaken to meet the requirements of PS1, ESS1 and National Water policy 2012.
Pollution prevention and management	**ESS 3:** Avoid the release of pollutants or, when avoidance is not feasible, minimize and control the concentration and mass flow of their release using the performance levels and measures specified in national law or the EHSGs.	**PS 3:** Consider the ambient Environmental condition. Apply Technically and financially feasible pollution prevention principle and techniques. This applies to the release of pollutants to air, water, and land due to routine, non-routine, and accidental circumstances with the potential for local, regional, and transboundary impacts.	In India, there are several statutes (Acts and Rules) which are enforced with respect to the pollution abatement. These include: **Air (Prevention and Control of Pollution) Act, 1981 amended in 1987;** Section 21: Restriction on use of certain industrial plant requires consent to establish or operation of any plant from the board. Section 22: Persons carrying on industry not to allow emission of air pollutants in excess of the standards laid down by the board under section 17(1)(19) Central Pollution Control Board (CPCB) order Dated 18th November, 2009: The regulation sets limits for air quality parameters for Residential, Commercial, ecologically sensitive area. **Water (Prevention and Control of Pollution) Act, 1974 amended in 1988;** **Water (Prevention and Control of Pollution) Rules, 1975 amended in 2011.** Section 24: Prohibition on use of stream or	No Gaps Exist. Pollution prevention and management will be undertaken to meet the requirements of National regulations and WB ESS.

Aspect	World Bank Standards	IFC Performance Standards	Government of India Regulations	Gaps and Action
			well for disposal of polluting matter. Section 25 & 26: Require consent to establish or operation of any industry from the board. **Noise Pollution (Regulation and Control) Rules, 2000 amendment in 2010.** Rule 3 & 4: The ambient Air quality Standards in respect of noise for different areas. Rule 5 sub rule 5 A: Restriction on the use of horns, sound emitting construction equipments during night time in the residential area. **Solid Waste Management Rules, 2016 and its amendments** Section 4: Segregate and store the waste in suitable bins and handover segregated waste to authorized waste picker or collector. **Hazardous and Other Wastes (Management and Trans boundary Movement) Rules, 2016, 2019:** Schedule-I , activity Drill cuttings excluding those from water based mud , Used or spent oil **Construction and Demolition Waste Management Rules, 2016 and its amendments:** Section 4, Clause 3 and 5: Duties of the waste generator. Waste generators who generate more than 20 tons or more in one day or 300 tons per project in a month shall have to segregate the waste, submit waste management plan. **Plastic Waste Management (amendment) Rules 2018.** Rule 8, Responsibility of the waste generators.	

Aspect	World Bank Standards	IFC Performance Standards	Government of India Regulations	Gaps and Action
			Storage of waste at source and handover the segregated waste to urban local body. **E waste (Management) Rules 2016.** Rule 9, Responsibility of Consumer or Bulk Consumer. **Central Motor Vehicle Act, 1988 and Central Motor Vehicle Rules, 1989:** Chapter V: Construction Equipment and maintenance of Motor Vehicle	
Assessment of Risk and Impacts on Biodiversity and Habitat. Conservation of Biodiversity and Habitats	**ESS6:** Identify the potential project related risks and impacts on habitats and the biodiversity that they support. For the protection and conservation of habitats and the biodiversity they support, the mitigation hierarchy includes biodiversity offsets. Offsets will be considered as a last resort, only if significant residual adverse impacts remain after all technically and financially feasible avoidance, minimization, and restoration measures have been considered.	**PS6:** Direct and indirect project-related impacts on biodiversity and ecosystem services and identify any significant residual impacts. For the protection and conservation of biodiversity, the mitigation hierarchy includes biodiversity offsets, which may be considered only after appropriate avoidance, minimization, and restoration measures have been applied	There are number of act and Rules are enforced with respect to the Biodiversity and Habitat. These are as follows: **Forest (Conservation) Act 1980 amended in 1988;** **Forest (Conservation) Rules 2003, 2004** Section 2, Restriction on the use of forest land for non forest purpose. Chapter 2, Clause 2.1, Sub clause (iv), Re diversion of forest land **Indian Wildlife Protection Act, 1972, amended in 2002** Section 27, Restriction on entry in sanctuary Section 28, Grant of permit Section 33, Control of sanctuaries **Wetlands (Conservation and Management) Rules 2017** Section 3, Applicability of Rules. Section 4, Restriction of activities in wetlands. **Saurashtra Felling of trees (infliction of Punishment) Act, 1951 & amendment Rule 2008.** Section 3, clause 1, sub clause (a), no person can fell tree without the written permission of	No Gaps Identified. Assessment and conservation of Biodiversity will be undertaken in a manner to meet the requirements of National regulations and WB ESS.

Aspect	World Bank Standards	IFC Performance Standards	Government of India Regulations	Gaps and Action
			the collector. Section 3, Clause 2, felling of babul trees. Section 1, Clause E, of Amendment Section 4 , and section 5 for transit pass **Bombay Forest Rules 1942, Bombay Forest (Gujarat Amendment) Rules, 2018,** Rule 66 of 1942, for transit pass.	
Identification & Protection of Cultural Heritage in Project Design and Execution	**ESS8**: Identification of the legally protected heritage area affected by the project. If project lies within the legally protected area or buffer zone. Comply with local, national, regional, or international cultural heritage regulations and the protected area management plans.	**PS8:** Identify and protect cultural heritage by ensuring that internationally recognized practices for the protection, field-based study, and documentation of cultural heritage are implemented.	The Ancient Monuments and Archaeological sites and Remains Act,1958 amended in 2010 with Ancient Monuments and Archaeological Sites and Remains (Amendment) Act, 2017 which is yet to be passed by Rajya Sabha. An Act to provide for the preservation of ancient and historical monuments and archaeological sites and remains of national importances.	No Gaps Identified. Identification and protection of legally protected heritage area will be undertaken to meet the requirements National regulations and WB ESS.
Stakeholder Identification and Analysis	**ESS10:** identify the different stakeholders; both project affected parties and other interested parties.	**PS1:** Identify the stakeholder	As project is not covered under the EIA notification 2006 and its amendments, hence, there is no provision exists for Stakeholder Identification and Analysis. However, The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 is an Act that regulates land acquisition and lays down the procedure and rules for Stakeholder Identification and Analysis.	Gaps Exist. Stakeholder Identification and Analysis will be undertaken to meet the requirements of ESS10, PS1 and National regulations.
Information Disclosure	**ESS10:** Disclose project Information to allow	**PS1:** Disclosure of relevant project information helps	As project is not covered under the EIA notification 2006 and its amendments, hence,	Gap Exist. Information Disclosure will be

Aspect	World Bank Standards	IFC Performance Standards	Government of India Regulations	Gaps and Action
	stakeholder to understand the risk and Impacts of the project.	Affected Communities and other stakeholders understand the risks, impacts and opportunities of the project	no provision exists for Disclosure of Information to stakeholder. However, The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 is an Act that regulates land acquisition and lays down the procedure and rules for Information Disclosure.	undertaken to meet the requirements of ESS 10, PS 1 and National regulations.
Consultation with Stake holders	**ESS10 & ESS5**: Carryout consultations with displaced persons, host communities and concerned NGOs. Inform all displaced persons of their entitlements and resettlement options	**PS1:** process of consultation in a manner that provides the Affected Communities with opportunities to express their views on project risks.	As project is not covered under the EIA notification 2006 and its amendments, hence, no provision exists for Consultation with Stake holders. However, The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 is an Act that regulates land acquisition and lays down the procedure and rules for Consultation with Stake holders.	Gap Exist. Stakeholder consultation will be undertaken to meet the requirements of ESS 10, PS 1 and National regulations.
Grievance mechanism	**ESS10:** Mechanism shall be in place for redressal of Grievance	**PS1**: Establish a grievance mechanism to receive and affected communities concern and grievance	Department of Administrative Reforms & Public Grievances is the nodal agency in respect of policy initiatives on public grievances redress mechanism and citizen centric initiatives. Each Ministry/ Department/ Public Sector Undertaking/ Autonomous Organisation may designate a full time Grievance Officer, known as the Director of Grievances.	Gap Exist. Grievance Redressal will be undertaken to meet the requirements of ESS 10, PS 1 and National regulations.
Organizational Capacity and Commitment	**ESS10:** Establish, Maintain and designate specific personnel to be responsible for the implementation and monitoring of stakeholder engagement activities and	**PS1:** Establish, maintain, and strengthen as necessary an organizational structure that defines roles, responsibilities, and	As project is not covered under the EIA notification 2006 and its amendments, hence, no provision exists for Organizational Capacity and Commitment.	Gaps Exist. Organizational Capacity and Commitment be undertaken to meet the requirements of ESS 10 and PS 1.

Aspect	World Bank Standards	IFC Performance Standards	Government of India Regulations	Gaps and Action
	compliance with this ESS.	authority to implement the ESMS.		
Stakeholder Engagement	**ESS10:** Emphasizes the importance of open engagement with stakeholders and providing opportunities for stakeholders views to be considered in the project design and during implementation.	**PS1:** Stakeholder analysis and planning, disclosure and dissemination of information, consultation and participation, grievance mechanism, and ongoing reporting to Affected Communities.	As project is not covered under the EIA notification 2006 and its amendments, hence, no provision exists for Stakeholder Engagement. However, The Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act, 2013 is an Act that regulates land acquisition and lays down the procedure and rules for Stakeholder Engagement.	Gap Exist. Stakeholder engagement will be undertaken to meet the requirements of ESS 10, PS 1 and National regulations.

Chapter 3 : Project Description

3.1. Existing Transportation System in Surat City

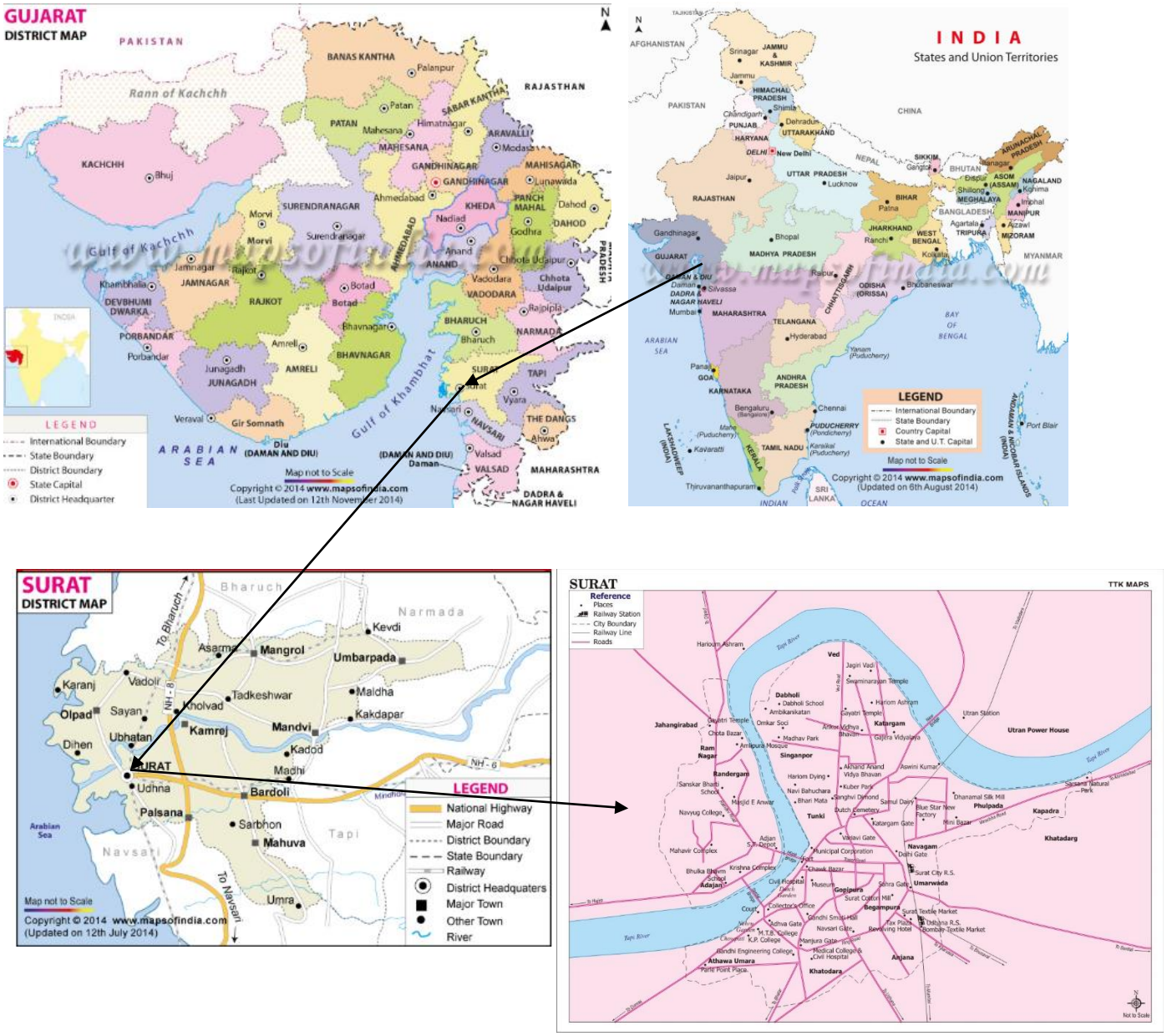
Surat is one of the cleanest city of India and is also known by several other names like "THE SILK CITY", "THE DIAMOND CITY", "THE GREEN CITY", etc. Surat railway station in early 2016, the Indian Railway Catering and Tourism Corporation rated the facility the best large station in India based on cleanliness. The Sitilink or Surat BRTS is a bus rapid transit system in the city. It is operated by Surat Municipal Corporation and as of August 2017, had a network of 245 buses connecting major localities. Surat International Airport located in Magdalla, 11 kilometres southwest of Surat. It is the second busiest airport in Gujarat in terms of both aircraft movements and passenger traffic. Currently airlines such as Air India, Alliance Air, Air Asia India, Spice Jet, IndiGo Airlines, Air Odisha, Ventura AirConnect provide flight services from the Surat to various major cities like New Delhi, Mumbai, Kolkata, Chennai, Bengaluru, Hyderabad, Goa and Jaipur. Air India Express is international flight on Surat Sharjah route. Apart from main city, Surat Airport also caters to various localities of south Gujarat including Navsari, Bardoli, Valsad, Bharuch, Ankleshwar.

In the absence of adequate full fledged public transport system, the use of individual modes such as cycles, 2- wheelers, cars etc. and un-organized intermediate public transport (IPT) has increased in Surat City. Cycling and walking constitute about 45% of the total trips and this feature needs to be integrated within the upcoming public transport initiatives. As per the records available, the total road length as on the year ending on 31-03-2016 is 3575 kms out of which 624 Kms is un-surfaced. The total vehicles registered in Surat during 2013-14 & 2014-15 was 131384 & 177161 respectively as per the figures indicated by Regional transport Office (RTO). The yearly growth rate of vehicles for all modes was 34.84%.

3.2. Study Area

With a view of developing effective and efficient mass transit system in addition to the existing public transportation, the Gujarat Metro Rail Corporation (GMRC) Ltd intends to develop the proposed Surat Metro Rail Phase-I Project having two Corridors. The location map is shown in **Figure 3.1.**

Figure 3.1: Location Map



3.3. Proposed Metro Rail System in Surat City

Two corridors have been finalized for implementation of Metro Rail Project in Surat. The details are given below:

- Corridor 1: Sarthana to Dream City (via Nana Varachha, Rly station, Majuragate, Exibition centre)
- Corridor 2: Bhesan to Saroli (via L P Savani, Majura Gate, Mithikhadi)

The proposed metro rail alignment is shown in Figure 3.2 and on google map is shown in Figure 3.3. All the components of project description for proposed Metro Corridors in respect of alignment details, rolling stock, power supply, traction system and signalling are summarised in DPR and reproduced in this chapter.

Figure 3.2: Proposed Metro Corridors in Surat City

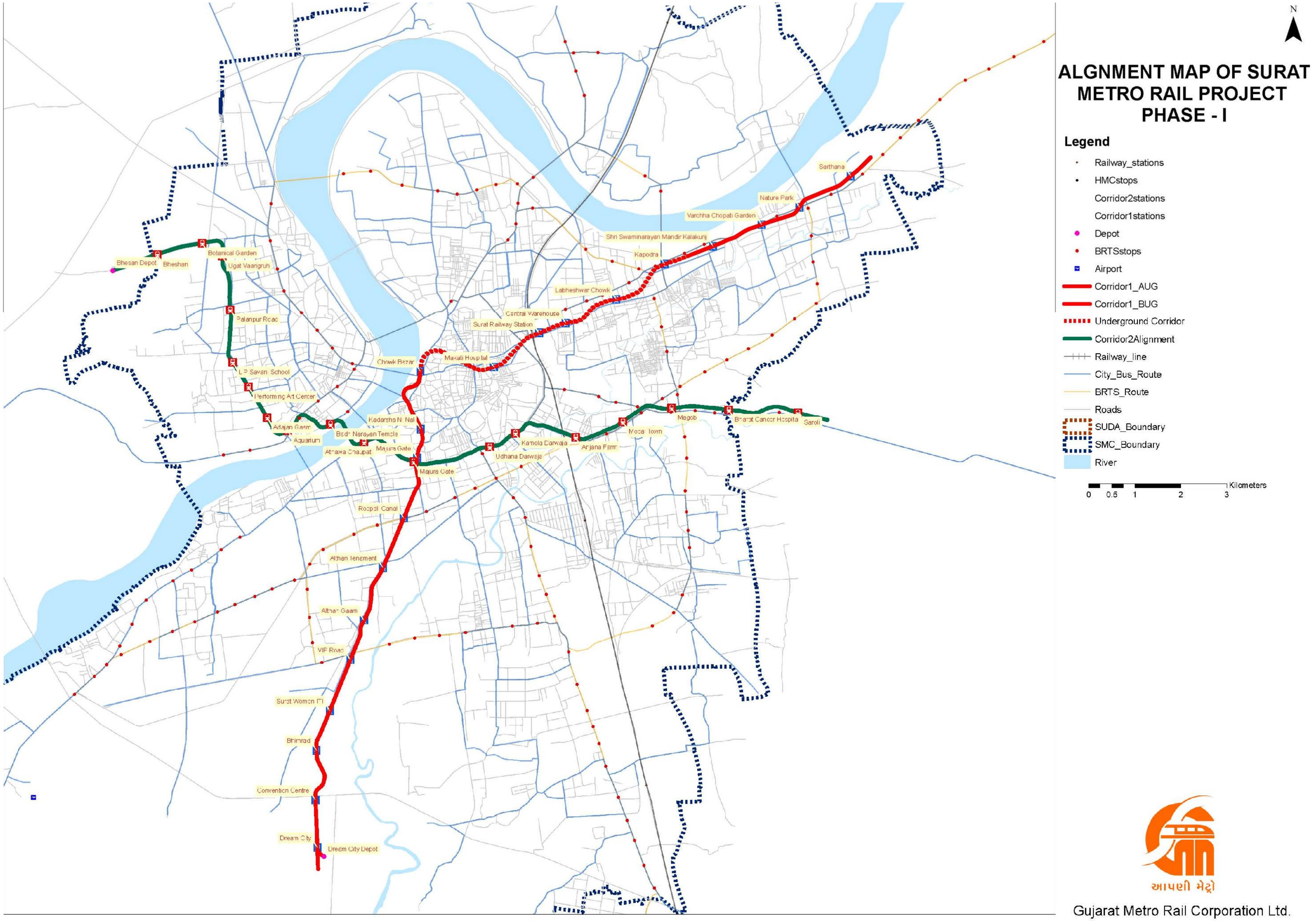


Figure 3.3: Google Map showing Proposed Metro Corridors

BHESAN-SAROLI CORRIDOR- 1)Bhesan, 2)Botanical Garden, 3)Ugat Vaarigruh, 4)Palanpur Road,5)L P Savani School,6)Performing Art Centre, 7)Adajan Gaan, 8)Aquarium, 9)Badri Narayan Temple, 10)Althawa Chaupati, 11) Majura Gate, 12) Udhana Darwaja, 13) Kamela Darwaja, 14)Anjana Farm, 15) Modal Town,16) Magob, 17) Bahrat Cancer Hospital,18) Saroli

SARTHANA – DREAMCITY CORRIDOR- 1)Dream City, 2) Convention Centre, 3)Bhimrad, 4)Surat Women ITI,5) VIP Road, 6) Althan Gaan, 7)Althan Tenement, 8) Roopali Canal, 9)Majura Gate, 10) Kadarsha Ni Nal 11) Chowk Bazar, 12)Makati Hospital, 13) Surat Railway Stations, 14)Central Warehouse, 15)Labheshwar Chowk, 16)Kapodra, 17) Shri Swaminaryan Mandir Kalakunj, 18)Varcha Chopati Garden ,19)Nature Park, 20)Sarthana

The proposed Phase I metro has a total length of 40.35 km having 38 stations. The salient features of the corridors are summarised in the following sections. These corridors will provides connectivity to all congested, important and densely populated areas of the city. Details of the length of corridors, elevated/underground length and number of stations is given in Table 3.1.

Table 3.1: Salient Features of the Corridors

S. NO	ROUTE	LENGTH IN KM			STATIONS
1.	Sarthana to Dream City	Elevated	15.14	21.61	14
		Underground	6.47		6
2.	Bhesan to Saroli	Elevated		18.74	18
			TOTAL	**40.35**	**38***

*Majura Gate Metro Station - Interchange Station

Corridor 1: Sarthana to Dream City

The proposed alignment of Corridor-1 starts from Sartthana as elevated section and heads towards Dream City as UG section. Total 20 stations have been proposed along the corridor of which 14 stations are elevated and 6 stations are UG. Summary list of stations along the corridor is given in Table 3.2.

Table 3.2: Proposed Stations along Corridor-1

S. No	Station Name	Chainage (m)	Inter Station Distance (m)	Elevated/ Underground
0	DEAD END	(-) 450	450.00	ELEVATED
1	Sarthana	0.0	1238.10	ELEVATED
2	Nature Park	1238.1	870.40	ELEVATED
3	Varcha Chopati Garden	2108.5	1088.10	ELEVATED
4	Shri Swaminarayan Mandir	3196.6	1040.40	ELEVATED
5	Kapodara	4237.0	1269.60	U/G
6	Labheshwar Chowk	5506.60	1162.30	U/G
7	Central Warehouse	6668.9	586.70	U/G
8	Surat Railway Station	7255.6	1231.40	U/G

S. No	Station Name	Chainage (m)	Inter Station Distance (m)	Elevated/ Underground
9	Maskati Hospital	8487.00	1802.80	U/G
10	Chowk Bazar	10289.8	1431.90	U/G
11	Kadarshna Ni Nal	11721.7	592.10	ELEVATED
12	Majura Gate	12330.8	1297.70	ELEVATED
13	Rupali Canal	13611.5	1084.70	ELEVATED
14	Althan Tenament	14696.2	1147.30	ELEVATED
15	Althan Gam	15843.5	855.60	ELEVATED
16	VIP Road	16699.1	1109.30	ELEVATED
17	Surat Women ITI	17808.4	851.10	ELEVATED
18	Bhimarad	18659.5	1169.90	ELEVATED
19	Convention Centre	19829.4	880.90	ELEVATED
20	Dream City	20710.3	450.00	ELEVATED
	DEAD END	21160.3	450.00	ELEVATED

Source: DPR

Corridor-2: Bhesan to Saroli

The proposed alignment of Corridor-2 starts from Bhesan and heads towards Saroli, this section is fully elevated. Total 18 stations have been proposed along this corridor. Summary list of stations along the corridor is given in Table 3.3.

Table 3.3: Proposed Stations along W-E Corridor

S. No	Station Name	Chainage (m)	Inter Station Distance (m)	Elevated/ Underground
0	DEAD END	-949.6		
1	Bheshan	0	949.60	ELEVATED
2	Botanical Garden	950.6	950.60	ELEVATED

3	Ugat Vaarigruh	1617.5	666.90	ELEVATED
4	Palanpur Road	2738.9	1121.40	ELEVATED
5	L.P.Savani school	3831.6	1092.70	ELEVATED
6	Performing Art Centre	4421.8	590.20	ELEVATED
7	Adajan Gam	5155	733.20	ELEVATED
8	Aquarium	5770.1	615.10	ELEVATED
9	Badri Narayan Temple	6891.5	1121.40	ELEVATED
10	Athwa Chaupati	7863.5	972.00	ELEVATED
11	Majura Gate	9019.2	1155.70	ELEVATED
12	Udhana Darwaja	10646.2	16270.00	ELEVATED
13	Kamela Darwaja	11247.2	601.00	ELEVATED
14	Anjana Farm	12593.9	1346.70	ELEVATED
15	Model Town	13636.3	1042.40	ELEVATED
16	Magob	14744.00	1107.70	ELEVATED
17	Bharat Cancer Hospital	15899.4	1155.40	ELEVATED
18	Saroli	17341.8	1442.40	ELEVATED
	DEAD END	17791.8	450.00	ELEVATED

Source: DPR

3.3.1. Stations

Station Design is dependent on the peak hour traffic load for each station. Accordingly maximum capacity required at any station for emergency evacuation has been adopted. Typical station drawings of Elevated, under ground and Interchange Stations are shown in Figure 3.4 to Figure 3.6.

Figure 3.4: Typical Drawing of Elevated Station

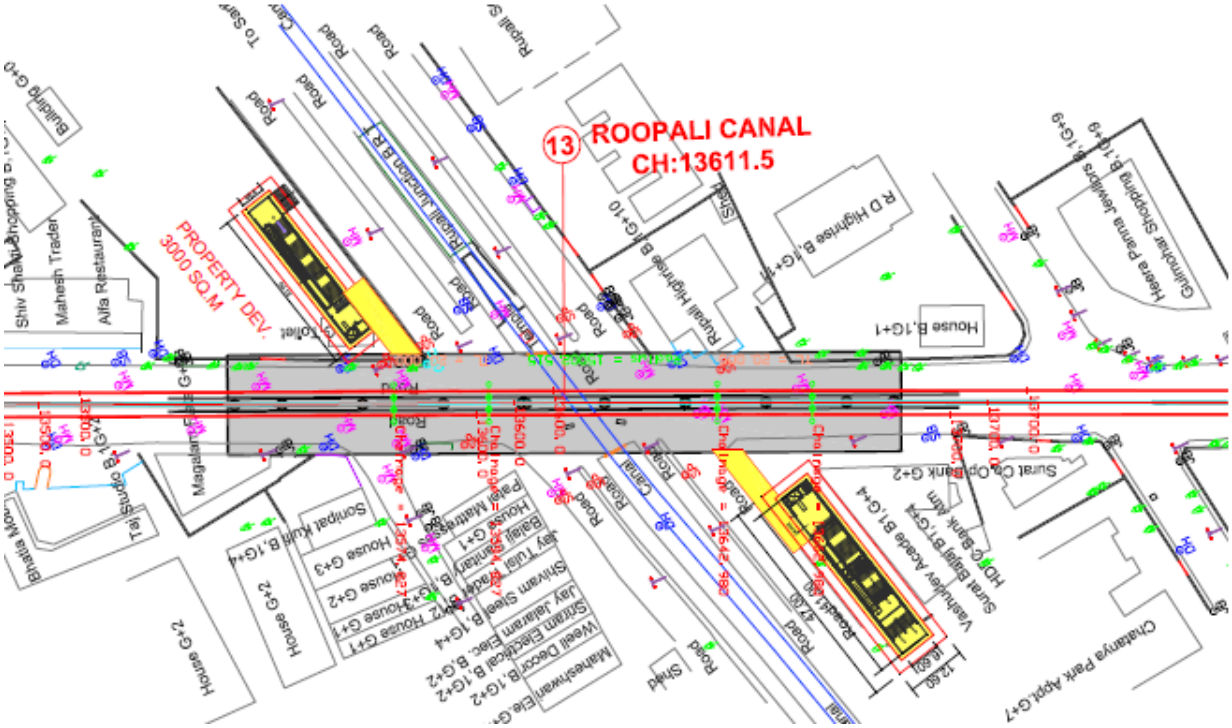
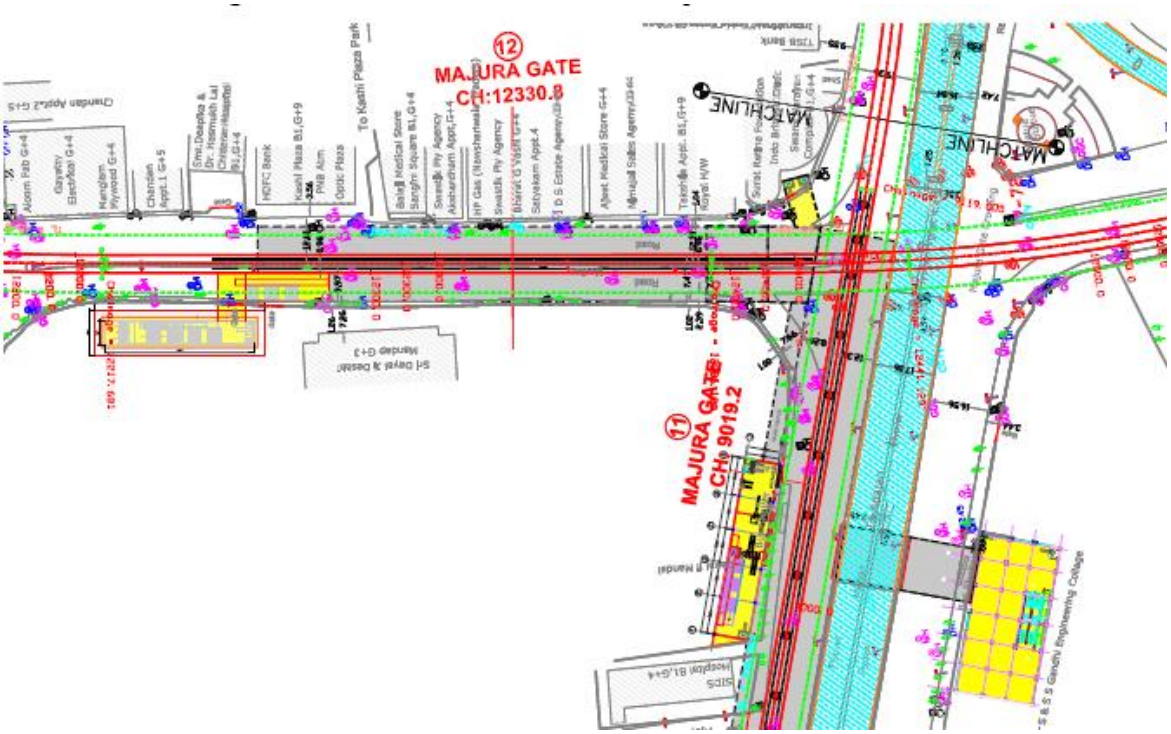


Figure 3.5: Typical Drawing of Underground Station

Figure 3.6: Typical Drawing of Interchange Station



The station is generally located on the road median. Total length of the station is ~140-m. All the stations are two-level stations. The concourse is planned along the whole length of the platform with staircases leading from either side of the road. The maximum width of the station at concourse is ~21-m. Passenger facilities like ticketing, information, etc as well as operational areas are provided at the concourse level.

Structure of the underground station (type-1) is essentially a concrete box about 20~21m wide, 14.6-m high and 190 or as per design long with an intermediate slab. Sides of the box are made of 1.2 m thick RCC. The roof of station box is about 1.2 m thick. Structure of this type-2 of underground station is essentially a concrete box about 10~11m wide, 20.5 ~ 22.0 m high and 190 or as per design long with an intermediate slab separating two Platforms.

3.3.2. Permanent Way

Gauge: Standard Gauge which is adopted for this metro railway permits sharper curves, which is advantageous for metro alignment in urban scenario and results in minimized property demolition and property acquisition. The Land requirement for the maintenance depots is also lower in Standard Gauge. Standard Gauge rolling stock results in recurring saving in energy consumption during operation as for the same passenger carrying capacity, gross weight of a metro coach is lower.

Formation: Ballast less track is proposed for elevated and underground stretches and depots so as to optimize maintenance and risk to road vehicles. This will help reduce fugitive dust emissions during operation.

Welding: To minimize noise and vibrations, track joints are proposed to be welded.

3.3.3. Rolling Stock, Traction and Signalling

The required transport demand forecast is the governing factor for the choice of the Rolling Stock. The forecasted Peak Hour Peak Direction Traffic calls for a Medium Rail Transit System (MRTS).

- A short train consisting of 3 cars which can be increased for increasing the Passenger Carrying Capacity of Trains with the consideration of matching the growing traffic demand.
- The rolling stock shall be Standard 1435 mm track gauge section having maximum width of 2.9 m, Axle load of 16 tonnes and capacity of 3 coach unit as 764 passengers. Seating arrangement will be longitudinal and AC class accommodation will be provided.
- Total traction and auxiliary power requirement is 30.25 MVA and 11.99 MVA to fulfil the power demand of Sarthana to Dream city and Bhesan to Saroli corridors in 2021 respectively.
- The system under normal operating conditions will be an automatically operated system utilizing Automatic Train Control and Automatic Train Protection (ATP) under the overall control of a train driver and OCC operators.
- Computer Based Interlocking (CBI) signalling and continuous automatic train control with Automatic Train Protection (ATP) is proposed while telecommunication system is integrated with Optical Fibre Cable, LED/LCD based boards, Mobile Radio, Mobile system etc.
- Fare collection system is provided with automation in association with Contactless Smart Card and Retractable Flap Type Control Gates, Ticket Office Machine, TR, PTD etc.

3.3.4. Ventilation and Air-Conditioning System

The underground stations of the corridor are built in a confined space. A large number of passengers occupy concourse halls and the platforms, especially at the peak hours. The platform and concourse areas do not have adequate natural ventilation. It is therefore, essential to provide forced ventilation in the stations and inside the tunnel for the purpose of:

- Supplying fresh air for the physiological needs of passengers and the staff
- Removing body heat, obnoxious odours and harmful gases
- Removing large quantity of heat dissipated by the train equipment/fixtures
- Removing fumes and heat emitted by station equipment/fixtures

3.3.5. Maintenance Depots

It is proposed to establish one depot- cum- workshop for each Corridor with following functions:

(i) Major over hauls of all the trains.

(ii) All minor schedules and repairs.

(iii) Lifting for replacement of heavy equipment and testing thereafter.

(iv) Repair of heavy equipment's.

The depots are to be developed with full/light/heavy repair facilities, stabling and light inspection facilities.

3.3.6. Power Requirements

Power supply is required for the operation of Metro system for running of trains, station services, workshop, depot and other maintenance works within the premises of metro system. The power requirement is for peak hour demand for traction and auxiliary application. Some of assumptions to estimate the power supply are :

- Specific energy consumption of rolling stock 60 kWh/1000 GTKM in case of 750V dc Traction at Pantograph level is considered as per MOUD guideline
- Elevated/at –grade station load – initially 250 kW, which will increase to 300 kW in the year 2046.
- Underground station load – initially 2000 kW, which will increase to 2200 kW in the year 2046.
- Depot & Metro Bhawan auxiliary load - initially 2000 kW (each), which will increase to 2200 KW in the year 2046.

Keeping in view of the train operation plan and demand of auxiliary and traction power, power requirements projected for the year 2021, 2026, 2036 and 2046 are summarized in Table 3.4.

Table 3.4: Power Demand in MVA

Corridor	Input Source	Peak demand - Normal		Peak demand** - Emergency	
		Year (2021)	Year (2046)	Year (2021)	Year (2046)
Sarthana to Dream City Corridor-1	**RSS near Nature Park/ Kapodara Station**				
	Traction	2.53	9.39	6.16	20.44
	Auxiliary	9.57	10.75	24.09	26.93
	Sub-total (A)	**12.10**	**20.14**	**30.25**	**47.37**
	RSS at Dream City Depot/ Near Althan Tenement				
	Traction	3.63	11.05	6.16	20.44
	Auxiliary	14.51	16.18	24.09	26.93
	Sub-total (B)	**18.14**	**27.23**	**30.25**	**47.37**
Bhesan to Saroli Corridor-2	**RSS at Bhesan Depot**				
	Traction	2.53	6.41	3.96	11.16
	Auxiliary	5.25	6.05	8.03	9.39
	Sub-total (C)	**7.78**	**12.46**	**11.99**	**20.55**

Corridor	Input Source	Peak demand - Normal		Peak demand** - Emergency	
		Year (2021)	Year (2046)	Year (2021)	Year (2046)
	RSS Near Magob Station				
	Traction	1.43	4.75	3.96	11.16
	Auxiliary	2.78	3.34	8.03	9.39
	Sub-total (C)	**4.21**	**8.09**	**11.99**	**20.55**

3.3.7. Sub Station

Keeping in view the reliability requirements, Four Receiving Sub-stations (RSS) are proposed to be set up for corridor – 1 (2 RSS) and corridor – 2 (2 RSS). This is an economical solution without compromising reliability. Based on the discussions in meeting with M/s Torrent Power, M/s GETCO & Surat Municipal Corporation, it is proposed to avail power supply for traction as well as auxiliary services from the following grid sub-stations at 66 kV voltage through cable feeders as given in Table 3.5.

Table 3.5: Source of Power Supply

S. No.	Corridor	Grid sub-station of Power Supply Authority (Input voltage)	Location of RSS of Metro Authority	Approx. length of cables
1.	Sarthana to Dream City Corridor-1	Two no. 66 kV bays from 220/66 kV Uttaran Grid Sub Station of M/s GETCO **or** Two no. 66 kV bays from 220/66 kV Puna Grid substation of Torrent	Near Nature Park Station **or** Near Kapodra station	5 km or 4km
2.		Two no. 66 kV bays from Khajod Grid Sub-station (Proposed) (M/s GETCO)	At Dream City Depot	2 km **or**
		Two no. 66 kV bays from Bhatar Grid Sub-station (M/s Torrent Power)	Near Althan Tenement	2 km
3.	Bhesan to Saroli Corridor-2	Two no. 66 kV bays from 220/66 kV Ved Grid Sub Station (M/s Torrent Power) **or** Two no. 66 kV bays from 220/66 kV Ichchapore Grid substation of GETCO	At Bhesan Depot	7 km or 8 km
4.		Two no. 66 kV bays from 220/66 kV Puna Grid substation of Torrent	Near Magob Station	2 km

3.3.8. Construction Activities

Illustrative list of construction activities is placed below. These activities typically involve movement of earth and construction material, movement and placement of pre-cast elements:

- In-situ open foundations and piles of columns
- In-situ casting of columns
- Pre-cast segments or pre-cast non-segmental girders
- Boring of tunnels by Tunnel Boring Machine or open cut and cover / NATM.

- Cut and cover or NATM and blasting for construction of underground stations
- In-situ earth retaining structures like diaphragm walls, sheet piles, secant piles etc

3.4. Cost Estimate

The overall Capital Cost for the Surat Metro rail network under Phase-I at March 2017 price level works out to **Rs.12,492 Crores** including applicable Taxes & Duties of **1950.81 Crores** as tabulated hereunder in Table 3.6.

Table 3.6: Corridor-wise Details of Capital Cost

Sr. No.	Name of the corridor	Capital Cost (Rs. Crore)	Taxes & Duties (Rs. Crore)	Total (Rs. Crore)
1.	Sarthana – Dream City	6554	1211.91	7765.91
2.	Bhesan–Saroli	3987	738.90	4725.90
	Total	**10,541**	**1950.81**	**12,491.81 =12,492**

3.5. Analysis of Alternatives

3.5.1. Alignment Planning

The principal objectives which were considered while selecting the metro-alignment are minimization of the impact on the environment & social, optimization of the functionality of the alignment, minimization of construction time, minimization of construction and operational cost and maximization of the results of the economic investment. The Ministry of Housing and Urban Affairs, Government of India announced the 'Appraisal Guidelines for Metro Rail Project Proposals' in September 2017. Comprehensive Mobility Plan (CMP) – 2017 of Surat City related to MRTS Project as provided by Centre of Excellence in Urban Transport,(CEPT), Ahmedabad following approach for this study was developed:

- Delineation of Study Area,
- Growth Dynamics,
- Demand assessment along the corridor,
- Characteristics of the corridor
- Integration with other mass transit corridors,
- Transportation Characteristics,
- Construction feasibility, and
- Environment and social aspects.

A Comprehensive Mobility Plan (CMP) aims to retain the compact structure of the city of Surat and proposes intensification of development along the rapid transit corridors. The report recommends the following corridors as potential metro network:

1. Dream city – Kamrej (28.9 km)
2. Bhesan – Umbhel (26.3 km)
3. Majuragate to Karamala (15.8 km)

This section of the chapter presents demand analysis and network characteristics, in order to assess suitability of a rapid transit mode for these three corridors.

3.5.2. Demand Assessment Along The Corridor

An assessment of potential ridership on the three CMP metro corridors for the future years of 2026 and 2046 has been undertaken. An analysis of future Public Transport (PT) riderships important to assess suitability of a metro or another rapid transit mode. Below **Table 3.7** shows model output of assigned passenger flows for year 2046 and ridership by mode.

Table 3.7: Model Outputs of Public Transport Assignments (2046)

Mode	Boardings (peak hour)	% Boardings
Bus	3,61,782	33%
BRT	2,74,896	25%
RTC	4,61,220	25%
Total	**10,97,898**	**100%**

** Peak hour is 10% of full day*

Legend

City bus

BRT

Rapid Transit Corridor

WITH TOD (CMP SCENARIO) 2046

Source: Comprehensive Mobility Plan Surat -2046, CoE- UT, CEPT University

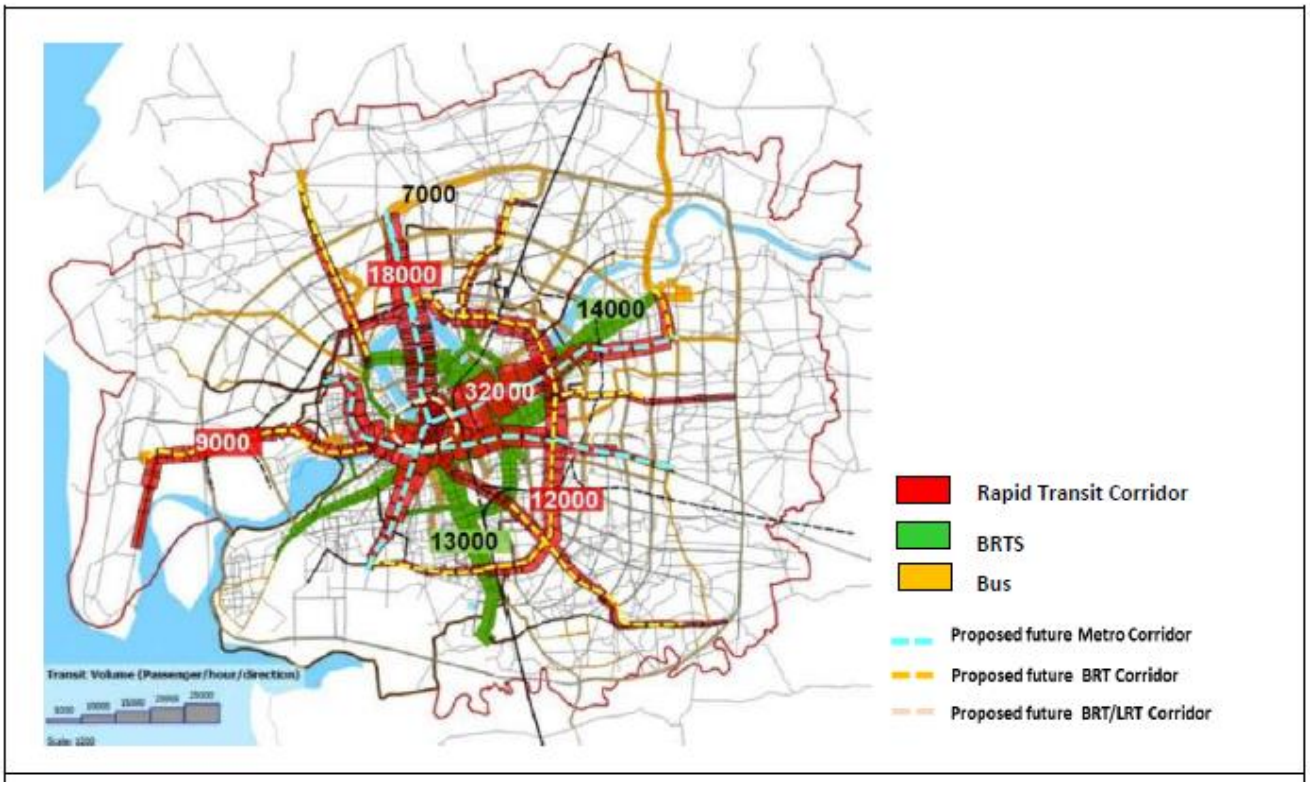
This network was analysed and the demand outputs in terms of peak passenger flows or peak hour peak direction traffic (PHPDT) for the year 2026 and horizon year 2046 has been presented in the Table 3.8. It can be seen that Dream City to Kamrej has the highest passenger flow of 16000 PHPDT in 2026 going up to 46000 PHPDT in 2046. The other two corridors - Bhesan – Umbhel and Majuragate to Karamala has a demand of around 24000 PHPDT and 19500 PHPDT in 2046 respectively.

Table 3.8: Mode Selection based on Daily Ridership and PHPDT

S. No.	Corridor Name	Via	Km	Year 2026		Year 2036		Year 2046		Potential mode
				PHPDT	Ridership	PHPDT	Ridership	PHPDT	Ridership	
1.	Dream city – Kamrej /Vav	Sarsana- Exhibition Centre, Bhatar Char rasta and Sarsana, Majura gate, Rly station and Sarthana	31	16,298	431,040	27,966	812,050	34,268	1,032,120	Metro
2.	Bhesan - Kadodara	Adajan, Majura gate, Kamela Darwaja, Saroli and Sabargam	26	6,500	202,140	16,945	546,390	24,032	782,330	Metro
3.	Majuragate - Karamala	Ved	16	4,618	109,590	15,999	356,720	24,386	533,140	Metro
4.	Sherdi to Unn (half ring)	Kapodara, Saroli, Devadh, Saniya Kande	29	2,821	80,760	9,614	278,960	14,503	421,910	BRT
5.	Railway Station to Railway station	Sahara Dharwaja,Udhana, Adajan, Katargam	12	3,961	98,830	5,359	176,720	8,144	219,600	BRT/LRT
6.	Jahangirpura to Olpad	Talad	11	1,664	28,480	6,194	115,640	9,497	179,880	BRT
7.	Pal to Hazira	Icchapur	15	2,927	43,940	6,725	112,420	9,241	158,610	BRT
8.	Kosad to Sayan	Gothan	10	1,510	25,830	4,499	82,450	6,601	122,730	BRT
9.	Punagam to Valthan	Kosmada	7	686	15,610	1,968	46,330	2,860	67,890	BRT
10.	Unn to Khajod	-	6	380	11,530	966	32,100	1,408	46,300	BRT
11.	Kamrej to Valthan	Vav	2	1,212	23,930	2,631	51,030	3,559	68,030	BRT
12.	Udhana teen rasta to Palsana	Dindoli	19	182	3,600	504	10,530	725	15,370	BRT

Hence based on the passenger flows Figure 3.7 shows the potential Metro and BRT corridor of about 73 km and 213 km for the future years respectively.

Figure 3.7: Prospective Future Metro And BRT Corridor (2046)



Source: Alternative Analysis Report, CEPT University, Ahmedabad

3.5.3. Corridor Characteristics

A description of the corridors in terms of the road widths and adjacent land uses has been presented below:

RTC 1 (Rapid Transit Corridor): Dream City to Kamrej/ Vav (29 km)

Dream city to Kamrej/ Vav corridor is about 29 km long cutting across the city centre and also connecting two far ends of the city. This corridor can be divided into three parts, i.e. Dream City to Majura Gate, Majura Gate to Surat railway station, Railway station to Sarthana and Sarthana to Kamrej/ Vav based on the adjacent land use characteristics. The section from proposed Dream city to Majura Gate is about 8km, of which Majura Gate to Althan is fairly densely developed with mixed uses- industries, educational institutions, commercial and lower and middle income residential land uses. The section from Althan to Dream City is currently developing with residential and commercial land uses coming up in this area. Recently, the Southern Gujarat Chamber of Commerce and Industry (SGCCI) has developed a huge Exhibition Centre on this corridor towards proposed Dream City, which is likely to be a major attraction node in near future. Dream city is the one of the prestigious projects of Surat for which master planning is completed and the construction is underway. Envisaged as the diamond hub of India, it is expected to generate huge employment in

future. Majura Gate to Railway station corridor passes through the city centre area which is very densely built up. The development consists of commercial, retail local markets, government buildings etc. The inner city is the major employment and attraction area in the city. The area around Railway station is also a major node. Gujarat State Road Transport Corporation (GSRTC) and the existing city bus terminals are also located in close proximity to the station, owing to which this node is already been planned and developed as the multimodal transit hub.

There is already an operational BRT from Railway station to Kamrej via Sarthana and Valak. The section from Railway station to Sarthana is highly dense with mostly commercial and retail markets. It also provides direct access to Hirabaug and surrounding area which is the existing diamond hub of the city. There are many power loom clusters situated along the corridor which operate 24x7.

The section from Sarthana to Kamrej is currently developing; Kamrej is already connected to the railway station directly through BRT. There is an alternative node -Vav, 3 kms south of Kamrej on the national highway, which can also be considered for connecting through a rapid transit corridor in future.

RTC 2: Bhesan to Umbhel (28 km)

Bhesan to Umbhel corridor of around 28km length passing through city center via Adajan, Majura Gate, Kamela Darwaja and Saroli. This corridor can be divided into three parts; i.e. Bhesan to Adajan, Adajan to Saroli and Saroli to Umbhel.

Bhesan to Adajan is the main street of the western part of the city with existing RoW ranging from 24m to 36m. A very small segment of around 3km from Bhesan to L P Savani has 24m RoW which is proposed to be widened to 30m in CMP proposal. Roads with 30m and above are appropriate for development of any kind of rapid transit system. Bhesan to Adajan is the section with predominant upper and middle income group residential development. Along the corridor, mixed land use with retail shops and market on the road abutting properties can be seen. Many schools, hospitals and recreational activities are also located along this corridor. This corridor provides direct connectivity with city center area.

The corridor from Adajan to Saroli via Athwa gate, Majura gate, Kamela Darwaja, is predominantly commercial. There are many government offices, colleges and major hospitals in the section from Adajan to Majura Gate. Civil Hospital and B D Mahavir Heart Hospital are the two big hospitals located along this section. Majura Gate and Udhana are major destinations because of textile and ancillary industries lying on this corridor. Udhana junction is on the BRT corridor connecting Sachin.

Udhna to Kamela Darwaja is a high employment zone because of location of textile markets and industries along this section. Kamela Darwaja to Parvat Patiya through Mithi khadi is

again predominantly lined with textile markets and industries on both side. From Parvat Patiya to Saroli, a lot of new textile markets are coming up on both sides. The existing RoW on Ring Road is 60m; Kamela Darwaja to Canal corridor, it is 24m and again from Canal corridor to Saroli, it is 60m. There is a proposal of widening from 24m to 30m the section from Kamela Darwaja to Canal corridor in CMP.

As per initial discussions with SMC, there is also a proposal of station for High Speed Rail (HSR) near Umbhel, owing to which major external demand is expected at this node in future.

Saroli to Umbhel corridor via Chedcha and Ladvi is about 8km long. It passes through the vacant area currently but there is a proposal of continuous road from Saroli to Umbhel with 45m of RoW. In future, this corridor would provide connectivity to HSR station and vacant land along this corridor may have an opportunity to be developed as high density commercial and residential land use.

RTC 3: Majura Gate to Karmala (15 km)

The corridor from Majura Gate to Karmala via Ved has a length of about 15 km. Currently the city does not have direct connection across the river, and there is no development on the other side of the river. There is a proposal of connection across the river from Ved to Karmala with 45m of RoW that will also attract development to happen across the river and will also provide direct connection to city center from other side of the river.

The Majura Gate to Ved via Chowk and Katargam has length of about 8km. A lot of government offices, local and informal markets are located along this corridor. The Katargam area has existing textile industries on one side and the other side of the corridor has predominantly residential land use with middle and upper middle income group of societies. Ved is an existing gamtal having predominantly residential development. There are also many educational institutes are located along the corridor.

3.5.4. Capacity of existing transport system (With and without project Assessment)

An analysis of vehicular traffic along the corridor with and without the rapid transit corridor has been undertaken, in order to ascertain the importance of the transit corridor. This would help assess the need for the project - what will be the status if the alternative is not implemented. Two indicators have been looked at:

1. Congestion relief because of metro: % corridor congested with VCR more than 0.9 in with metro and without metro scenario
2. Travel time benefits: Travel time along the corridor and time savings in with metro and without metro scenario

The results of this analysis have been presented in the summary Table 3.9.

Table 3.9: Analysis of Vehicular Traffic along the Corridors

Rapid Transit Corridors (RTC)	Approx. Length (km)	Passenger demand on the corridor (PHPDT)		Existing Predominant Land use characteristics along the corridor	Proposed Road widths (m) in CMP	% of corridor congested corridor (2046)		Pvt. Veh. Time in Min		Vehicular travel time saving (with versus without metro)
		2026	2046			With metro	Without metro	With metro	Without metro	
1. Dream City to Kamrej/Vav	**29**	**16000**	**34000**			**29%**	**43%**	**78.24**	**88.01**	**11%**
A. Dream city to Majura Gate	8	7300	19500	Residential and commercial	Existing is 18 - 24 and proposed to be widened to 30m in CMP	55%	69%	23.0	22.9	0%
B. Majura Gate to Railway Station	4	16000	33500	Commercial and historically developed local markets	Existing is 15 - 24 m	61%	76%	18.74	22.53	17%
C. Railway Stn to Sarthana	9	15000	34000	Diamond industries, Textile looms & Industries & commercial	Existing is 36 - 60 and proposed to be widened to 45 - 60 m in CMP	15%	38%	19.53	24.21	19%
D. Sarthana to Kamrej/ Vav	7	4000	11000	Under development phase, opportunity to develop as TOD	Existing Sarthana to Kamrej - 60 m Proposed from Sarthana to Vav - 45 m	0%	2%	16.97	18.37	8%
2. Bhesan to Umbhel	**28**	**6500**	**24000**			**28%**	**42%**	**81.1**	**88.9**	**9%**
A. Bhesan to Adajan	7	5800	16500	Residential and mixed land use with recreational activities	Existing 24 - 36 m and proposed to be widened to 30 - 36 m in CMP	12%	18%	15.6	17.5	11%

Rapid Transit Corridors (RTC)	Approx. Length (km)	Passenger demand on the corridor (PHPDT)		Existing Predominant Land use characteristics along the corridor	Proposed Road widths (m) in CMP	% of corridor congested corridor (2046)		Pvt. Veh. Time in Min		Vehicular travel time saving (with versus without metro)
		2026	2046			With metro	Without metro	With metro	Without metro	
B. Adajan to Saroli	11	6500	24000	Commercial, Textile markets & industries and public buildings	Existing 36 - 60 m and proposed to be widened to 60 m in CMP throughout	69%	88%	43.5	48.6	10%
C. Saroli to Umbhel	8	3000	10000	Mostly vacant	45 m RoW proposed in CMP	0%	15%	22.0	22.9	4%
3. Majura Gate to Karmala	**15**	**4400**	**19500**			**25%**	**44%**	**46.1**	**59.6**	**23%**
A. Majura Gate to Ved	7	4400	19500	Residential and Katargam textile power looms	Existing 18 - 36 m Proposed to 36 m (except section in old city)	48%	64%	26.2	34.3	24%
B. Ved to Karmala	9	4300	19200	Mostly vacant	Existing - No connection across river 45 m proposed in CMP with river bridge connection	0%	22%	19.9	25.4	22%

3.5.5. Observations

Rapid Transit Corridor 1: Dream City to Vav

Rapid Transit Corridor 1 section from Majura Gate to Sarthana section has the maximum PHPDT about 34000 in year 2046 which is considerably high. In the year 2026 also, the demand is 15000 PHPDT. In order to accommodate passenger flow of this magnitude, a metro system is more appropriate. The section from Dream City to Majura Gate is not fully developed as yet and has the proposed diamond hub 'Dream City' coming up in the future. The demand along this section is therefore expected to increase from 7300 PHPDT in 2026 to 19500 PHPDT in 2046. The rapid transit corridor in this section would support the proposed Dream City development and hence can be considered for metro along with Majura Gate to Sarthana section.

The demand build up from Sarthana-Kamrej/Vav is comparatively slower as this section is currently vacant and the development of the area is expected to happen post 2026. This section can therefore considered for metro in Phase II. With and without project analysis shows that this corridor if planned as a metro, would reduce the congested network length to 30% in comparison to 44% in the 'Without metro' scenario. The travel time saving along this corridor would be around 10% in with metro scenario.

Rapid Transit Corridor 2: Bhesan to Umbhel

Bhesan –Umbhel rapid transit corridor connects western part of the city to the city centre and commercial area coming up along Saroli corridor. As presented in the summary above, the section from Bhesan to Saroli (i.e. A & B) has 16500-24000 PHPDT in 2046 according to demand assessment. In the year 2026, the passenger demand is around 5800-6500 PHPDT. Saroli-Umbhel section demand in 2026 is 3000 PHPDT increasing to 10000 PHPDT. However, as and when the High Speed Rail corridor and station comes up, external passenger demand would also get added to this corridor. With and without project scenario shows an overall 10% benefit in terms of private vehicular travel time along the corridor.

Since the passenger flow from Bhesan-Saroli section is more than 15000 PHPDT in 2046, it is proposed to consider metro as the mode for Bhesan-Saroli section. Extension of this corridor to Umbhel may be considered in Phase II based on HSR project implementation and demand analysis.

Rapid Transit Corridor 3: Majura Gate to Karmala

This corridor has a passenger demand of around 19000 PHPDT in year 2046. Majura Gate to Ved section is about 7km, which is predominantly residential in the Ved area and has textile industries and power looms towards Katargam. Looking at the demand, this corridor can be recommended for metro system, however owing to road width constraints in the Majura Gate to Katargam section, underground metro may have to be considered. The section from Ved to Karmala is completely vacant and currently there is no connection across the river. This proposed growth towards north of the city may take time to build up. The 2026

demand for both the sections is not very high (around 4400 PHPDT). This corridor may hence be considered in Phase II. In terms of travel time, significant savings to the tune of 25% is observed.

3.5.6. Economic Effects (Social Cost Benefit analysis)

Economic analysis have been undertaken to assess economic effects of alternative transit system options for identified corridors. Considering the ridership potential of the identified two corridors, two different transport system alternatives namely (i) Metro and (ii) BRT have been analyzed through costs and benefits perspectives as given in Table 3.10.

Table 3.10: Ridership Potential

Year	Ridership (in lakh)	
	Metro	BRT
2024	2.8	2.7
2028	7.4	4.1
2038	14.3	7.2
2048	18.3	9

This section attempts to estimate the incremental benefits of metro project over BRT Project for two identified corridors. The incremental social benefits can be divided into direct and indirect benefits as given in Table 3.11.

Table 3.11: Ridership Quantifiable and Non Quantifiable Benefits

Sr. No	Benefits	Incremental Direct Benefit due to Metro Project	Indirect benefits due to decongestion on other modes/routes owing to Metro Project
1	Lower Vehicle Operating Cost (VOC) (including Capex in Vehicle, Maintenance cost and Fuel Saving)	The Metro Project is likely to contribute to a diversion of a larger number of passengers from private and other public modes to the metro as compared to the BRT alternative. This will lead to more pronounced savings in vehicle operating costs (VOC).	VOC would also get reduced for vehicles plying on the mixed traffic zone due to decongested roads. The effect would be more pronounced in case of a metro alternative.
2	Passenger time saving	Speed of metro is higher in case of a metro leading to higher savings in terms of Vehicle Operating Time (VOT) as compared to BRTS alternative. The diversion of passengers is higher in case of metro as compared to BRT. Thus, aggregate VOT benefits would be higher, Reduction of waiting time for trips diverted to Metro from Intermediate Para Transit, two wheelers, cars as compared to BRT owing to higher frequency, more reliability and speed.	Time savings to mixed lane vehicles due to decongestion. Decongestion benefits likely to be higher in case of metro hence enhancing the VOT per vehicle plying on mixed lane VOT benefits will be enhanced at the aggregate level due to higher metro ridership v/s in a BRT.

Sr. No	Benefits	Incremental Direct Benefit due to Metro Project	Indirect benefits due to decongestion on other modes/routes owing to Metro Project
3	Reduction in Pollution	Decreased carbon emissions from higher number of diverted vehicles on Metro as compared to BRT alternative. Also, BRT Buses would also generate emissions. All such emissions would not happen in case of Metro.	Number of total vehicles would be reduced due to Public Transport. This reduction would be higher in case of metro, leading to higher speed and high fuel efficiency in mixed lane and hence Lower Carbon emissions.
4	Reduction in accidents	Lower accidents due to decrease in no, of vehicles of on the project corridor owing to trips diverted to Metro	-
5	Better access to workplace due to TOD	Shorter trip distances for workers employed on TOD stations, employment in Metro case etc as compared to BRT base.	
6	Econ. Impetus to micro region	Better and faster accessibility due to Metro Service may enhance labour pool and skill availability with multiplier benefits	Improved accessibility due to decongested roads/other modes may enhance labour pool and skill availability with multiplier benefits.
7	Overall increased mobility	Better quality of life to citizens, particularly to daily commuters, women, students, elderly and disabled.	Benefits resulting from reduced congestion captured.
8	Benefits to City Image	Would improve city image attracting higher investments and businesses, could decrease outmigration and increase immigration.	
9	Better Comfort Level to Passengers commuting through City Bus Service	Improved quality of services, ease, reduction in crowding owing to higher frequency and speed. These factors enhance Comfort Level to Passengers.	
10	Indirect benefits of Reduce Pollution to Population leaving around project corridor	Diversion of Traffic will also contribute to reduced congestion and pollution thereof.	

Source: Alternative Analysis Report, CEPT University,Ahmedabad

3.5.7. Conclusions

Based on the demand analysis, it can be concluded that all three corridors may be considered for metro. However, in the first phase following corridor /corridor sections are recommended:

- Dream City- Sarthana (21.61 km)
- Bhesan-Saroli (18.74 km)

For Phase II, Majura Gate- Ved can be considered. The extension to Karamala based on road network development and demand build can be considered after demand assessment.

Similarly, extensions from Sarthana-Kamrej/Vav and Saroli-Umbhel can be taken up in the subsequent phase.

3.5.8. Environmental and Social Considerations for Alternative Analysis

As per provisions of the EIA Notification, 14th September 2006 and its amendments, any person who desires to undertake any new project in any part of India or the expansion or modernization of any existing industry or project listed in Schedule-I of the said notification shall submit an application to the MOEFCC, Government of India in accordance with the guidelines issued by the Central Government in the MoEFCC from time to time. Metro Rail project is not included in the Schedule-I of the EIA Notification, 2006. Thus, the project does not require an environmental clearance certificate from the MoEFCC.

However, Appraisal Guidelines for Metro Rail Project Proposals, issued by Ministry of Housing & Urban Affairs (MoHUA), Government of India (September 2017); recommends a preliminary environmental analysis of the project, to identify environmentally sensitive areas early on, so that these areas can be avoided if possible during design.

For the potential metro corridors, a preliminary environmental analysis of the project has been done. This is to identify environmentally sensitive areas early on, so that these areas can be avoided if possible during design. During the DPR preparation the section finalized based various design considerations and alternative construction techniques to reduce the impact on environmental and social aspects. These details are described as under:

A. Design Considerations

I. Corridor-I: Sarthana to Dream City

Underground Section: This corridor runs from North-Eastern parts of Surat to Central to Southern parts of Surat. From chainage 3805.00m to chainage10270m, section has been envisaged as Underground section due to inadequate space available on road in the areas like Kapodra, Labheshwar Chowk, Central Warehouse, Surat Railway Station, Muskati Hospital and Gandhi Baug and Row is less than 20m through which alignment is passing and planning elevated metro alignment on such narrow road will involve demolition of large number of structures. It is also required to provide metro stations at interval of approximately 1.00 Km, in this stretch to serve inhabitants, which further requires more width and resulting in more demolitions.

The alignment is critical from alignment point of view near to Labheshwar Chowk Station and Moskati Hospital Station due to space constraints and heavy congestion on road areas and to save dismantling of large number of structures. These two underground stations have been planned on side platform pattern for UP line and Down line at different levels i.e. one-over other. To accommodate these two level stations up line and down line tunnels have been planned one above other by tapering the inter tunnel distance on both sides of the stations.

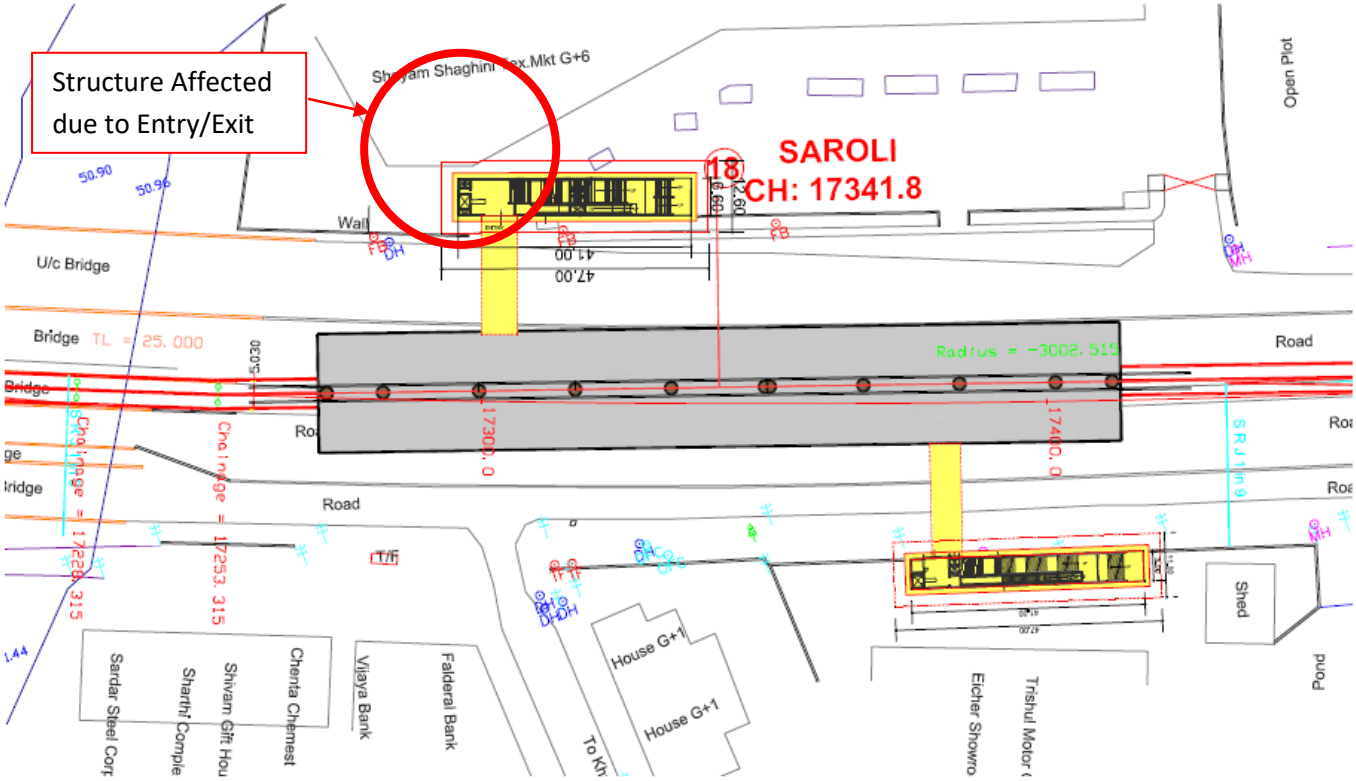
Elevated Section: In order to minimize land acquisitions and to provide good accessibility from either direction, the metro alignments are located mostly along the center of the roads

with single pier supporting the viaduct so that the existing roads remain in use as usual. But, at some locations the geometrics of the roads especially at road turnings may not match with geometric parameters required for metro rail systems. In such cases, either the alignment will be off the road or some properties abutting the road would get affected.

II. Corridor 2: Bhesan to Saroli

This Corridor runs from North-Western parts of Surat to Central area and finally terminates towards Eastern parts of Surat. The entire corridor is elevated corridor. As per Surat DPR-2018, Saroli Station's entry/exit S-1 (Area -595.90 sq.m) is proposed in area of Shyam Sangini Textile Market, which is G+6 with basement Parking. During the Site visit by GMRCL team from 05/12/2019 to 07/12/2019 with RITES & SMC representatives for SIA survey, it is found that Entry/exit of the station is partially affecting Shyam Sangini Textile Market and it may be needed to partially demolish the building which may affect 48 shops. The location of Entry/Exit of Saroli Station is shown in Figure 3.8. In order to reduce the demolition of building and to reduce R&R issues, it is being proposed to relocate the exit as there is enough open land available and road width is sufficient to build Entry/exit structures.

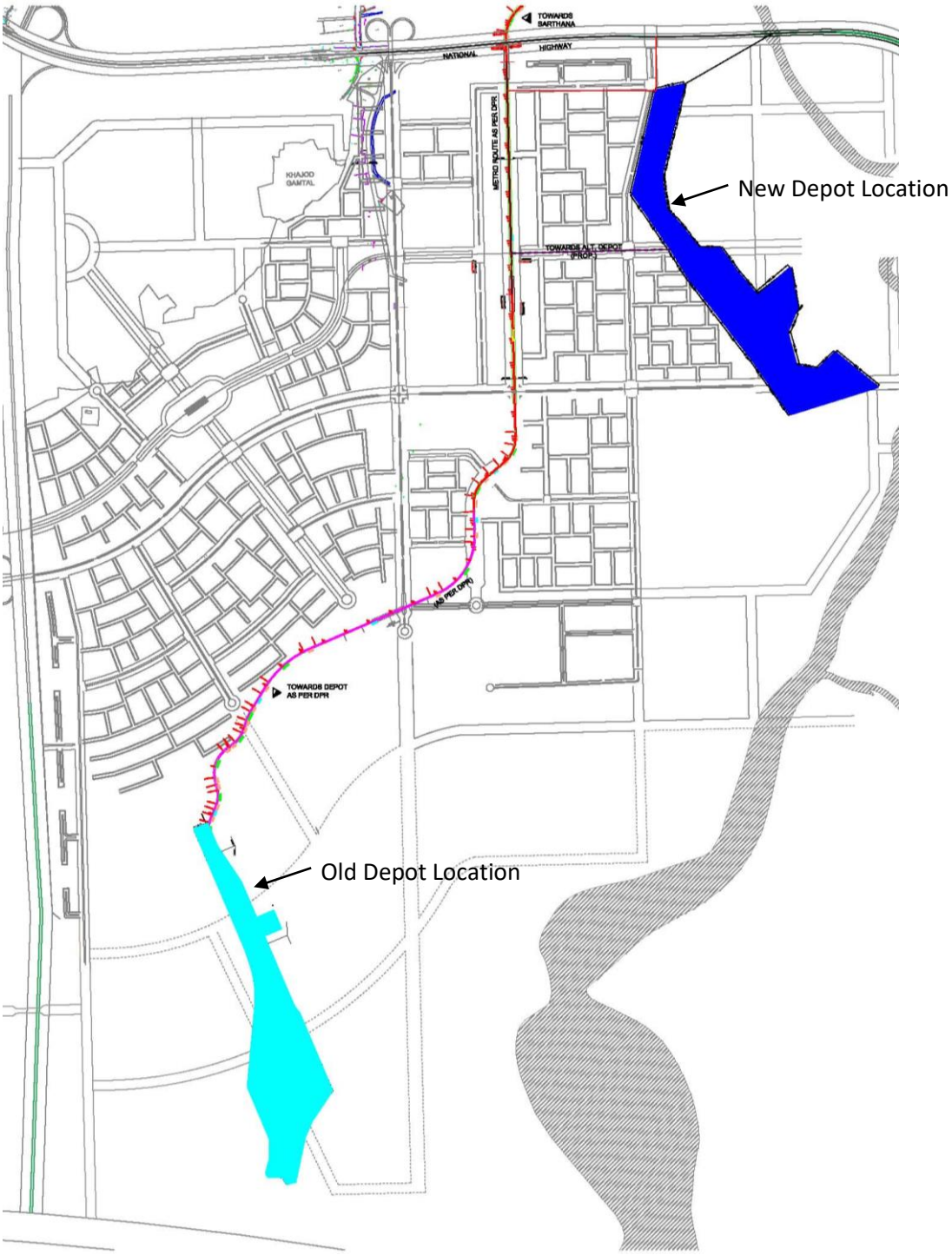
Figure 3.8: Entry/Exit of Saroli Station



III. Depot Planning

Design Interventions carried out at Dream City Depot and along the alignment in roder to reduce the impact on ecology, PAFs and land acquisition. The depot proposed in the DPR was approximately 3 km away from Dream City Metro stations and is located in low laying marshy land. The new depot proposed is approximately 700 mts away from the Metro Station and easy to approach from the main road. The drawing showing the proposed new depot location is given at Figure 3.9. The new depot location is identified at a location, where there is minimal marshy land and no land acquisition is involved.

Figure 3.9: Dream City Depot Location



B. Alternative Construction Techniques

Underground Alignment: For this stretch, cut and cover method of the underground construction can be employed for the construction of the underground sections. However keeping in view the inconvenience to the traffic movement and land acquisition, it is proposed to use Tunnel Boring Machine (TBM) or New Austrian Tunneling Method (NATM) in the overburden soil mass. Tunnel excavation for a major length of underground section is expected to be carried out by Tunnel Boring Machines.

During tunneling, the use of precast concrete tunnel segments has been widely used due to its efficient and economical installation process compared to that of normal cast in-situ lining practice and also reduces the construction time at site and less exposed the pollution during construction.

Underground Stations: No land at surface is required permanently for underground section, except for small areas for entry/exit structures, traffic integration, chilling plant and ventilation shafts at stations. To the extent possible these will be located either on footpath edge or in front marginal open setback of the building along the road. All the 06 underground stations have been proposed to be constructed by cut and cover with top-down or bottom up methods. The diaphragm walls for such station constructions would be 80 to 120 cm. thick and will function as a permanent side wall of the station. By resorting to top-down method the surface could be restored quickly and further excavations and construction of the station will not hamper the surface activities which provide the early traffic restoration and less exposed the pollution during construction.

Elevated Section: The choice of superstructure has to be made keeping in view the ease of Constructability and the maximum standardization of the formwork for a wide span ranges in order to reduce the traffic congestion at work sites, to reduce air and noise pollutions. The segmental construction technology is proposed for construction of the corridors due to the following advantages:

- Segmental construction is an efficient and economical method for a large range of span lengths and types of structures. Structures with sharp curves and variable super elevation can be easily accommodated.
- Segmental construction permits a reduction of construction time as segments may be manufactured while substructure work proceeds and assembled rapidly thereafter.
- The area required is very less for this technology in compared to cast in situ technology. The superstructure is manufactured at a place away from busy areas and placement of superstructure is done with the system erected from piers at heights.
- It is easier to transport smaller segments by road trailers on city roads.
- It is easy to incorporate last minute changes in span configuration if the site situation so warrants.
- Interference to the traffic during construction is significantly reduced.
- Segmental construction contributes towards aesthetically pleasing structures and good finishes.
- The overall labour requirement is less than that for conventional methods.
- Better quality control is possible in the casting yard

- During construction, the technique shows an exceptionally high record of safety.
- The method is environment friendly as no concreting work is carried at site for the superstructure.

Elevated Stations: Elevated station is generally proposed with elevated concourse so that land is required only for locating the entry/exit structures. To the extent possible the Entry and Exit points of stations (underground and elevated) were planned on the foot paths. Traffic integration facilities are provided wherever required and, but no land is proposed for acquisition. In view of the constraints on space on ground floor, it is proposed to provide the concourse area exactly below the Station Building at mezzanine level. All the stations in elevated stretch including terminal station are planned with side platforms. The staircase giving access to concourse area from ground will be located at the edge of footpaths or in front marginal open setback of the buildings in the as far as possible in the open space. Nevertheless it is not possible to find open space at all the locations therefore acquisition of certain private structures is inevitable.

In the cross section of station structure, viaduct column will be located on the median/footpath and other supporting columns/piers shall be provided on other sides of the road within ROW. In some the cases where space is major constraint, the cross section is supported by a single viaduct column and cantilever arms to eliminate the columns in the right of way.

Chapter 4 : Environmental Baseline Data

4.1. General

Environmental baseline data describes the existing environmental settings in the study area. The objective of the Environmental Impact Assessment (EIA) is to ascertain the baseline environmental conditions and then assess the impacts as a result of the proposed project during various phases of the project cycle. The environmental baseline data has been compiled for:

- Land Environment (Physiography, Soils, Geology and Minerals)
- Water Environment (Water resources, water use and quality)
- Ambient Environment (Meteorology, Ambient Air Quality and Noise Quality)
- Ecological Environment (Flora and Fauna)

Environmental baseline data includes the physical, biological and socio-economic data. The data collection was carried out in the months from April 2019 to June 2019. A scoping matrix along with the frequency adopted for data collection for environmental attributes is summarized in Table 4.1. Based on environmental scoping matrix and project settings the attributes likely to be affected were identified for baseline data generation.

Table 4.1: Environmental Attributes and Frequency of Monitoring

S. No	Attribute	Parameter	Frequency	Source
LAND ENVIRONMENT				
1.	Soil	Soil Characteristics	One Season	Field studies and Detailed project report
2.	Geology	Geological Status	---	Secondary Data
3.	Seismology	Seismic Hazard	---	Secondary Data
WATER ENVIRONMENT				
4.	Water Quality	Physical, Chemical and Biological parameters	One Season	Field studies
AIR, NOISE AND METEOROLOGY				
5.	Ambient Air Quality	PM2.5, PM10, SO_2,, NO_2, and CO	One Season	Field Studies
6.	Meteorology	Temperature, RH, Rainfall, wind direction and speed	Last five years	India Meteorological Department (IMD)
7.	Vibration	Peak Particle Velocity	One season	Field Studies
8.	Noise	Noise levels in dB (A)	One Season	Field Studies
ECOLOGY				
9.	Trees	Number/species	Once	Filed Studies

4.2. Land Environment

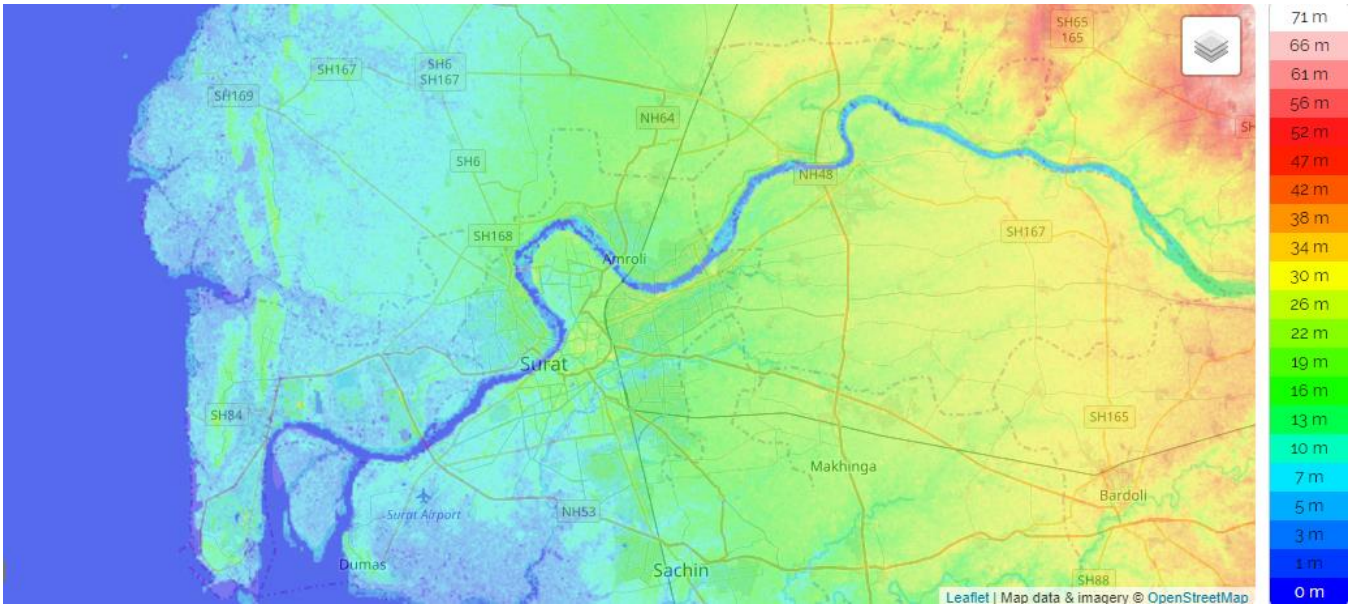
Field studies were carried out towards collection of baseline data with respect to physical environment viz. physiography, geology, soils, minerals, drainage, land use pattern and

seismicity. The data on physical environment was collected from existing literature and from field observations.

4.2.1. Physiography

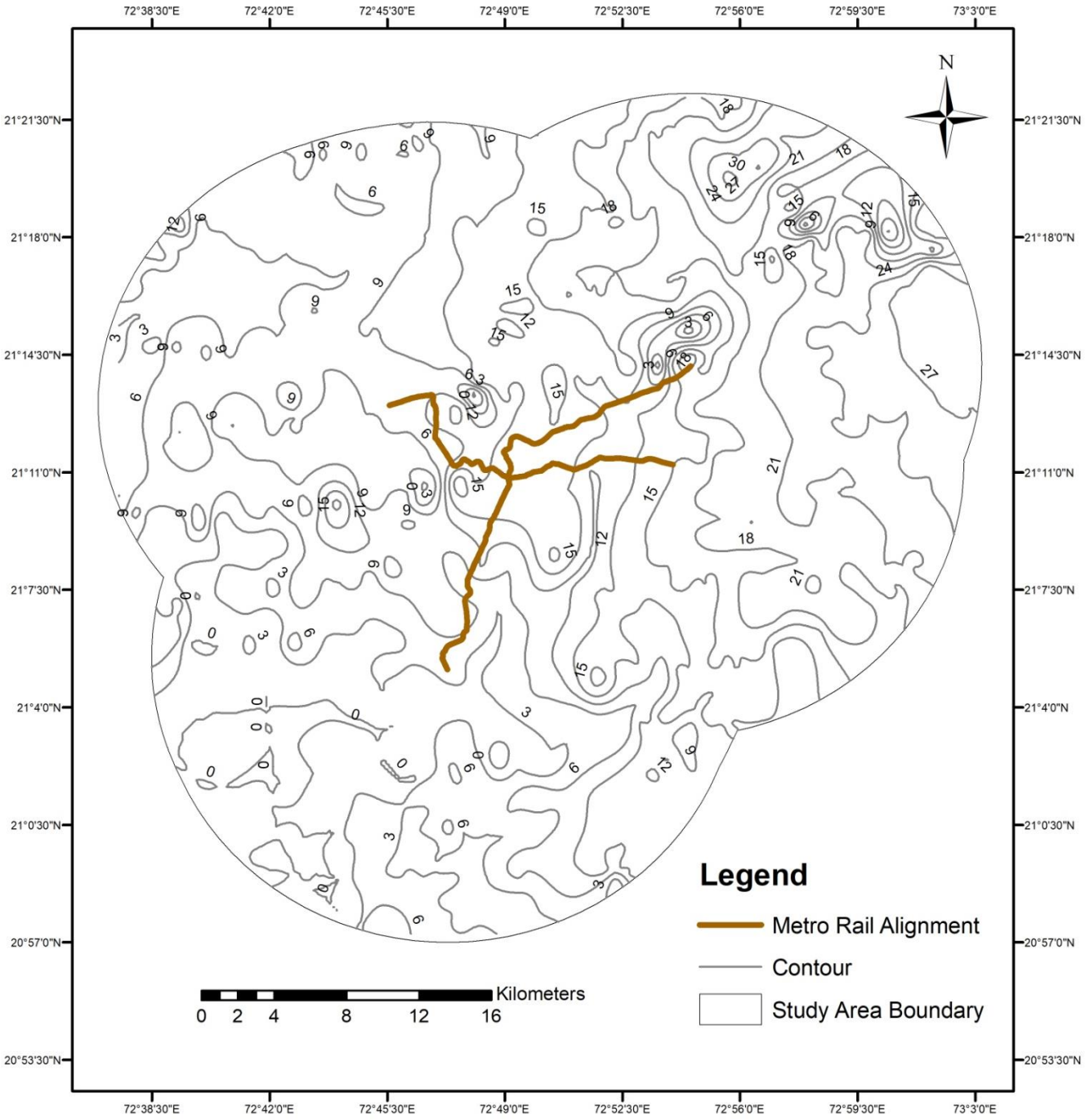
Surat is a port city situated on the banks of the Tapti River which was formally known as Tapi River. It is a major river in peninsular India with length of 724 km. It runs from east to west. The river rises in the Satpura range of Madhya Pradesh and flows outwards and empties itself into the gulf of Cambay. Damming of the Tapi caused the original port facilities to close; the nearest port is now in the Magadalla and Hazira area of Surat Metropolitan Region. The city is located at 21°10'N and 72°50'E. It has an average elevation of 13 metres. The Arebian Sea is only 16 km away from the city and Magadalla port is near vicinity. The physiographical map of the study area is shown in Figure 4.1. The Surat district is surrounded by the Bharuch, Narmada, Navsari, to the west is the Gulf of Cambay and the surrounding districts. According to the Bureau of Indian Standards, the town falls under seismic zone-III.

Figure 4.1: Physiographic Map of Study Area



Contour map of the study area with a 15 km radius along proposed Metro Rail Corridors is shown in Figure 4.2 with the contour lines having 3 m interval. From the figure it is depicted that the project area is with elevation range of 0m to 30m. However as per the current ground levels, the proposed metro alignment is passing through 6m to 18m elevations.

Figure 4.2: Contour Map of the Study Area



4.2.2. Geology and Hydrogeology

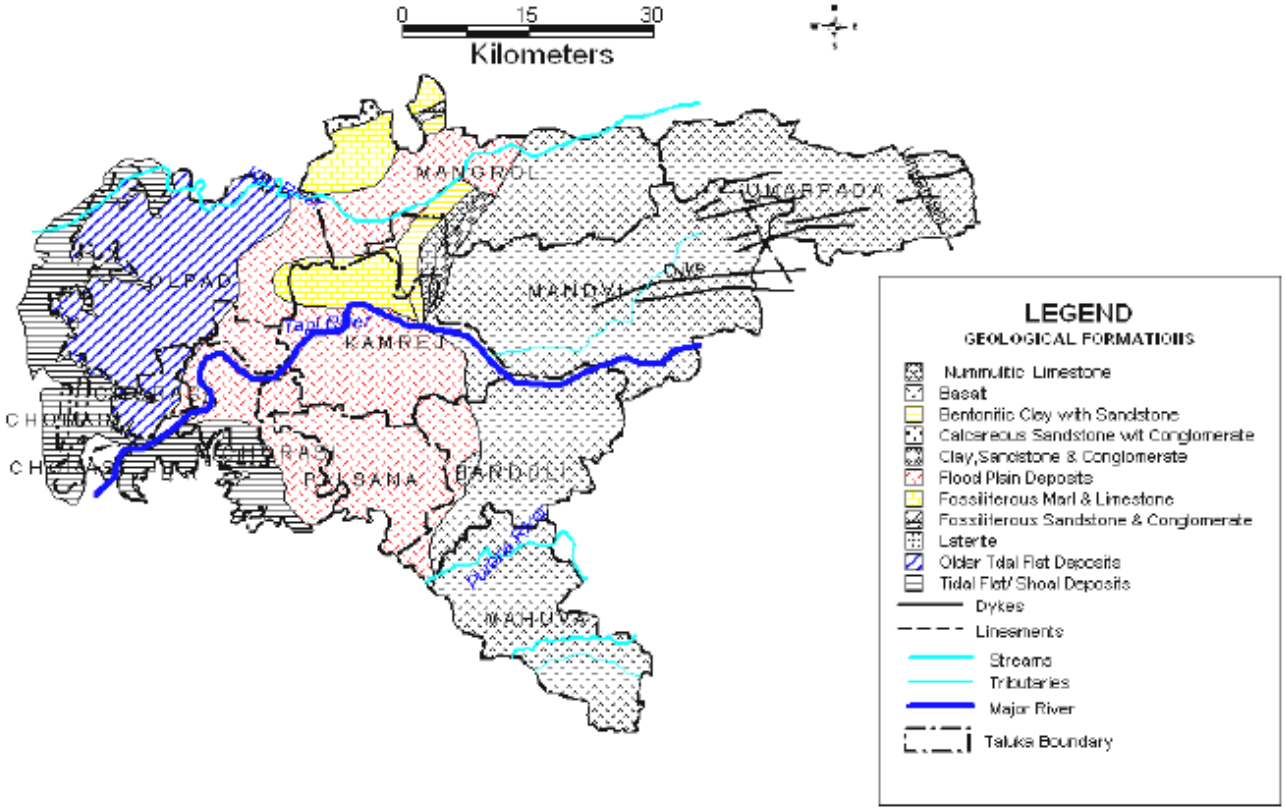
Major geological formations exposed in the district are Quaternary alluvium, Tertiary limestone and sandstones and Deccan Trap basalt. The geology of the district and the succession of geological formations in the district, as given by Geological Survey of India in their brochure titled " Know Your District-Surat" is given in Table 4.2 and depicted in Figure 4.3. Surat comes under cenozoic era of tertiary period of Palaeocene series/epoch.

Table 4.2: Geological Succession of Surat District

Era	Period	Epoch /Series	Description
Cenozoic	Quaternary	Holocene	Soil, Younger alluvium Blown and Fluvio-marine deposits.
	Tertiary	Pleistocene	Older alluvium, conglomerate
		Miocene	Ferruginous sandstone (Gaj)
		Eocene	Numulitic limestone
		Palaeocene	Sub-Numulitic limestone, gypseous shale,Supra- Trappean sediments lateritised and having bauxite
Mesozoic	Lower Tertiary to Upper Cretaceous	Early Eocene To Upper	Deccan Trap basalt with dykes

Source: CGWB, Surat District

Figure 4.3: Geological Map of Surat District



Source: CGWB, Surat District

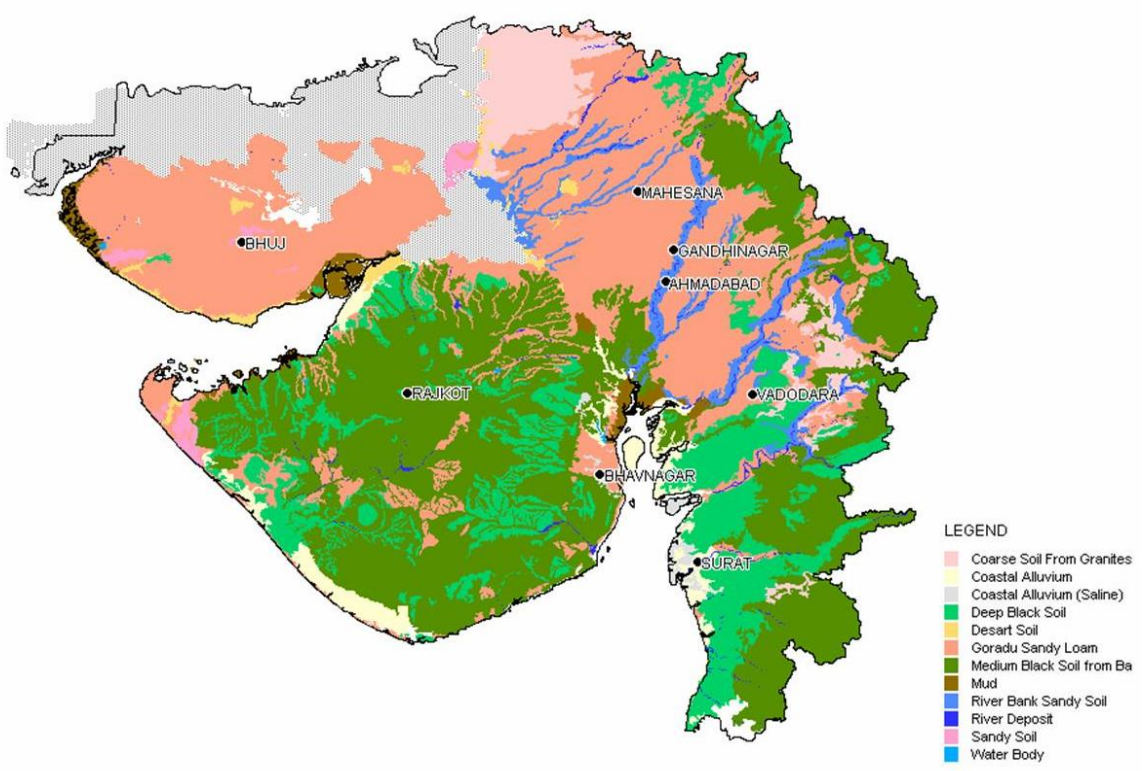
The hydro geological frame work of the area is essentially governed by geological setting, distribution of rainfall fall and facilities of circulation and movement of water through inter connected primary and secondary porosity of the geological units forming the aquifers. The major aquifers in the district are formed by alluvium and Deccan Trap basalt with Tertiary formations occupying a small patch. The alluvium occurs in the western part of the district and along the streams. In Alluvial areas the depth of the tube wells range from 36 to 169m with piezometric levels varying between 2 to 10m. The specific capacity of the wells ranges from 19 to 1658 m3/day/m.

4.2.3. Soils

Soil is the material found on the surface of the earth that is composed of organic and inorganic material. Soil varies due to its structure and composition. The physical and chemical characters of the parent rock, physiography, altitude, climatic condition and plants and animals of the surrounding region influence the process of soil formation.

In coastal region the soil is deep to very deep dark grey to black colour. The texture is clay loam to silty loam. The area is affected by tide as well as leaching of salts from up land forming saline alkali soils. The content of clay in this is high and permeability is low (mud flats). To have the general idea soil map of Gujrat state is shown in Figure 4.4. Figure dipects that Surat having costal alluvium and costal alluvium (saline) soils.

Figure 4.4: Soil Map of Gujarat



Source: Narmada, Water Resources, Water Supply and Kalpsar Department (Water Resources Division)

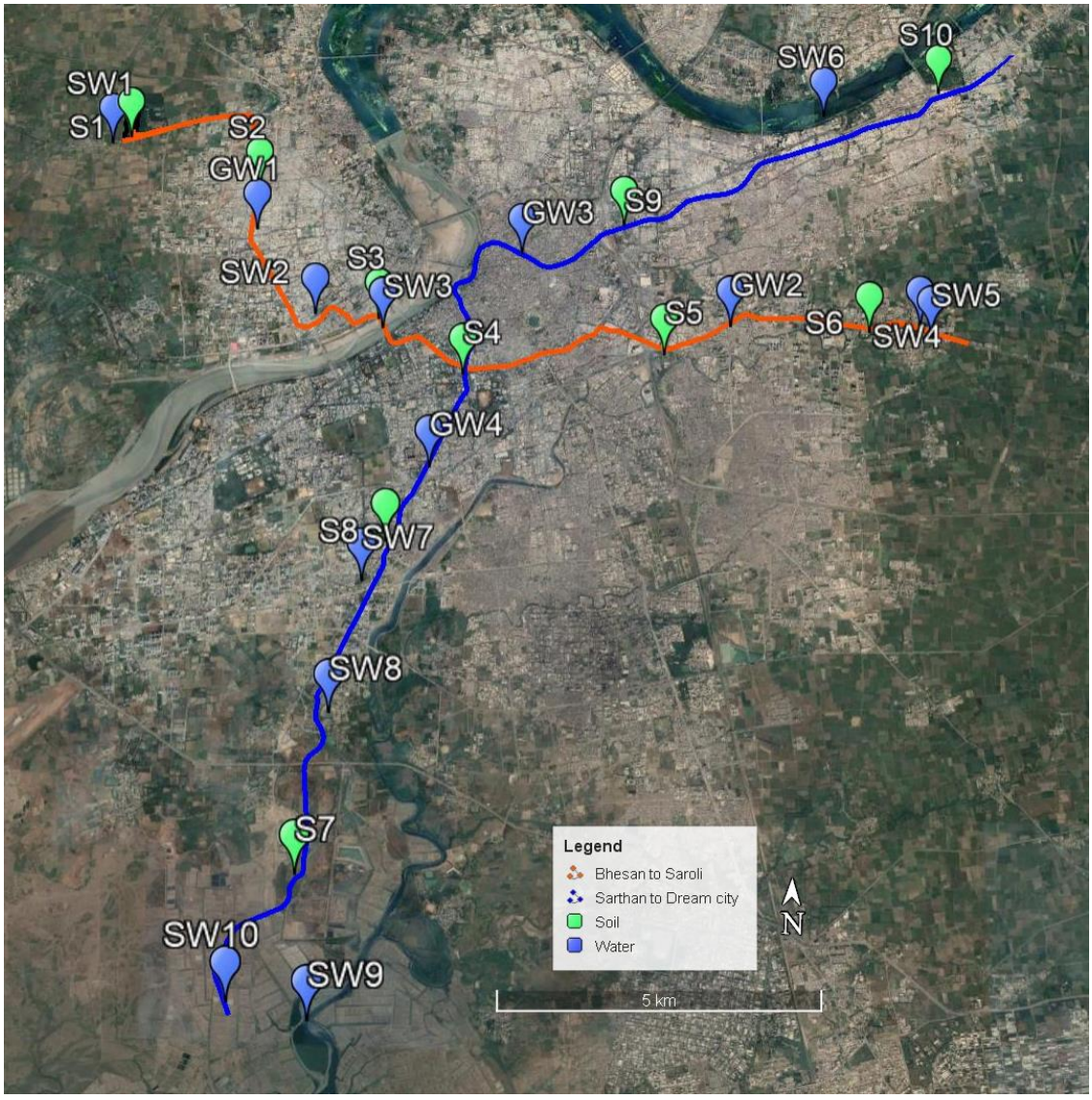
To get the actual characteristics of soil in the project area, soil samples from various locations were collected and analyzed to understand the soil characteristics of the area with respect to its productivity potential. Selection of location for soil sampling is done on the basis of land use/land cover. Project is proposed in built up and or paved area however there is an open spaces available. Soil samples have been collected from these open spaces near office complexes and agricultural land etc. Ten representative soil samples have been collected along the proposed metro alignment during 29/5/2019 to 7/6/2019. The sampling locations are given in Table 4.3 and sampling location map for soil along with water is shown in Figure 4.5. The laboratory analysis results so obtained are summarized in Table 4.4.

Table 4.3: Description of Soil Quality Monitoring Locations

Location Number	Location Details	Latitude	Longitude
Bhesan to Saroli			
S1	Near Water Treatment Plant, Bhesan	21°13'3.02"N	72°45'38.78"E
S2	Near Palanpur Jakatnaka	21°12'34.34"N	72°46'58.67"E
S3	Near Badrinath temple (Tapi River)	21°11'17.96"N	72°48'14.67"E
S4	Near Gandhi college	21°10'47.13"N	72°49'5.53"E
S5	Ambedkar Nagar	21°11'2.02"N	72°51'2.35"E
S6	Transport Nagar	21°11'18.55"N	72°53'2.12"E
Sarthana to Dream City			
S7	Dream City	21°9'12.53"N	72°48'26.10"E
S8	Althan Forest, Althan	21°6'17.18"N	72°47'47.36"E
S9	Near Railway Station	21°12'17.18"N	72°50'37.16"E
S10	Near Nature Park and Zoo	21°13'3.02"N	72°45'38.78"E

S- Soil

Figure 4.5: Water And Soil Sample Monitoring Locations



SW- Surface Water, GW – Ground Water, S- Soil

Table 4.4: Soil Quality Data

S. No.	Parameters	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
1.	Texure Sand, %	55.00	56.00	42.00	52.00	56.00	57.00	57.00	35.00	58.00	38.00
	Texture Silt, %	13.00	16.00	8.00	16.00	18.00	18.00	18.00	11.00	10.00	15.00
	Texure Clay, %	32.00	28.00	50.00	28.00	26.00	25.00	25.00	54.00	32.00	47.00
	Texture class	Sandy clay loam	Sandy clay loam	Clay	Sandy clay loam	Sandy clay loam	Sandy clay loam	Sandy clay loam	Clay	Sandy clay loam	Clay
2.	pH	7.36	7.23	7.3	9.4	7.99	7.60	6.98	7.90	8.20	7.40
3.	Electrical Conductivity, dS/m	0.41	1.10	26.6	0.61	0.25	0.43	0.87	0.41	0.58	0.54
4.	Chloride, mg/100g	88.85	106.6	2355	37.32	46.2	31.9	44.4	37.3	55.0	30.2
5.	Available Phosphorus, mg/100gm	0.32	0.06	0.37	0.21	0.22	0.77	0.71	0.14	0.04	0.13
6.	Magnesium, Meq/100gm	6.8	13.5	23.4	8.2	5.9	6.4	8.8	6.3	7.2	7.2
7.	Sodium, Meq/100gm	2.3	2.9	58.4	1.1	1.4	1.0	1.8	1.3	2.2	2.2
8.	Potassium, Meq/100gm	0.4	0.3	0.1	0.4	0.3	0.35	0.46	0.4	0.37	0.37
9.	Organic carbon, %	0.39	0.67	0.67	0.50	0.11	0.38	0.35	0.94	0.46	0.73
10.	Calcium, Meq/100gm	12.3	15.4	18.7	9.4	7.6	8.1	9.2	7.4	8.6	8.6
11.	Arsenic as As mg/100gm	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL
12.	Calcium ,meq/100gm	12.3	15.4	18.7	9.4	7.6	8.1	9.2	7.4	8.6	8.6
13.	Available Nitrogen, %	0.10	0.06	0.09	0.08	0.07	0.07	0.08	0.1	0.14	0.14
14.	Boron as B, mg/100gm	BDL	0.01	0.21	BDL	0.01	0.02	0.01	BDL	0.005	0.01
15.	Mercury as Hg, mg/100gm	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL
16.	Molybdenum as Mo, mg/kg	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL
17.	Total Sulphur as SO_3, mg/100gm	4.25	4.03	75.01	4.13	1.81	2.15	3.47	2.74	3.35	3.74
18.	Total Zinc, mg/100gm	0.26	0.31	0.22	0.15	0.15	0.13	0.65	0.13	0.1	0.14
19.	Manganese as Mn, mg/100gm	0.05	0.04	0.77	0.49	0.10	0.04	1.16	1.08	0.53	0.81
20.	Total Lead as Pb, mg/100gm	0.74	0.62	0.39	0.72	0.80	0.40	2.08	0.42	0.31	0.18
21.	Total Copper, mg/100gm	0.84	0.34	0.52	0.23	0.45	0.22	0.82	0.53	0.19	0.25
22.	Nickel, mg/100gm	3.09	3.68	3.32	3.66	4.13	3.29	2.5	3.25	4.35	2.95
23.	Cadmium, mg/100gm	0.09	0.3	0.55	0.54	0.24	0.36	1.71	0.03	0.22	0.04
24.	Iron, mg/100gm	1.71	0.56	0.71	0.65	0.24	0.28	0.94	1.35	0.8	0.62

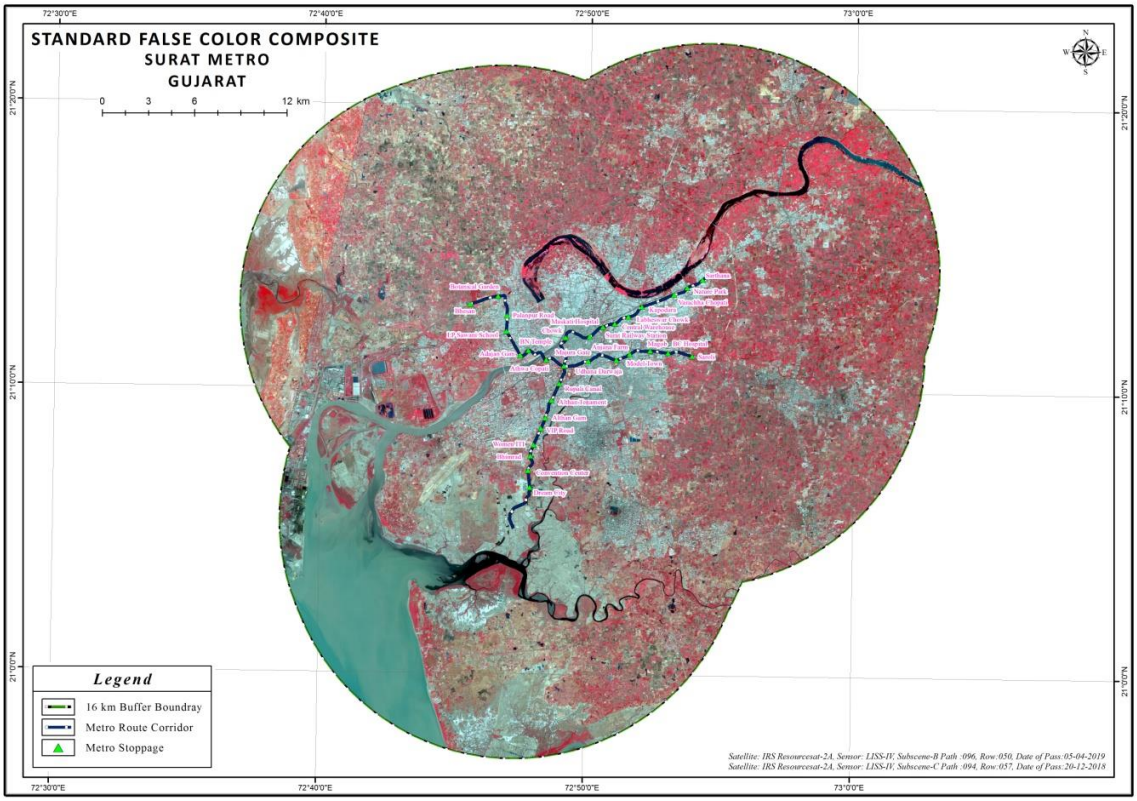
From the laboratory analysis report, following observations noted down in the view of fertility of soils; Soil in the study area is normal soil and some places its saline soil to alkaline soil. Texture class of the soil is clay and sandy clay laom. Sulhur content at one place is very high and it is limit for rest of the places. Available Fe content in the soil ranged from 2.08 – 4.02 mg/kg. All the soil samples were found in sufficient amount of available iron considering 4.5 mg/ kg as a critical limit suggested by Lindsay and Norvell (1978).

4.2.4. Land Use Pattern

The term “Land Use” relates to the human activity or economic function associated with a specific piece of land, while the term “Land Cover” relates to the type of feature present on the surface of the earth (Lillesand and Kiefer, 2000). The satellite-remote-sensing technology has found its acceptance worldwide for rapid resource assessment and monitoring, particularly in the developing world. All these advancements have widened the applicability of Remotely Sensed data in various areas like forest cover, vegetation type mapping and their changes in regional and seasonal scale.

The Area of Interest (AOI) is of 1607 sqkm which comprises of 15 km buffer from the proposed Metro Rail Corridor. The latest satellite imagery of IRS data of LISS-IV Image of Resourcesat-2 having Row 057 and Path 094 and date of pass as 20-12-2018 (Sub-scene-C) and Row 050 and Path 096 and date of pass as 05-04-2019 (Sub-scene-B). Satellite imagery was obtained from the National Remote Sensing Centre (NRSC), Hyderabad (Figure 4.6).

Figure 4.6: Satellite Imagery of the Study Area



The data was in Universal Transverse Mercator (UTM) projection system with spheroid and datum as WGS84 and Zone as 43N. Before image classification satellite images were pre-processed. All the processing and classification of the satellite images were performed in the ERDAS Imagine 9.1 software. The data were delivered in separate bands of 2, 3 and 4.

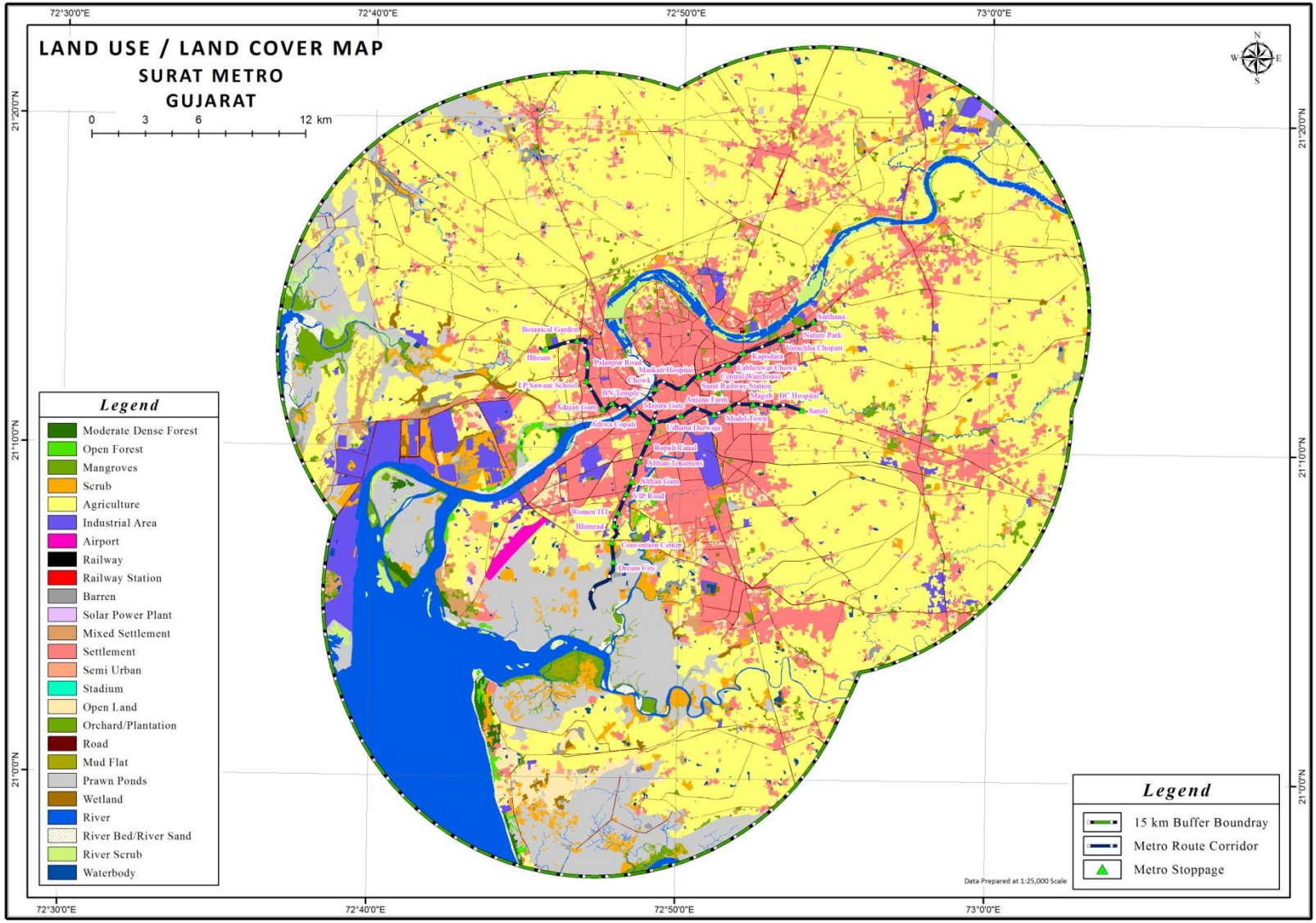
The first step was to stack together to get a composite. The scenes were then mosaic together since the study area covered parts of both the scenes. Once a satisfactory mosaic was done whereby the joins were no longer visible and the radiometric resolution of both the scenes were matched, this subset of the original images was then run through the supervised classification tool of the software to finally create the grouping of classes present in the study area.

The landuse/landcover map was developed based on the image colour, texture, tone and ground truth verification. Ground truth verification was carried out to validate the results of the classified image and reconciliation was carried out with actual location of land mark features such as water bodies, forest land, agriculture land, etc. Statistics of class wise land use land cover is given in Table 4.5 and the land use/ land cover map is shown in Figure 4.7.

Table 4.5: Landuse Class Statistics of Study Area

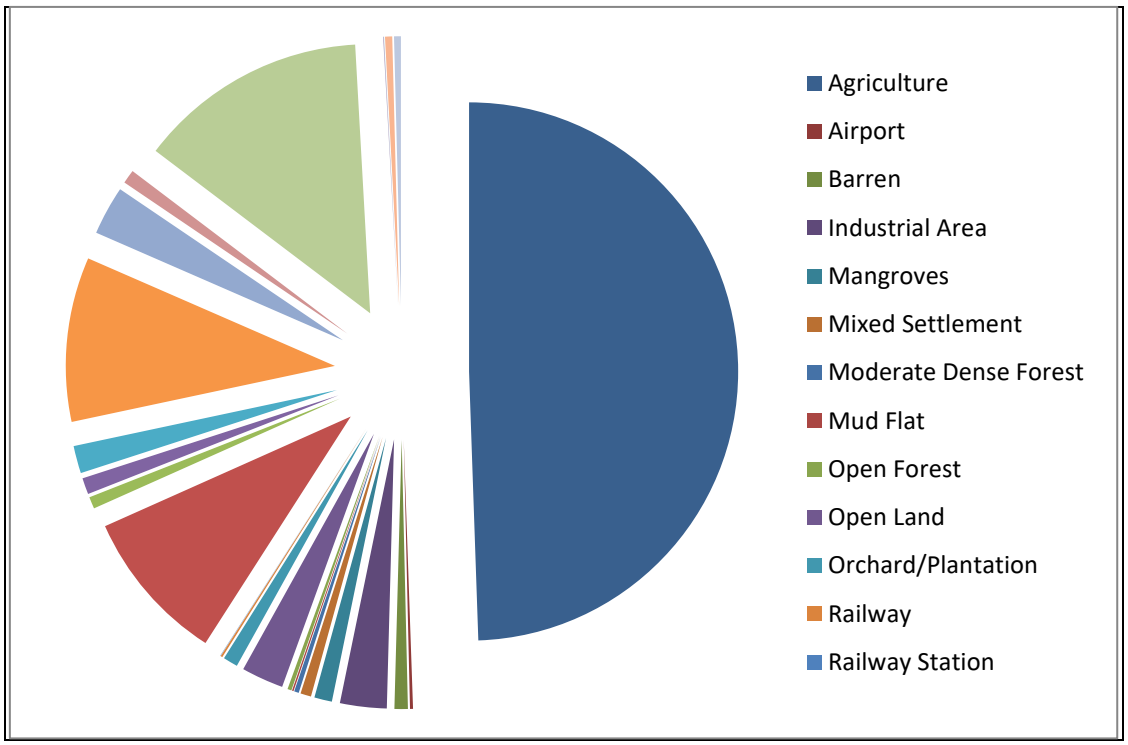
Sl. No	Description	Area (Sq.km)	Area (%)
1	Agriculture	795.04	49.45
2	Airport	2.85	0.18
3	Barren	12.99	0.8
4	Industrial Area	45.04	2.8
5	Mangroves	17.01	1.06
6	Mixed Settlement	10.37	0.64
7	Moderate Dense Forest	4.46	0.28
8	Mud Flat	1.43	0.09
9	Open Forest	3.82	0.24
10	Open Land	40.69	2.53
11	Orchard/Plantation	14.25	0.88
12	Railway	1.75	0.11
13	Railway Station	0.29	0.02
14	River	148.67	9.24
15	River Bed/River Sand	11.49	0.71
16	River Scrub	15.73	0.98
17	Road	26.52	1.65
18	Prawn Ponds	158.88	9.88
19	Scrub	47.21	2.94
20	Semi Urban	13.33	0.83
21	Settlement	221.82	13.8
22	Solar Power Plant	0.7	0.04
23	Stadium	0.1	0.01
24	Water body	6.88	0.43
25	Wetland	6.53	0.41
Grand Total		**1607.84**	**100**

Figure 4.7: Land use/Landcover Map of the Study Area



Graphical Presentation of the land use classification within a 15 km radius of the proposed project is shown in Figure 4.8.

Figure 4.8: Land use Classification of the Corridors



Built Up Area: Built-up area covered about 18.14% of the area within a 15 km radius of the study area. The surrounding villages around the project site are well developed with road electricity and water.

Agricultural Land: Agricultural land covers about 49.45% of the total study area.

Scrubs and Shrubs: It is found that the some part of study area is also covered with scrub covering about 2.94% of total study area.

Forest Area: 0.52% of the total study area is covered by forests.

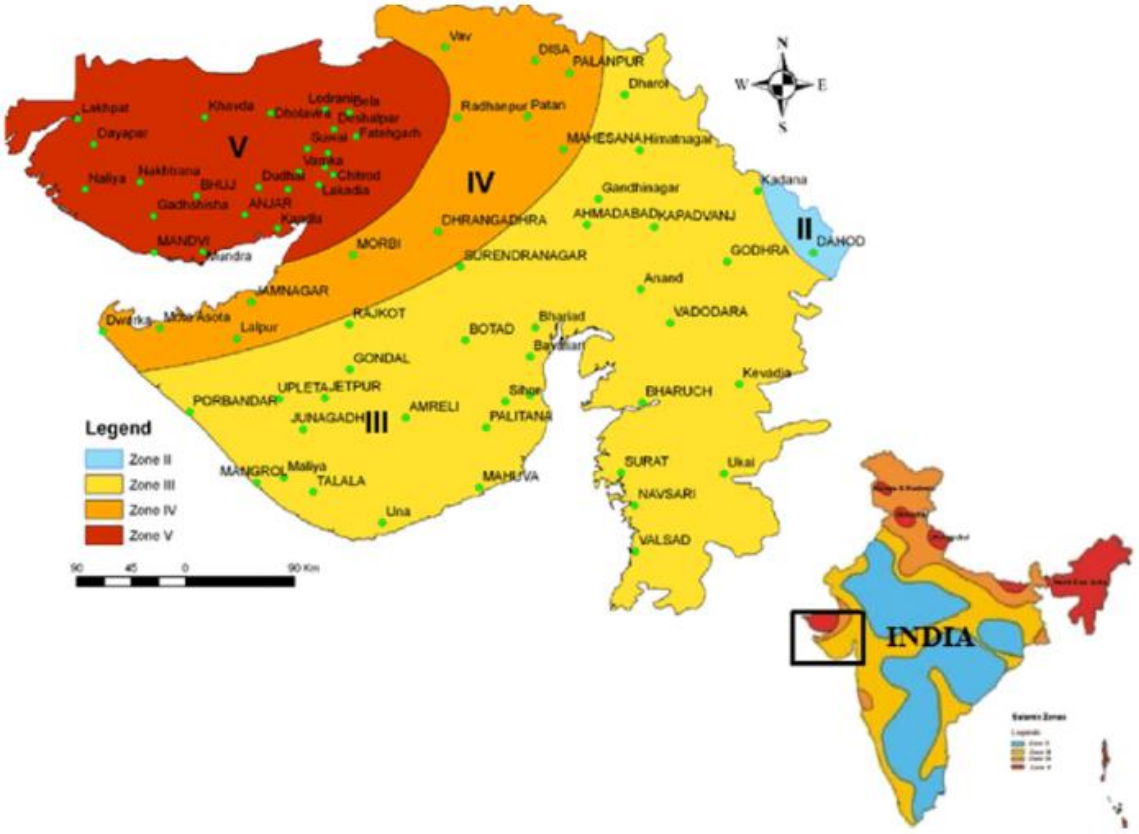
Water Bodies: There are many small water bodies in the study area and contributes to 9.67% of total area.

All the nearby villages are well connected with the roads.

4.2.5. Seismicity

The state of Gujarat falls in all five zones i.e. of very small portion of Low Damage Risk Zone (Zone II), maximum portion of Moderate Damage Risk Zone (Zone III), High Damage Risk Zone (Zone IV) and western portion is in Very High Damage Risk Zone (Zone V) as per revised seismic zoning map of India. The proposed project corridors in Surat city are falls in Zone III as shown in Figure 4.9.

Figure 4.9: Seismic Zoning Map of Gujarat



4.3. Water Environment

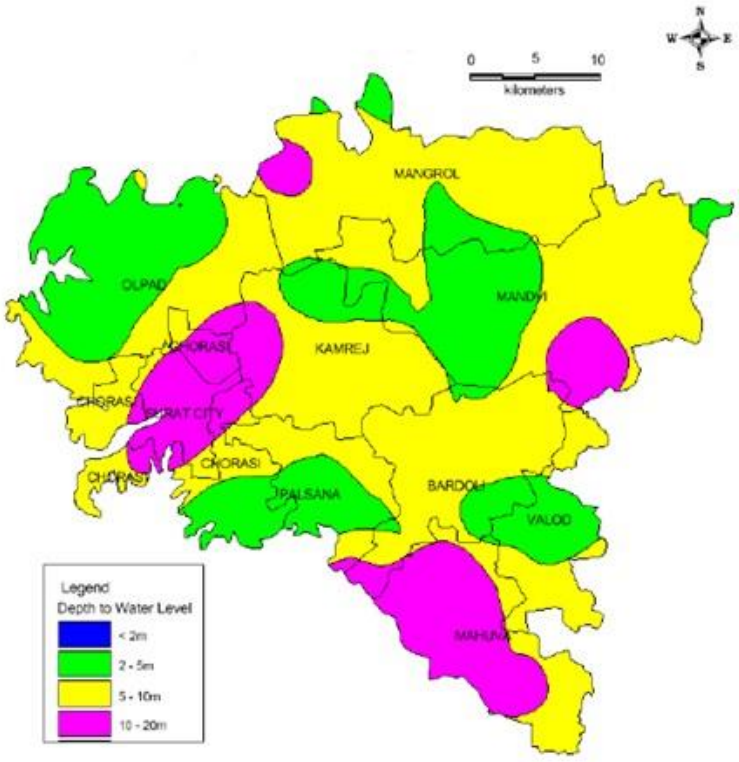
Water environment consists of water resources and its quality. Its study is important from the point of view to assess the sufficiency of water resources for needs of the project in its various stages and the impact of the project on water environment.

4.3.1. Water Resources of the District

The western and north western parts of the district are covered by Alluvium. This aquifer can broadly be demarcated into two zones namely newer alluvium and older alluvium. The newer alluvium is present along the river courses and comprises fine to coarse grained sand trap wash with cay intercalations. The sand is unconsolidated but shows some degree of cohesion at places. Water levels are in general deeper in newer alluvium. Older alluvium is present in inter river plains and comprises sand, clay, kankar, grave and silt. The ground water occurs mostly under unconfined conditions but at places semi-confined conditions are also observed, probably due to presence of clay lenses. As per the CGWB report of Surat district, the depths of the wells in alluvium generally range from 3 to 30 m with some of the wells having bores down to 50m below bottom of the wells. The maximum number of wells, about 80% is 5 to 20m deep. Depths to water levels in the wells range from 0.5 to 15m bgl about 90% wells shows water levels less than 10m bgl. Yields of the dug wells and dug-cum-bored wells range between 100 and 450 m^3/day.

The Figure 4.10 depicting the depth to water level of the Surat district has been prepared based on water level data for May and November 2012 collected from National Network of Hydrograph Stations (NNHS). The depth to water level in Surat city varies from 10m to 20m but at some places it goes up to 5m.

Figure 4.10: Depth to Water Level Map for Surat District



Source: CGWB Report, Surat District

The Tapi River runs through the city of Surat, is one of the major rivers of peninsular India with a length of around 715 km. It is one of only three rivers in peninsular India that run from east to west. The river covers 3601 ha of the total land under Surat Urban Development Authority (SUDA) along with small portion of Mindhola estuary as well on southern part of SUDA area. SUDA region have about 898 no. of small and large size lakes covering about 548 ha of land area under lakes and pond. Notable examples include Gaviyar Lake, Lake at Subhash Garden, Dummas Beach, Sultanabad – Dummas, Umbher lake, Gopi Talav; Out of total, 13 lakes have been interlinked as of today for managing storm water during the monsoon.

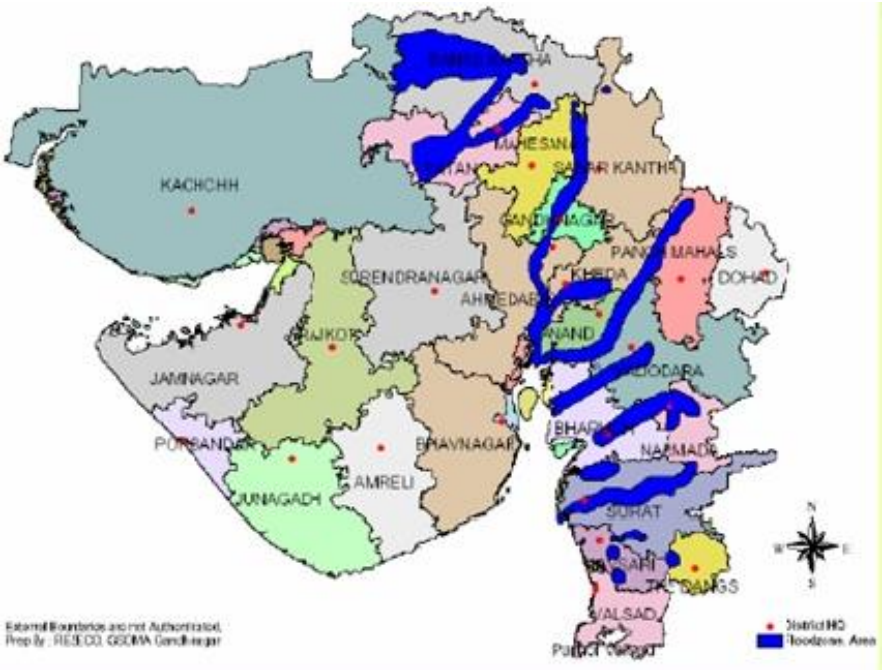
Flood Prone Areas:

Surat lies in the flood plain and also lies in an estuarial region. There are number of creeks in the south- western part of the city. The city has been historically susceptible to flooding. Development of Surat in the past has been affected a number to times due to the high flood in Tapi.

Flood risk exposure and vulnerability of the city has increased over the past few decades. Flood in the year 2006 covered almost the entire city and affected nearly two-third of the Surat city population. Since the altitude of Surat is less than 13 m above mean sea level, even with a sea level rise of one meter, the area under high tide zone particularly western parts of the city which are less than 10 m above mean sea level may be highly impacted. Now with the Ukai dam and the flood embankment scheme undertaken by the State Government, this area is safe from low intensity floods. The height of Nehru bridge is however one of the controlling point for further strengthening of flood embankment.

The walled city area has no good drainage system. There are few pockets of low lying area, where water gets logged in the rainy season. It is observed that the area of Navasari Bazar, Khapatia Chakla, Panini Bhit, Vada Chauhta area get flooded in rainy season. The other affected area is across the railway line towards the southern side of Kamrej which is also low lying area having the natural drain towards Khakra Khadi. Flood prone areas of the state are shown in Figure 4.11.

Figure 4.11: Flood Prone Areas of Gujarat State



Source: DDMP 2019, Surat

4.3.2. Drainage

The Tapi is the major river which passes through the central parts of the district and flows towards the west. The river is perennial in nature. It originates in Madhya Pradesh near Betul and has about 62225 sq. Km. of catchment area. The average width of the river upstream of Kathor Bridge in about 500m; and 700m downstream of the bridge. Pickup weir was constructed on the Tapi River in 1954 at Kakarapar about 56km west of Surat. Ukai dam, constructed in 1965, is situated about 25km upstream of Kakarapar weir. Other prominent rivers draining the district are Kim, and Purna.

The drainage map of the study area was mapped using the field data and Topographical map. The drainage map shows that there are rivers, small stream and water bodies in the study area and water flow is from the North East direction to South West Direction of the study area. Drainage map of study area is shown in Figure 4.12.

4.3.3. Water Quality of the Project Area

The water sampling was done on the basis of availability of ground water and surface water source along the proposed metro alignments. There were no water in the canal during the field study (May 19). Hence 10 surface water samples from small water ponds were collected and other 4 samples were collected from ground water source. Water quality includes the physical, chemical and biological characteristics of water. An understanding of the various factors influencing water quality is thus very important as human health is largely dependent on the quality of water available for use.

In order to assess the baseline water quality status of the study area, 14 representative samples along the metro corridors were collected during May - June 2019. Details of water sampling locations are given in Table 4.6 and location map in Figure 4.5.

The physiochemical and biological laboratory analysis of water samples is summarised in Table 4.7. Zooplankton, Phytoplanktons and Benthos for surface water samples is given in Table 4.8, Table 4.9 and Table 4.10 respectively.

Table 4.6: Details of Water Sampling Locations

Location No	Particulars	Latitude	Longitude
Bhesan to Saroli			
SW1	Pond near Bhesan village (near water treatment plant)	21°12'58.96"N	72°45'27.84"E
SW2	Pond in Kavi Kalpai Garden	21°11'19.67"N	72°47'37.26"E
SW3	Tapi river	21°11'12.80"N	72°48'17.35"E
SW4	Canal water near Saroli	21°11'22.76"N	72°53'31.87"E
SW5	Pond near Tekra Falia	21°11'17.07"N	72°53'38.19"E
GW1	Near L P Sawani Bus Stop	21°12'8.96"N	72°46'59.23"E
GW2	Near Model Township	21°11'19.59"N	72°51'40.85"E
Sarthana to Dream City			
SW6	Tapi river near Chawpatty	21°13'28.37"N	72°52'36.24"E
SW7	Pond at Siddhraj Jaysing Park	21° 8'50.43"N	72°48'13.96"E
SW8	Bhimrad Town Lake	21° 7'39.87"N	72°48'0.12"E
SW9	Creek near Prawn ponds	21° 5'5.98"N	72°47'59.13"E
SW10	Prawn Pond water	21° 5'13.70"N	72°47'15.25"E
GW3	Near Bhagal/Lal Gate	21°11'58.76"N	72°49'37.14"E
GW4	Near Ishtar Nagar	21° 9'54.09"N	72°48'48.73"E
SW-surface Water, GW- Ground Water			

Figure 4.12: Drainage Map of the Study Area

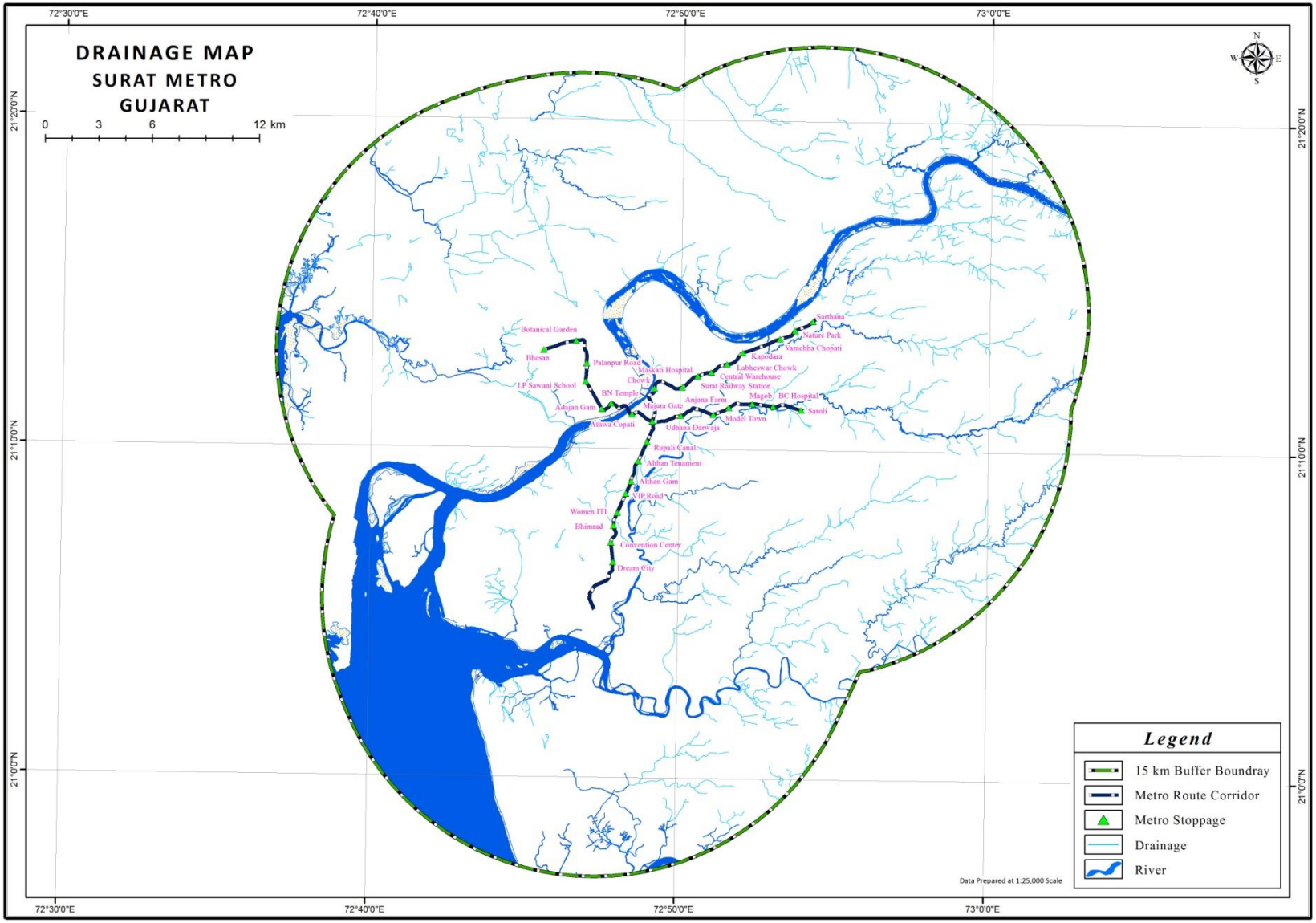


Table 4.7: Physico-Chemical Analysis of Water Samples

Parameters	Unit	SW1	SW2	SW3	SW4	SW5	SW6	SW7	SW8	SW9	SW10	GW1	GW2	GW3	GW4	Acceptable/ Permissible Limit*
pH @ 25^{0}C		7.89	8.24	7.54	7.75	7.62	7.63	8.12	7.27	7.02	7.6	8.2	8.43	7.87	8.13	6.5-8.5/NR**
Total Hardness as CaCO3	mg/l	750	490	3700	580	510	140	1340	250	680	3800	560	186	128	300	200/600
Calcium	mg/l	140	24	257	106	84	20	144	76	120	281	40	14	29	24	75/200
Magnesium	mg/l	97	104	744	77	73	22	377	15	92	753	112	37	14	58	30/100
Total Dissolved Solids (TDS) at 180 oC	mg/l	2204	1956	18716	1624	1084	365	4090	910	1688	34820	3090	812	286	2040	500/2000
Total Suspended Solids (TSS)	mg/l	6	12	462	68	12	6	377	15	65	27	BDL	BDL	BDL	BDL	-
COD	mg/l	18	24	24	145	42	8	8	24	145	32	<4	BDL	BDL	BDL	-
BOD (5day at 200C)	mg/l	7	9	8	28	10	5	BDL	10	32	5	<4	BDL	BDL	BDL	-
Chloride	mg/l	666	765	10747	490	190	52	1340	275	538	19994	1172	92	26	682	250/1000
Total Alkalinity as CaCO3	mg/l	456	580	328	618	690	260	710	314	620	400	510	462	185	430	200/600
Sulphate	mg/l	572	131	1512	170	63	24	736	89	140	1229	412	80	25	212	200/400
Fluoride*	mg/l	BDL	0.58	1.1	0.55	0.34	0.18	0.62	BDL	0.73	0.86	1.5	0.36	0.25	0.51	1/1.5
Nitrate	mg/l	0.01	0.01	0.16	0.20	0.12	BDL	0.07	BDL	0.01	1.63	0.01	BDL	BDL	0.01	45/NR
Total Phosphate as PO4	mg/l	0.01	0.72	0.70	0.86	0.58	0.18	3.1	0.07	0.64	0.01	1.2	0.7	BDL	0.8	0.05/1.5 Max.
Phenolic Compound	mg/l	BDL	BDL	BDL	0.04	BDL	BDL	BDL	BDL	0.02	BDL	BDL	BDL	BDL	BDL	0.001/0.002
Sodium	mg/l	400	525	4900	375	225	100	945	250	390	12100	940	250	52	640	-

Parameters	Unit	SW1	SW2	SW3	SW4	SW5	SW6	SW7	SW8	SW9	SW10	GW1	GW2	GW3	GW4	Acceptable/ Permissible Limit*
Potassium	mg/l	25	35	275	15	12	4	68	30	18	250	44	12	9	22	-
Aluminium*	mg/l	0.44	0.49	149.6	0.48	0.38	0.1	BDL	1.05	0.34	0.13	0.23	BDL	0.3	0.24	0.03/0.3
Arsenic*	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	-
Cadmium	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.003/NR
Manganese*	mg/l	0.03	0.04	3.56	0.12	0.04	BDL	0.02	0.11	0.1	0.01	0.02	BDL	0.02	0.01	0.1/0.3
Total Chromium	mg/l	BDL	BDL	BDL	0.05	BDL	BDL	BDL	BDL	0.04	BDL	<0.03	BDL	BDL	BDL	-
Iron	mg/l	0.82	0.20	0.83	1.05	0.77	0.90	0.06	0.05	0.35	2.29	0.07	0.05	0.08	0.01	0.3/NR
Copper	mg/l	0.05	0.06	0.12	0.1	0.2	0.06	BDL	0.02	0.04	0.06	BDL	BDL	BDL	BDL	0.05/1.5
Zinc	mg/l	0.6	0.14	0.36	0.35	0.4	0.24	0.1	0.06	BDL	0.3	0.3	BDL	BDL	0.2	5/15
Lead	mg/l	BDL	BDL	BDL	0.2	BDL	BDL	BDL	BDL	0.1	BDL	BDL	BDL	BDL	BDL	0.01/NR
Mercury	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.001/NR
Nickel	mg/l	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.02/NR
Total Coliform (MPN)	MPN/ 100ml	35	46	42	125	76	42	8	8	118	26	Absent	Absent	Absent	Absent	Absent
Dissolved oxygen	mg/l	5.4	5.2	5.3	4.2	4.6	5	6.2	4.8	4.1	5.8	-	-	-	-	-

*SE-Surface water, GW-Ground water, BDL-Below Detectable Limit, *(Acceptable limit/Permissible limits as per IS 10500-2012), **No Relaxation*

Table 4.8: Zooplankton Analysis of Surface Water Samples

Location No	Total Count	Percentage Composition of		
		Rotifera	Copepoda	Cladocera
SW1	910	50	32	18
SW2	1040	45	32	23
SW3	890	48	30	22
SW4	1410	38	32	30
SW5	1115	42	40	18
SW6	1045	45	33	22
SW7	712	52	45	3
SW8	1080	45	43	12
SW9	1422	38	36	26
SW10	1105	41	37	22

Species Observed (Zooplanktons): Rotifera :Keratella sp. , Brachionus sp., Lecane Sp.
Copepoda :Nauplius larva, Cyclops sp. **Cladocera** : Brachionus sp. Camptocercus sp.

Table 4.9: Phytoplankton Analysis of Surface Water Samples

Location No	Total Count (No/ml)	Percentage Composition date Total Count of Phytoplankton		
		Chlorophyceae	Bascillariophyceae	Cynophyceae
SW1	290	45	30	25
SW2	310	36	47	17
SW3	165	44	38	18
SW4	412	27	35	38
SW5	192	42	36	22
SW6	315	40	38	22
SW7	152	46	35	19
SW8	212	40	36	24
SW9	478	45	18	37
SW10	250	41	32	27

Species Observed (Phytoplankton): 1. Chlorophyceae: Ankistrodesmus sp., Chlorococcum sp, Scenedesmus sp., Chlorella sp., Cosmarium sp., Closterium sp., Chlamydomonas sp.
2. Bacillariophyceae : Navicula sp., Nitzschia sp. Cyclotella sp., Synedra sp., Pinnularia sp.
3. Cynophyceae : Oscillatoria sp., Gomphosphaeria sp.

The samples were analyzed for physical and chemical constituents for the purpose of domestic and irrigation use. The results of water parameters are compared with IS 10500:2012 of drinking water standards. Tolerance Limits for Inland Surface Waters subject are compared with CPCB standards.

Parameters like pH, calcium, floride and ntrate are found within permissible limits in both surface and ground water samples. Hard water found in 3 locatons, TDS is very high i.e. 18716 mg/l in Tapi river due to salt water intrusion in high tide from creek. Also It is found high in 2 other locations. High sulphate, cholirede, magnesium, sodium and aluminium content found in Tapi river sample. It is generally considered that the levels of DO in the estuarine water should not fall below 3.0 mg/l for prolonged periods under tropical conditions. DO content in all surface water is good but some coliform found in all surface water samples.

Table 4.10: Qualitative Benthos Group Wise Analysis

Systematic Group	SW1	SW2	SW3	SW4	SW5	SW6	SW7	SW8	SW9	SW10
Polychaetes	+	+	+	+	+	+	+	+	+	+
Mysids	_	_	_	+	_	_	_	_	+	+
Ostracods	_	_	_	_	_	_	_	_	_	_
Isopodes	_	_	_	+	_	_	_	_	+	+
Amphipodes	_	_	_	+	_	_	_	_	+	+
Brachyurans	_	_	+	_	_	+	_	_	_	_
Insects	-	-	+	-	+	+	-	-	+	-
Gastropodes	-	-	-	-	-	-	-	-	-	-
Pelecypodes	-	-	-	-	-	-	-	-	-	-

+ indicates present and – indicate absent

Remark: Benthos sample has been collected using Benthos sampler (Beckmann grab sampler) & Scrapper.

High total count to phytoplanktones found in creek water and one of canal water due to coastal water pushing in more phytoplankton population towards mouth of the estuary. Zooplankton standing stock was studied in terms of biomass, population. Highest count was found in creek water sample. The Copepods, Keratella sp. , Brachionus sp., Lecane Sp., Nauplius larva, Cyclops sp., Brachionus sp and Camptocercus sp were the major groups of zooplankton recorded during the study.

4.4. Meteorology and Ambient Air Environment

The ambient environment is responsible for the health of human beings, animals, wildlife and vegetation. The ambient environment covers climate, atmospheric pollution and noise pollution. All air pollutants emitted by point and non-point sources are transported, dispersed or concentrated by meteorological and topographical conditions. The meteorological parameters regulate the transport and diffusion of pollutants into the atmosphere. In order to assess the impact on existing ambient environment due to the project, it is necessary to have baseline status of ambient environmental parameters. Meteorological data on rainfall, wind, humidity, and temperature were collected from regional office (Ahmadabad) of Indian Meteorological Department (IMD) for last five years (2014 to 2018).

4.4.1. Meteorology

Meteorology is an important parameter in environmental impact assessment study. It is responsible for the movement of air and air pollutants. All air pollutants emitted by point and non-point sources are transported, dispersed or concentrated by meteorological and topographical conditions. The meteorological parameters regulate the transport and diffusion of pollutants into the atmosphere. Meteorological data like monthly total rainfall, maximum & minimum temperature and relative humidity of the Surat for the period of Jan 2014 to Dec 2018 are given in Table 4.11 to Table 4.15.

Table 4.11: Temperature, Rainfall & Humidity for Year 2014

Month	Mean Temperature (deg c)		Monthly Total Rainfall (mm)	Mean of Relative Humidity (%)	
	Maximum	Minimum		0830 Hrs	1730 Hrs
JANUARY	28.9	16.4	013.4	075	047
FEBRUARY	30.7	18.4	001.8	070	040
MARCH	34.3	21.3	000.0	070	032
APRIL	36.5	25.0	000.0	070	044
MAY	35.8	27.7	000.4	072	060
JUNE	34.2	28.9	022.7	076	069
JULY	31.9	26.8	398.9	086	080
AUGUST	31.2	26.3	158.5	087	079
SEPTEMBER	31.8	25.8	216.8	087	075
OCTOBER	36.3	25.1	004.5	072	051
NOVEMBER	34.8	22.4	000.0	069	047
DEDCEMBER	30.8	16.7	000.0	064	042

Table 4.12: Temperature, Rainfall & Humidity for Year 2015

Month	Mean Temperature (deg c)		Monthly Total Rainfall (mm)	Mean of Relative Humidity (%)	
	Maximum	Minimum		0830 Hrs	1730 Hrs
JANUARY	29.0	14.9	000.0	079	043
FEBRUARY	32.8	18.1	000.0	077	037
MARCH	34.6	21.5	000.2	064	037
APRIL	34.6	25.5	008.2	076	057
MAY	36.5	28.2	000.0	074	058
JUNE	32.9	26.9	525.8	083	075
JULY	31.7	27.3	154.2	084	080
AUGUST	31.9	26.7	057.3	084	074
SEPTEMBER	32.3	25.4	402.3	085	070
OCTOBER	36.4	25.6	000.2	073	051
NOVEMBER	35.0	22.4	000.0	068	047
DEDCEMBER	31.7	16.1	000.0	065	040

Table 4.13: Temperature, Rainfall & Humidity for Year 2016

Month	Mean Temperature (deg c)		Monthly Total Rainfall (mm)	Mean of Relative Humidity (%)	
	Maximum	Minimum		0830 Hrs	1730 Hrs
JANUARY	31.2	15.8	000.0	076	034
FEBRUARY	32.1	18.2	000.0	072	038
MARCH	36.5	22.9	003.4	075	033
APRIL	37.1	25.5	000.0	069	040
MAY	35.5	28.6	000.0	075	066
JUNE	34.4	28.6	107.3	080	072
JULY	30.9	26.5	489.1	088	082
AUGUST	30.3	26.3	183.7	087	083
SEPTEMBER	31.4	25.5	145.3	085	076
OCTOBER	32.7	23.3	177.4	083	063
NOVEMBER	33.9	18.2	000.0	064	039
DEDCEMBER	32.4	16.9	000.0	065	040

Table 4.14: Temperature, Rainfall & Humidity for Year 2017

Month	Mean Temperature (deg c)		Monthly Total Rainfall (mm)	Mean of Relative Humidity (%)	
	Maximum	Minimum		0830 Hrs	1730 Hrs
JANUARY	30.7	16.0	000.0	074	041
FEBRUARY	33.6	18.7	000.0	065	034
MARCH	35.6	22.0	000.0	062	031
APRIL	36.3	25.1	000.0	067	047
MAY	35.8	28.1	000.5	073	061
JUNE	33.8	27.5	302.6	082	072
JULY	30.8	26.3	355.2	087	084
AUGUST	30.9	26.2	383.8	087	080
SEPTEMBER	32.7	26.5	074.8	087	070
OCTOBER	35.3	24.2	032.7	078	053
NOVEMBER	33.7	19.4	000.0	065	041
DEDCEMBER	29.6	18.1	012.9	069	049

Table 4.15: Temperature, Rainfall & Humidity for Year 2018

Month	Mean Temperature (deg c)		Monthly Total Rainfall (mm)	Mean of Relative Humidity (%)	
	Maximum	Minimum		0830 Hrs	1730 Hrs
JANUARY	31.2	17.1	000.0	071	037
FEBRUARY	33.4	19.5	000.0	068	031
MARCH	36.4	22.9	000.0	064	028
APRIL	36.7	25.8	000.0	075	040
MAY	35.7	28.5	000.0	074	064
JUNE	34.0	28.1	172.4	079	071
JULY	29.3	26.1	865.8	091	086
AUGUST	30.2	25.8	233.5	088	082
SEPTEMBER	31.8	25.1	066.6	081	066
OCTOBER	37.1	24.5	000.0	068	046
NOVEMBER	35.5	20.5	000.0	066	043
DEDCEMBER	30.5	15.9	000.0	063	038

Monthly wind rose diagram for last 10 years i.e. 2009 to 2018 at 8:30 hrs and 17:30 hrs are shown in Figure 4.13 and Figure 4.14 respectively. From the Figure, the predominant direction is from NW and SW for 08:30 hrs and N, NW and SW direction for 17:30 hrs.

Figure 4.13: Wind Rose Diagram for Year 2009-2018 08:30 Hrs

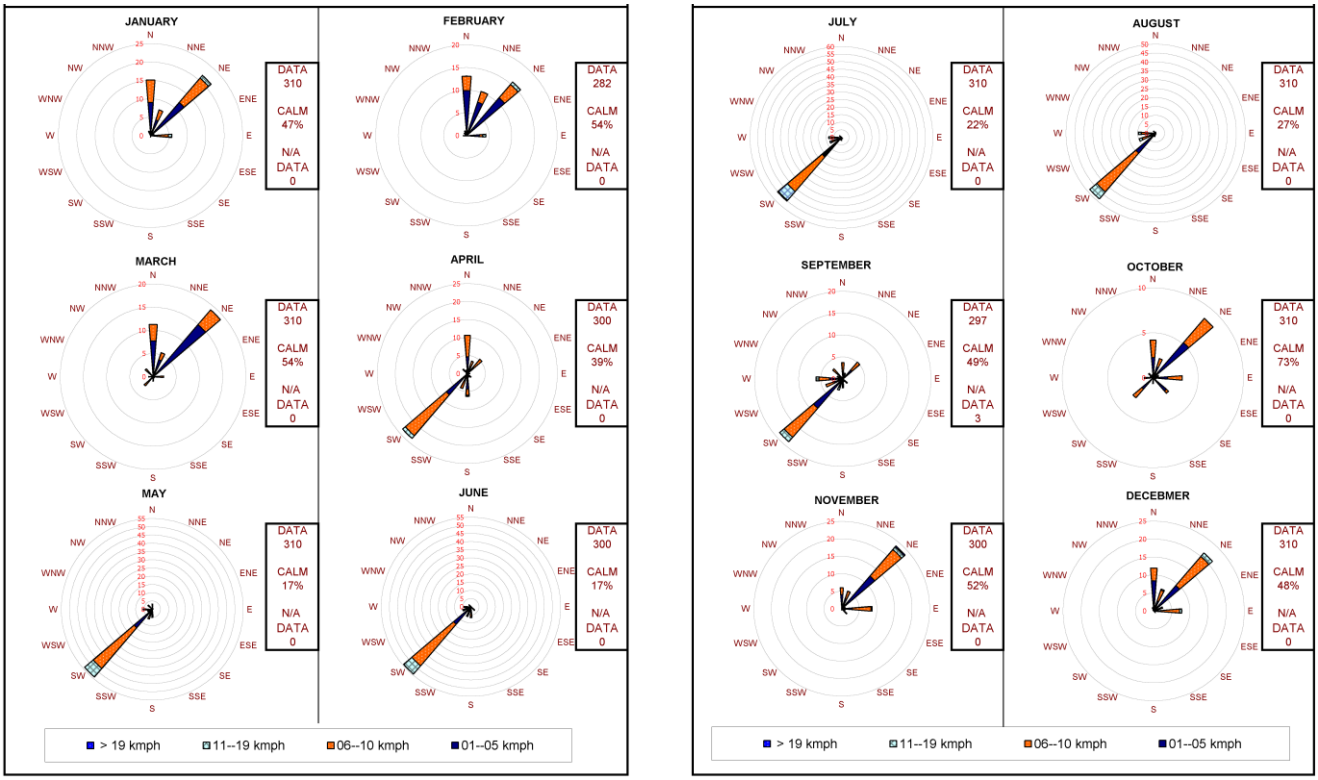
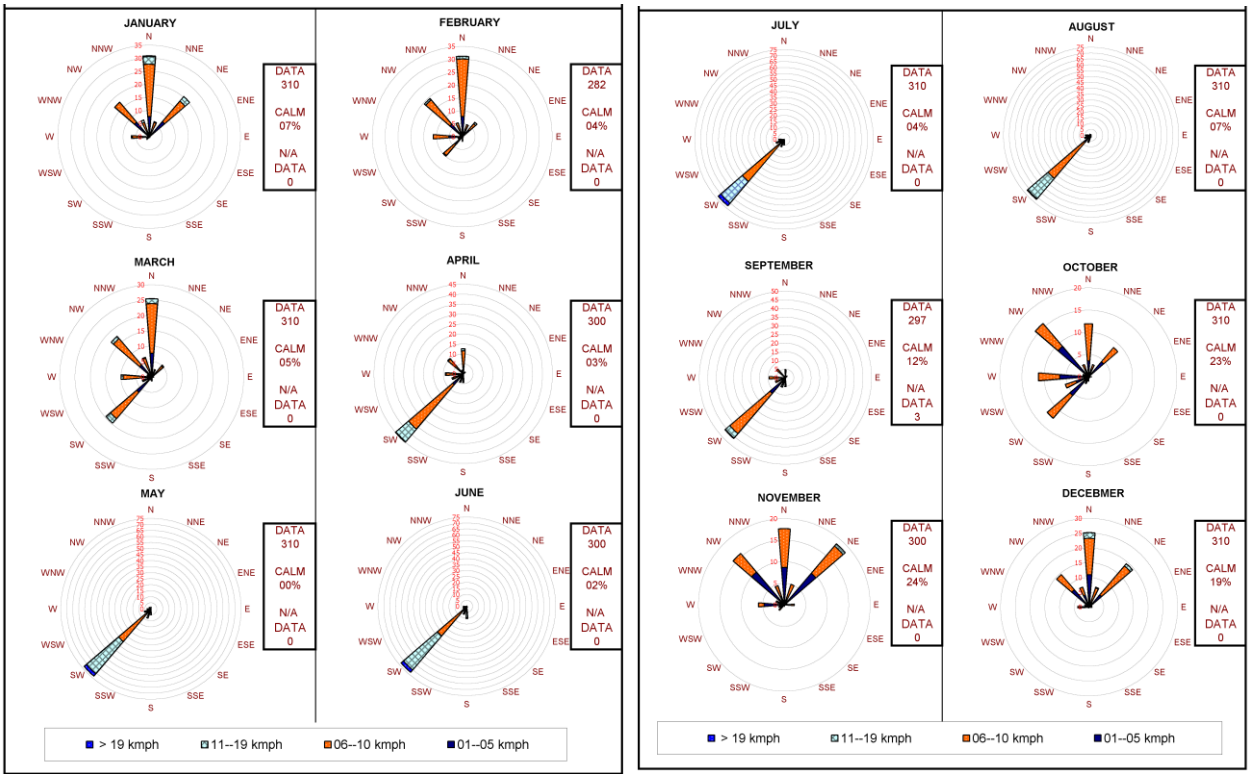


Figure 4.14: Wind Rose Diagram for Year 2009-2018 17:30 Hrs



4.4.2. Air Quality

The prime objective of baseline air quality survey was to assess conformity to standards of ambient air quality. The ambient air quality monitoring stations have been selected strategically based on the following criteria;

- Where maximum ground level concentrations are expected like crossing of major roads, commercial areas etc.
- Where pollution problem exists or is expected i.e. mainly in industrial areas, urban areas, traffic intersections etc.
- Area where population density is high is considered for locating monitoring stations.

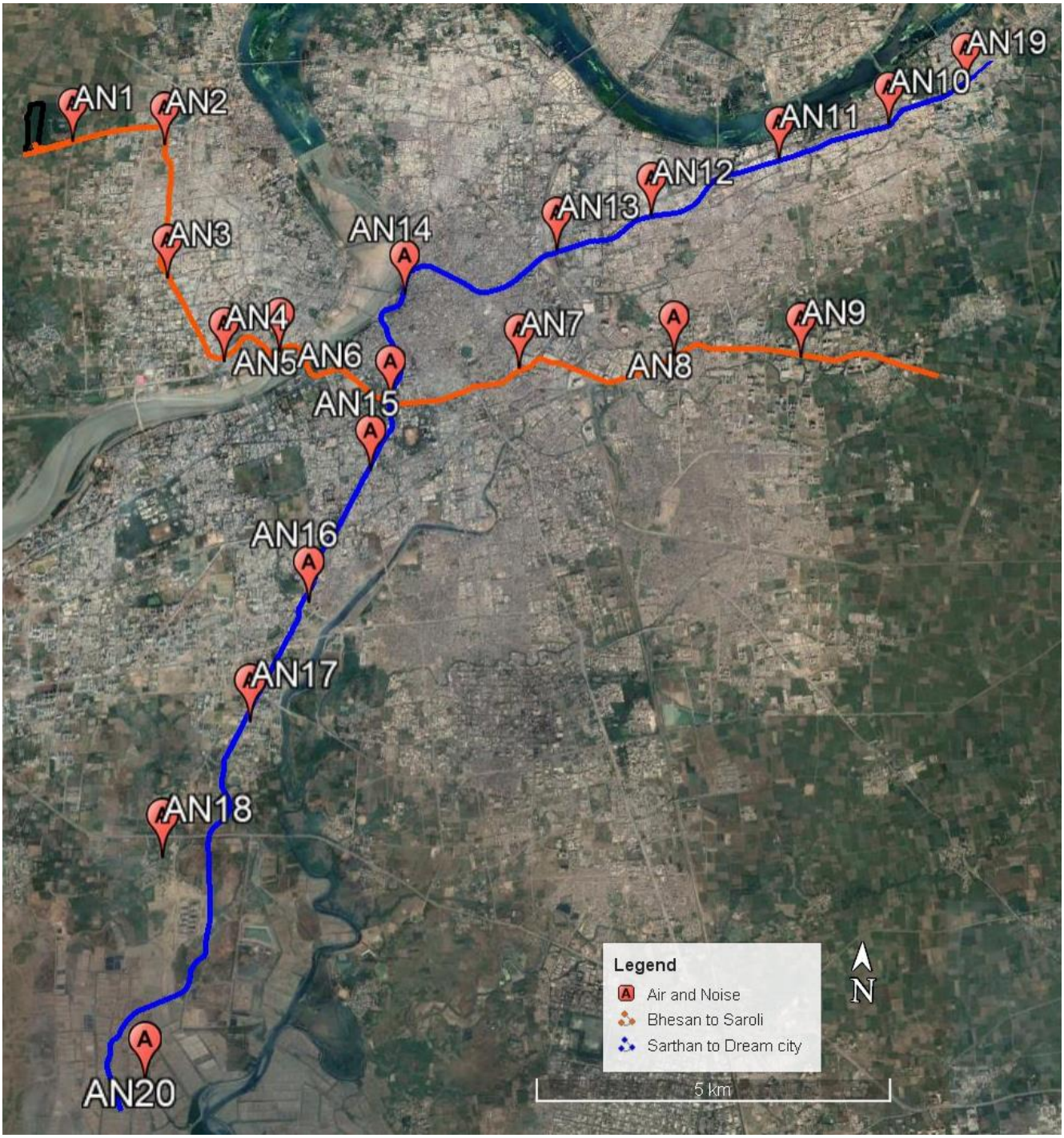
The location is selected nearby sensitive places like institutional area, public places, worship places etc. On above basis 20 ambient air monitoring stations were selected strategically to generate the representative samples for ambient air quality covering residential, institutional and industrial area along the corridors. Two days continuous air quality monitoring at 20 locations along the metro corridors has been done in the month of May – June 2019. Monitoring location details are provided in Table 4.16 and location map is in Figure 4.15. The monitoring results for ambient air quality are summarised in Table 4.17 and national ambient air quality standards as per MoEFCC are given in Table 4.18.

Table 4.16: Air Quality Monitoring Locations

Location No	Location Name	Latitude	Longitude
Bhesan to Saroli			
AN 1	Near Treatment Plant, Bhesan Gam	21°13'8.69"N	72°46'0.23"E
AN 2	Near Ganga Nagar Society	72°46'0.23"E	72°46'51.64"E
AN 3	Near Bhagat Party Plot	21°11'53.74"N	72°47'0.40"E
AN 4	Near Imperial Maal/Cinepol multiplex	21°11'9.80"N	72°47'35.35"E
AN 5	Badri nath temple (Maansarovar Society)	21°11'15.77"N	72°48'4.75"E
AN 6	Majura Gate chawk	21°10'52.08"N	72°49'6.35"E
AN 7	Kinnery Cinema Stop	21°11'11.04"N	72°50'14.50"E
AN 8	At crossing of Modal Town	21°11'20.40"N	72°51'37.59"E
AN 9	Near Bhawani Industrial Estate, Transport Nagar	21°11'21.61"N	72°52'45.94"E
Sarthana to Dream City			
AN 10	Near Bajrang Nagar society	21°13'31.60"N	72°53'34.12"E
AN 11	Mamta Park Society, Kapodra	21°13'9.95"N	72°52'33.17"E
AN 12	Near Hans Society	21°12'36.12"N	72°51'23.27"E
AN 13	Hanuman Temple Near Station	21°12'15.22"N	72°50'32.02"E
AN 14	Near Andrew Library, Chawk Bazar	21°11'49.28"N	72°49'10.33"E
AN 15	Near Fame Raj Empire Multiplex	21°10'15.31"N	72°48'58.22"E
AN 16	Near Shri Dayalji Kasanji Bhatarkar Vidya Sankul	21° 9'7.39"N	72°48'30.63"E
AN 17	Residential area of Bhimrad	21° 8'8.02"N	72°48'5.03"E
AN 18	Khanjod	21° 7'2.44"N	72°47'26.44"E
AN 19	Subhash Nagar	21°13'55.66"N	72°54'17.46"E
AN 20	Near Prawn ponds, Near Dream City (after Bhimrad)	21° 5'20.54"N	72°47'27.84"E

A-*Air, N-Noise*

Figure 4.15: Air Quality and Noise Level Monitoring Locations



A-Air, N-Noise

Table 4.17: Ambient Air Quality Monitoring Results

PARAMETER	UNIT	LOCATION NUMBER																			
		A1		A2		A3		A4		A5		A6		A7		A8		A9		A10	
		DAY		DAY		DAY		DAY		DAY		DAY		DAY		DAY		DAY		DAY	
		1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2
PM10	µg/m3	72.3	68.4	83.9	81.6	91.3	89.3	90	93.5	81.8	87.5	78	78	97.8	96.4	80.4	76.1	77.01	80.01	110	80.04
PM2.5	µg/m3	47	37.2	41.8	39.5	45.9	44.4	49	47.2	44.2	40.4	38.96	40.21	52.2	50.9	42.07	38.11	40.12	42.15	50.94	38.02
SO2	µg/m3	7.7	8.3	14.2	15.6	15.2	16	18.1	16.9	15.5	14.4	20.2	16.6	21.8	19.4	19.7	20.2	23.5	18.4	14.33	13.33
NOx	µg/m3	17.43	17.07	26.46	25.74	26.56	26.88	28.33	26.56	23.95	23.61	28.7	26.2	32.38	28.97	30.5	30.6	32.2	28.1	23.61	21.79
CO	mg/m3	0.45	0.68	0.61	0.43	0.56	0.55	0.54	0.49	0.48	0.58	0.53	0.57	0.4	0.6	0.42	0.51	0.46	0.44	0.44	0.57

.....Continued

PARAMETER	UNIT	LOCATION NUMBER																			
		A11		A12		A13		A14		A15		A16		A17		A18		A19		A20	
		DAY		DAY		DAY		DAY		DAY		DAY		DAY		DAY		DAY		DAY	
		1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2
PM10	µg/m3	88	88.4	91.8	90.81	84.12	84.16	82	88	74	78.99	73.4	77.99	78.3	70	75.5	81.86	86	91.1	73.2	86.17
PM2.5	µg/m3	49	46.5	46.01	45.02	42.16	40.85	40	43.9	40.2	42.06	38.7	39.8	23.7	38.1	40	43.83	44	48.2	37.4	40.89
SO2	µg/m3	18.5	16.6	12.2	17.2	16.3	13.5	15.7	14.1	19.7	20.2	18.5	18.4	20.3	22.1	20.6	17.9	16.3	18.3	18.7	20.6
NOx	µg/m3	28.6	26.2	20.3	25.2	22.3	22.3	24.2	24.1	23.6	30.6	22.4	27.8	24.3	32.2	26.7	29.4	30.2	28.6	24.9	33.2
CO	mg/m3	0.43	0.59	0.49	0.6	0.58	0.42	0.43	0.51	0.42	0.55	0.61	0.44	0.39	0.57	0.45	0.6	0.56	0.52	0.53	0.51

Table 4.18: National Amibent Air Quality Standard (MOEFCC)

Parameters				
PM10 µg/m³	PM2.5 µg/m³	SO2 µg/m³	NOx µg/m³	CO mg/m³
100	60	80	80	2

Five major air pollutants viz. particulate matter (PM_{10} & $PM_{2.5}$), Sulphur Dioxide (SO_2), Nitrogen Dioxide (NO_2) and Carbon Monoxide (CO) representing the air quality were monitored. The air quality monitoring results indicate that all parameters were within the permissible limits of National Ambient Air Quality Standards, GOI.

These parameters were compared with IFC standards and it was observed that PM_{10} is exceeding the permissible limits at all locations; $PM_{2.5}$ exceeding permissible limits except on Day 1 at Location A17; and SO_2 exceeding the permissible limits on one day at Locations A6, A7, A8, A9, A15, A18, A20 & both days at location A17.

4.4.3. Noise Environment

Noise exposure can lead to adverse effects on health. The impacts of noise sources on surrounding community depend upon:

- Characteristics of noise sources (instantaneous, intermittent or continuous in nature). It can be observed that steady noise is not as annoying as one, which is continuously varying in loudness.
- Time of day at which noise occurs, for example high noise levels at night in residential areas are not acceptable because of sleep disturbance.
- Location of noise source, with respect to noise sensitive land use, which determines the loudness and period of exposure.

The noise levels were collected over 24 hours in the month of May - June 2019 at hourly interval during morning, afternoon and evening such that peak and off peak hours are covered. The monitoring locations are listed in Table 4.16 and marked in Figure 4.15. The monitoring results for noise levels of the study area are summarised in Table 4.19.

Table 4.19: Noise Level Monitoring Results

Location No	Day	Leq	L10	L50	L90	Lmax	Lmin	Lday	Lnight	LDN
N1	1	66.2	45.6	61.2	70.7	72.3	45.1	65.3	48.5	71.4
	2	65.8	46.1	60.2	70.4	71.3	45.2	64.6	48.9	72.6
N2	1	70.7	51.9	66.2	75.7	77.4	49.3	68.4	57.5	75.8
	2	70.4	52.2	63.9	75.4	77.3	48.9	68.7	54.2	77.2
N3	1	71.6	51.3	66.6	76.4	77.8	49.8	69.6	58.5	80.6
	2	69.9	54.6	66.3	75	76.3	52.8	69.5	57	79.2
N4	1	68.8	53.7	65.3	73.6	74.8	50.6	68.1	57.1	78
	2	71.5	53.4	68.3	76	78.2	50.6	70.6	55.9	80.7
N5	1	64.4	45.9	64.1	67.8	68	44.6	64.5	50.1	72
	2	62.1	48	58.9	67.2	68	44.1	61.2	51.9	71.4
N6	1	71.6	46.5	66	76.1	78	45.7	69.8	50.4	76.4
	2	65.7	45.8	59.4	70.4	71.3	44.6	64.3	48.7	71.5
N7	1	69.7	55.6	66.7	73.9	76.2	53.1	69	59.5	79
	2	72.8	56.4	70.2	76.9	78.1	55	72.5	60.9	80.9
N8	1	64.4	45.9	64.1	67.8	68	44.6	64.2	50.7	71.3
	2	62.1	48	58.9	67.2	68	44.1	61.2	51.9	71.4

Location No	Day	Leq	L10	L50	L90	Lmax	Lmin	Lday	Lnight	LDN
N9	1	66.5	54.3	64.9	69.8	72.4	52	66.3	59.4	75.2
	2	69.9	52.7	67.7	73.9	75.4	50	69.4	57	78
N10	1	65.3	45.2	61	69.5	72.5	44.5	64.3	46.7	73.2
	2	69.7	54.6	66.5	73.5	74.4	51.8	69.2	57.3	78.6
N11	1	65.7	48	66.6	68	70.1	44.1	65.9	52.4	74.1
	2	65.4	50	64.4	68.6	71.8	48.7	65.3	54	73.9
N12	1	68.5	54.9	66	72.7	74.3	52.8	68.6	57.1	77.4
	2	64.7	45.5	64.5	68.7	70.3	43.6	64.1	47	72
N13	1	64.7	45.9	63.7	67.3	70.8	45.1	63.3	52.7	72.9
	2	68.9	46.4	61.8	73.8	74.9	44.5	65.7	53.8	78.5
N14	1	68.4	45.9	64.9	72.8	76.8	44.6	66.4	48.5	76.6
	2	67.3	56.7	65.7	70.4	75.2	53.1	67.1	57.2	76.1
N15	1	69.2	45.4	67.8	73.1	74.9	45	69	49.4	77.6
	2	67.4	45.1	61.7	72.2	74.6	43.6	65.6	47.3	74.8
N16	1	64.7	51.9	64.1	68	68	50.4	65.3	54.6	71
	2	62.6	52.2	59.1	66.9	69	50.4	61.6	54.9	68.8
N17	1	64.7	50	64.1	67.9	69.6	48.7	65.2	53.2	69.6
	2	64.9	47.3	65.8	67.9	68	46.3	65.4	49.4	68.3
N18	1	64.2	49.1	59.9	68.3	70.5	44.5	63.7	51.5	67.9
	2	64.5	45.6	60	68.7	70.4	45	63.1	49.7	67.4
N19	1	67.8	56.1	64.9	72.2	74.8	54.9	67.1	59.6	76.7
	2	69	52.8	67	73.8	76.4	52	68.5	55.6	78.2
N20	1	70.4	51	64.9	74.3	79.7	47.3	68.3	51.6	78.2
	2	68.6	48	67.3	73	74.9	47.2	68.2	48.8	76.1

Noise monitoring results shows that noise level at all locations is exceeding the noise level standards prescribed by CPCB (Table 4.20) either day or night or both for Residential Zone, Commercial Zone. The main source of noise in the project area is the traffic movement on the road.

Table 4.20: Ambient Noise Standards Criteria

Area Code	Category of Area	Limits in dB (A) Leq	
		Day time*	Night time
A	Industrial area	75	70
B	Commercial area	65	55
C	Residential area	55	45
D	Silence Zone**	50	40

Source: CPCB guideline(as per The Noise Pollution (Regulation and Control) Rules, 2000)

** Day timeisfrom 6.00 AM to 9.00 PM, *****Silence Zone*** *is defined as an area up to 100m around premises of Hospitals, Educational Institutions and Courts.*

The noise monitoring results were compared with IFC standards also and it was observed that Day and Night time monitoring results within the permissible limits of commercial zones at A3, A4, A6, A8, A9, A14, A15, A17 locations; and at remaining locations exceeding the permissible limits.

4.4.4. Vibration

Human response to vibration is subjective and will be different for different people. When the vibrations reach the floors and walls it may result in perceptible vibration depending on the amplitude and frequency of the vibrations. Rattling of windows, dishes, and similar parts may also result in audible noise which is called ground-borne noise. People may be more annoyed if they are exposed to both noise and vibration compared to when only vibration is felt.

Ground-borne vibration can be a major concern for nearby neighbours of a transit system route or maintenance facility. Some common sources of ground-borne vibration are trains, buses on rough roads, and construction activities such as blasting, pile- driving and operating heavy earth-moving equipment.

Vibration monitoring was carried out at 10 locations for 24 hrs continuous along the corridors in the month of May & June 2019. The monitoring locations were selected so as to cover the stucutres having historical importance, Multispeciality and or Superspeciality hospitals, old and densely populated areas and important religious places which are nearly close to the alignment. The location map of the monitoring stations is shown in Figure 4.16 and a location detail is given in Table 4.21. Vibration monitoring was carried out using the Nomis Seismograph equipment which can measure the radial, transverse and vertical vibration of ground-borne vibrations.

Figure 4.16: Vibration Monitoring Locations

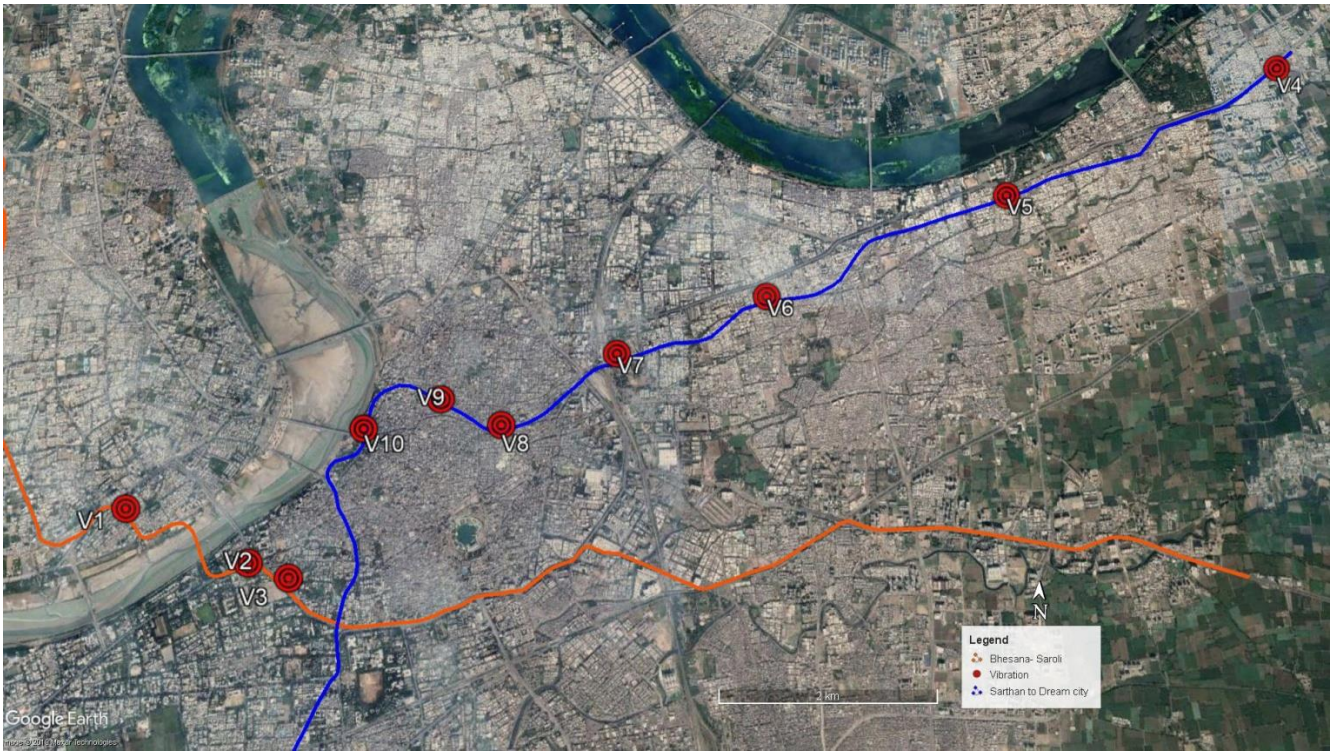


Table 4.21: Vibration Monitoring Locations

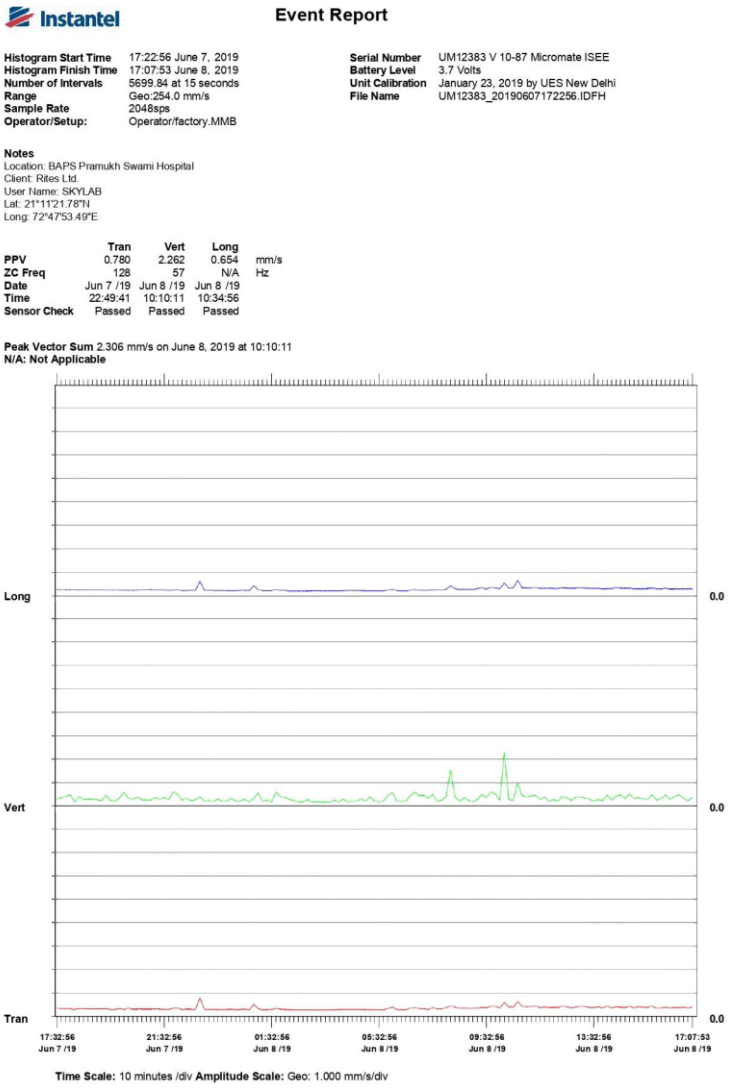
Location No	Particulars	Latitude	Longitude
Bhesan to Saroli			
V1	BAPS Pramukh Swami Hospital	21°11'21.78"N	72°47'53.49"E
V2	Tristar Hospital, Athwa Gate	21°11'6.56"N	72°48'35.85"E
V3	G D Mehta Mahavir Heart Hospital	21°11'2.50"N	72°48'49.45"E
Sarthana to Dream City			
V4	Near Nirmal Nagar Society	21°14'1.98"N	72°54'25.96"E
V5	Swami Narayan Temple	21°13'13.07"N	72°52'46.49"E
V6	Shri Harihar Mahadev Temple	21°12'35.96"N	72°51'22.79"E
V7	Lambe Hanuman Temple, near Railway station	21°12'15.55"N	72°50'32.34"E
V8	Clock Tower	21°11'51.42"N	72°49'54.87"E
V9	Rani Talab Masjid	21°11'59.02"N	72°49'33.95"E
V10	Andrew Library, Chawk Bazar	21°11'48.90"N	72°49'9.05"E

Monitoring is conducted at the road near building. Monitoring area comprises of all types of premises e.g. residential, commercial, industrial, institutional etc. Three dimensional vibration readings are taken and recorded 24 Hrs continuous. Instrument used for the measurement of tri-axial vibration is Instantel Micromate. Vibrations in terms of PPV (mm/s) in the longitudinal, transversal and vertical directions are monitored. Monitoring results at each locations are discussed below.

1. BAPS Pramukh Swami Hospital

Surrounding of the Baps Pramukh Swami Hospital location is comprises of commercial premises and mainly residential buildings. Main source of the vibrations at the location is due to vehicular traffic. Average transversal vibration is observed below 1 mm/s. However, peak observed is 0.8 mm/s. Though average vertical vibration is observed below 1 mm/s, peak of 2.76 mm/s was observed, which is quite normal due to vehicular traffic. Average longitudinal vibration is observed below 1 mm/s with peak of 0.65 mm/s. Graphical representation of the vibration is given in Figure 4.17.

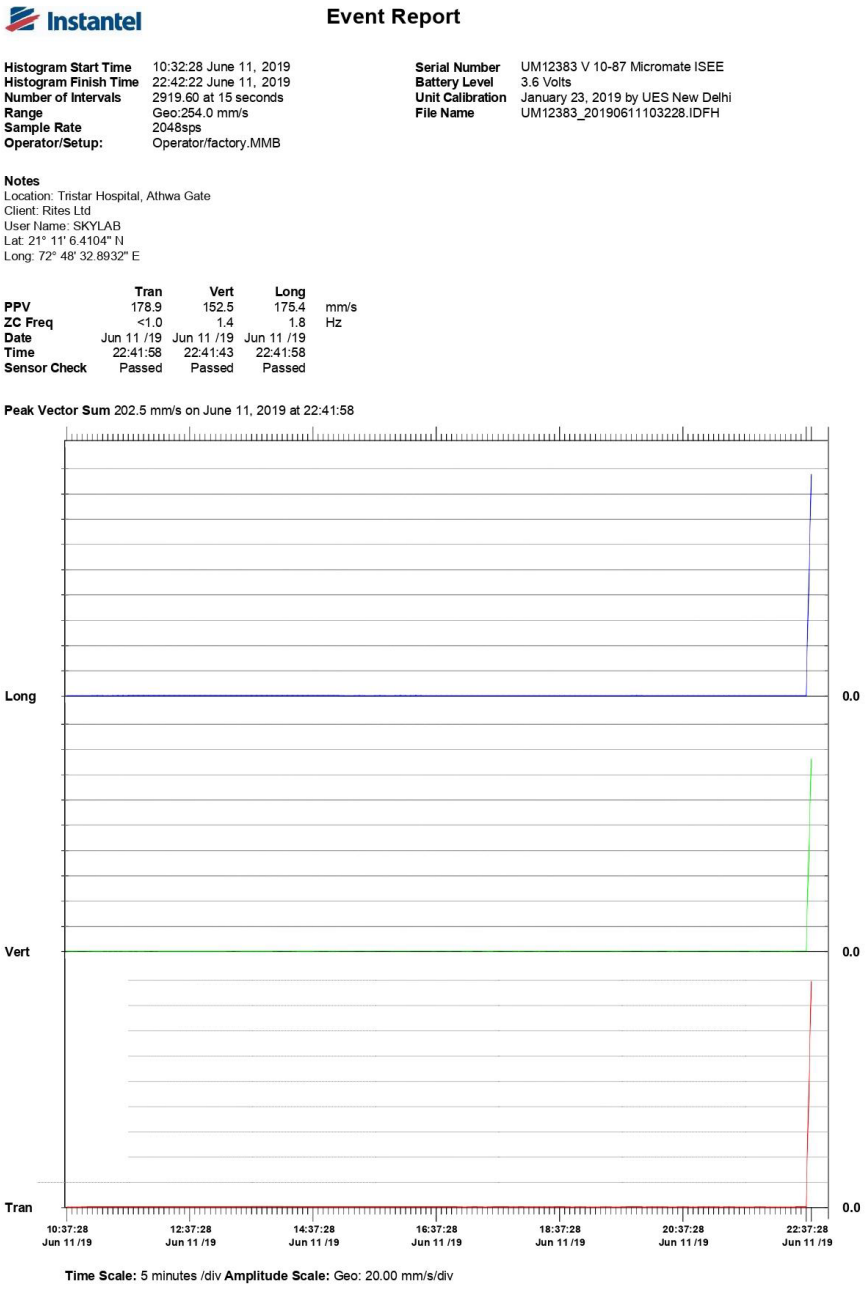
Figure 4.17: Vibration Monitoring Results at BAPS Pramukh Swami Hospital



2. Tristar Hospital, Athwa Gate

Location is observed to be surrounded by the dense population. Heavy traffic Road, Commercial premises and residential buildings are observed. Main source of the vibrations at the location is observed due to vehicular traffic. Location is near Aeroplane Circle. Overall transversal vibration is observed below 0.4 mm/s. Peak in the transversal direction is observed as 178 mm/s. Overall vertical vibration is observed 0.6 mm/s and peak as 152.5 mm/s. Overall longitudinal vibration is observed below 0.4 mm/s with peak of 175.4 mm/s. These high peaks in the vibration in all directions are due to rain. To avoid these rain interruptions, monitoring was stopped and resumed. Graphical representation of the vibration is given in Figure 4.18.

Figure 4.18: Vibration Monitoring Results at Tristar Hospital, Athwa Gate



3. Near G D Mehta Mahavir Heart Hospital

Overall transversal vibration is observed below 0.62 mm/s. Overall vertical vibration is observed below 0.9 mm/s. Overall longitudinal vibration is observed 0.44 mm/s. These vibrations are quite normal for usual traffic. Graphical representation of the vibration is given in Figure 4.19.

Figure 4.19: Vibration Monitoring Results Near G D Mehta Mahavir Heart Hospital

Instantel

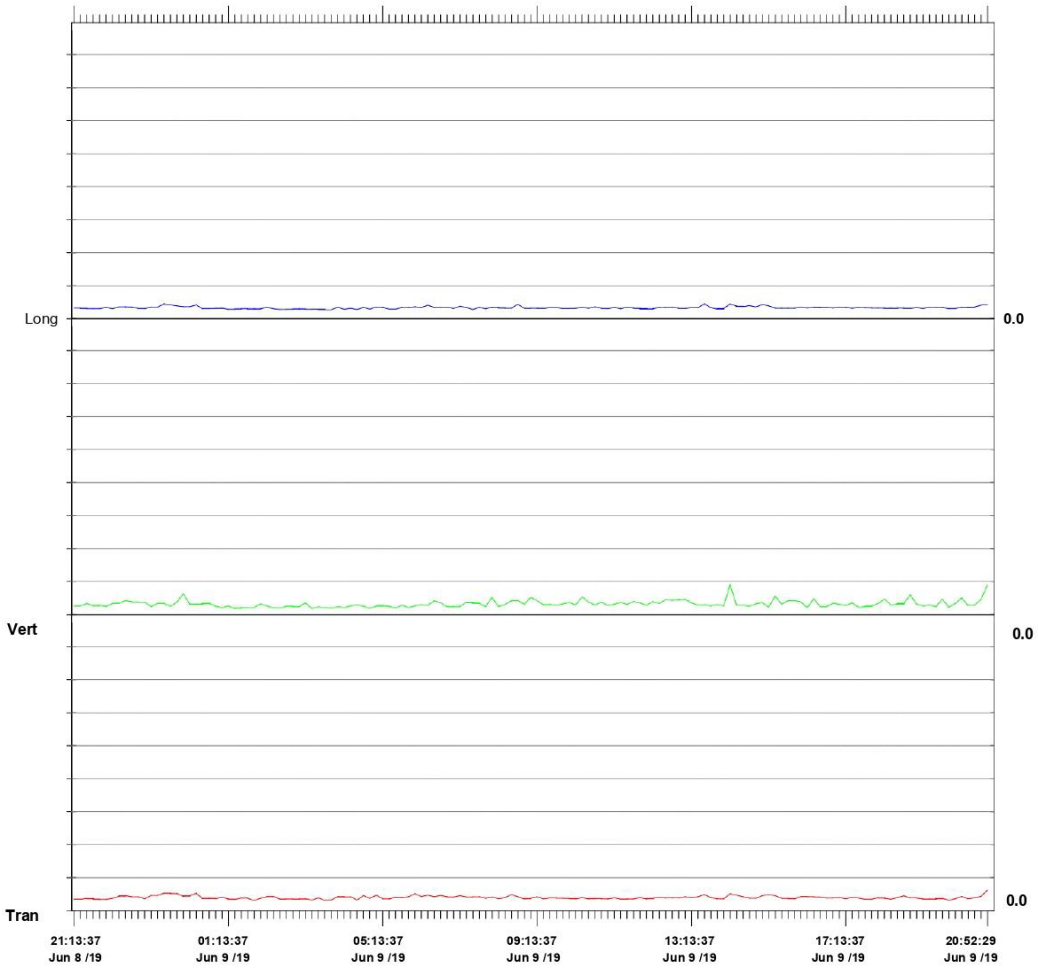
Event Report

Histogram Start Time	21:03:37 June 8, 2019	**Serial Number**	UM12383 V 10-87 Micromate ISEE
Histogram Finish Time	20:52:29 June 9, 2019	**Battery Level**	3.7 Volts
Number of Intervals	5715.48 at 15 seconds	**Unit Calibration**	January 23, 2019 by UES New Delhi
Range	Geo:254.0 mm/s	**File Name**	UM12383_20190608210337.IDFH
Sample Rate	2048sps		
Operator/Setup:	Operator/factory.MMB		

Notes
Location: D Mehta Mahavir Heart Hospital
Client: Rites Ltd
User Name: SKYLAB
Lat: 21°11'2.50"N
Long: 72°48'49.45"E

	Tran	Vert	Long	
PPV	0.615	0.906	0.441	mm/s
ZC Freq	32	171	1.9	Hz
Date	Jun 9 /19	Jun 9 /19	Jun 8 /19	
Time	20:50:52	14:03:52	23:27:22	
Sensor Check	Passed	Passed	Passed	

Peak Vector Sum 0.918 mm/s on June 9, 2019 at 14:03:52



Time Scale: 10 minutes /div **Amplitude Scale:** Geo: 1.000 mm/s/div

4. Near Nirmal Nagar Society

Location is mainly residential area. Busy traffic is observed on the road. Location is near Surat-Kamrej Highway. At night time, Heavy traffic of trucks, multi axial vehicles etc. are observed. Overall transversal vibration is observed below 1.5 mm/s. Peak in the transversal direction is observed as 1.4 mm/s. Overall vertical vibration is observed 6.5 mm/s and peak

as 6.4 mm/s, which may be due to heavy vehicle passage. Overall longitudinal vibration is observed below 1.5 mm/s with peak of 1.25 mm/s. Heavy vehicular traffic was observed during night time. Graphical representation of the vibration is given in Figure 4.20.

Figure 4.20: Vibration Monitoring Results Near Nirmal Nagar Society

Instantel

Event Report

Histogram Start Time	11:45:18 June 5, 2019	**Serial Number**	UM12383 V 10-87 Micromate ISEE
Histogram Finish Time	11:35:14 June 6, 2019	**Battery Level**	3.8 Volts
Number of Intervals	5719.72 at 15 seconds	**Unit Calibration**	January 23, 2019 by UES New Delhi
Range	Geo:254.0 mm/s	**File Name**	UM12383_20190605114518.IDFH
Sample Rate	2048sps		
Operator/Setup:	Operator/factory.MMB		

Notes
Location: Near Nirmal Nagar Society
Client: Rites Ltd
User Name: SKYLAB
Lat: 21°14'1.98"N
Long: 72°54'25.96"E

	Tran	Vert	Long	
PPV	1.403	6.376	1.253	mm/s
ZC Freq	128	128	>200	Hz
Date	Jun 6 /19	Jun 6 /19	Jun 6 /19	
Time	01:04:33	11:23:18	00:11:48	
Sensor Check	Passed	Passed	Passed	

Peak Vector Sum 6.476 mm/s on June 6, 2019 at 11:23:18

Time Scale: 10 minutes /div **Amplitude Scale:** Geo: 1.000 mm/s/div

5. Swami Narayan Temple

Swami Narayan Temple is at inner road near Surat-Kamrej Highway. Location is mainly surrounded by residential buildings. Though the peak for the particle velocity for transversal direction is observed as 177.2 mm/s, overall transversal velocity is observed below 3 mm/s. Overall vertical vibration is observed below 3 mm/s. Peak in the vertical direction is

observed as 58.96 mm/s. Overall longitudinal vibration is observed below 2 mm/s with peak of 91.89 mm/s. Sudden peaks in the vibration is for very short duration. Movement of heavy goods was transported thereby resulting unusual high peaks. Graphical representation of the vibration is given in Figure 4.21.

Figure 4.21: Vibration Monitoring Results at Swami Narayan Temple

6. Shri Harihar Mahadev Temple

Shri Harihar Mahadev Temple is located in densely populated area at Lambe Hanuman Road. Location is surrounded by residential buildings, Hospitals etc. Overall transverse velocity is observed below 1 mm/s, and peak as 0.867 mm/s. Overall vertical vibration is

observed below 1.25 mm/s. Peak in the vertical direction is observed as 2.743 mm/s. Overall longitudinal vibration is observed below 0.75 mm/s with peak of 1.371 mm/s. Peak value observed is during the start of monitoring and may be due to sweeping work by the worker. Graphical representation of the vibration is given in Figure 4.22.

Figure 4.22: Vibration Monitoring Results at Shri Harihar Mahadev Temple

Instantel

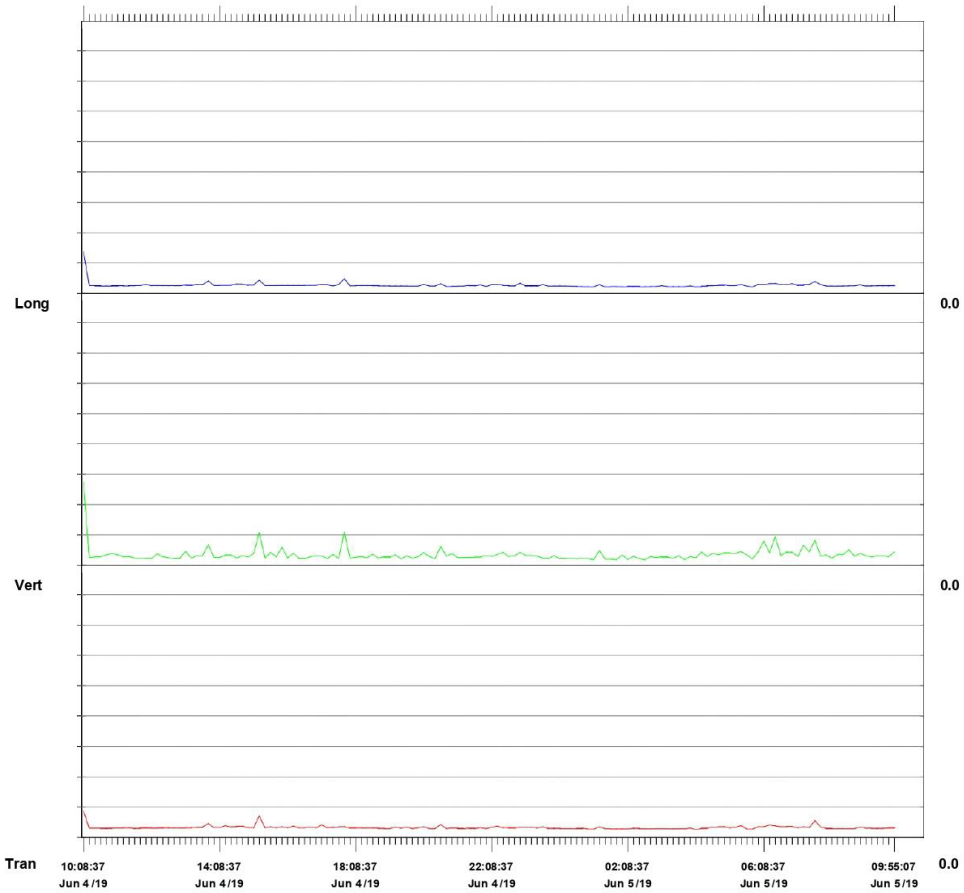
Event Report

Histogram Start Time	09:58:37 June 4, 2019	**Serial Number**	UM12383 V 10-87 Micromate ISEE
Histogram Finish Time	09:55:07 June 5, 2019	**Battery Level**	3.8 Volts
Number of Intervals	5746.02 at 15 seconds	**Unit Calibration**	January 23, 2019 by UES New Delhi
Range	Geo:254.0 mm/s	**File Name**	UM12383_20190604095837.IDFH
Sample Rate	2048sps		
Operator/Setup:	Operator/factory.MMB		

Notes
Location: Shri Harihar Mahadev Temple
Client: Rites Ltd
User Name: SKYLAB
Lat: 21°12'35.96"N
Long:72°51'22.79"E

	Tran	Vert	Long	
PPV	0.867	2.743	1.371	mm/s
ZC Freq	85	<1.0	>200	Hz
Date	Jun 4 /19	Jun 4 /19	Jun 4 /19	
Time	10:01:07	10:01:07	10:01:07	
Sensor Check	Passed	Passed	Passed	

Peak Vector Sum 2.751 mm/s on June 4, 2019 at 10:01:07



Time Scale: 10 minutes /div **Amplitude Scale:** Geo: 1.000 mm/s/div

7. Lambe Hanuman Temple

Lambe Hanuman Temple is located just at a distance of 100 meters from the Surat Railway Station. Location is characterized by commercial premises, vehicular traffic etc. Source of vibration at the location is mainly vehicular traffic. Vibration in the transversal direction was observed less than 5.25 mm/s. However, peak of 45.2 mm/s was observed. Vibration in the vertical direction was observed below 5 mm/s with peak of 4.02 mm/s. Though the sudden peak of 13.37 mm/s was observed for short period, overall longitudinal vibration was observed less than 5.25 mm/s. Sudden increase in the vibration for short duration was due to loading and unloading of the goods of fruits, vegetable etc. sellers at late night, early morning observed at location. Tri-axial graph is shown in Figure 4.23.

Figure 4.23: Vibration Monitoring Results at Lambe Hanuman Temple

Instantel

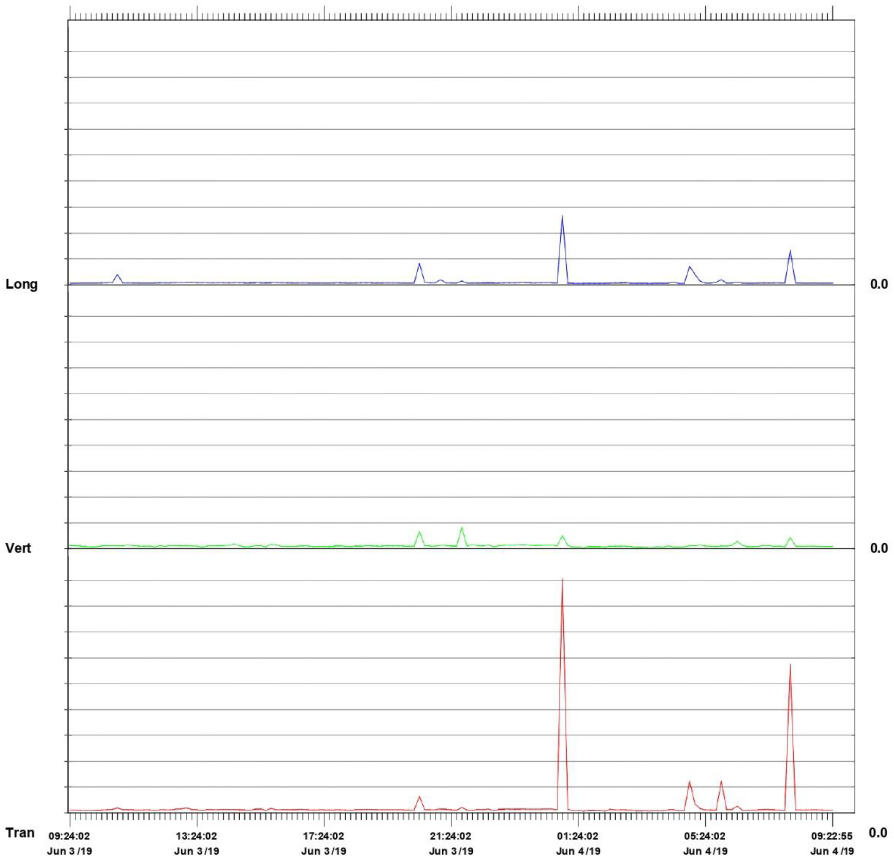
Event Report

Histogram Start Time	09:14:02 June 3, 2019	**Serial Number**	UM12383 V 10-87 Micromate ISEE
Histogram Finish Time	09:22:55 June 4, 2019	**Battery Level**	3.8 Volts
Number of Intervals	5795.52 at 15 seconds	**Unit Calibration**	January 23, 2019 by UES New Delhi
Range	Geo:254.0 mm/s	**File Name**	UM12383_20190603091402.IDFH
Sample Rate	2048sps		
Operator/Setup:	Operator/factory.MMB		

Notes
Location: Lambe Hanuman Temple, near Railway station
Client: Rites Ltd
User Name: SKYLAB
Lat: 21°12'15.55"N
Long: 72°50'32.34"E

	Tran	Vert	Long	
PPV	45.20	4.051	13.37	mm/s
ZC Freq	5.4	>200	2.2	Hz
Date	Jun 4 /19	Jun 3 /19	Jun 4 /19	
Time	00:52:32	21:43:02	00:52:32	
Sensor Check	Passed	Passed	Passed	

Peak Vector Sum 46.13 mm/s on June 4, 2019 at 00:52:32



Time Scale: 10 minutes /div **Amplitude Scale:** Geo: 5.000 mm/s/div

8. Clock Tower

Clock Tower location is surrounded by the various public places and road having heavy vehicular traffic. Location is at a distance of just 5 mins from Surat railway station. This is mainly Bazar area comprising various small as well as big shops, commercial premises, shops etc. Source of vibration at the location is mainly due to vehicular traffic. Vibration in transversal direction was observed less than 1.21 mm/s with peak for the velocity for transversal direction of 1.20 mm/s. Overall vertical vibration is observed to be less than 1.9 mm/s. Peak in the vertical direction is observed as 1.82 mm/s. Overall longitudinal vibration is observed less than 0.9 mm/s with peak of 0.84 mm/s. These vibration peaks are quite normal for busy vehicular traffic road. Graphical representation of the vibration is given in Figure 4.24.

Figure 4.24: Vibration Monitoring Results at Clock Tower

Instantel

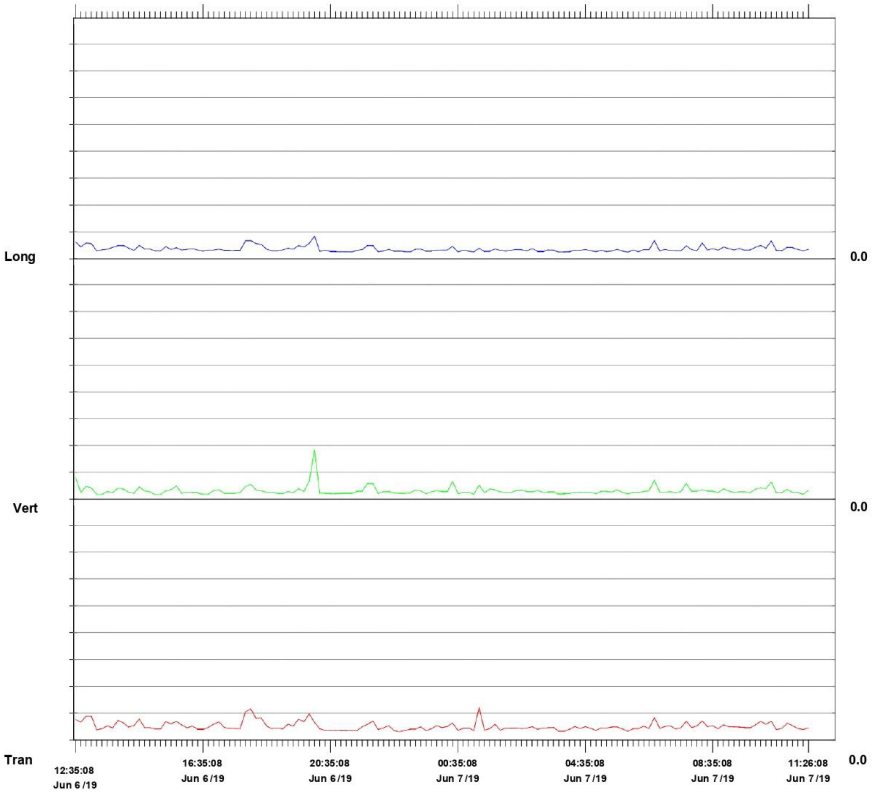
Event Report

Histogram Start Time	12:25:08 June 6, 2019	**Serial Number**	UM12383 V 10-87 Micromate ISEE
Histogram Finish Time	11:26:08 June 7, 2019	**Battery Level**	3.8 Volts
Number of Intervals	5523.98 at 15 seconds	**Unit Calibration**	January 23, 2019 by UES New Delhi
Range	Geo:254.0 mm/s	**File Name**	UM12383_20190606122508.IDFH
Sample Rate	2048sps		
Operator/Setup:	Operator/factory.MMB		

Notes
Location: Clock Tower
Client: Rites Ltd.
User Name: SKYLAB
Lat: 21°11'51.42"N
Long: 72°49'54.87"E

	Tran	Vert	Long	
PPV	1.206	1.821	0.835	mm/s
ZC Freq	23	>200	>200	Hz
Date	Jun 7 /19	Jun 6 /19	Jun 6 /19	
Time	01:08:08	19:56:23	19:56:23	
Sensor Check	Passed	Passed	Passed	

Peak Vector Sum 1.860 mm/s on June 6, 2019 at 19:56:23



Time Scale: 10 minutes /div **Amplitude Scale:** Geo: 1.000 mm/s/div

9. Rani Talab Masjid

Surrounding of Rani Talab Masjid monitoring location is comprises of commercial premises, Hospitals and residential buildings. Main source of the vibrations at the location is observed due to vehicular traffic. Overall vibration in the transversal direction is observed to be less than 1 mm/s and peak was observed as 0.938 mm/s. Peak vibration in the vertical direction was captured as 0.914 mm/s where as overall vibration observed to be less than 1 mm/s. Overall vibration in the longitudinal axis was observed to be less than 0.8 mm/s and peak as 0.785 mm/s. Graphical representation of the vibration is given in Figure 4.25.

Figure 4.25: Vibration Monitoring Results at Rani Talab Masjid

Instantel

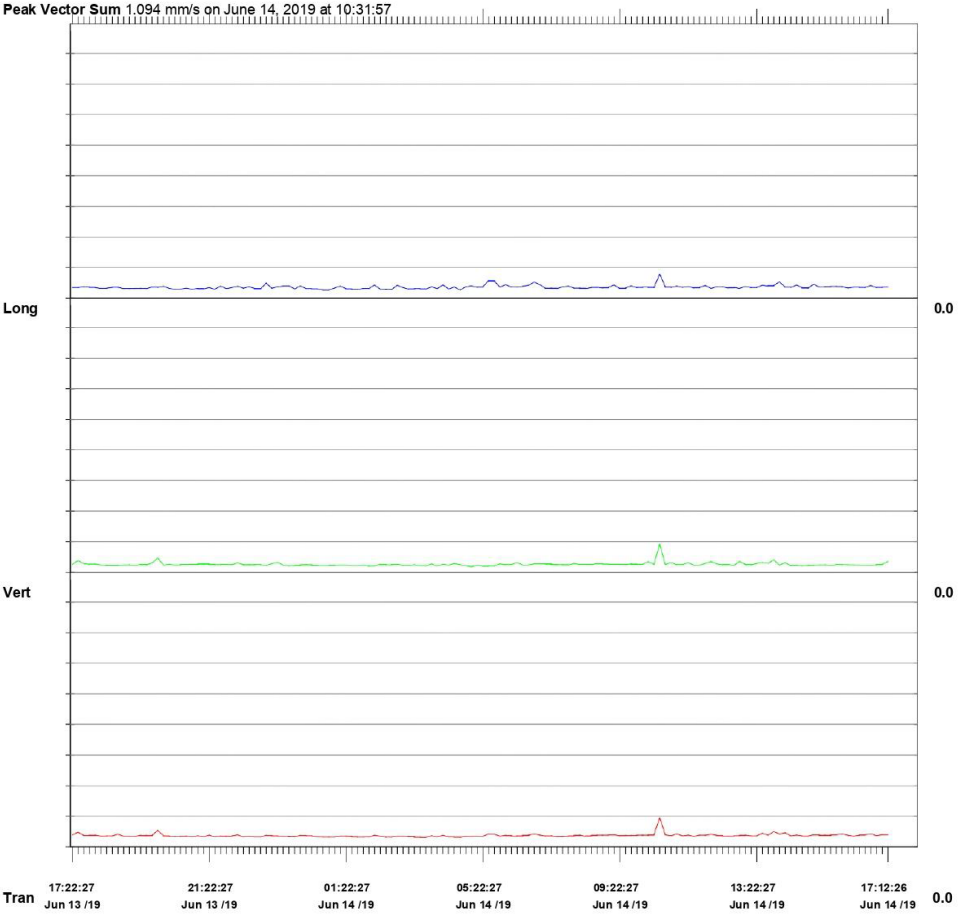
Event Report

Histogram Start Time	17:12:27 June 13, 2019	**Serial Number**	UM12383 V 10-87 Micromate ISEE
Histogram Finish Time	17:12:26 June 14, 2019	**Battery Level**	3.7 Volts
Number of Intervals	5759.97 at 15 seconds	**Unit Calibration**	January 23, 2019 by UES New Delhi
Range	Geo:254.0 mm/s	**File Name**	UM12383_20190613171227.IDFH
Sample Rate	2048sps		
Operator/Setup:	Operator/factory.MMB		

Notes
Location: Rani Talab Masjid
Client: Rites Ltd.
User Name: SKYLAB
Lat: 21°11'59.02"N
Long: 72°49'33.95"E

	Tran	Vert	Long	
PPV	0.938	0.914	0.780	mm/s
ZC Freq	51	>200	51	Hz
Date	Jun 14 /19	Jun 14 /19	Jun 14 /19	
Time	10:31:57	10:31:57	10:31:57	
Sensor Check	Passed	Passed	Passed	

Peak Vector Sum 1.094 mm/s on June 14, 2019 at 10:31:57



Time Scale: 10 minutes /div **Amplitude Scale:** Geo: 1.000 mm/s/div

10. Andrew Library

Andrew Library monitoring location is near the one of the end of Nehru Bridge. Surrounding of the location is comprises of institutions, recreational centres, residential buildings etc. Main source of the vibrations at the location is observed due to vehicular traffic. Overall vibration in the transversal direction is observed to be less than 2.2 mm/s and peak was observed of 78.2 mm/s. Peak vibration in the vertical direction was captured as 174 mm/s whereas overall vibration observed was less than 1.8 mm/s. Overall vibration in the longitudinal axis was observed as 2.4 mm/s and peak as 25.0 mm/s. Sharp peaks for the vibration was observed at the start of the monitoring, are may be due to the sudden pin time impact and not any recurring source impact. Graphical representation of the vibration is given in Figure 26.

Figure 4.26: Vibration Monitoring Results at Andrew Library

Instantel

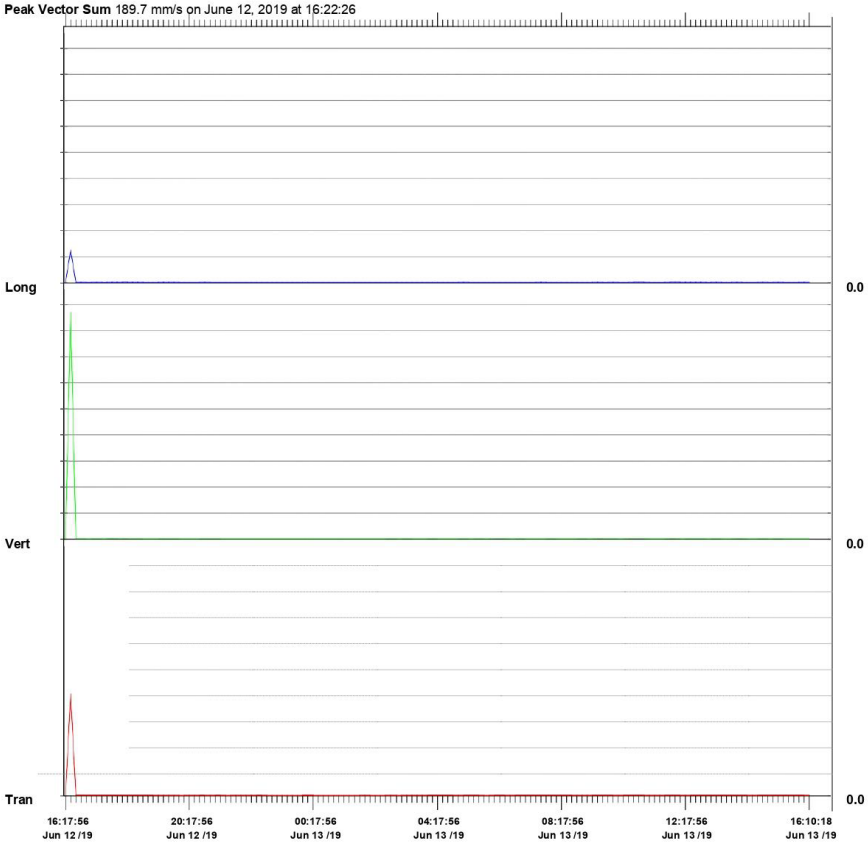
Event Report

Histogram Start Time	16:07:56 June 12, 2019	**Serial Number**	UM12383 V 10-87 Micromate ISEE
Histogram Finish Time	16:10:18 June 13, 2019	**Battery Level**	3.7 Volts
Number of Intervals	5769.48 at 15 seconds	**Unit Calibration**	January 23, 2019 by UES New Delhi
Range	Geo:254.0 mm/s	**File Name**	UM12383_20190612160756.IDFH
Sample Rate	2048sps		
Operator/Setup:	Operator/factory.MMB		

Notes
Location: Andrew Library, Chawk Bazar
Client: Rites Ltd.
User Name: SKYLAB
Lat: 21°11'48.90"N
Long: 72°49'9.05"E

	Tran	Vert	Long	
PPV	78.16	174.0	25.00	mm/s
ZC Freq	9.6	64	13.5	Hz
Date	Jun 12 /19	Jun 12 /19	Jun 12 /19	
Time	16:22:26	16:22:26	16:22:26	
Sensor Check	Passed	Passed	Passed	

Peak Vector Sum 189.7 mm/s on June 12, 2019 at 16:22:26



Time Scale: 10 minutes /div **Amplitude Scale:** Geo: 20.00 mm/s/div

4.5. Ecological Environment

An ecological study of the ecosystem is essential to understand the impact due to project development activities on the existing flora and fauna of the area. The present study was undertaken to predict changes as a result of project activities and to suggest measures for maintaining the conditions to the maximum possible extent. This section describes the ecology of the area based on field study and information compiled from secondary data available for the study/project area.

4.5.1. Forest

The geographic area of the Surat district is 7657 sq km. Forest cover in different canopy density classes, shrub and change in forest cover in comparison to 2009 assessment of Forest Survey of India is given in Table 4.22. The forest in the district is of moderately dense to open dense. The proposed alignment is passing through two parks named 'Gandhi Park' and 'Jawaharlal Nehru Park'.

Table 4.22: Forest Cover Area of Surat District

Type	Area in Sq. Km
Geographic Area	7657
Forest	
Very dense forest	84
Moderately dense forest	778
Open dense forest	454
Total Forest Area	**1316**
Percent of GA	17.19%

Source: Indian State Forest Report 2011, Forest Survey of India

4.5.2. Flora

Trees survey was carried out in the month of May - June 2019 along the proposed metro alignments and at station locations. About 2093 trees observed within ROW along the corridors and depots. Summary of tree enumeration is given in Table 4.23 and station wise inventory of trees is given in Table 4.24 and Table 4.25. The height and girth of the trees ranges from 3m to 14m and 30cm to 400cm respectively. The predominant tree species with their botanical name and local name along the corridor are given in Table 4.26.

Table 4.23: Summary of Tree Enumeration

S. No	Description	Number of Trees
1	Bhesan to Saroli Corridor	762
	Bhesan Depot (16.95 Ha)	319
2	Sarthana to Dream City Corridor	836
	Dream city Depot (24.09 Ha)	176
	Total	**2093**

Table 4.24: Detailed Tree Inventory For Bhesan To Saroli Corridor

Sr no	Place	Number	Cut	Transplant
1.	Bhesan depot (16.95 ha)	319	277	42
2.	Start to Bhesan	11	8	3
3.	Bhesan station	2	1	1
4.	Bhesanst to botanical garden	4	1	3
5.	Botanical garden station	0	0	0
6.	Botanical garden st to Ugatvarigruha	5	0	5
7.	Ugatvarigruha station	12	0	12
8.	Ugatvarigruhast to palampur road station	68	41	27
9.	Palampur road station	21	9	12
10.	Palampur road station to L P Savani school	91	11	80
11.	L P Savani school station	12	1	11
12.	L P Savani school to Performing art centre	26	1	25
13.	Performing art centre station	14	5	9
14.	Performing art centre stn to Adajaan gam	101	1	100
15.	Adajaan gam station	16	0	16
16.	Adajaan gam to Aquarium	15	1	14
17.	Aquarium station	15	2	13
18.	Aquarium to Bhadrinarayan temple	30	8	22
19.	Bhadrinarayan temple station	6	5	1
20.	Bhadrinarayan temple st to Ahathwachopati	67	38	29
21.	Ahathwachopati station	31	11	20
22.	Ahathwachopatist to Majura gate	19	4	15
23.	Majura gate station	7	1	6
24.	Majura gate st to Udhanadarwajast	33	6	27
25.	Udhanadarwaja station	10	2	8
26.	Udhanadarwajast to KamelaDarwaja	6	0	6
27.	KamelaDarwaja station	5	1	4
28.	KamelaDarwajast to Anjana farm	14	0	14
29.	Anjana farm Station	13	0	13
30.	Anjana farm st to Model town st	51	0	51
31.	Model town station	43	5	38
32.	Model town st to Magob	10	4	6
33.	Magob station	0	0	0
34.	Magobst to Bharat cancer hospital	0	0	0
35.	Bharat cancer hospital station	2	0	2
36.	Bharat cancer hospital st to saroli	1	0	1
37.	Saroli station	1	0	1
38.	Sarolist to end	0	0	0
	Total	**1081**	**444**	**637**

Table 4.25: Detailed Tree Inventory for SarthanaTo Dream City Corridor

Sr no	Place	Number	Cut	Transplant
1.	Start to Sarthana	10	1	9
2.	Sarthana station	2	0	2
3.	Sarthana to nature park station	6	1	5
4.	Nature park station	1	1	0
5.	Nature park station to Varachachopati garden	28	2	26
6.	Varachachopati garden station	18	0	18

7.	Vrachachopati garden st to sriswamirnarayanmandirkalakunj	**37**	20	17
8.	Sri swaminarayanmandirkalakunj station	0	0	0
9.	Sri swaminarayanmandirkalakunjst to start of the ramp	3	1	2
10.	Ramp and end of ramp	5	3	2
11.	**Kapodra station**	**6**	**0**	**6**
12.	**Labheshwar road station**	**9**	**3**	**6**
13.	**Central warehouse station**	**56**	**31**	**25**
14.	**Surat railway station**	**19**	**4**	**15**
15.	**Maskati hospital station**	**6**	**2**	**4**
16.	**Chowk bazaar station**	**10**	**4**	**6**
17.	Ramp	50	12	38
18.	Ramp end to kadarshahi mall	8	1	7
19.	kadarshahi mall station	27	3	24
20.	Kadarshahi mall to majura gate	23	1	22
21.	Majura gate station	13	2	11
22.	Majura gate station to roopali canal	70	42	28
23.	Roopali canal station	8	0	8
24.	Rooplai canal station to althantenanmnet	44	9	35
25.	Althantenanment station	27	5	22
26.	Althantenanment to Althangaamst	24	3	21
27.	Althangaam station	24	1	23
28.	Althangaamst to VIP road	73	3	70
29.	VIPA road station	22	8	14
30.	VIP road to surat women IIT	69	33	36
31.	Surat women IIT station	7	2	5
32.	Surat women IIT station to Bhimrad	10	6	4
33.	Bhimrad station	28	16	12
34.	Bhimradst to Convention centre	28	28	0
35.	Convention centre station	4	3	1
36.	Convention centre st to Dream city	17	17	0
37.	Dream city station	20	20	0
38.	Dream city to end	24	24	0
39.	Connecting line to depot	0	0	0
40.	Dream city depot (25.66 ha)*	176	176	0
	Total	**1012**	**488**	**524**

Text in bold are UG stations; * Trees are estimated from the maps

Table 4.26: Predominant Tree Species Along The Corridors

Botanical Name	**Local Name**	**Botanical Name**	**Local Name**
Ficus benghalensis	Bargad	*Cassia siamea*	cassia
Azadirachta indica	Neem	*Hammelia patens*	Hammelia
Peltoforam pteroocarpum	Peltoforam	*Cassia fistula*	Amaltash
Alstonia scholaris	Devils tree	*Moringa tinctorea*	Moringa
Pongamia pinnata	Karanj	*Samania saman*	Indian rain tree
Ficus religiosa	Pipal	*Ziziphus jujube*	Beri
Neolomarkia cadamba	Kadam	*Acacia nilotica*	Babul
Melia azedarach	Bakain	*Plolyalthia logifolia*	Asoka
Kigelia africana	Balam kheera	*Adansonia digitata*	African Baobab

Two giant African Baobab (*Adansonia digitata)* trees are observed within ROW opposite Croma Shopping Complex & Jodiyar Mata Mandir near proposed Adajan Gam-station along Bhesan to Saroli corridor. Each Baobab tree having about 14 m height and 11.2m and 10.2m of girth as shown in Figure 4.27. These tree species are native of Africa which is widely used by African Tribal community as medicinal plant. Out of these two Giant biobab tree, one tree is collapsed due to weather disturbance on 02-05-2020. The 450 year old tree is hallow inside and infected with termites. Surat Municipal Corporation proposed to preserve the felled tree at the same place.

Figure 4.27: Giant African Baobab Trees

The Besan depot is proposed on 16.95 ha. The predominant trees species present in proposed depot land are *Eucalyptus sp, Borassus flaberiformis, Leucaena leucocephala, Azadirachta indica, Acacia nilotica and Cassia aruculata*. The total number of tree species found in filed survey is 319. Significant number of shrub like *Prosopis julifera* are occupied in the entire depot area.

The *Prosopis julifera* is invasive species in many countries including India. However eradication of weed species like *P. julifera* is important as the species not having any medicinal value and absorbs ground water, macro & micro nutrients from the soil. It will create intra-specific competition between native species. The *Prosopis julifera* possesses abundant hybrid-vigour, and it can sustain harsh environment.

The fauna present in the depot are common and no REET species found in the depot area. The exotic and native environmental friendly species are given for transplantation and rest are removed. The *Prosopis julifera* covers monotonous plantation. Most of the feral fauna found shelter at the base of bushy & thorny habitation *Prosopis julifera*. There is no diversity within species, between species, and of ecosystems.

The present habitat in depot location is modified habitat. Modified habitats are areas that may contain a large proportion of plant and/or animal species of non-native origin, and/or where human activity has substantially modified an area's primary ecological functions and species composition. No loss of native species due to proposed Bhesan depot except exotic species. The habitat will be restored at Compensatory plantation and trans-plantation sites.

Mangrove: The *Avicennia marina* is the predominant mangrove species found at the left bank of Tapi river. Four mangrove shrubs of about 3m height and 30cm girth are found at the left bank of river Tapi near Chowpati Garden (Jawaharlal Nehru Park) is shown in Figure 4.28.

Figure 4.28: Avicennia Marina Mangroves Near Chowpati Garden

Coastal Regulatory Zone (CRZ): As per the CRZ maps of Surat prepared by National Centre for Sustainable Coastal Management, Chennai & Forest and Environment (Government of Gujarat), the portion of alignment passing from BAPS Swaminarayan temple to Chopati Garden comes under CRZ –II and CRZ-IVB. CRZ clearance needs to be taken from MoEFCC. The proposed new depot site backside of Dream city is devoid of CRZ.

Agriculture: As per the landuse/landcover map of study area (15km on either side of alignment) agriculture is the major landuse class i.e. 49% of total area. But the proposed alignment is passing through the road in urban area. It is not passing through the agriculture land. Surat is primarily an agricultural district with sugarcane and paddy as the

predominant crops. The other major crops cultivated are jowar, wheat, groundnut, banana, vegetables etc. About 65 % of land holdings are with small and marginal farmers and the average size of the holdings is 2.02 ha.

4.5.3. Fauna

The avi-fauna of the district is rich and varied; the mammalian fauna is possibly the poorest in the state because of the fact that much of the area is now under intense agriculture and settlements. Common birds observed in the vicinity of project area are owls, peacock, pigeons, crows, and cranes on consultation with the local people. It is learnt that no endangered bird species observed in the project area. The domestic animals in the project area are cows, bullocks, sheep, goats and dogs etc. Commercial cultivation of Tiger Shrimp is observed from Google Maps in some part of the land proposed for depot at Dream City.

Flamingo: The lesser flamingo is a local migrant in India. The flamingo (Phoeniconaias minor) was classified “Near Threatened” in the 2006 IUCN Red List of Threatened Species. However, as per Schedule of Indian Wildlife Protection Act 1972, lesser flamingo comes under Schedule-IV. The Lesser Flamingo is adapted to respond to changes in local environmental conditions by moving among wetlands, and thus depends on a network of suitable sites (Childress et al., 2007). Lesser Flamingos were largely feeding on algae and diatoms and probably on insect larvae and copepods as reported by Ali and Ripley (1987) and Bird-Life International (2009). Diets of Lesser Flamingos in India include diatoms from the surface of tidal mud.

Tapi River in Surat city is natural habitat of lesser flamingo for feeding & roosting. The flamingo forming flocks constantly move between feeding sites. Flamingos usually come to Surat for feeding and roosting in the months of November to February. However, they were spotted by local people in other months also. During the field study conducted by RITES team, the flock of lesser flamingo is observed between Sardar Bridge to Pt. Deendayal Upadhyay Bridge in Tapi River (Figure 4.29), which is feeding & roosting zone for flamingo.

Figure 4.29: Flock of Lesser Flamingo In Tapi River

4.5.4. Protected Areas in the region

No National Parks, Wildlife Sanctuary and Biosphere reserve etc are found within 5 km on either side of the metro corridors.

4.6. Sensitive Receptors

From the alignment drawings made available by GMRCL, the sensitive receptors were noted down corridor wise. Sensitive receptors such as educational institutes (schools, colleges, universities etc), hospitals, and religious places within 100 m from the central line of the proposed metro rail corridor has been demarcated. In the field study about 188 sensitive receptors found within 100 m from the centre line of proposed metro corridors. The list along with their geographical locations (latitude and longitude) is given in the Table 4.27 (A & B) and shown in the Figure 4.30.

Table 4.27: Sensitive Receptors within 100m from The Centre Line

(A) For Sarthana to Deream City Corridor

S. No.	SENSITIVE RECEPTOR	LONGITUDE	LATITUDE	DISTANCE (m)	DIRECTION
EDUCATIONAL					
1.	SGE College	72.89995	21.23030	16	LHS
2.	Sardar Patel Vidhya Bhawan	72.89187	21.2250211	26	LHS
3.	Tapovan Science And Commerce Academy	72.88722	21.2239525	13	
4.	Suman School	72.88604	21.22295	53	LHS
5.	Little Flower High School	72.87277	21.21764	66	LHS
6.	NM Jhaveri School (Union School)	72.8279	21.19901	-	below center line
7.	Ks Joshi School, Smt. Km Jariwala Girls Engineering College	72.82272	21.20125	-	below center line
8.	Anglo Urdu High School	72.82302	21.20094	82	LHS
9.	IP Mission School	72.82126	21.20095	93	RHS
10.	J J Training College	72.81883	21.19633	47	LHS
11.	Andrew Library	72.81942	21.19705	14	RHS
12.	Ramkrishna Vidhya Bhavan	72.85081	21.20696	21	RHS
13.	Gandhi College	72.81720	21.17957	98	RHS
14.	ICFAI University	72.81730	21.17493	52	RHS
15.	Institute Of Design And Center	72.81517	21.17001	25	
16.	Kinder Dream Pre School	72.81539	21.16996	33	RHS
17.	Ishwarlal Gulabbhai Desai Vidyalaya	72.81114	21.16129	51	RHS
18.	School	72.80832	21.15302	13	LHS
19.	Symga High School	72.82055	21.18659	100	RHS
HOSPITAL					
20.	Shree Physioteraphy And	72.90905	21.23593	18	LHS

S. No.	SENSITIVE RECEPTOR	LONGITUDE	LATITUDE	DISTANCE (m)	DIRECTION
	Fitness Clinic				
21.	Shree Krishna Dental Hospital	72.90153	21.2299	16	RHS
22.	Patel Hospital And Health Center	72.8976	21.22904	12	RHS
23.	New Life Hospital	72.89807	21.22872	32	LHS
24.	Vedanta Hospital & ICU	72.88453	21.22341	45	RHS
25.	Vatsalya Hospital & Apple Skin Care	72.88086	21.22124	8	RHS
26.	Vatsalya Hospital	72.86975	21.21639	75	LHS
27.	Santosh Hospital	72.85644	21.20937	72	LHS
28.	Women Hospital And Baby Tube Center	72.85644	21.21028	74	RHS
29.	Radhey Hospital And Maternity Home	72.85576	21.20948	40	LHS
30.	Akshar Hospital	72.83918	21.2006	92	LHS
31.	Rainbow Children Hospital	72.83683	21.20073	32	RHS
32.	Burhani Hospital	72.83434	21.19804	42	LHS
33.	Dumaswala Hospital	72.83315	21.19847	100	RHS
34.	Maskati Dharmarth Hospital	72.8326	21.19789	30	RHS
35.	Eye Hospital	72.81787	21.19427	73	LHS
36.	Kapse Children Hospital	72.8171	21.19308	25	LHS
37.	Paavan Orthopedic Hospital	72.81707	21.19297	27	LHS
38.	District TB Center	72.81956	21.18507	23	RHS
39.	Girish Group Of Hospital	72.81955	21.18508	25	
40.	Rahul Hospital And Clinic	72.81967	21.18516	26	
41.	Swagat Hospital	72.81929	21.18471	27	RHS
42.	Ketan Children Hospital	72.82089	21.18498	100	LHS
43.	Leela Health Care	72.81941	21.18404	18	LHS
44.	Sharda Hospital	72.8193	21.18367	13	LHS
45.	Vasan Eye Care Hospital	72.81894	21.18447	26	RHS
46.	Baby Beat Heart Center	72.81897	21.18355	34	RHS
47.	Harihar ENT Hospital	72.81886	21.18235	12	LHS
48.	Smt. Deepika & Dr. Hasmukhlal C Vankawala Children Hospital	72.81876	21.18209	42	LHS
49.	Arora Surgical Hospital	72.81887	21.18137	28	LHS
50.	Tulsi Hospital	72.81892	21.18092	46	LHS
51.	Pooja Hospital	72.81765	21.17512	11	RHS
52.	Shree Guru Nanak Dharmrath Hospital	72.81803	21.17442	75	LHS
53.	Park Nursing Home	72.81776	21.17398	68	RHS
54.	Rainbow Hospital	72.81682	21.17206	32	LHS
55.	Pal Hospital	72.8139	21.16787	13	RHS
56.	Sanjivani Hospital & ICCU	72.8143	21.16715	78	LHS

S. No.	SENSITIVE RECEPTOR	LONGITUDE	LATITUDE	DISTANCE (m)	DIRECTION
	Center				
57.	Amruta Hospital	72.81393	21.1663	54	RHS
58.	Sanjivani Hospital	72.81317	21.16473	16	RHS
59.	Jugal Hospital	72.81048	21.16001	9	RHS
60.	Sonal Hospital and Diabetes Clinic	72.82628	21.19948	95	LHS
61.	Bombay Multi Speciality Hospital	72.80471	21.1434	32	LHS
RELIGIOUS PLACES					
62.	Shyam Dham Mandir	72.90834	21.23641	54	RHS
63.	Shyam Temple	72.90381	21.23212	47	RHS
64.	Lambe Hanuman	72.88789	21.22303	9	RHS
65.	Kalakunj Swami Naryan Mandir	72.88047	21.22043	41	LHS
66.	Shri Harihar Mahadev Temple	72.85655	21.21004	1	RHS
67.	Shri Sambanath Jain Temple	72.85204	21.20796	54	RHS
68.	Ramkrishna Vidhya Bhavan	72.85081	21.20696	21	RHS
69.	Tarani Devi Temple	72.84563	21.20521	30	LHS
70.	Lambey Hanuman Temple	72.84306	21.2046	33	RHS
71.	Shri Goverdhan Nathji Bethakji	72.84199	21.20414	27	RHS
72.	Momin Masjid	72.84347	21.20446	5	RHS
73.	Gurukrpa Mandir	72.83543	21.19888		LHS
74.	Jagnat Mahadev Mandir	72.83438	21.19853	91	RHS
75.	Shree Santinath Bhagwan Jain Derasar	72.83213	21.19726	100	LHS
76.	Aebi Gebi Hanuman Temple	72.83112	21.19746	76	RHS
77.	Patalia Hanuman	72.83072	21.19798	31	RHS
78.	Minara masjid	72.83099	21.19889	15	RHS
79.	Bhagal masjid	72.83067	21.19853	1	RHS
80.	Chand sahid masjid	72.82778	21.19894	60	RHS
81.	Hussain masjid	72.82659	21.19862	50	LHS
82.	Markaz Masjid	72.82829	21.19924	51	RHS
83.	Badi Masjid	72.82644	21.2002	67	RHS
84.	Rani Talab Masjid	72.82656	21.19968	-	Below Centre Line
85.	Masjid e Kadim	72.82541	21.20059	6	LHS
86.	Sabiri Masjid	72.82234	21.20151	53	RHS
87.	Champa Masjid	72.82081	21.20018	-	Below Centre Line
88.	Chruch Of North India	72.82185	21.20119	73	RHS
89.	Bethany House of Worship	72.82115	21.20094	70	RHS
90.	Masjid e Hamayu Sarai	72.82077	21.19963	77	LHS
91.	Baxi Masjid	72.81957	21.19933	-	Below

S. No.	SENSITIVE RECEPTOR	LONGITUDE	LATITUDE	DISTANCE (m)	DIRECTION
					Centre Line
92.	Shri Adinath Jain Derasar	72.81978	21.19916	55	LHS
93.	Raj Rajeshwar Mahadev	72.8202	21.19918	5	RHS
94.	Masjid e Nagina	72.81954	21.19856	13	LHS
95.	CNI Christ Chruch	72.81992	21.19788	1	RHS
96.	Bakriwali Masjid	72.81989	21.19713	23	LHS
97.	Jaghnath Mahadev	72.81767	21.195	57	RHS
98.	Mubarak Mosque	72.81811	21.19473	10	RHS
99.	Botwala Masjid	72.81754	21.19354	32	LHS
100.	Chandraprabhu Swamis Jain Temple	72.81704	21.19263	62	LHS
101.	Mulla Hashim Masjid	72.81691	21.19179	41	RHS
102.	Hatimi Masjid	72.81693	21.19206	62	RHS
103.	Aksa Masjid	72.81697	21.19135	43	RHS
104.	Tulja Maa Mandir	72.81747	21.19117	40	LHS
105.	Sayed Jafer Sahid Bawa Dargah	72.81673	21.19041	78	RHS
106.	Khankah E Chishtiya Nizamiya Rehmaniya	72.81799	21.18882	50	RHS
107.	Ahle Sunnat Wal Jamaat Chhoti Masjid	72.81774	21.18882	35	RHS
108.	Hazrat Murad Shah Bava Rahmatullahi Alaihi	72.81888	21.18812	32	LHS
109.	Ahle Sunnat Wal Jamat Hazrat Jamalshah Masjid	72.81893	21.18821	21	RHS
110.	Sai Mandir	72.8193	21.18597	37	RHS
111.	Hazrat Peer Syed Hasan Shahid Rahmatullahi Alaihi	72.81958	21.18579	21	LHS
112.	Shree Ananddayak Vasu Pujay Swami Jain Sang	72.81812	21.17514695	21	LHS
113.	Shri Ram Temple	72.81365	21.16604	10	LHS
114.	Shankeshwar Da jain Temple	72.81967	21.18518	62	LHS
115.	Baba Amarnath Panchmukhi Temple	72.81138	21.16128	39	RHS
116.	Khodiyar Mata Ji Temple	72.80826	21.1531	24	LHS
117.	SAI Baba Temple	72.81343	21.16575	43	LHS
118.	Dade Elahi Masjid	72.83646	21.2004	27	LHS
119.	Temple	72.81625	21.19357	26	LHS
120.	Temple	72.81510	21.17089	17	LHS
121.	Masjid	72.817595	21.192022	73	LHS
122.	Temple	72.816471	21.192824	-	ROW
123.	Temple	72.85911	21.188131	32	RHS
124.	Kahar Machi Samaj Temple	72.81695	21.19475	68	RHS
125.	Mosque	72.81672	21.19458	12	RHS

S. No.	SENSITIVE RECEPTOR	LONGITUDE	LATITUDE	DISTANCE (m)	DIRECTION
126.	Mastana Shah Dargah	72.84227	21.20423	28	RHS
127.	Masjid	72.84265	21.20432	14	RHS
128.	Temple	72.87204	21.21782	-	ROW
129.	Temple	72.89387	21.22628	6	RHS
130.	Temple	72.85721	21.20865	-	ROW

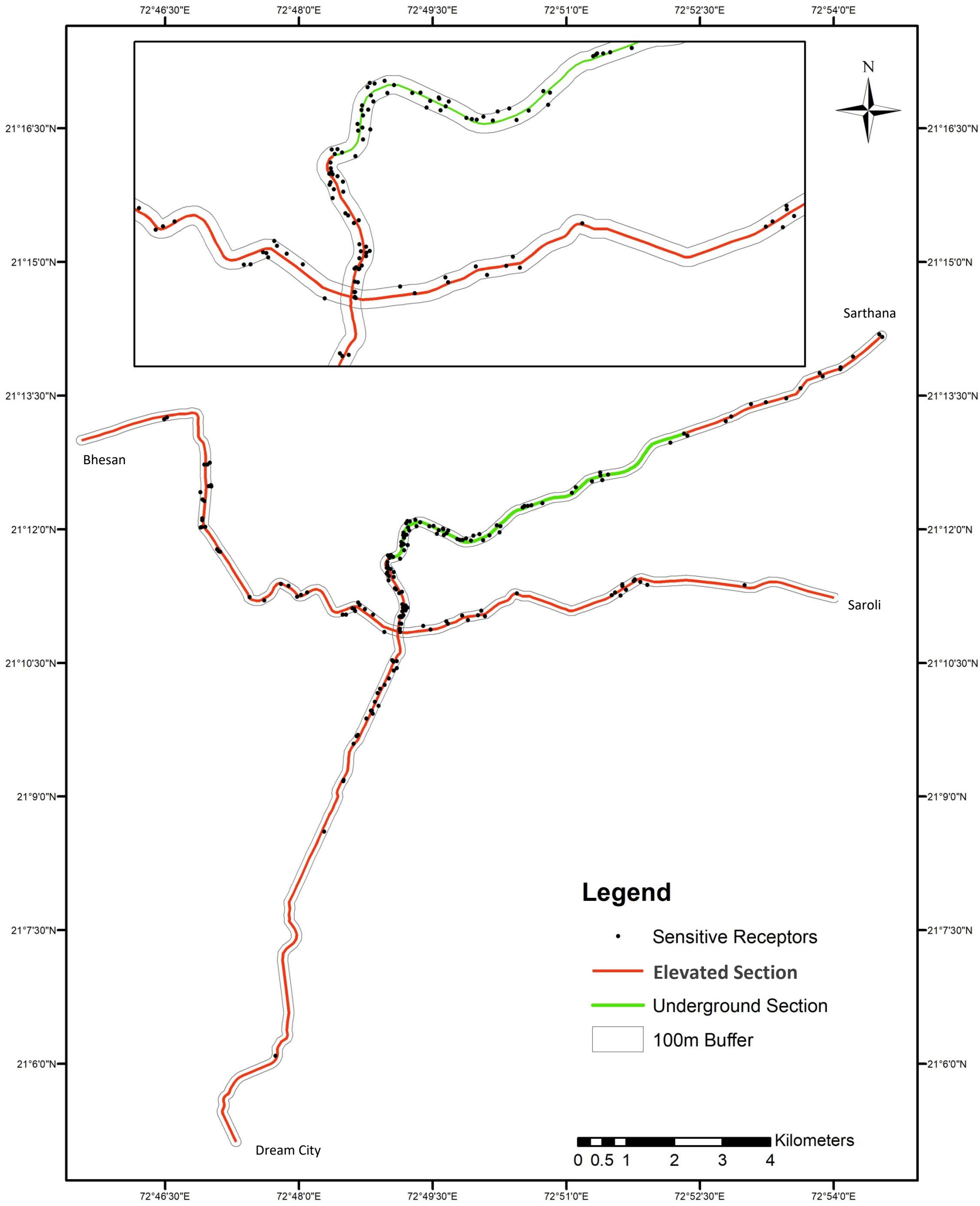
Note: Receptors in Blue color for U/G section and in Black color for Elevated Section

(B) For Bhesan To Saroli Corridor

S. No.	SENSITIVE RECEPTOR	LONGITUDE	LATITUDE	DISTANCE (m)	DIRECTION
EDUCATIONAL					
1.	South Gujarat Medical College	72.76666643	21.22066543	62	RHS
2.	Perfact Brilliant Classess	72.78230876	21.21128404	17	RHS
3.	Lord Krishna School	72.78265616	21.20906462	93	LHS
4.	Vidhya Kunj School	72.78233706	21.20550967	54	RHS
5.	L P Savani School	72.78248532	21.2009322	37	RHS
6.	Friends Classess	72.7845439	21.19704768	37	RHS
7.	Dakshina	72.79549574	21.19009697	11	RHS
8.	Nutan Vidhyalay Madhyamik Schoo	72.79987775	21.18775958	15	RHS
9.	KP College Of Commerce	72.80856401	21.18438719	77	LHS
10.	INFID	72.80984603	21.18710885	89	LHS
11.	Dr S.S. Gandhi Engineering College	72.81596737	21.18074611	100	RHS
12.	G & G. V. Kadilwal	72.82161701	21.18118238	92	LHS
13.	RMG Maheswari Public School	72.86166309	21.19195255	10	RHS
14.	Rose Buds Nursery School	72.82816276	21.18259729	26	LHS
15.	Lord Krishna International School	72.86116369	21.1888764	100	RHS
16.	Kidzee School	72.86116369	21.1888764	64	RHS
RELIGIOUS					
17.	Sati Mata Temple	72.77444346	21.22034034	42	RHS
18.	Shree Sidhi Vinayak Temple	72.7755008	21.22093562	37	RHS
19.	Shanidev Temple	72.78287924	21.21207484	35	LHS
20.	Shree Ram Temple	72.78300546	21.21227406	80	LHS
21.	Shigontar Mata Mandir	72.78302518	21.20812638	47	LHS
22.	Shri Rameshwar Neelkanth Temple	72.78315762	21.20783574	97	LHS
23.	Amba Mata Temple	72.78225298	21.20509665	17	RHS
24.	Jalaram Temple	72.78497268	21.19588041	32	LHS
25.	Bhrugudeep Jalram Temple	72.78493823	21.19588081	16	LHS
26.	Aaj Shree Khodiyar Mata Ji Mandir	72.79063177	21.1875468	15	LHS
27.	Badrinarayan Temple	72.79989308	21.18800927	10	LHS
28.	Bholenath Temple	72.80122145	21.18832396	20	LHS
29.	Surat Yatim Khana Masjid	72.81184907	21.18504562	65	LHS
30.	Gajjarwadi Jain Temple	72.81066031	21.18485716	90	RHS
31.	Udhwana Darwaja Masjid	72.8300858	21.18283373	41	LHS

S. No.	SENSITIVE RECEPTOR	LONGITUDE	LATITUDE	DISTANCE (m)	DIRECTION
32.	Bileshwar Mahadev Mandir	72.83475209	21.1837045	50	RHS
33.	Masjide Aaisha	72.83988701	21.18712083	11	LHS
34.	Shanti Nath Swetambari Mandir	72.86156471	21.18826416	67	RHS
35.	Sai Baba Temple	72.86263787	21.18956446	17	RHS
36.	Masjid/Temple	72.83327	21.18371	5	RHS
HOSPITAL					
37.	Palanpor Urban Health Center	72.78137241	21.20705015	99	RHS
38.	Sai Medical Hospital & Maternity Home	72.78233337	21.20197401	30	RHS
39.	Arpan Children Hospital	72.78236191	21.20200374	23	RHS
40.	Shubham Hospital	72.78230233	21.20060231	37	RHS
41.	Divyesh Eye Hospital	72.78231619	21.20038815	25	LHS
42.	Dhameliya Kidney Hospital & Lithotripsy Centre	72.79214018	21.18714477	15	RHS
43.	BAPS Pramukh Swami Hospital	72.79850884	21.18901748	43	LHS
44.	Adventist Work Hardt Hospital	72.80835753	21.18434609	50	RHS
45.	Care Hospital	72.809875	21.18533239	27	RHS
46.	Bridge Cross Hospital	72.86555737	21.18949527	62	RHS
47.	Kalpa Hospital	72.8611607	21.18867375	20	RHS
48.	Tri Star Hospital	72.81006483	21.18519469	46	RHS
49.	Tender Touch Children Hospital	72.8115287	21.18481086	55	LHS
50.	G D Mehata Mahavir Heart Hospital	72.8125946	21.18405599	38	LHS
51.	Shreyas Ano-Rectal Hospital & Research Centre	72.81914338	21.18108108	10	
52.	Arora Surgical Hospital	72.81914338	21.18108108	47	LHS
53.	Nirmal Hospital	72.82477335	21.18146528	11	LHS
54.	Navsarjan Hospital	72.82747503	21.18254744	80	LHS
55.	Apple Hospital	72.83120844	21.1830755	48	RHS
56.	Vatsalya Children Hospital	72.83367936	21.18459388	72	LHS
57.	Chirayu Hospital	72.86492805	21.19001557	60	RHS
58.	Bharat Cancer Hospital	72.88334386	21.18933534	26	LHS

Figure 4.30: Map of Sensitive Receptors



Chapter 5 : Negative Environmental Impacts

5.1. General

Environmental impacts could be positive or negative, direct or indirect, local, regional or global, reversible or irreversible. This chapter discusses the potential negative impacts on the environment due to project activities. The negative impacts due to proposed project activities identified on the basis of project description (Chapter 3) and environmental baseline data (Chapter 4). Impacts on ambient air quality, noise and vibration level have been quantified while impact on water, soil and archaeological / heritage assets are presented in terms of `alert` in case norms and standards are likely to be violated during construction and operation of the project. Impact on ecology has been assessed quantitatively in terms of trees to be felled / relocated and in terms of adverse impact on aquatic species or their habitats. Other impacts have been flagged. Significant impacts pre-construction, during construction and during operation of the project have been assessed.

Negative impacts likely to result from the proposed development have been described in this Chapter under the following headings while recommendations for mitigating measures have been elaborated in Chapter 7.

- Impacts due to Project Location;
- Impacts due to Project Design;
- Impacts due to Construction; and
- Impacts due to Project Operation.

For each of these headings, potential impacts have been considered, while recommendations for mitigating measures have been stated in Chapter 7.

Impacts due to the proposed Metro project and their brief description along with rating are given in Table 5.1.

Table 5.1: Rating of Environmental Impacts

S. No	Parameter	Impact Rating	Short Term / Long Term
A.	**IMPACTS DUE TO PROJECT LOCATION AND DESIGN**		
1.	Displacement and loss of livelihood of Project Affected People (PAPs)	To be assessed as part SIA report	Permanent, negative, irreversible, can be mitigated, scale to be evaluated.
2.	Change of Land use	R2	Permanent, negative, irreversible, can be mitigated, small scale.
3.	Loss of trees and impact on ecology	R1	Permanent, negative, irreversible, can be mitigated, small scale.
4.	Drainage and Utilities: Diversion/shifting	R2	Short term and/or permanent, negative, irreversible can be mitigated; small scale.

S. No	Parameter	Impact Rating	Short Term / Long Term
5.	Impact on Archaeological Monuments and Heritage Assets	R4	Permanent, negative, irreversible, can be mitigated, small scale.
6.	Right of Way: Optimal choice of length of elevated and underground sections.	R1	Permanent, negative as well as positive, irreversible, can be mitigated, small scale.
7.	Alignment: • Route and design of alignment • Stations, Track design and Architecture	R1	Permanent, negative as well as positive, irreversible & can be mitigated, small scale.
8.	Inter Modal Integration	R1	Long term, negative as well as positive, irreversible, can be mitigated, small scale. Negative Impact on environment around station will be significant if integration of metro with other modes is not implemented.
9.	Use of Energy and Water at stations and depots	R1	Permanent, negative, irreversible & can be mitigated, small scale.
10.	Risk Due to Natural Hazards	R4	Permanent, negative, irreversible & can be mitigated, small scale.
B.	**IMPACTS DUE TO PROJECT CONSTRUCTION**		
1.	➢ Air pollution: • Particulate air pollution due to activities like excavation; • emissions due to transportation of muck and material ➢ Noise, Vibration	R1	Temporary, negative, irreversible can be mitigated, small scale.
2.	Disposal of muck, C&D waste and hazardous waste; pre-casting and material yards	R1	Permanent, negative, irreversible, can be mitigated, small scale.
3.	Water demand and water quality	R1	Temporary, negative, reversible, can be mitigated, small scale.
4.	Soil erosion and land subsidence	R2	Temporary, negative, irreversible, can be mitigated, small scale.
5.	Traffic diversions	R2	Temporary, negative, reversible, can be mitigated, small scale.
6.	Labor camp and on-site labour safety/ welfare	R1	Temporary, negative, reversible, can be mitigated, small scale.
7.	Supply of construction material	R2	Temporary, negative, irreversible, can be mitigated, small scale.
C.	**IMPACTS DUE TO PROJECT OPERATION**		

S. No	Parameter	Impact Rating	Short Term / Long Term
1.	Noise and Vibration	R1	Permanent, negative, irreversible, can be mitigated, small scale.
2.	Energy and water supply at stations and depots	R1	Permanent, negative, irreversible, can be mitigated, small scale.
3.	Traffic congestions around stations	R1	Permanent if inter modal integration is not done, negative, reversible, can be mitigated, small scale.
4.	Impacts due to Depots: Water supply, Waste water disposal, Oil Pollution, Noise Pollution, Solid Waste disposal, Loss of trees.	R1	Permanent, negative, reversible except in case of ecology, can be mitigated, small scale.
D.	**POSITIVE IMPACTS DUE TO PROJECT**		
1.	Employment Opportunities	Positive impact	Permanent, positive, small scale.
2.	Benefits to Economy: access, reduced costs of road infrastructure, vehicle operating & time, accidents.	Positive impact	Permanent, positive, large scale.
3.	Reduction in road traffic, fuel consumption and air pollution, GHG emission	Positive impact	Permanent, positive, large scale.

Rating: R1: Significant negative impact is expected. R2: Some negative impact is expected.
R3: Extent of impact is unknown R4: No impact is expected

5.2. Environmental Impacts

This section identifies and appraises the negative impacts on various aspects of the environment likely to result from the proposed development. The environmental aspects are:

- Land Environment
- Water Environment
- Air Environment
- Noise Environment
- Ecological Environment

The impacts on the above environmental components have been further assessed during various phases of project cycle namely project location, project design, construction and operation.

5.3. Impacts due to Project Location

- Displacement and loss of livelihood of Project Affected People (PAPs)
- Change of Land use
- Impact on/loss of wildlife/trees/forest
- Utility/Drainage Problems

- Impact on Archaeological Monuments and Heritage Assets

5.3.1. Displacement and loss of livelihood of Project Affected People (PAPs)

The project affected people are addressed in a separate Social Impact Assessment (SIA) report.

5.3.2. Change of Land Use

Land will be required permanently for stations, and running sections. Both government and private land will be acquired for the project and the details are given in civil engineering in the DPR. No forest land is required for the proposed project corridors.

5.3.3. Impact on Ecological Environment

The proposed corridors are in urban/city area. An inventory of trees along the project corridor was carried out. Number of trees likely to be affected due to the proposed project is about 2093. Most of the trees affected at median of the road along the corridor and on either side of the road at metro stations. The inventory of trees within corridor is given in the Table 5.2 and details from station to station and depot are presented in Annexure 5.1. No rare or endangered tree species have been observed during field studies.

Table 5.2: Number of Trees along the Corridor/Depots

S. No	Alignment/Depots	Number of Trees	Cut	Transplant
1.	Bhesan - Saroli	762	167	595
2.	Sarthana – Dream City	836	312	524
Sub-Total along the Alignment (A)		**1598**	**479**	**1119**
3.	Bhesan Depot	319	277	42
4.	Dream City Depot	176	176	0
Sub-Total at Depot (B)		**495**	**453**	**42**
Total Trees (A+B)		**2093**	**932**	**1161**

With removal of these trees, the process for CO_2 conversion will get affected and the losses are reported below:

Total number of Trees : 932
Decrease in CO_2 absorption @ 21.8 Kg/year/ tree : 20,318 kg/year
Decrease in Oxygen production @ 49 Kg/year/ tree : 45,668 kg/year

It is observed during field survey most of the tree species are exotic. No rare endangered endemic and threatened (REET) species are observed during the field study.

Bhesan Depot is located on SMC land and the area is covered largely with shrubs of Prosopis Julifera (Kejdi). About 319 trees have been identified in the Depot area, out of which 277 trees are to be cut and 42 trees are to be transplanted. The compensatory Afforestation has been planned in line with Saurashtra Felling of trees (infliction of Punishment) Act, 1951 & amendment Rule 2008, Forest Conservation Act 1980 and ESS 6 of World Bank. Deputy Conservator of Forests, Surat in consultation with the GMRCL has identified 15 ha land near village Anaval for Afforestation.

5.3.4. Utility/Drainage Problems

The proposed Metro corridors are planned to run through the urban area above the ground (elevated) in less densely populated and underground in populated and sensitive areas. The alignment will cross drains, large number of sub-surface, surface and utility services, viz. sewer, water mains, storm water drains, telephone cables, overhead electrical transmission lines, electric pipes, traffic signals etc. These utilities/ services are essential and have to be maintained in working order during different stages of construction by temporary/ permanent diversions or by supporting in position. Plans and cost of such diversions are covered in the section on Civil Engineering.

5.3.5. Impact on Archaeological Monuments/ Heritage Sites

No legally protected Archaeological Monuments / Heritage structures are affected due to the proposed metro rail project. Therefore, ESS 8 of World Bank and The Ancient Monuments and Archaeological sites and Remains Act, 1958 amended in 2010 are not triggered.

5.4. IMPACTS DUE TO PROJECT DESIGN

Impacts due to project design are seen in following ways:

- Consumption of energy and water at stations and noise impact of underground line in trade off with visual intrusion resulting from elevated line.
- Measures relevant to both underground and elevated alignment

5.4.1. Right of Way

Impact due to project design shall be vary depending upon whether the alignment is located underground, elevated or at-grade. In case of underground metro the space at ground level and above can continue to be used for roads and light structures and visual intrusion will be minimize. Visually less-intrusive viaduct and stations can be constructed subject to construction cost on account of specialized formwork and high strength materials.

Underground metro also allows the advantage of low noise as compared to elevated line during construction and operation. However energy consumption in underground metro is higher compare to elevated metro.

5.4.2. Alignment, Stations, Track design and Architecture

An alignment with less number of curves and radius - which is desirable rather than minimum, optimal station spacing, track with elastic fittings result in decrease in energy consumption, wear & tear and noise & vibration. Elevated metro with sleek structural elements provides aesthetically appealing structures. The spatial design of station has significant impact on safely of passengers, time spent in ingress & egress from station and energy consumption in stations.

5.4.3. Inter Modal Integration

Physical and operational integration of metro with other modes especially walk, public transport and intermediate public transport (hired modes) is found to increase ridership and decrease congestion inside and outside the stations. Negative Impact on environment around station will be significant if integration of metro with other modes is not implemented.

5.4.4. Use of Energy and Water at stations and depots

Consumption of energy for climate control, lighting and other facilities at stations is significantly reduced by proper design of passenger flow inside stations, space & facilities inside stations and multimodal integration facilities outside stations.

5.4.5. Risk Due to Natural Hazards

The project area lies in Zone III as per revised Seismic Zoning Map of India corresponding to moderate seismic hazard. Engineering construction shall be done so as to meet codal provisions. No other natural hazards such as due to climate change are foreseen.

5.5. IMPACT DUE TO PROJECT CONSTRUCTION

Negative environmental impacts are:

- Air pollution
- Noise Pollution
- Vibration and risk to existing buildings
- Impact due to Muck disposal
- Impact due to construction/demolition waste
- Impact due to Hazardous waste
- Impact due to Pre-casting yards and Material stockpiling

- Increased water demand
- Impact on ground and surface water quality
- Soil erosion and land subsidence
- Impacts due to traffic diversions
- Impacts due to Labour Camp
- Welfare of labour on site
- Safety of labour
- Impacts due to supply of construction material
- Impacts on human health
- Impact on Flamingo

5.5.1. Air pollution

Air pollution occurs due to excavation, loading and unloading of construction materials, and emissions from vehicles, construction equipment and DG sets etc. It also occurs in sites of muck disposal, debris disposal and pre-casting yards.

Air pollution due to transportation of construction material and excavated/fill material is quantified as follows by considering the BS III vehicles. Trucks and cranes are required to transport civil construction material from pre-cast yards and batching plants to construction site and between construction site and soil disposal site. Emission during the period of construction due to truck movement on account of transportation of civil construction material and disposal/backfill of earth is estimated as summarized in Table 5.3.

Table 5.3: Total Emissions due to Truck Movement during Construction

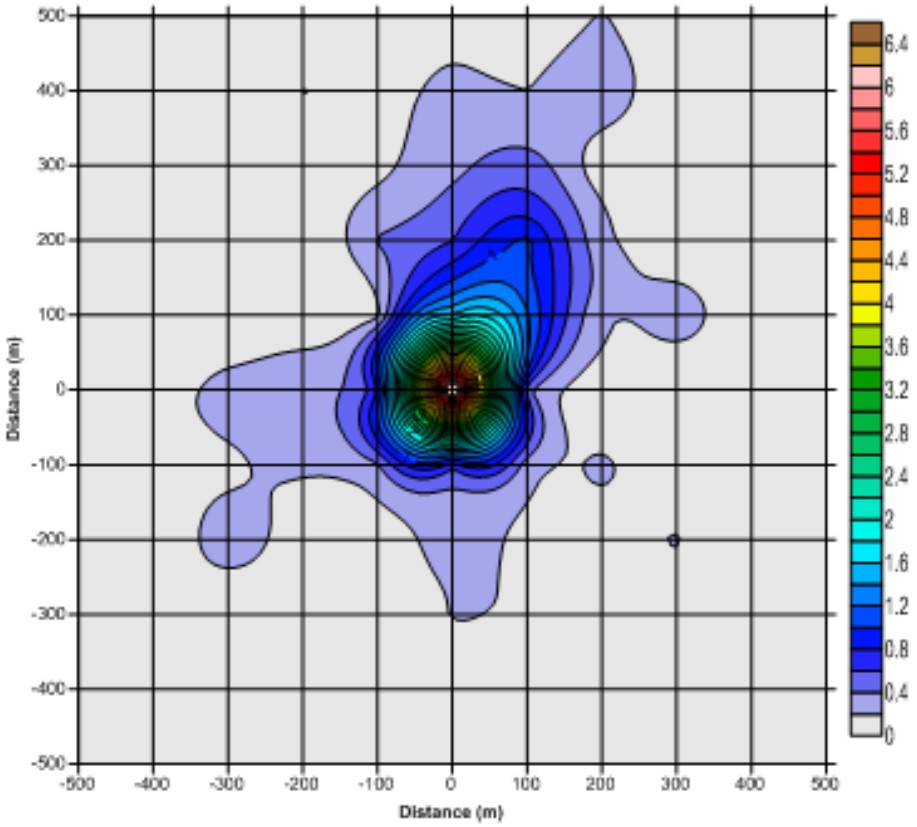
Pollutant	Quantity (ton)	
	Bhesan to Saroli	Sarthana to Dream City
Carbon Monoxide	13.48	12.68
Particulate Matter 2.5	0.60	26.33
Hydro-Carbons	0.60	0.00
Nitrogen Oxide	28.00	12.68
VOC	4.40	4.14
Carbon dioxide	1740.18	1636.43
Fugitive dust	3.3	3.1

During the construction phase excavation activity particulate matter PM10 and PM2.5 are expected to be the critical pollutants associated with the construction activity. In this case, pollution emission sources shall be distributed throughout the project site and shall fall under the category of area source. The land acquired is fairly flat, so extensive formation work is not expected during this phase.

For the atmospheric dispersion modelling study, United States Environmental Protection Agency (USEPA) approved and also recommended by the Ministry of Environment and Forests and Climate Change (MoEFCC), Government of India, regulatory air quality model i.e. Industrial Source Complex Short Term (ISCST3-version 3) is applied to predict ground level incremental concentrations (GLCs) of concerned critical pollutants. Prior to air quality modelling exercise, meteorological condition during one season over project site is extensively studied. Concentrations are estimated for the critical pollutants assessed over appropriate averaging times (i.e. 8 hours and 24 hours) based on the applicability of the National Ambient Air Quality Standards (NAAQS).

In the absence of information regarding the quantity and type of construction equipment to be deployed at any particular time, emission factor $1.81X10^{-5}$ g/m^2/sec (EPA 2006) for general construction activities is used for PM10 emissions estimates. However, for PM2.5, 10% of PM10 is assumed. Considering the phase-wise modelling area of 100 m x 100 m of the project site it is assumed that the excavation activity will occur in the daytime between 0800 IST and 1800 IST. The cumulative impact due to baseline and construction activites are within the permissible limits of Indian and IFC guideline.

Figure 5.1: Spatial Distribution of PM2.5 (μg/m3) during Construction: Without Control



Impact due to DG SET: Due to the confined nature of heavy construction activity during limited period, tailpipe emissions from construction equipment are assumed to be negligible but DG set of small capacity of 50 KV will be used for power back-up supply may be considered as air pollution point source. Estimated emission rate of criteria pollutants

PM2.5, PM10, SO_2 and NO_x along with source characteristic used for atmospheric dispersion modelling is provided in the Table 5.4.

Figure 5.2: Spatial Distribution of PM10 (µg/m3) during Construction: Without Control

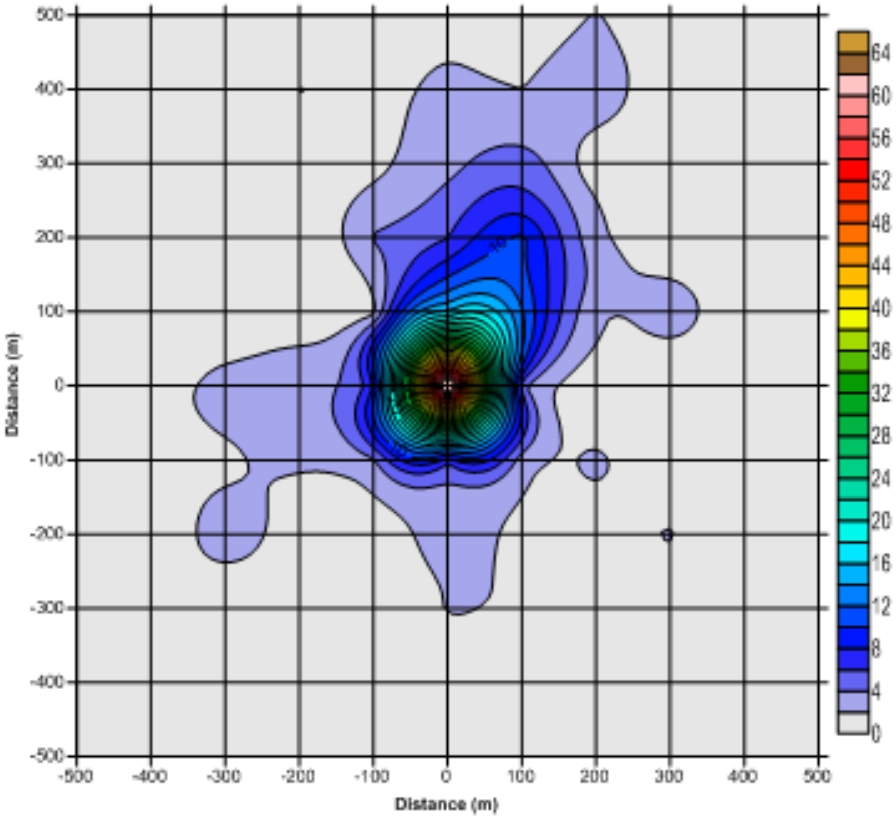


Table 5.4: Emission Source Characteristics

Stack No.	DG (KVA)	Point Source Characteristics								
		$PM_{2.5}$	PM_{10}	SO_2	NO_x	CO	Height (m)	velocity (m/s)	Temp (k)	Diameter (m)
1	1x50	0.001	0.002	0.01	0.1	0.04	5	10.0	708	0.10

Air quality modelling under the observed meteorological condition, the maximum incremental load on applicable averaging hour 24 hours for criteria pollutants PM2.5, PM10, SO2 & NOx, and 8 hours average for CO due to the operation of DG set during daytime construction activity between 0800 IST and 1800 IST are predicted to be about 0.12 µg/m3, 0.24 µg/m3, 1.19 µg/m3, 11.87 µg/m3 and 10.69 µg/m3 respectively and occurs within 100 m from the DG set location. Spatial variations of incremental load of PM2.5, PM10, SO2, NOx, and CO on modelled 1 km x 1 km impact zone on grid size of 100 m x 100 m are shown in Figure 5.3 to Figure 5.7. Cumulative emissions due to baseline and DG sets are within the permissible limits of Indian standards; however these emissions are exceeding IFC permissible limits for PM10 & PM2.5; and SO2 exceeding at A6, A7, A8, A9, A15, A17, A18, A20. Modelling results shows that minimal short impact on the ambient air environment can be envisaged during construction phase. Even though the air quality impact will be

temporary but adequate environment management plan can be suggested during the construction phase.

Figure 5.3: Spatial distribution of 24-hours average PM2.5 conc. (µg/m3):DG Set

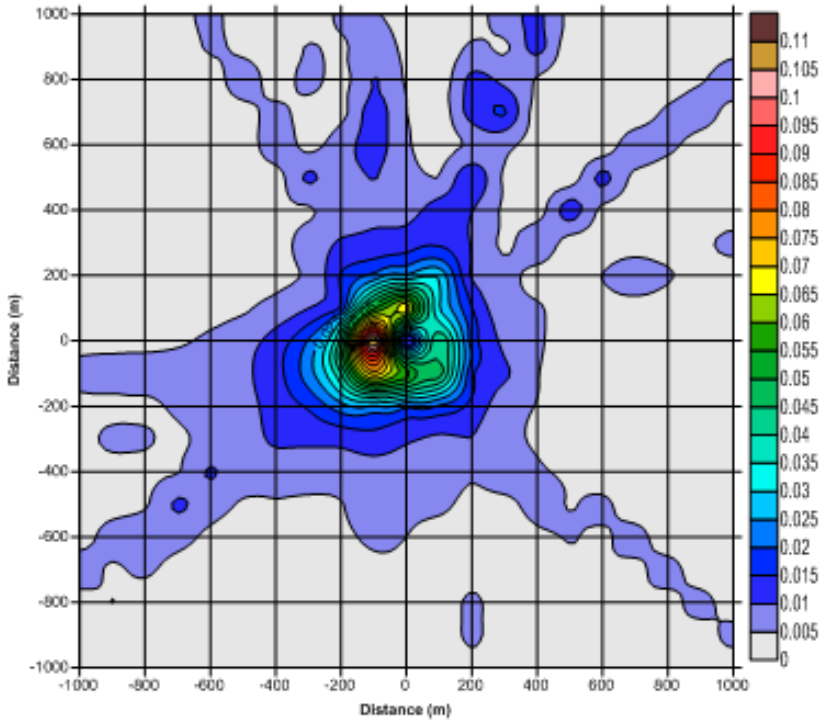


Figure 5.4: Spatial distribution of 24-hours Average PM10 conc. (µg/m3): DG Set

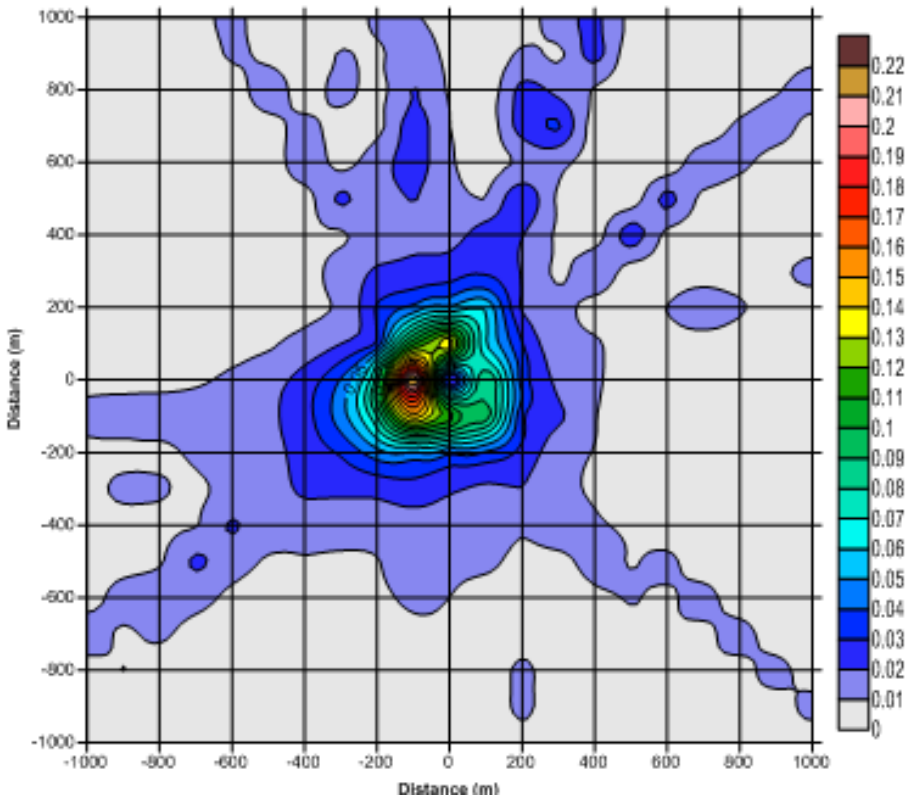


Figure 5.5: Spatial distribution of 24-hours average SO2 conc. (µg/m3): DG Set

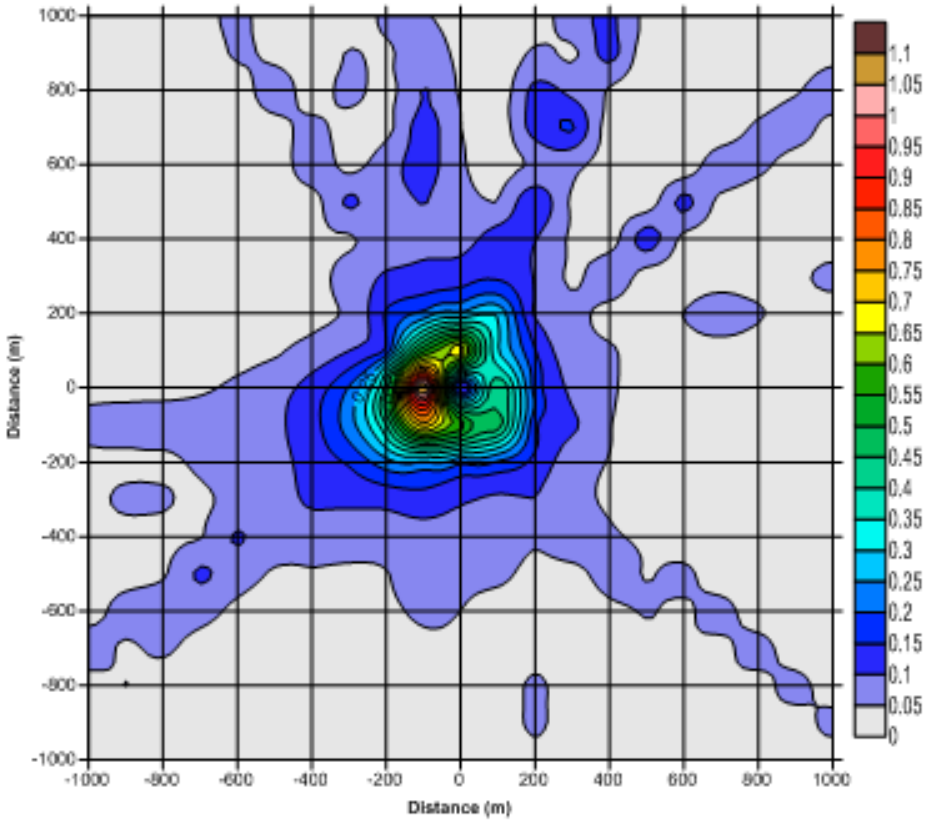


Figure 5.6: Spatial distribution of 24-hours average NOx conc. (µg/m3): DG SET

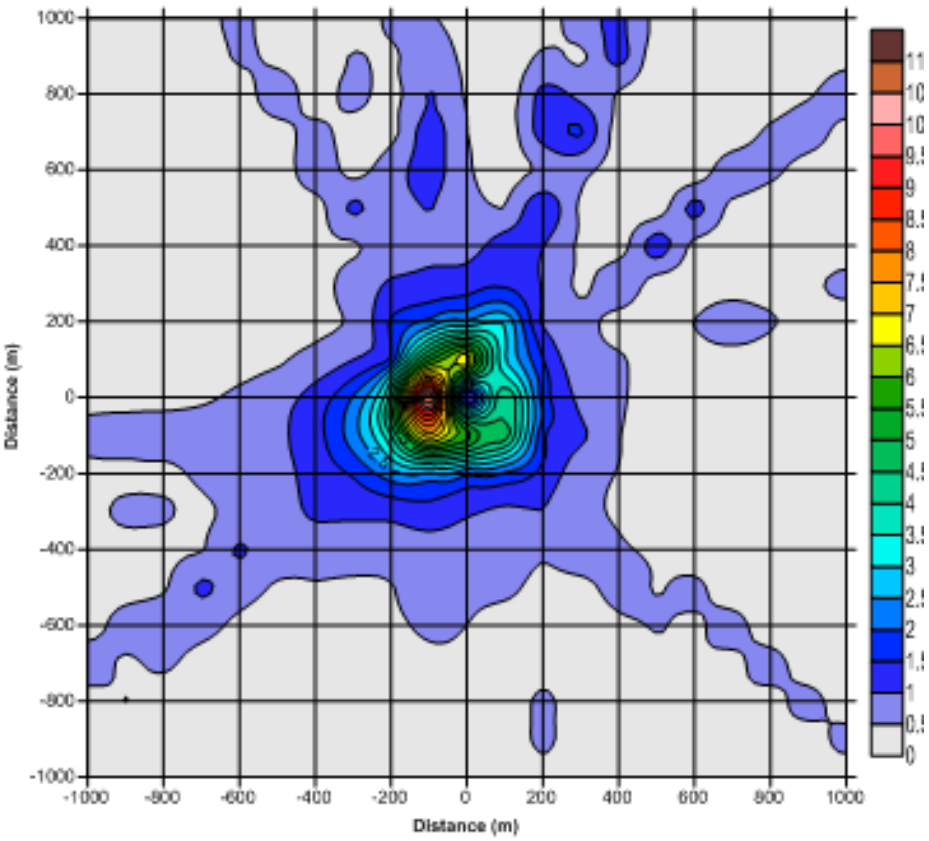
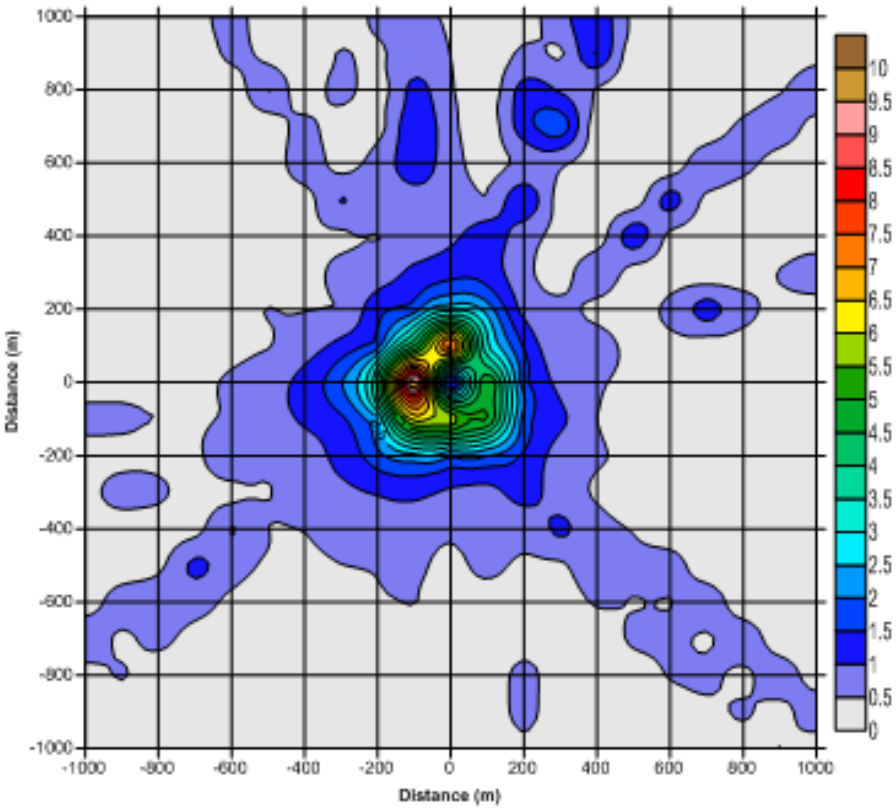


Figure 5.7: Spatial distribution of 8-hours average CO conc. (µg/m3): DG SET



5.5.2. Noise Pollution

Noise is a contributing factor to degradation of human health. The major sources of noise pollution during construction are movement of vehicles for transportation of material and equipment. The assessment of the impacts of noise on the surrounding community depends upon:

- Characteristics of noise source (instantaneous, intermittent, or continuous in nature)
- Time of day at which noise occurs
- Location of noise source with respect to noise receptor

For modelling purpose, terrain is considered flat and there are no sound absorptive materials are present in the direction of the sound wave propagation so as to formulate the worst-case scenario. For predicting noise emissions impacts due to proposed project, the noise emission sources are examined during both construction and operational phases.

Noise during the construction period could be bothersome to nearby residences. Construction workers would also be subject to construction noise while working on the site. Construction would take approximately three to four years to complete. Construction would be carried out in several discrete steps along one or more construction areas (i.e. construction would not take place along the entire alignment at one time), each with its own mix of equipment and, consequently, its own noise characteristics. Metro Rail

construction would involve cut-and-fill activities, placing ballast, and installing new track and paving. For the Project, the highest construction noise levels would likely be associated with any pile driving typically associated with wall and bridge structure construction. Construction noise is temporary and would vary widely both spatially and time-wise over the course of the project's construction.

The most prevalent noise source at Project construction sites would be internal combustion engines. Earth-moving equipment, material-handling equipment, and stationary equipment are all engine-powered. Mobile equipment operates in a cyclical fashion, but stationary equipment (e.g. generators and compressors) operate at sound levels that are fairly constant over time. Because trucks would be present during most phases and would not be confined to the Project site, noise from trucks could affect more receptors. Other noise sources would include impact equipment and tools such as pile drivers, if pile foundations are needed. Impact tools could be pneumatically powered, hydraulic, or electric.

Construction noise would be intermittent, occurring during construction at various locations in the Project area. Construction noise levels would depend on the type, amount, and location of construction activities. The type of construction methods would establish the maximum noise levels of construction equipments used. The amount of construction activity would quantify how often construction noise would occur throughout the day. The location of construction equipment relative to adjacent properties would determine any effects of distance in reducing construction noise levels.

Because the means and methods of construction would not be known until a contractor is selected for the Project, analysis of construction noise would be based on typical activities and equipment used for demolition, excavation, and erection works. Both daytime and night time construction activities will be analyzed, since it is possible that construction work would occur 24 hours a day. Maximum Environmental Noise Levels (MENL) that would be applicable to construction noise is associated with the Project. However, sound originating from temporary construction sites as a result of construction activity can be exempted except in so far as such provisions relate to the reception of noise which include residential properties, recreational and entertainment facilities [e.g. camps, parks, camping facilities, and resorts], and community service facilities [e.g. orphanages, homes for the aged, hospitals, health and correctional facilities] between the hours of 10:00 p.m. and 7:00 a.m.

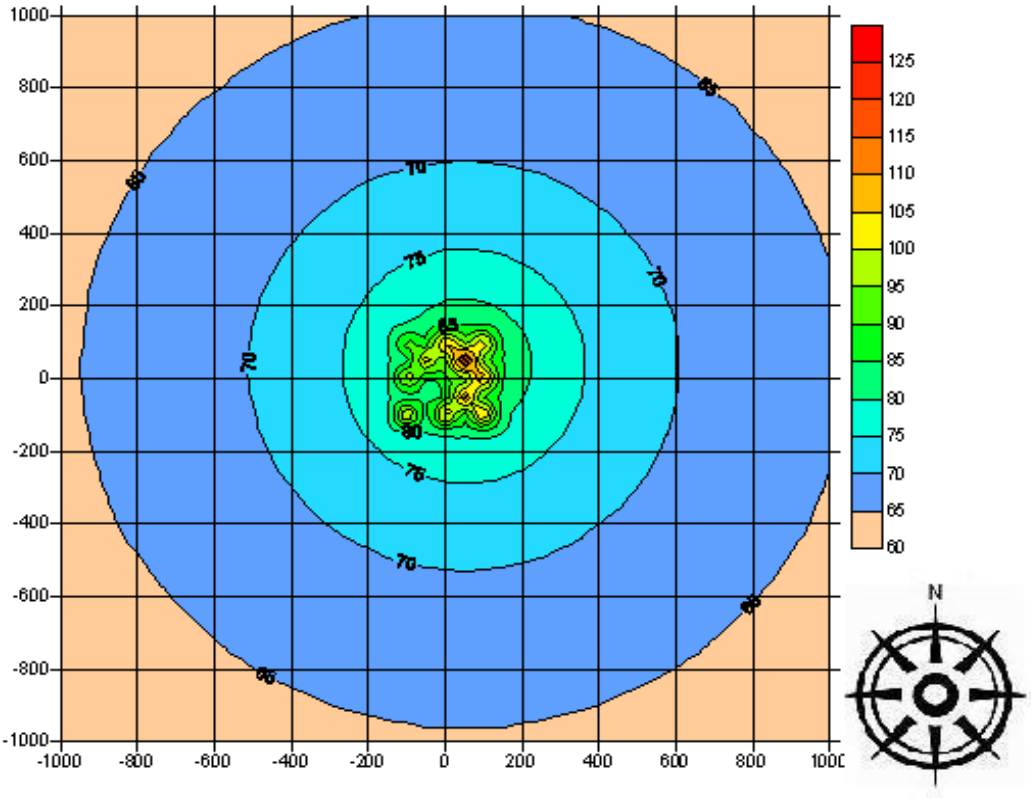
In absence of noise generating construction equipment details during large construction activities, the typical major noise generating sources envisaged during construction phase and their typical noise level value considered for impact assessment are presented in Table 5.5.

Noise modelling during construction phase was carried out using noise model "DHAWANI" assuming that all the above equipment emit noise considering as worst-case scenario. The spatial variation of the predicted noise levels at an interval of 5 dB(A) without control around the project site on the area of 1 km x 1 km are shown in Figure 5.8.

Table 5.5: Typical Noise Levels Generated from Construction Equipment

Name of Source	Noise Level at 1 m from source (calculated) in dB (A)
Air Compressor	111
Back Hoe/Loader	105
Concrete Mixer Truck	109
Concrete Pumper	94
Concrete Vibrators	101
Cranes - mobile	105
Dump Truck	107
Generator	75
Hammering	110
Jackhammer	112
Pile Driver	124
Radial Arm Saw	104

Figure 5.8: Spatial Variation of Construction Equipment Noise Levels dB(A)



Modelling result shows that noise level meets the CPCB Noise Quality Standards of 65 dB (A) at a distance of about 900 m for commercial area; similarly IFC noise quality standard of 70 dB(A) at a distance of 600 m for commercial area. Uncontrolled noise levels generated from construction equipment, in the range of 94-124 dB (A) have been considered for prediction purpose. However, the CPCB standards specified for limited construction equipment reflect

that noise emission specifications for such equipment should not exceed 75 dB (A). The noise levels predicted here are without mitigation measures.

It is assumed that with the adoption of the mitigation measures noise levels will be further restricted within very short distances from the source. With respect to occupational exposure, the permissible threshold is 90 dB (A). Thus, based on the modelling results it can be concluded that all sensitive receptors (i.e. labour colonies) should be located beyond 125 meters from the noise generating source location during construction activities.

5.5.3. Vibration Impacts and Risk to Existing Buildings

If significant impacts due to vibration are expected, mitigation measures have to be implemented and building condition survey have to be conducted before, during and after construction. Transportation and Construction Vibration Guidance Manual, Caltrans, September 2013) specified threshold criteria for various structures are listed in Table 5.6. These criteria for monuments are more stringent than those prescribed in UK, Germany, Switzerland and Japan. Vibration source levels for typical construction equipment are listed in Table 5.7.

Table 5.6: Guideline Vibration Damage Threshold Criteria

Structure and Condition	Maximum PPV (in/sec)	
	Transient Sources	Continuous/Frequent Intermittent Sources
Extremely fragile historic buildings, ruins, ancient monuments	0.12	0.08
Fragile buildings	0.2	0.1
Historic and some old buildings	0.5	0.25
Older residential structures	0.5	0.3
New residential structures	1.0	0.5
Modern industrial/commercial buildings	2.0	0.5

Note: Transient sources create a single isolated vibration event, such as blasting or drop balls. Continuous / frequent intermittent sources include impact pile drivers, pogo-stick compactors, crack-and-seat equipment, vibratory pile drivers, and vibratory compaction equipment.
Source: Transportation and Construction Vibration Guidance Manual, Caltrans, September 2013

Table 5.7: Vibration Source levels for Construction Equipment

Equipment		PPV at 25 ft (in/sec)	Approximate L_v# at 25 ft
Pile Driver (impact)	upper range	1.518	112
	Typical	0.644	104
Pile Driver (sonic)	upper range	0.734	105
	Typical	0.170	93
Calm shove drop (slurry wall)		0.202	94

Equipment		PPV at 25 ft (in/sec)	Approximate L_v# at 25 ft
Hydromill (slurry wall)	in soil	0.008	66
	in rock	0.017	75
Large bulldozer		0.089	87
Caisson drilling		0.089	87
Loaded trucks		0.076	85
Jackhammer		0.0345	79
Small bulldozer		0.003	58
# RMS velocity in decibels (VdB) re 1 µinch/sec			

Source: Transit Noise and Vibration Impact Assessment, Federal Transit Administration, May 2006

Construction activities involve various sources of vibrations such as blasting, pile driving, dynamic compaction of weak soils, and operating heavy machines. Dynamic effects of these sources may create substantial vibration problems for surrounding buildings influencing structures, sensitive devices, and people. The level of structural vibrations caused by construction work depends mostly on interaction of three major factors: dynamic sources, geology, and structures. Each of them affects structural vibrations. Only dynamic sources can be modified in certain degree to comply with vibration limits. The rest of the two cannot be changed.

The factors which affect the level of vibration at in the ground, arising from any activity, are: the amount of energy transmitted into the ground by the source; the rate of attenuation of the energy as it propagates through the ground; and the distance of the observation point from the location at which the energy enters the ground. Piling methods differ from many other vibration sources in that the position of the source which transfers energy into the ground continually changes as piling progresses, since the tip gets progressively deeper and encounters different soil and the length of the pile shaft in contact with the ground increases as driving progresses. A further difference is that the actual energy source does not come into direct contact with the ground, except in the case of end-driven piles. Therefore, factors which may need to be considered are the details of the pile and piling hammer or driver, the nature of the ground into which the pile is being driven and the distance from the pile to the measurement location.

A. Vibration Impact due to Vibratory Pile Driver

Vibratory pile generates continuous vibration. The vibro-driver clamps to the top of the pile and operate by means of one or more pairs of horizontally opposed contra-rotating eccentric weights. Operating frequencies are typically between 25 and 50Hz. The vibration reduces the shear strength of the soil close to the pile, thereby reducing the friction at the pile-soil interface. The combined weight of the pile and the driver cause the pile to be driven into the ground. With reference to various studies and research carried out for the vibratory pile driving by TRL and FTA, following simulation model for predicting the resultant vibration in PPV (Peak particle velocity) (PPV in mm/s):

$$Vres = kv / x^{ß}$$

Where,
Vres = is vibration in mm/s at a distance
kv = Constant related to geology structure
x = is the distance measured along the ground surface from the pile
ß = constant varying subject to attenuation factor

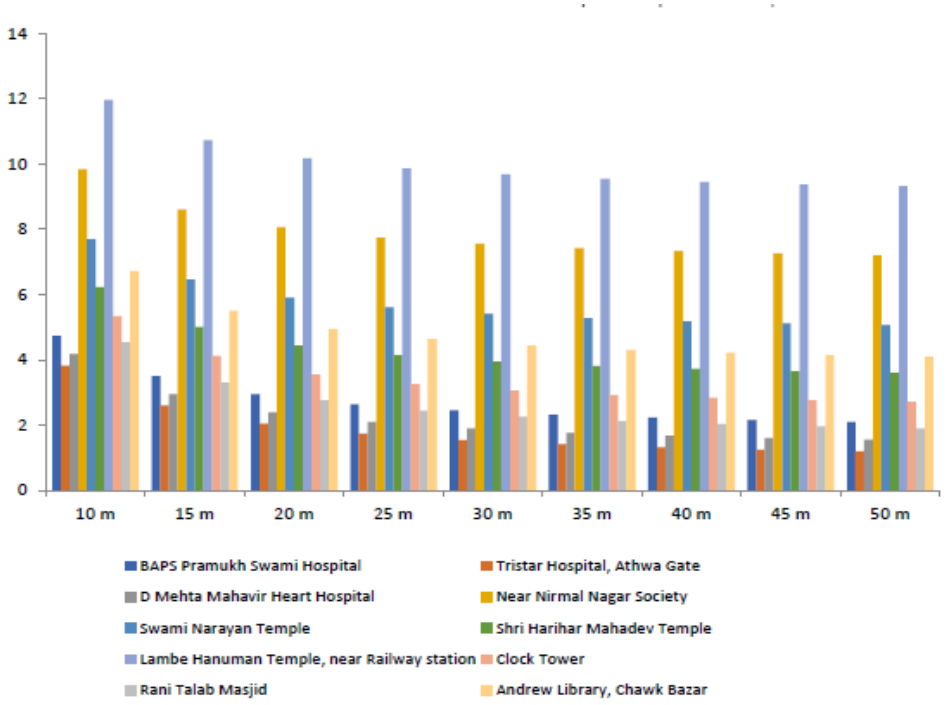
Hence, by considering extreme condition at monitored sites, predictable vibrations are as given in Table 5.8 and graphically shown in Figure 5.9.

Table 5.8: Vibration due to Vibratory Pile Driver

Sr No.	Location Name	PPV (mm/s)	Vibration at horizontal distance from source in PPV (mm/s)								Permissible
			10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m	
1	BAPS Pramukh Swami Hospital	1.73	4.74	3.51	2.95	2.65	2.45	2.32	2.23	2.16	2.54
2	Tristar Hospital, Athwa Gate	0.82	3.83	2.60	2.05	1.74	1.55	1.41	1.32	1.25	2.54
3	D Mehta Mahavir Heart Hospital	1.18	4.19	2.95	2.40	2.09	1.90	1.77	1.67	1.60	2.54
4	Near Nirmal Nagar Society	6.84	9.84	8.61	8.06	7.75	7.56	7.43	7.33	7.26	7.62
5	Swami Narayan Temple	4.69	7.70	6.47	5.91	5.60	5.41	5.28	5.19	5.12	6.35

Note: Standards for Fragile Buildings, Historic and Old Buildings, Older Residential Structures as per Transportation and construction Vibration Guidance Manual, Caltrans, September 2013

Figure 5.9: Graphical Representation of the Vibrations due to Vibratory Pile



Vibration generated due to vibratory piles will be within the permissible limits within a distance of 30m at BAPS Pramukh Swami Hospital; 20m at Tristar Hospital, Mahavir Heart

Hospital & Swami Narayan Temple; and 25m at Nirmal Nagar Society. Figure 5.10 to Figure 5.14 shows the Vibration Contours at monitored locations due to Vibratory Pilling.

Figure 5.10: BAPS Pramukh Swami Hospital

Figure 5.11: Tristar Hospital, Athwa Gate

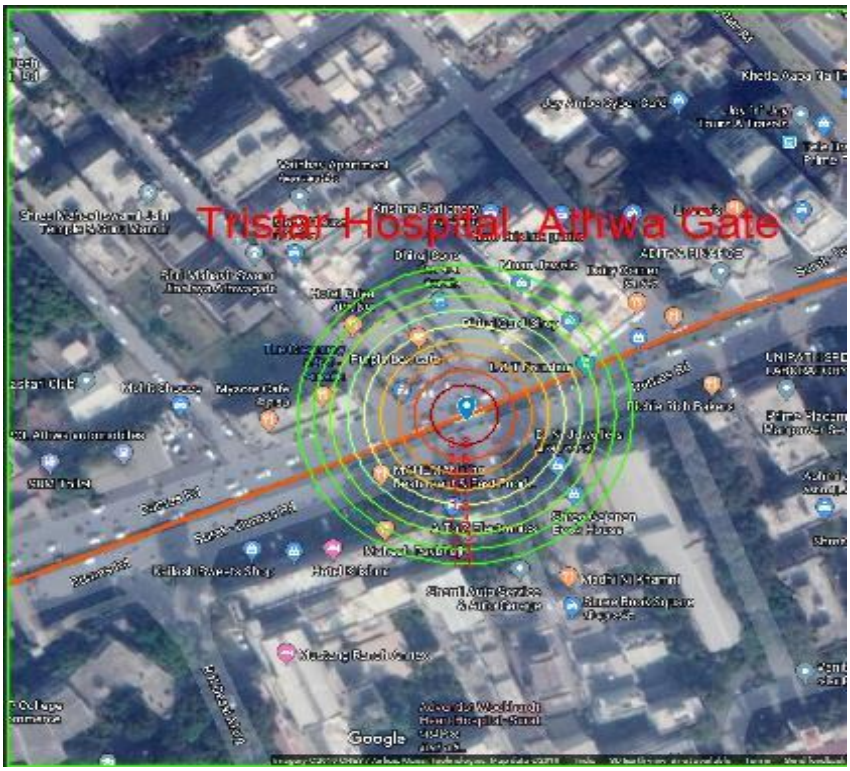
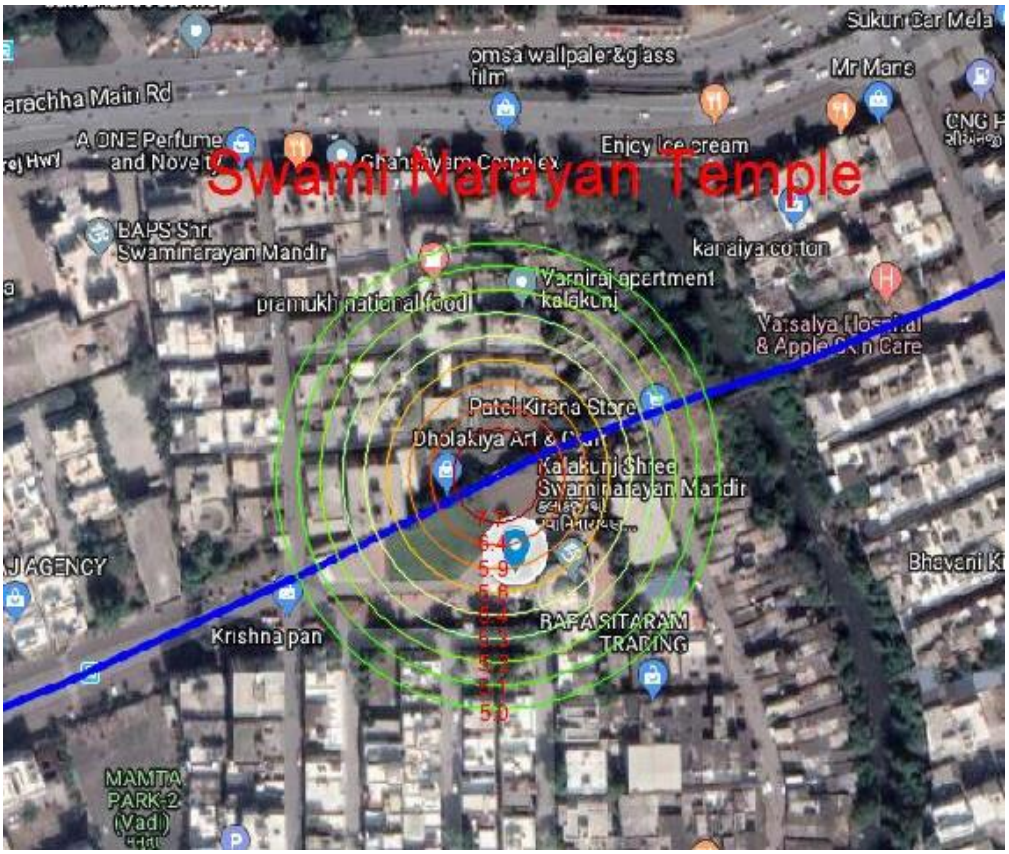


Figure 5.12: D Mehta Mahavir Heart Hospital

Figure 5.13: Near Nirmal Nagar Society

Figure 5.14: Swami Narayan Temple



B. Vibration Impact due to Impact Pile Driving

During impact piling, the hammer impact initiates a stress pulse in the pile which travels along the pile until it reaches the pile toe, at that point part of the energy is reflected and part transmitted into the ground. At the vertical pile/soil interface along the length of the pile, the angle of incidence will be close to 90^{o} and so almost all energy is reflected and very little is transmitted into the ground. Consequently, even when the pile has been driven to considerable depth, the major part of the energy can be considered to enter the ground from the pile tip of a impact driven pile (Attewell and Farmer, 1973). However, energy may also be transferred to the ground along the pile shaft through friction as the pile moves through the soil. This would generate a vertically polarised shear wave. It has also been suggested that flexure of the pile shaft may occur during driving, which can initiate vibration (Selby, 1991).

The different driving methods interact with the ground in different ways. The energy transmitted into the ground by impact piling techniques gives rise to transient or intermittent vibration which is propagated at frequencies governed primarily by the characteristics of the ground. Soils do not possess resonant frequencies but generally behave as bandpass filters, possessing a limited range of frequencies within which most vibration energy propagate.

$$Vres = kp[\sqrt{W} / r\ 1.3]$$

Where as

Vres = is vibration in mm/s at a distance

r = is the slope distance from the pile toe (m)

[r = √ (L2 + x2)], where L = is the pile toe depth (m)

x = is the distance measured along the ground surface (m)

W = Nominal energy of Hammer in J.

Kp = depending on soil type

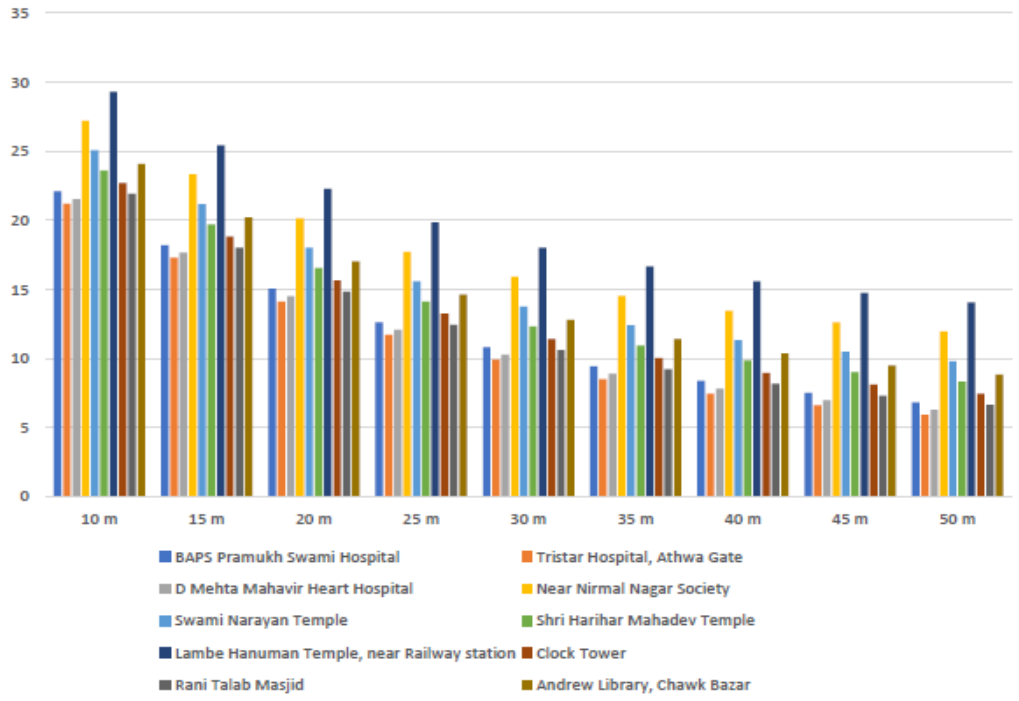
Hence, by considering extreme condition at monitored sites, predictable vibrations are given in Table 5.9 and graphically shown in Figure 5.15.

Table 5.9: Vibration due to Impact Pile Driver

Sr No.	Location Name	PPV (mm/s)	Vibration at horizontal distance from source in PPV (mm/s)								Permissible
			10 m	15 m	20 m	25 m	30 m	35 m	40 m	45 m	
1	BAPS Pramukh Swami Hospital	1.73	22.10	18.22	15.05	12.64	10.82	9.44	8.37	7.52	2.54
2	Tristar Hospital, Athwa Gate	0.82	21.20	17.31	14.14	11.73	9.91	8.53	7.46	6.62	2.54
3	D Mehta Mahavir Heart Hospital	1.18	21.55	17.67	14.50	12.08	10.27	8.88	7.81	6.97	2.54
4	Near Nirmal Nagar Society	6.84	27.21	23.33	20.16	17.74	15.93	14.54	13.47	12.63	7.62
5	Swami Narayan Temple	4.69	25.06	21.18	18.01	15.60	13.78	12.40	11.33	10.48	6.35

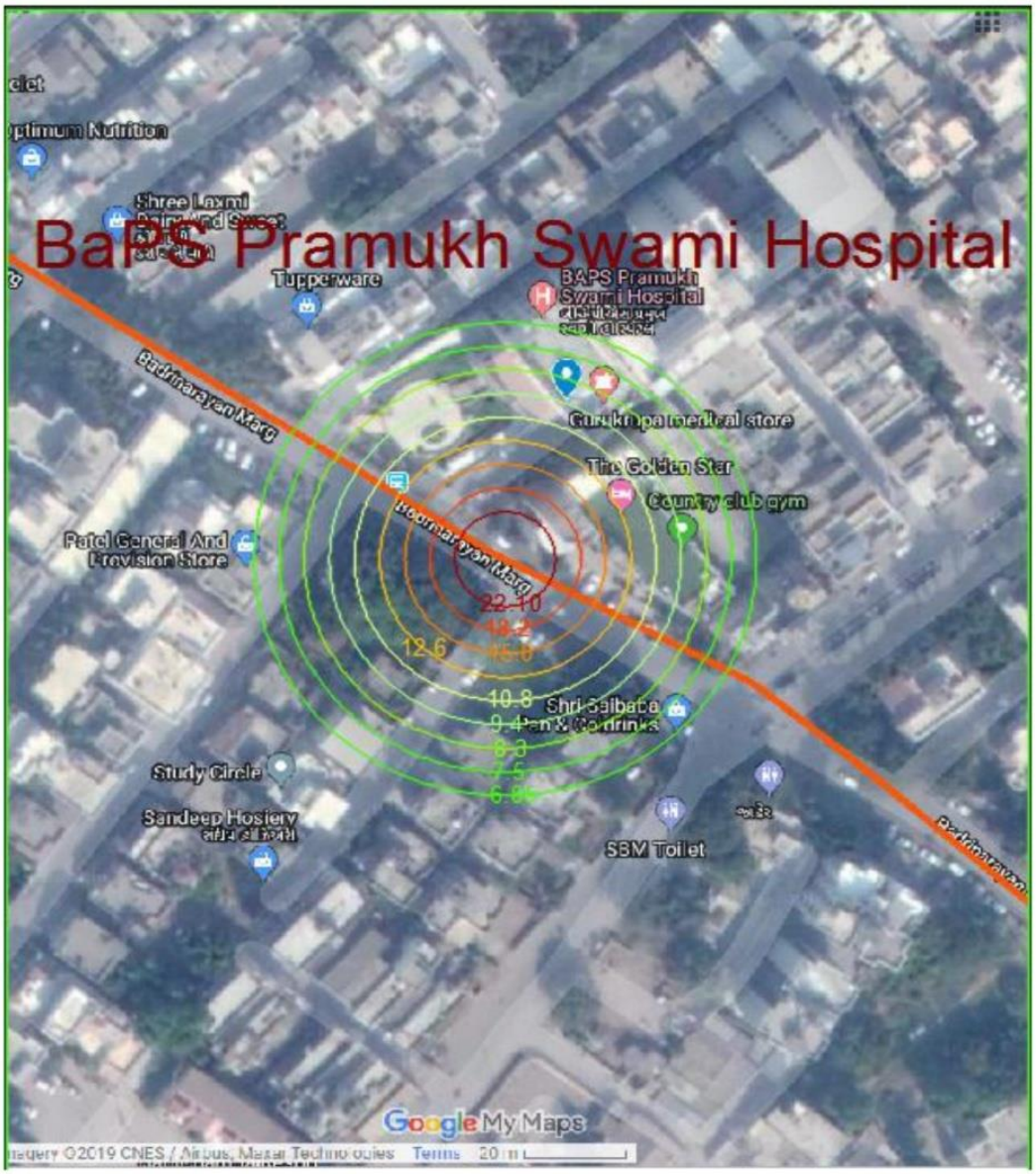
Note: Standards for Fragile Buidlings, Historic and Old Buildings, Older Residential Structures as per Transportation and construction Vibration Guidance Manual, Caltrans, Secptember 2013

Figure 5.15: Graphical Representation of the Vibrations due to Impact Pile Driver



It is understood that the predicted PPV from the Impact pile driver is very higher than other type of pile drivers. Auger Displacement Piling (ADP) system offers a quick, quiet and virtually vibration free alternative to driven type piles[1]. Hence, Impact Piles should be used in case of hard rock only and for other areas Auger displacement piling system should be used. Figure 5.16 to Figure 5.20 shows the Vibration Contours for the particular location due to impact pilling.

Figure 5.16: BAPS Pramukh Swami Hospital



[1] Auger Displacement Piles, Cementation Skanska, 2010

Figure 5.17: Tristar Hospital, Athwa Gate

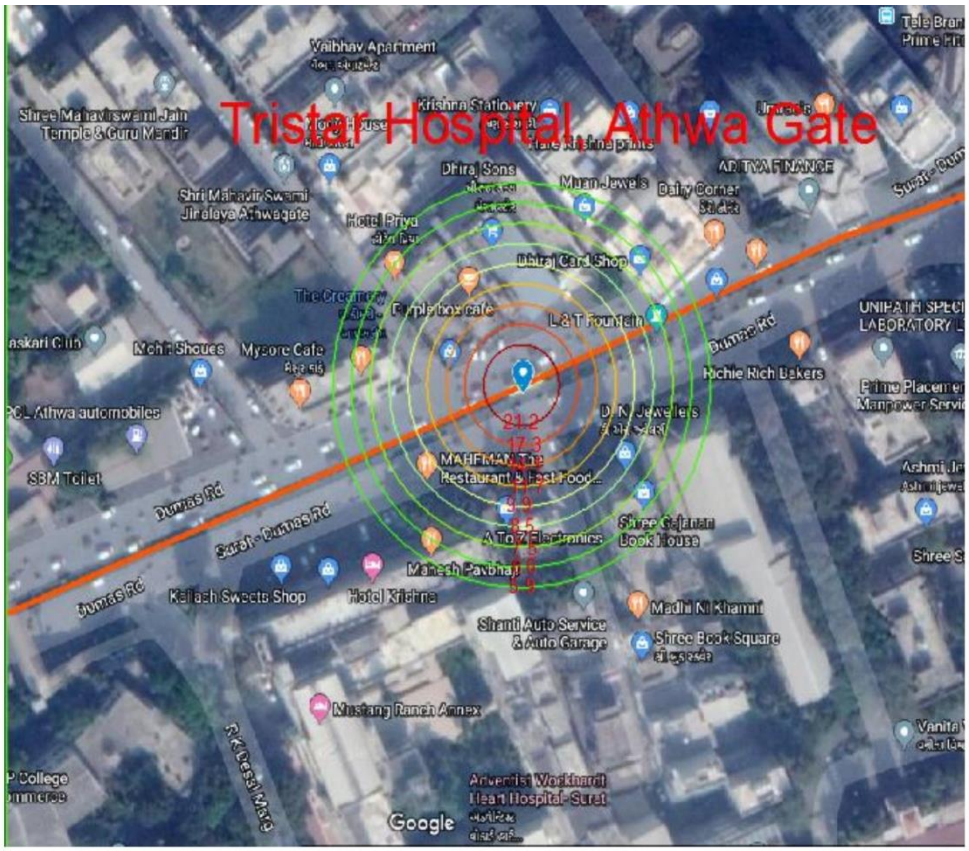


Figure 5.18: D Mehta Mahavir Heart Hospital

Figure 5.19: Near Nirmal Nagar Society

Figure 5.20: Swami Narayan Temple

C. Vibration Impact due to Tunnel Boring Machine (TBM)

Summary of tunneling methods are given in Table 5.10 (Priyadarshi Hem, Norman B. Keevil Institute of Mining Engineering - University of British Columbia, September 2015, www.technology.infomine.com).

Table 5.10: Tunneling Methods

Tunnelling Methods	Advance Rate (m/day)	Cross-section Area (m^2)/Dia (m)	Rock Properties	Comments
Pipe Jacking	5 – 20	12 mm (0.5") 3.6 m (12')	Best suited for cohesive soil; can also be applied to non-cohesive and dry conditions and in hard rocks.	Primarily for laying pipelines; small cross-section and limited capability in hard rock's show little scope of its use in underground mining
Hammer Tunnelling	2 – 10	Up to 30 m^2	Low to medium strength; RQD: 0 - 100	Breaking, mucking and reinforcement can go together; low investment
Road headers	5 – 15	Up to 45 m^2	Up to 140 MPa (UCS); can adapt to changing rock mass conditions	Low investment, 15% - 30% of TM investment for same cross-section; can be rented; can be delivered and ready for operation in 3-6 months; free space and accessibility; can handle a variety of subsections
TBMs	15 – 50	Up to 160 m^2	From 20 to 140 MPa (UCS); can be fabricated to a variety of ground conditions	Fabrication period of around 12 months; Onsite set up time 3-6 months
Drill & Blast	3.5 m - 5 m & 7 - 15 m (without rock support installation)	-	-	-

MPa ~ 10.20 kg/sq cm

TBM typically consists of a large rotating cutting wheel in front of large metal cylinder(s) known as shields as well as trailing control and ancillary mechanisms. Behind the cutting wheel is a chamber where the spoil is removed using conveyors to the rear of the machine. The cutting wheel is moved forward by hydraulic jacks supported off the finished tunnel walls. When the cutting wheel has reached maximum extension the TBM head is braced against the tunnel walls and the rear section of the TBM is dragged forward. Ground borne vibration caused by tunneling operations have the potential to cause significant community annoyance because, in general the operations are continuous. Vibration from operating TBM machines is influenced by distance, structural rock conditions, physical rock properties, ground water and depth of overburden. Vibration magnitude in term of peak particle velocity (PPV) from the TBM has been predicted using following empirical formula (Godio et al., 1992 cited in Hillar & Crabb, 2000) and Code of Practice for Noise and Vibration control on construction and open sites - Part 2: BS5228-2 Annex E as given below.

$$PPV\ (mm/s) = A\ D\text{-}1.3$$

Where:

A is a constant whose value depends on the ground condition (assumed hard ground; A=180) and
D is the direct distance (m) from the vibration source to the measurement location.

Using the above empirical propagation equation, peak particle velocity (PPV) due to TBM has been predicted and presented in Table 5.11. Maximum predicted PPV of 4.84 mm/sec is estimated at Shri Harihar Mahadev Temple; however this vibration is within the permissible limits.

Table 5.11: Vibration due to TBM

S. No	Location	Radial Distance* from UG Alignment in m	PPV (mm/sec)	
			Vibration due to TBM	Permissible**
1.	Shri Harihar Mahadev Temple	16	4.84	6.35
2.	Lambe Hanuman Temple, near Railway station	27	2.43	6.35
3.	Clock Tower	17	4.53	6.35
4.	Rani Talab Masjid	27	2.53	6.35
5.	Andrew Library, Chawk Bazar	17	4.36	6.35

***** Radial distance assuming depth of top of tunnel is 15m below ground
** Standards for Historic and Old Buildings as per Transportation and construction Vibration Guidance Manual, Caltrans, Secptember 2013

Spatial variation of resultant PPV shows that predicted PPV attenuates from the source location to permissible levels at all monitored locations without considering any mitigation measures. However, prior building condition survey should be taken up before commencing the TBM operation.

D. Sensitive Receptors

Vibration levels are estimated at the sensitive receptors identified along the both corridors as given at Annexure 5.2. All these receptors are classified as fragile, Historic and Old Buildings, and Older Residential Structures as per the categories given in Transportation and Construction Vibration Guidance Manual. The analysis of the results indicates the following:

- Impact piles shall be used only when any structure comes under Older Residential and Historic & Old Buildings is located at a distance more than 50 m.
- Vibration levels generated due to TBM are within the permissible limits at all sensitive receptors located along the UG section of Sarthana-Dream City Corridor.
- Vibration levels generated due to Vibratory piles are exceeding at 1 educational, 10 Hospitals along Sarthana-Dream City Corridor.

- Vibration levels generated due to Vibratory piles are exceeding at 5 Hospitals along Bhesan-Sarthana Corridor.

5.5.4. Impact due to Muck Disposal

The muck generated due to proposed project activities cannot be avoided. Therefore the generated muck will be reused and remaining will be disposed in a manner that will not affect the human health and environment in line with ESS 3 of World Bank.

Out of the two metro corridors, one metro corridor is a mix of elevated and underground right of way and other corridor is fully elevated. The construction activity involves cut and cover, tunneling, excavation and fill. All these activities will result in excavation of about 6,89,520 m^3 and fill of about 35,739 m^3 with net quantity to be disposed of about 6,53,780 m^3. Muck disposal if not properly done can result in environmental pollution.

Obstruction to Natural Watercourses: The materials if not disposed-off properly would be taken by the surface run-off creating huge obstruction to free flow of natural stream water. If people are residing nearby that would affect their life by way of flooding or by spoiling the premises.

Siltation in Surface Water Reservoirs: Most of the materials would be taken down stream through rivers and ultimately depositing in to reservoirs leading to heavy siltation. This in turn would reduce the reservoir capacity substantially within a very short span. This could incur huge losses to the exchequer. De-siltation is also expensive and normally carried out after many years of operation of reservoirs in the natural circumstances.

Soil Erosion: Massive soil erosion is the most direct impact at muck disposal site and debris excavation at the portals of the tunnel. The precipitation and the consequent run off would erode the loose materials by way of suspension and solution.

5.5.5. Impact due to construction/demolition waste Disposal

Debris disposal can result in air and water pollution, noise, diversion of green parks and temporary displacement. The C&D waste generated will be reused and remaining will be disposed in a manner that will not affect the human health and environment in line with C&D Waste Management Rules, 2016 & its amendments and ESS 3 of World Bank.

5.5.6. Impact due to Hazardous Waste

Hazardous waste would mainly arise from the maintenance of equipment which may include used engine oils, hydraulic fluids, waste fuel, spent mineral oil/cleaning fluids from mechanical machinery, scrap batteries or spent acid/alkali, spent solvents etc. Unsafe disposal can result in water and soil pollution, diversion of green parks and temporary displacement. The Hazardous waste management should be handled by authorised/licensed agent as per guidelines of Hazardous and Other Wastes (Management and Trans boundary Movement) Rules, 2016 & its amendments and ESS 3 of World Bank.

5.5.7. Increased Water Demand

The water demand will increase during construction phase. The water demand is estimated to be 670 kilolitres per day (kld).

5.5.8. Impact due to Pre-casting yards and Material stockpiling

Sites for casting of structural concrete elements and material stockpiling can result in air and water pollution, noise, diversion of open areas like green parks and temporary displacement.

5.5.9. Impact on Ground and Surface Water Quality

Ground water contamination can take place if sewage at labour camps or chemical substances from construction site or dumped muck or construction/demolition waste or used water from the RMC plant get leached by precipitation of water and percolate to the ground water table. The construction activities should be planned in such a way that the water resources should not be affected in accordance with Water (Prevention and Control of Pollution) Act, 1974 & its amendments and ESS 3 of World Bank.

5.5.10. Soil Erosion and land subsidence

Run off from unprotected excavated areas can result in excessive soil erosion, especially when the erodability of soil is high. Land subsidence is anticipated at stations which will be constructed by cut and cover method.

5.5.11. Traffic Diversions

During construction period, complete/partial traffic diversions on road will be required, as most of the construction activities are on the road.

5.5.12. Impacts due to Labour Camps

Improper disposal of municipal solid waste generated by labour camps can pollute surface water bodies and groundwater. Burning of waste can cause air pollution. Construction workers are more prone to infectious diseases due to unsafe sexual activity and lack of sanitation facilities (water supply and human waste disposal) and insect vectors. Problems could arise due to cultural differences between workers from outside and local residents. About 2856 persons are likely to work during peak construction activity in both the corridor. Six labour camps will be proposed at appropriate locations. The water requirement at all labour camps will be 385 KLD, waste water generation 308 KLD & Municipal solid waste generation 856 Kg per day.

5.5.13. Welfare of Labour on construction site

Facilities such as shelter at workplace, canteen, first aid and day crèche are statutory requirement and essential to productivity. Labour camp will be established by the contractor according to the Section 32 of BOCWA. Labour camp will be setup by the contractor at suitable location.

5.5.14. Safety of Labour

Safety of labour during construction on elevated and underground sections is a statutory requirement and also has impact on progress of work.

5.5.15. Impact due to Supply of Construction Material

Construction material such as aggregate and earth are sourced from approved quarries such that environmental impacts as well as wastage of natural resources are minimized and mitigated.

5.5.16. Impacts of Air Pollution on Human Health

Air pollution from road based vehicles especially particulates are found to cause diseases of brain, heart, lungs and kidneys:

- Higher levels of exposure to ambient PM are associated with worse cognitive decline[2].
- Increased risk of fatal CHD associated with each 10 µg/m^3 increase in annual PM2.5 exposure[3].
- Significant association between exposure to PM2.5 and risk of incident CKD, eGFR decline, and ESRD[4].
- The mortality rate advancement attributable to traffic pollution was similar to that associated with chronic respiratory and pulmonary diseases and diabetes[5].
- Exposure to ambient PM2.5 air pollution is associated with a lower level of sperm normal morphology and a higher level of sperm concentration[6].
- In addition to noise induced hearing loss, stress hormone increases and increase in the risk of myocardial infarction are caused by noise[7].

[2] *Exposure to Particulate Air Pollution and Cognitive Decline in Older Women, Jennifer Weuve et al, JAMA Internal Medicine, February 2012*
[3] *Chronic fine and coarse particulate exposure, mortality, and coronary heart disease in the Nurses' Health Study, Puett RC et al , Environ Health Perspect. November 2009*
[4] *Particulate Matter Air Pollution and the Risk of Incident CKD and Progression to ESRD, Benjamin Bowe et al, Journal of the American Society of Nephrology, September 2017*
[5] *Traffic Air Pollution and Mortality Rate Advancement Periods, Murray M. Finkelstein et al, American Journal of Epidemiology, July 2004*
[6] *Exposure to ambient fine particulate matter and semen quality in Taiwan,* Xiang Qian Lao et al,, Occupational and Environmental Medicine, November 2017

5.5.17. Impact on Flamingo

The proposed metro alignment section crosses the Tapi River near BAPS Swaminarayan temple to Jawaharlal Nehru (Chowpati garden) as shown in Figure 5.21

Figure 5.21: Proposed alignment Crossing Tapi River

Construction activities of the proposed metro corridor may disturb feeding & roosting areas of the flamingos in the immediate vicinity of the development. Due to construction the river water may be muddy. As the flamingos are feeding on the diatoms, because of poor visibility and rise in temperature, the feeding habitat of lesser flamingo may be disturbed. However, as construction activities are temporary and carried out during lean season, the negative effect on the water quality would be short term and the original condition would be attained as soon as the construction activates of pillars are terminated.

Studies reveal that noise pollution has been shown to have a negative impact on flamingo productivity. During construction, the predicted noise level near the Tapi River is almost equal to the baseline noise due to heavy traffic on Sardar Bridge.

Bright lighting during construction at nights may disturb flamingos, which mostly migrate during night (WII, 2010). These artificial lights may disturb their development, orientation, activity pattern, natural behavior and hormone related processes such as internal clock mechanism.

[7]*Health effects caused by noise: Evidence in the literature from the past 25 years, H Ising, B Kruppa, Noise & Health, 2004*

Since the flamingos have the habit of moving between sites, multiple feeding sites are available from downstream of the Ukai dam, hence, the flamingos may be able to successfully forage in these nearby locations. Therefore, no impact on the flamingos due to the construction of proposed metro are anticipated.

5.6. IMPACTS DUE TO PROJECT OPERATION

Along with many positive impacts the project may cause the following negative impacts during operation of the project:

- Noise Pollution
- Vibration
- Energy Consumption at stations
- Water supply and Sanitation at Stations
- Traffic Congestion around stations
- Impacts due to Depot
- Impact on Flamingo

5.6.1. Noise Pollution

Airborne noise is radiated from at-grade and elevated structures, while ground-borne noise and vibration are of primary concern in underground operations. During the operation phase, the main source of noise will be from running of metro trains. Basic sources of wayside airborne noise are:

i. Wheel / Rail Noise : Due to wheel /rail roughness
ii. Propulsion Equipment: Traction motors, cooling fans for TM, reduction gears etc.
iii. Auxiliary Equipment: Compressors, motor generators, brakes, ventilation systems, other car mounted equipment
iv. Elevated Structure Noise
    - At low speed(<15 km/h) auxiliary equipment may predominate
    - At speeds up to approx. 50 km/h, W/R noise predominates
    - At speeds greater than 50 km/h, the propulsion equipment noise predominates
    - For light weight steel elevated structures, the structure noise can predominate at all speeds above 15 km/h

The information about Metro Rail operations considered to predict noise level at all stations of both the corridors as mentioned in DPR are given at Table 5.12. Predicted noise levels at all metro stations were modelled as per guidelines specified in “Transit Noise and Vibration Impact Assessment, May 2006 by Federal Transit Administration (FTA)”. The data considered for the modelling are as follows:

- SELref: 82 dB (A) for fixed guide-way Rail cars @ 50mph at 50 feet distance,
- Ground Factor G for ground attenuation is considered as zero,
- Attenuation due to Shielding between source and receptor is considered as zero,
- Adjustment Factor of +4 is considered for Aerial Structure with Slab Track,
- Performance Parameters considered for the project as per DPR are
  - Max Acceleration 1.0 m/s^2
  - Max Deceleration 1.1 m/s^2

Table 5.12: Inputs to Noise Modelling

Corridor	Train Speed (Kmph)	No. of Cars	Track Configuration	Trains per day in Both Directions							
				Year 2021		Year 2026		Year 2036		Year 2046	
				Day	Night	Day	Night	Day	Night	Day	Night
Sarthana to Dream City	80	3	Viaduct/ Underground	182	8	294	16	402	16	498	16
Bhesan to Saroli	80	3	Viaduct	140	8	174	16	276	16	324	16

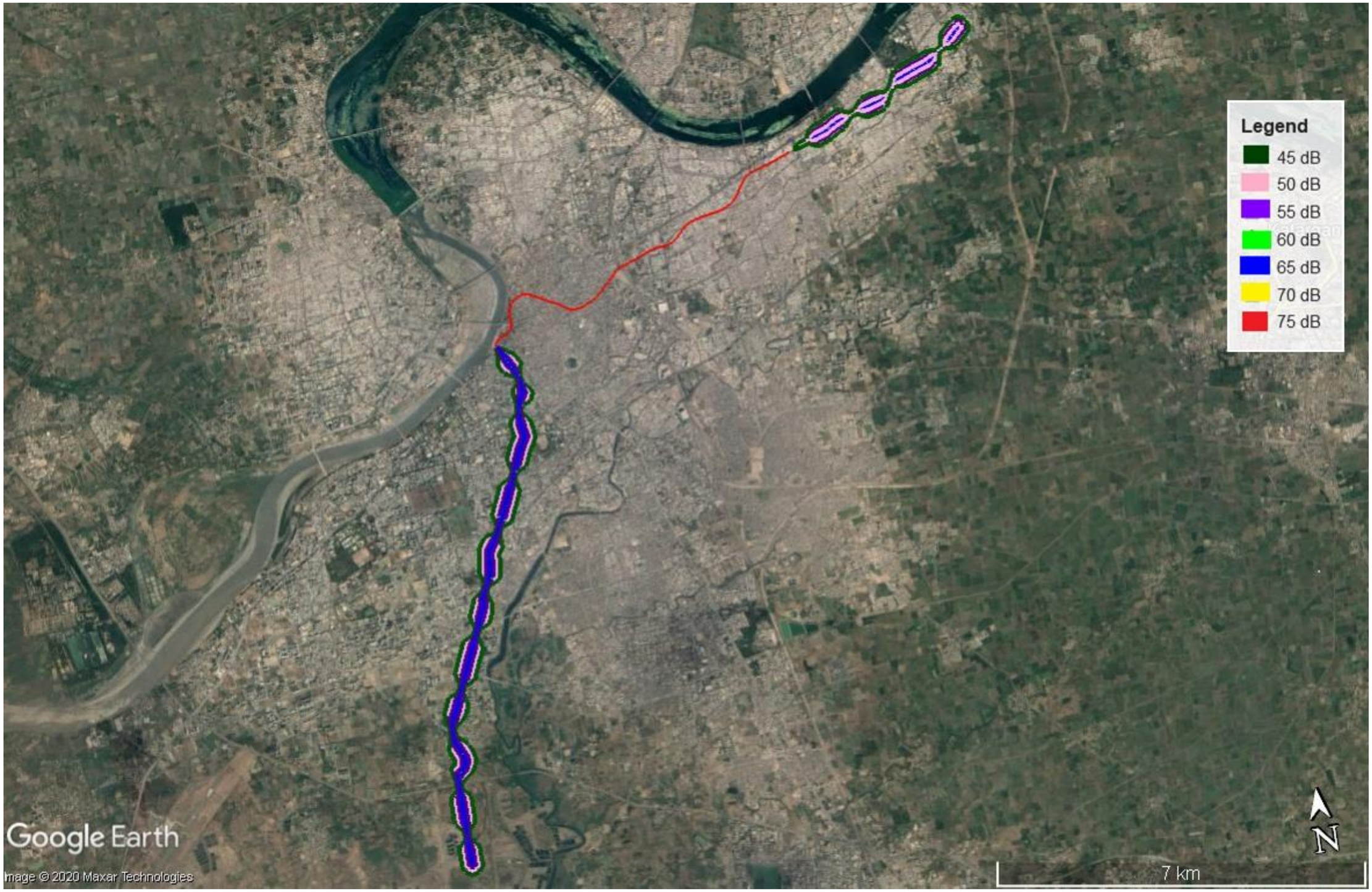
Note: Day: 06.00-22.00 hrs and Night 22.00-06.00 hrs
Source: DPR for Surat Metro Rail Project, June 2018

Maximum acceleration and deceleration are used to calculate the distance between stations, where the metro rail attains the maximum speed of 80 KMPH. It is also assumed that the metro enters and leaves the station at a speed of 20 KMPH. Accordingly, predicted noise levels at 15 m distance are given at Table 5.13. Noise contours for Leq are generated along the proposed metro rail corridors for year 2021 and given in Figure 5.22.

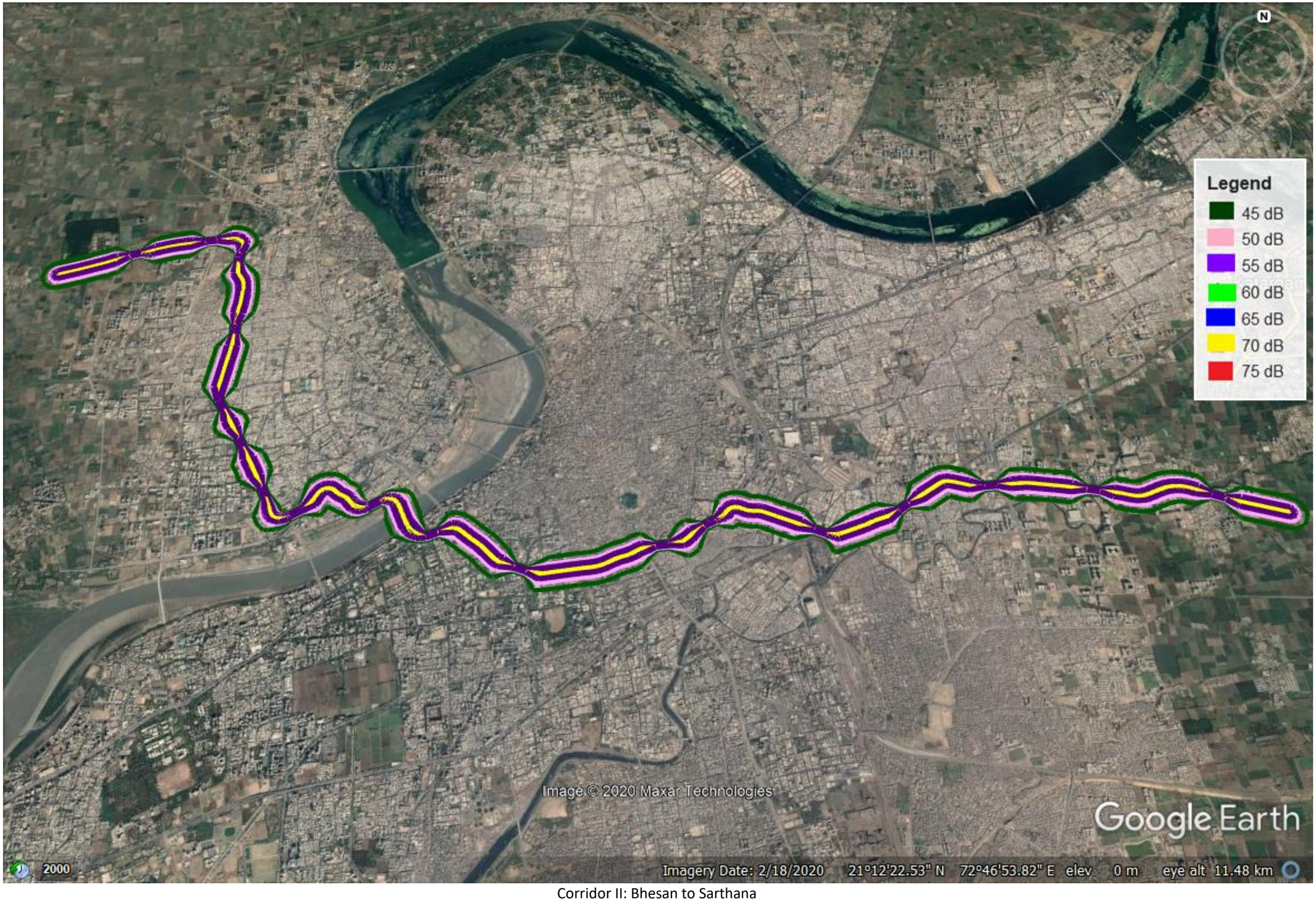
Table 5.13: Predicted Noise Levels due to Metro Operations

Year	Predicted Noise Levels in dB (A) at 15 m distance			
	20 KMPH		80 KMPH	
	Ld	Ln	Ld	Ln
Corridor I: Sarthana to Dream City				
2021	53.64	47.34	65.68	59.38
2026	55.72	50.35	67.76	62.39
2036	57.08	50.35	69.12	62.39
2046	58.01	50.35	70.05	62.39
Corridor II: Bhesan to Saroli				
2021	52.50	47.34	64.54	59.38
2026	53.44	50.35	65.48	62.39
2036	55.45	50.35	67.49	62.39
2046	56.14	50.35	68.18	62.39

Figure 5.22: Noise Contours of Proposed Corridors



Corridor I: Sarthana to Dream City



Corridor II: Bhesan to Sarthana

The predicted noise levels estimated due to metro operations for year 2021, 2026, 2036 and 2046 reveals the following:

- Lday at 15 m distance is exceeding the existing day time ambient noise levels measured at location N8, N18, for year 2021, 2026, 2036 and 2046; and exceeding at location N11 for year 2026, 2036 and 2046.
- Lnight at 15 m distance is exceeding the existing night time ambient noise levels measured at location N1, N3, N8, N10, N11, N18 for year 2021, 2026, 2036 and 2046; and exceeding at location N15 for year 2026, 2036 and 2046.
- Corridor I - Sarthana to Dream City: The noise levels generated due metro at maximum speed of 80 KMPH will be within the permissible limits prescribed for commercial areas at a distance of 20 m for year 2021; 27 m for year 2026; 29 m for year 2036; and 30 m for year 2046.
- Corridor II - Bhesan to Saroli: The noise levels generated due metro at maximum speed of 80 KMPH will be within the permissible limits prescribed for commercial areas at a distance of 19 m for year 2021; 24 m for year 2026; 26 m for year 2036; and 27 m for year 2046.

Cumulative future noise levels in the project area are modelled based on existing measured sound levels and predicted noise levels due to future daily metro rail operations. Cumulative noise levels at each noise monitoring location are given in Table 5.14.

- CPCB Standards: Cumulative day and night noise levels at all the base line monitoring locations exceeding the permissible levels prescribed by CPCB for Silence Zone, Residential Zone, and Commercial Zone; except night time noise levels for commercial zone permissible limits at locations N6, N15, and N17 for all the years.
- IFC Noise Standards: Cumulative noise levels are
  - exceeding the permissible limits for residential zones for all years at location N1, N2, N10, N11, N18 & N19;
  - exceeding the silence zone permissible limits at N5 & N16;
  - within the commercial zone permissible limits at remaining locations on elevated section for all years except at locations N4 and N7 where day time noise levels are exceeding the permissible limits for commercial zones.
- The difference between baseline noise values and cumulative noise values are exceeding the 3 dB(A) for day and night time at location N1 for all the years; only night time at location N10 & N11 for all the years; and day time at location N11 for year 2036 and 2046.
- Noise mitigations measures like noise barriers need to be provided at Location N1 to reduce noise impacts.

Table 5.14: Cumulative Noise Levels at Monitoring Locations due to Metro Train Operations

S. No	Noise of the Monitoring Locations	Ele/UG	Distance from Metro Centreline in m	Ambient Noise Levels in dB(A)		Predicted Noise Levels in dB(A) at Monitoring Locations								Cumulative Noise Levels in dB(A) at Monitoring Locations							
						2021		2026		2036		2046		2021		2026		2036		2046	
				Lday	Lnight	Lday	Lnight	Lday	Lnight	Lday	Lnight	Lday	Lnight	Lday	Lnight	Lday	Lnight	Lday	Lnight	Lday	Lnight
Corridor I: Sarthana to Dream City																					
N10	Near Bajrang Nagar society	Ele	19	69.2	57.3	63.6	57.3	65.7	60.3	67.1	60.3	68.0	60.3	70.3	60.3	70.8	62.1	71.3	62.1	71.7	62.1
N11	Mamta Park Society, Kapodra	Ele	19	65.9	54	63.6	57.3	65.7	60.3	67.1	60.3	68.0	60.3	67.9	59.0	68.8	61.2	69.5	61.2	70.1	61.2
N15	Near Fame Raj Empire Multiplex	Ele	32	69	49.4	47.1	40.8	49.1	43.8	50.5	43.8	51.4	43.8	69.0	50.0	69.0	50.4	69.1	50.4	69.1	50.4
N16	Near Shri Dayalji Kasanji Bhatarkar Vidya Sankul	Ele	85	65.3	54.9	38.6	32.3	40.7	35.3	42.0	35.3	42.9	35.3	65.3	54.9	65.3	54.9	65.3	54.9	65.3	54.9
N17	Residential area of Bhimrad	Ele	73	65.4	53.2	39.9	33.6	42.0	36.6	43.3	36.6	44.3	36.6	65.4	53.2	65.4	53.3	65.4	53.3	65.4	53.3
N18	Khanjod	Ele	668	63.7	51.5	32.7	26.4	34.8	29.4	36.1	29.4	37.1	29.4	63.7	51.5	63.7	51.5	63.7	51.5	63.7	51.5
N19	Subhash Nagar	Ele	45	68.5	59.6	44.1	37.8	46.2	40.8	47.5	40.8	48.5	40.8	68.5	59.6	68.5	59.7	68.5	59.7	68.5	59.7
Corridor II: Bhesan to Saroli																					
N1	Near Treatment Plant, Bhesan Gam	Ele	13	65.3	48.9	65.8	60.6	66.7	63.6	68.7	63.6	69.4	63.6	68.6	60.9	69.1	63.8	70.4	63.8	70.8	63.8
N2	Near Ganga Nagar Society	Ele	15	68.7	57.5	52.5	47.3	53.4	50.3	55.4	50.3	56.1	50.3	68.8	57.9	68.8	58.3	68.9	58.3	68.9	58.3
N3	Near Bhagat Party Plot	Ele	33	69.6	58.5	57.7	52.5	58.6	55.5	60.6	55.5	61.3	55.5	69.9	59.5	69.9	60.3	70.1	60.3	70.2	60.3
N4	Near Imperial Maal/Cinepol multiplex	Ele	80	70.6	57.1	38.0	32.8	38.9	35.8	40.9	35.8	41.6	35.8	70.6	57.1	70.6	57.1	70.6	57.1	70.6	57.1
N5	Badri nath temple (Maansarovar Society)	Ele	20	64.5	51.9	50.0	44.8	50.9	47.9	52.9	47.9	53.6	47.9	64.7	52.7	64.7	53.3	64.8	53.3	64.8	53.3
N6	Majura Gate chawk	Ele	24	69.8	50.4	48.4	43.3	49.4	46.3	51.4	46.3	52.1	46.3	69.8	51.2	69.8	51.8	69.9	51.8	69.9	51.8
N7	Kinnery Cinema Stop	Ele	33	72.5	60.9	45.7	40.5	46.6	43.5	48.6	43.5	49.3	43.5	72.5	60.9	72.5	61.0	72.5	61.0	72.5	61.0
N8	At crossing of Modal Town	Ele	16	64.2	51.9	64.0	58.8	64.9	61.8	66.9	61.8	67.6	61.8	67.1	59.6	67.6	62.3	68.8	62.3	69.3	62.3
N9	Near Bhawani Industrial Estate, Transport Nagar	Ele	41	69.4	59.4	43.8	38.6	44.7	41.6	46.7	41.6	47.4	41.6	69.4	59.4	69.4	59.5	69.4	59.5	69.4	59.5

- As per FTA guidelines, Noise Impact is moderate at N1, N3, N4, N6, N7, N8, N9, N10, N11, N15, N19 monitoring locations and no impact at remaining locations.

Noise impact on sensitive receptors along the project corridors are modelled and given at Annexure 5.3. Background noise levels at the receptor are considered as noise levels of the nearest baseline noise monitoring location. The analysis of the results indicates the following:

- The cumulative noise levels at all these receptors are exceeding the permissible limits issued by CPCB and IFC.
- Noise impact criteria on these receptors is categorised as follows.
  - No Impact - difference between baseline noise levels and cumulative noise levels are less than 3 dB (A)
  - Severe Impact - difference between baseline noise levels and cumulative noise levels are less than 3 dB (A)

Following noise mitigation measures need to be provided wherever noise impact criteria are severe at the receptors.

- Noise barriers need to be provided wherever noise impact criteria are severe at all the receptors and at hospitals where noise impact criteria is severe during night time.

5.6.2. Vibration

Ground vibrations by railway traffic are generated by two prime excitation mechanisms (i) the quasi-static displacement caused by the axle load as the wheel moves along the track, and (ii) the inertia forces due to the acceleration of the unsprung mass of the train as it rolls over the irregular profile of the railhead. Rail irregularity profiles, have generally an erratic nature. Therefore, the associated ground vibrations tend to display a stochastic character.

Most of the studies on train-induced ground vibration conducted until a few years ago have dealt primarily with vibration from the second mechanism because moving-load contribution to the total vibration was considered secondary. However, actual observations and measurements with high-speed trains have in the past several years revealed that one might observe remarkably large vibrations in soft soil sites from the moving-load excitation mechanism (e.g. Woldringh & New, 1999). Irregularity of the railhead and the wheel tread are important forms of track imperfection that have long been recognized as major sources of vibration by railway traffic. Railhead irregularity may be considered to cover wave-lengths from a few millimetres to 50 meters and more. Irregularities corresponding to short wavelengths, commonly referred to as corrugation and roughness, are essentially responsible for high frequency vibration and noise.

Human response to vibration is subjective and will be different for different people. When the vibrations reach the floors and walls it may result in perceptible vibration depending on the amplitude and frequency of the vibrations. People may be more annoyed if they are exposed to both noise and vibration compared to when only vibration is felt. According to the U.S. Department of Transportation, (1998) the perception threshold of humans for peak particle velocity is about 0.04 mm/s (65VdB with reference 1e-6 inch/sec).

For a person in their residence, the lower threshold for annoyance is 72 VdB (FTA 2006). The vibration affects human health by causing fatigue, increased pulse & respiration rates, dizziness & loss of balance, anger and irritation.

An empirical model is based on empirical observations of a phenomenon. Empirical models are often simple and can for metro-induced ground vibrations often be used as prediction models at an early stage of planning. Norwegian Geotechnical Institute (NGI) developed a semi-empirical model based on data from measurements and statistical analysis of the results. It can be used to predict vibration levels and is based on six general parameters: ground conditions; train type; line quality and embankment design; train speed; distance from track to nearby structure; the building in terms of foundation, structure etc. The model is given as a mathematical function and is adjusted using several factors according to below equation,

$$V = FV * FR * FB = [VT * FS * FD] * FR * FB$$

where FV is the basic vibration function, consisting of a train type specific vibration level VT for a reference speed and reference distance from track. Furthermore, FS is a speed factor, FD is a distance factor, FR is a track quality factor and FB is a building amplification factor. Considering the variation in the speed, vibration due to rail- wheel interaction, frequency etc during the projection period. Predicted vibration level due to Rapid Transit rail is given in Table 5.15.

Table 5.15: Predicted Vibration Level due to Rapid Transit Rail

Locations	Vibration Levels in mm/sec			
	Monitored	Predicted	Cumulative	Permissible
BAPS Pramukh Swami Hospital	1.73	0.16	1.89	2.54
Tristar Hospital, Athwa Gate	0.82	0.16	0.98	2.54
D Mehta Mahavir Heart Hospital	1.18	0.16	1.34	2.54
Near Nirmal Nagar Society	6.84	0.16	7.00	7.62
Swami Narayan Temple	4.69	0.16	4.85	6.35
Shri Harihar Mahadev Temple	3.23	0.16	3.39	6.35
Lambe Hanuman Temple, near Railway station	8.95	0.16	9.11	6.35
Clock Tower	2.34	0.16	2.50	6.35
Rani Talab Masjid	1.53	0.16	1.69	6.35
Andrew Library, Chawk Bazar	3.72	0.16	3.88	2.54

Predicted vibration levels reveals that most of the receptor locatins fall within vibration damage threshold criteria. vibration level can be significantly reduced by incorporating the following standard measures/factors:

i. Vehicles with NO stiff primary suspension
ii. NO Worn or Wheel with flats
iii. NO Worn/Corrugated Track
iv. With Floor-to-Floor Attenuation in receptor buildings

5.6.3. Energy Consumption at Stations

Energy is required at stations for facilities like lighting, passenger information, access, security, climate control, escalators/elevators etc.

5.6.4. Water Supply, Waste Water and Municipal Solid Waste disposal at Stations

The water demand at stations comprising drinking, toilet demands and solid waste generation for passengers and staff will be of the order of magnitude as indicated in Table 5.16 and Table 5.17 for corridor 1 and corridor 2 respectively. The demand is estimated on the following assumptions:

- Alighting passenger for Year 2021, 2026, 2036 & 2046 as given in DPR are considered for the calculations
- 15% of Alighting passengers at each station will use the toilets
- Water requirement for each user will be 5 lit/passenger/day
- 10% of Alighting passenger at each station will generate Solid Waste
- Solid Waste generation will be 100gm/passenger/day

The water demand during operation phase will be supplied by Surat Municipal Corporation. Daily sewage flow is considered as 90% of the water requirement at each station as given in the Table, which will be treated through Bio Digesters.

5.6.5. Traffic Congestion around Stations

Commencement of metro services results in passenger rush at stations which in turn results in congestion around stations if inter modal integration with other modes is not done.

Table 5.16: Water Demand, Waste Water and SW Generation for Dream City to Sarthana Corridor

Station Name	Water Demand (KLD)				Waste Water Generation (KLD)				Solid Waste Generation (Kg)			
	2021	2026	2036	2046	2021	2026	2036	2046	2021	2026	2036	2046
Dream City	0.87	8.24	39.33	61.09	0.78	7.41	35.40	54.98	11.60	109.82	524.39	814.47
Convention Center	0.41	0.55	2.32	3.49	0.36	0.49	2.09	3.14	5.40	7.32	30.98	46.58
Bhimrad	0.95	1.21	4.87	7.54	0.85	1.08	4.38	6.79	12.60	16.07	64.88	100.53
Woman ITI	0.38	0.54	2.05	3.06	0.34	0.48	1.84	2.75	5.00	7.14	27.32	40.79
VIP Road	23.58	32.46	70.57	104.45	21.22	29.22	63.52	94.00	314.40	432.86	940.98	1392.63
Althan Gam	8.16	11.34	28.94	38.59	7.34	10.21	26.05	34.73	108.80	151.25	385.85	514.47
AlthanTenament	6.77	10.75	21.09	12.73	6.09	9.68	18.98	11.46	90.20	143.39	281.22	169.74
Rupali Canal	5.67	9.88	18.99	23.47	5.10	8.90	17.09	21.12	75.60	131.79	253.17	312.89
Majura Gate	25.88	43.37	71.45	114.57	23.29	39.03	64.31	103.12	345.00	578.21	952.68	1527.63
Kadarsha Ni Nal	4.41	9.59	22.17	27.73	3.97	8.63	19.95	24.96	58.80	127.86	295.61	369.74
Gandhi Baug	33.66	59.12	134.87	193.52	30.29	53.20	121.38	174.17	448.80	788.21	1798.29	2580.26
Maskati Hospital	9.83	17.73	26.20	25.44	8.84	15.96	23.58	22.90	131.00	236.43	349.27	339.21
Surat Railway Station	26.22	46.38	118.92	164.53	23.60	41.74	107.03	148.07	349.60	618.39	1585.61	2193.68
Central Warehouse	10.67	38.46	25.48	32.45	9.60	34.62	22.93	29.20	142.20	512.86	339.76	432.63
Labheshwar Chowk	15.29	19.02	26.21	38.96	13.76	17.12	23.59	35.06	203.80	253.57	349.51	519.47
Kapodra	56.10	53.36	105.18	98.25	50.49	48.02	94.66	88.43	748.00	711.43	1402.44	1310.00
Shri Swaminarayan Mandir Kalakunj	13.83	45.79	50.36	47.12	12.45	41.21	45.32	42.40	184.40	610.54	671.40	628.20
VarchaChopati Garden	6.11	13.06	20.34	21.74	5.49	11.75	18.31	19.56	81.40	174.11	271.20	289.80
Nature Park	14.09	45.79	61.41	61.99	12.68	41.21	55.27	55.79	187.80	610.54	818.78	826.58
Sarthana	10.01	13.06	24.80	28.60	9.00	11.75	22.32	25.74	133.40	174.11	330.73	381.32

Table 5.17: Water Demand, Waste Water and SW Generation for Bheshan to Saroli Corridor

Station Name	Water Demand (KLD)				Waste Water Generation (KLD)				Solid Waste Generation (Kg)			
	2021	2026	2036	2046	2021	2026	2036	2046	2021	2026	2036	2046
Bheshan	1.81	6.29	24.65	38.49	1.63	5.66	22.18	34.64	24.17	83.87	328.61	513.25
Botanical Garden	3.96	10.11	26.70	36.76	3.56	9.10	24.03	33.08	52.78	134.84	355.95	490.13
UgatVaarigruh	9.08	21.05	51.80	69.27	8.18	18.94	46.62	62.35	121.11	280.65	690.63	923.64
Palanpur Road	8.81	15.48	24.53	27.41	7.93	13.94	22.08	24.67	117.50	206.45	327.09	365.45
L.P. Savani School	10.21	18.05	24.63	29.24	9.19	16.24	22.16	26.32	136.11	240.65	328.35	389.87
Performing Art Centre	5.02	10.06	21.46	27.41	4.52	9.06	19.31	24.67	66.94	134.19	286.08	365.45
Adajan Gam	5.79	11.69	15.09	18.68	5.21	10.52	13.59	16.81	77.22	155.81	201.27	249.09
Aquerium	7.77	13.74	21.00	24.90	6.99	12.37	18.90	22.41	103.61	183.23	280.00	331.95
Badri Narayan Temple	7.58	14.93	21.95	25.25	6.83	13.43	19.75	22.72	101.11	199.03	292.66	336.62
AthwaChopati	3.92	7.45	15.87	20.28	3.53	6.71	14.29	18.25	52.22	99.35	211.65	270.39
Majura Gate	29.77	54.60	74.34	92.08	26.79	49.14	66.90	82.88	396.94	728.06	991.14	1227.79
Udhna Darwaja	25.38	42.53	76.54	98.69	22.84	38.28	68.88	88.82	338.33	567.10	1020.51	1315.84
Kamela Darwaja	17.98	28.96	40.04	39.31	16.18	26.06	36.04	35.38	239.72	386.13	533.92	524.16
Anjana Farm	10.02	17.98	23.28	28.32	9.02	16.18	20.95	25.49	133.61	239.68	310.38	377.66
Model Town	16.69	24.07	38.92	43.93	15.02	21.67	35.03	39.54	222.50	320.97	518.99	585.71
Magob	46.46	83.32	163.08	200.49	41.81	74.99	146.77	180.44	619.44	1110.97	2174.43	2673.25
Bharat Cancer Hospital	1.42	3.63	8.65	13.88	1.28	3.27	7.78	12.49	18.89	48.39	115.28	185.00
Saroli	68.02	97.09	200.10	263.77	61.22	87.38	180.09	237.39	906.94	1294.52	2668.06	3516.94

5.6.6. Impacts due to Depot

Two depots have been proposed at each corridor i.e. one at Bheshan and another at Dream city. The depots will have following facilities:

- Washing Lines,
- Operation and Maintenance Lines,
- Workshop, and
- Offices.

The depot area will be levelled through cut and fill method within the depot. Impacts anticipated at depot sites are:

- Water supply
- Effluent Treatment
- Oil Pollution
- Noise Pollution
- Surface drainage changes
- Solid Waste disposal
- Loss of trees.

i. Water Supply

Water supply will be required for different purposes in the depot. As per the Indian Railway Works Manual, the water demand for train washing (Departments, workshop and Contractor office) is 3600 liter per day. A three day cycle is assumed for train washing at depot. The water demand at both the depots would be about 259 KLD for train washing and 46 KLD for domestic purpose for official staffs and staff quarters. Water requirement for other purposes like horticulture, flushing urinals/closet will be met from recycled water. Drinking water requirement will be met from supply water by Municipal Corporation.

ii. Effluent Generation

About 210 KLD of waste water will be generated from washing of trains and 41 KLD sewage will be generated at both the depots. The waste water will be treated and treated water will be recycled to use at depot horticulture and flushing purpose.

iii. Oil Pollution

Oil spillage during change of lubricants, cleaning and repair processes, in the maintenance Depot cum workshop for maintenance of rolling stock, is very common. The spilled oil should be trapped in oil and grease trap. The collected oil would be

disposed off to authorised collectors, so as to avoid any underground/ surface water contamination.

iv. Noise Pollution

The main source of noise from depot is the operation of workshop. The roughness of the contact surfaces of rail and wheel and train speed is the factors, which influence the magnitude of rail - wheel noise.

v. Surface Drainage

In case of filling in low-lying area of depot sites, the surface drainage pattern may change. Suitable drainage measures will be adopted to drain off the area suitably in the nearby water body.

vi. Loss of Trees

About 319 numbers of trees are observed at Bhesan depot site. These trees are tree likely to be affected; out of this 277 trees will be cut and 42 trees will be transplanted. About 176 numbers of trees are to be cut at Dream City Depot. Afforestation cost is given in the Environmental Management Plan.

5.6.7. Impact on Flamingo

During operation, the predicted noise level is less than the baseline noise level near Tapi River. Hence, no impact due to noise from proposed project is anticipated on flamingo productivity.

Chapter 6 : Positive Environmental Impacts

Various positive impacts have been listed under the following headings:

- Employment Opportunities
- Benefits to Economy
- Direct benefits to passengers
- Safety
- Traffic Noise Reduction
- Reduction of Traffic on Road
- Less Fuel consumption and
- Reduced Air pollution.

6.1. Employment Opportunities

During construction local skilled and unskilled labourers will have an opportunity for employment directly or indirectly. About 2856 persons are likely to work during peak period of construction activities. In operation phase of the project about 35 persons per kilometre length of the corridor is considered. In operation phase of the project about 1435 persons will be employed for operation and maintenance of the proposed system. Thus, the project would provide substantial direct employment; besides, more people would be indirectly employed in allied activities and trades.

6.2. Benefits to Economy

The project will facilitate movement of people from different parts of city. These corridors will yield benefits in terms of growth in economic activity due to better accessibility; reduction in vehicle operating costs, cost of road construction and maintenance, loss of productivity due to health disorders resulting from pollution and accidents, savings in travel time and improvement in quality of life due to reduction in road travel.

In this study only savings in fuel consumption and reduction in air pollution have been quantified.

6.3. Direct Benefits to Passengers

The project will result in direct benefits to users of Metro and other modes: reduction in vehicle operating costs, savings in travel time, improvement in quality of life, reduction in loss of productivity due to health disorders resulting from pollution and reduction in road accidents.

6.4. Safety

Substantial decrease in road accident has been noticed after 2007 with the implementation of Delhi Metro Phase I & II in Delhi. Hence, operation of Surat Metro Rail Phase I also provide improved safety and lower the number of accidental deaths. This may happened due to shifting of commuters from road to metro.

6.5. Traffic Noise Reduction

A 50% reduction of the traffic volume may result in a 3 dB reduction in noise levels, regardless of the absolute number of vehicles. Reduction in traffic volume of 10% & 50%reduces noise at the tune of 0.5 dB & 3.0 dB respectively[8].

6.6. Reduction of Traffic on Road

The basis of reduction of vehicle is shift of ridership from road vehicle to the metro railway. The reduction in number of vehicles gives benefits to economy by reduction in Vehicle Operating Cost (VOC), Fuel Consumption, Pollution Load, Accidents and Travel Time etc. On implementation of the project, the consumption of petrol and diesel will get reduced. As per Surat Metro DPR 2018 verious travel benefits like travel time saved, fuel saved, reduction in accidents and vehicle kilometre reduced due to modal shift are quatified in Table 6.1.

Table 6.1: Travel Benefits due to Mode Shift

Quantified Benefits in Horizon Years	2023	2024	2025	2026	2027	2028
Annual Time Saved by Metro Passengers in Cr. Hr.	2.94	3.37	3.86	4.43	4.70	5.00
Annual Fuel Saved by Metro Passengers in thousand Tons.	30.99	35.84	41.46	47.98	51.47	55.23
Daily vehicles reduced (off the road)	41720	47814	54797	62801	66708	70859
CO2 reduced in thousand tons	33.61	38.52	44.15	50.60	53.75	57.09
Other gases reduced in thousand tons	1.912	2.191	2.511	2.878	3.057	3.247
Reduced No of Fatal Accidents in Year	14.00	16.04	18.38	21.07	22.38	23.77
Reduced No of Other Accidents in year	92.21	105.68	121.11	138.80	147.44	156.61
Annual Vehicle km Reduced in Thousand Km.	19.42	22.26	25.51	29.24	31.06	32.99

Source: Surat Metro DPR 2018

[8]Relation between traffic volume & noise levels, Ellebjerg (2013)

6.7. Reduced Air Pollution

Compared to other modes of transport, the metro is least polluting and can be classified as an environment friendly technology since no air emissions are involved in running and operating the metro trains. The reduction of air pollutants corresponding to the reduction of daily vehicle kilo meters are presented in Table 6.2.

Table 6.2: Reduction in Air Pollutants due to Vehicle Reduction

Tons/Year	2023	2024	2025	2026	2027	2028
CO	1077	1235	1415	1622	1723	1830
HC	629	720	826	946	1005	1068
NOX	158	181	207	237	252	268
PM	45	52	60	68	73	77
SO2	2.91	3.33	3.82	4.38	4.65	4.94
CO2	33614	38524	44151	50600	53748	57092
Total Emission Saved	35526	40715	46662	53478	56805	60339

Source: Surat Metro DPR 2018

6.8. Carbon Credits

The Kyoto Protocol is an international agreement linked to the United Nations Framework Convention on Climate Change, which commits its Parties by setting internationally binding emission reduction targets. This set legally binding targets for 37 industrialised countries to limit or reduce overall GHG emissions by at least 5% below 1990 levels during the period 2008-2012. In the Doha Amendment to the Kyoto Protocol December 2012 new commitments to reduce GHG emissions by at least 18 percent below 1990 levels in the eight-year period from 2013 to 2020 were committed from 1 January 2013 to 31 December 2020.

Under the Protocol, countries must meet their targets primarily through national measures. However, the Protocol also offers additional means to meet their targets by way of three market-based mechanisms. These are:

- **International Emissions Trading** - the international transfer of emission allocations between industrialized (Annex 1) countries. E.g. a country that stays within its target can sell the surplus allowances to another country that has exceeded its limit.
- **Clean Development Mechanism (CDM)** - creates Certified Emission Reductions (CERs) through emission reduction projects in developing countries, regulated by the United Nations. Each CER is equivalent to one tonne of CO_2. These CERs can be traded and sold, and used by industrialized countries to a meet a part of their emission reduction targets under the Kyoto Protocol.

- **Joint implementation (JI)** - any Annex I country can invest in emission reduction projects in any other Annex I country as an alternative to reducing emissions domestically.

The emission reductions generated by these mechanisms are collectively referred to as 'carbon credits'. A carbon credit is a financial instrument that represents a reduction or the avoidance of one tonne of carbon dioxide equivalent (tCO_2e) from the atmosphere.

CDM Benefits: The clean development mechanism was designed to meet a dual objective:

- to help developed countries fulfil their commitments to reduce emissions, and
- to assist developing countries in achieving sustainable development.

Benefits of CDM projects include investment in climate change mitigation projects in developing countries, transfer or diffusion of technology in the host countries, as well as improvement in the livelihood of communities through the creation of employment or increased economic activity.

General Eligibility requirements for CDM projects

- Submission of the "prior consideration of the CDM" form within 6 months of the project starts date.
- Stakeholder consultation and addressing their observations
- Submission of documentation on environmental impacts of the project activity.
- The project activity is expected to result in a reduction in anthropogenic emissions by sources of greenhouse gases that are additional to any that would occur in the absence of the proposed project activity,
- Baseline and monitoring methodologies comply with requirements.
- All permissions/NOCs related to project obtained and project contribute to sustainable development criteria of Host country.

Transport projects that contribute to reductions in GHG emissions, and have potential for Carbon finance, include improved public transport (bus rapid transit, metros, light rail transit), electric and hybrid vehicles (e.g., electric scooters), inter-urban rail infrastructure (including double-tracking, new freight lines, new passenger lines, or rapid passenger trains), technologies like regenerating breaking system which generate electricity while breaking and electrification railways projects (depending on whether the country uses fossil fuels to generate power or not). The principles for reduction in GHG emission into transportation is as follows:

- Reduce emissions per km
  - Fuel switch e.g. bio-fuels
  - Behavioural management change e.g fleet management, driver training
  - Technology vehicle change. i.e. Energy efficiency improvements, regenerating breaking system & electric vehicles

- Reduce emission per unit transported
  - Passenger transport e.g. urban planning modal shift, use of large units, Improved occupation rate
  - Fright transport
- Reduce distance or number of units
  - Urban planning to induce behaviour change via government policies/measures
- Use of renewable energy and energy efficiency in stations and offices
  - Solar roof tops, led lighting, natural lighting, use of glare glasses, VAM etc.

6.8.1. Carbon Credits for Surat Metro Rail Project

The measures used to calculate Carbon Credit for the projects are classified into following categories:

- Reduce emissions per km by use of regenerating breaking system & electric vehicles
- Reduce emission per unit transported modal shift from two, three and four wheelers

A. Reduce Emissions per km by use of Regenerating Breaking System & Electric Vehicles

The project activity operates low GHG emitting rolling stocks having regenerative braking system in Surat Metro. The project replaces the conventional electro-dynamic rheo-static braking technology, with regenerative braking technology fitted rolling stocks. The regenerated electrical energy reduces the consumption of equivalent grid electrical energy required by the powering trains, thereby conserving electrical energy and subsequently leading to GHG emission reduction.

Therefore the emission reductions are additional as the above project activity is expected to result in a reduction in anthropogenic emissions by sources of greenhouse gases(GHG) that are additional to any that would occur in the absence of the proposed project activity.

The net annual Energy requirement for Traction and auxiliary power for both the corridors of Surat Metro is given in Table 6.3 and Table 6.4 for Bhesan to Saroli corridor and Sarthan to Dream city corridor respectively. Rolling stock proposed for the metro has regeneration features and it is expected that 30% of total traction energy. Hence the total regeneration energy for both the corridors in Year 2021 is 95.72 million units and in year 2046 is 136.39 million units.

CO_2 emissions due Energy consumption (Traction and Auxiliary) and reduction due to regeneration energy for year 2021 are estimated based on the CO_2 Intensity of Electricity from grid as given in CMP tool kit 2014 and results are given in Table 6.5.

Table 6.3: Traction Energy Requirement for Bhesan to Saroli Corridor

Sr No	Item	Unit	2021	2026	2036	2046
Traction Power Requirement						
1.	Total Energy required per Year	million units	11.46	14.59	22.60	26.36
2.	**Regeneration Energy (30% of total Traction Energy) per Year**	million units	**3.44**	**4.38**	**6.78**	**7.91**
3.	Net Traction Energy Consumption per Year with 30% regeneration energy	million units	8.02	10.21	15.82	18.45
Auxiliary Power Requirement						
4.	Yearly Auxiliary Energy Consumption	million units	18.93	19.6	20.53	22.14
Gross Annual Energy Consumption (Traction & Auxiliary)		million units	**26.95**	**29.81**	**36.35**	**40.59**

Source: Surat Metro DPR 2018

Table 6.4: Traction Energy Requirement for Sarthana to Dream City Corridor

Sr No	Item	Unit	2021	2026	2036	2046
Traction Power Requirement						
1.	Total Energy required per Year	million units	17.10	27.69	37.46	46.14
2.	**Regeneration Energy (30% of total Traction Energy) per Year**	million units	**5.13**	**8.31**	**11.24**	**13.84**
3.	Net Traction Energy Consumption per Year with 30% regeneration energy	million units	11.97	19.38	26.22	32.30
Auxiliary Power Requirement						
4.	Yearly Auxiliary Energy Consumption	million units	56.8	58.37	60.15	63.5
Gross Annual Energy Consumption (Traction & Auxiliary)		**million units**	**68.77**	**77.75**	**86.37**	**95.80**

Source: Surat Metro DPR 2018

Table 6.5: CO_2 Emissions from Energy Requirement of Metro Operations

Item	Sarthana to Dream City Corridor	Bhesan to Saroli Corridor
CO_2 emissions due to Gross Energy Consumption (Traction & Auxiliary) of Metro Operations for Year 2021 - Ton/day	53.90	137.54
Reduction in CO_2 Emissions due to Regeneration Energy (30% of total Traction Energy) for Year 2021 – Ton/day	6.88	10.26

Chapter 7 : Environmental Management Plan

7.1. Approvals/Clearances

Project Implementation Unit (PIU) will have to ensure that all necessary approvals/clearances are in place before start of implementation. Roles and responsibilities for securing necessary approvals and or clearances are listed in Table 7.1.

Table 7.1: Roles and Responsibilities - Securing Approvals/Clearances

S. No.	Issue	Provision of Laws & Regulations	Due Date	Approving Authority
Pre-Construction Phase				
1.	Permission for felling of trees and compensatory afforestation	Tree removal will be guided as per state government rules.	Before Construction	The Park and Garden Department of Surat Municipal Corporation (SMC), and Forest Department
2.	Prior Environmental Clearance: Metro Railway is not listed among activities requiring prior Environmental Clearance in EIA Notification 2006. In case of Nagpur Metro, MoEFCC clarified that construction of building for commercial purposes having built area equal to or more than 20,000 sqm shall require prior EC from SEIAA and that the project has to comply the norms such as green building features, rain water harvesting, water conservation etc.	EIA Notification 2006 and its amendments		SEIAA
3.	CRZ clearance as project is falling in CRZ –II & CRZ-IV-B	CRZ Notification 2019		MoEFCC
4.	Building Permissions for Depot, stations, property development	EIA Notification 2006 and its amendments		Municipal Corporation
5.	Utility / traffic diversion	Respective Acts and Rules		Local Offices of respective Agencies.
6.	Consent to Establish construction yards, labour camps, stations and depots (since non-residential)	Water (Prevention and Control of Pollution) Act 1974, Air (Prevention and Control of Pollution) Act, 1981		State Pollution Control Board; Development Authority for land use clearance
7.	Sites to establish labour camps, pre-casting and material yards	Land use Master Plan and DC&PR		Municipal Corporation

S. No.	Issue	Provision of Laws & Regulations	Due Date	Approving Authority
Construction Phase				
8.	• Consent to Establish and Operate hot mix plant, crushers, batching plant etc • Consent to Establish labour camps	Air (Prevention and Control of Pollution) Act, 1981	Before Construction	• State Pollution Control Board • Municipal Corporation
9.	Permission for drawal of groundwater for construction **(not recommended)**	Environment (Protection) Act, 1986	Before Construction	Regional Director, Central Ground Water Board and Municipal Corporation
10.	Authorization for Disposal of Hazardous Waste	Hazardous Waste (Management and Handling and trans boundary movement) Rules 2016	Before Construction	State Pollution Control Board
11.	Consent for disposal of waste water from construction sites and sewage from labour camps	Water (Prevention and Control of Pollution) Act 1974	Before Construction	State Pollution Control Board
12.	Labour employment, safety, welfare measures	The Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Act, 1996	Before Construction	District Labour Commissioner
13.	Permission for management of C&D waste and muck	Construction and Demolition Waste Management Rules 2016	Before Construction	Municipal Corporation and State Pollution Control Board
Operation Phase				
14.	Consent to Operate Depot	Environment Protection Act , 1986	After Construction	State Pollution Control Board
15.	Installation and operation of DG sets at stations	Air (Prevention and Control of Pollution) Act, 1981	After construction	State Pollution Control Board

7.2. Mitigation Measures

The main aim of mitigation measures is to protect and enhance the existing environment of the project. This section includes measures for:

- **Location and Design**
  - Compensatory Afforestation
  - Right of Way, Alignment, Track design and Architecture
  - Spatial Planning of stations and Inter-Modal Integration

- Robust Design
- Provision for Green Buildings and solar power
- Use of Energy and Water
- Utility Plan

- **During Construction**
  - Supply of Construction Material
  - Pre-casting yards and Material Stockpiling
  - Construction Material Management and Housekeeping
  - Hazardous Waste Management
  - Construction and Demolition Waste management
  - Muck Disposal
  - Energy Management
  - Labour Camp
  - Welfare of Labour
  - Safety of Labour
  - Air Pollution Control Measures
  - Noise Control Measures
  - Vibration Control Measures
  - Water pollution control measures
  - Traffic Diversion/Management
  - Soil Erosion and Land Subsidence Control
  - Dewatering of underground works
  - Management Plan for Flamingos

- **During Operation**
  - Noise and Vibration Management
  - Water Supply and Sanitation at stations
  - Management Plan for Depot
  - Management Plan for Flamingos
- **Training**
- **Establishment of Environment Division**
- **Disaster Risk Management**

7.3. EMP During Project Location and Design

7.2.1. Compensatory Afforestation

Removal of Air Pollutants: Particulate matter in the atmosphere is intercepted by tree canopy. The particulates are retained on the plant surface or washed off by rain or dropped to ground with leaf fall. Urban trees have been found to remove PM10 and PM2.5 from the

atmosphere. Removal of PM2.5 is lower than removal of PM10 but the health benefits are higher[9].

Increase in Groundwater Recharge : Quantity of rainfall percolating to a specified depth of soil was found to decrease with distance from canopy edge towards with minimum percolating quantity in open area. Soil infiltration is improved near trees due to litter and tree roots promoting activity of earthworms, insects etc. resulting in increased soil macro porosity. Under conditions where surface runoff of rain water is redistributed towards trees, net water stored in soil near trees increases. In case of trees in which at least 25% of their water intake from soil is from depth greater than 1.5m, 10 trees per hectare with canopy cover 5% provide the highest groundwater recharge: tree density greater than this optimal cover showed reduced groundwater recharge[10].

The compensatory Afforestation is planned in line with Saurashtra Felling of trees (infliction of Punishment) Act, 1951 & amendment Rule 2008, Forest Conservation Act 1980 and ESS 6 of World Bank. It has been estimated about 932 trees that are likely to be cut are exotic in nature and 1161 trees are likely to be transplanted from the two corridors and two depots. It is proposed to plant three saplings for each tree to be cut and one sapling for each tree transplanted; hence 3957 trees need to be planted. Estimated compensatory afforestation cost for the proposed corridors is given in Table 7.2. Plantation rate of Rs. 1200/- per sapling and transplantation rate of Rs. 9999/- per tree (Parks and Garden Department of SMC) has been assumed for estimation of cost.

Table 7.2: Cost of Compensatory Afforestation & Transplantation

S. No	Corridor/Depot	Cost in Rs. Lakh			
		Compensatory Afforestation	Transplantation	Add. tree for transplantation	Total Cost
1.	Sarthana to Dream City	11.23	52.39	6.29	69.91
2.	Bhesan to Saroli	6.01	59.49	7.14	72.65
3.	Dream City Depot	6.34	-	-	6.34
4.	Bhesan Depot	9.97	4.20	0.50	14.68
Total Cost		**33.55**	**116.09**	**13.93**	**163.57**

Afforestation

The number of trees to be transplanted depends on the site condition and root condition & health of tree. More stress should have been given for transplantation of the tree rather than removal. The trees which are not possible to transplant/Not-Straight or Tilted or diseased will only be cut. The total area required for afforestation of these trees comes out to about 3.59 ha. Precautionary measures proposed for the afforestation are as follows:

[9]Modeled PM2.5 removal by trees in ten US cities and associated health effects, David J Nowak, Satoshi Hirabayashi, Allison Bodine, Robert Hoehn, Elsevier, Environmental Pollution 178 (2013) 395-402

[10]*Intermediate tree cover can maximize groundwater recharge in the seasonally dry tropics, U.Llstedtetal, February 2016, www.nature.com*

- Plantation will be made in the monsoon months (July-August)
- The height of the plants should not be less than 30 cm and should be supplied in polythene
- bags are not to be removed until the moment of planting
- All plants supplied must be planted within three days after transported from the nursery
- Arrangements must be made to water in case of insufficient rains after planting
- Provide compost/manure suggested quantity for each pit before plantation

The Parks and Garden Department of SMC is responsible for the conservation and management of Afforestation in the project area. The native plant species and miscellaneous indigenous tree species recommended for afforestation.

The Baobab trees of heritage significance located near Adajan Gam metro station on Corridor-2 as mentioned in DPR, are to be preserved at the same place during implementation of for Surat metro rail project.

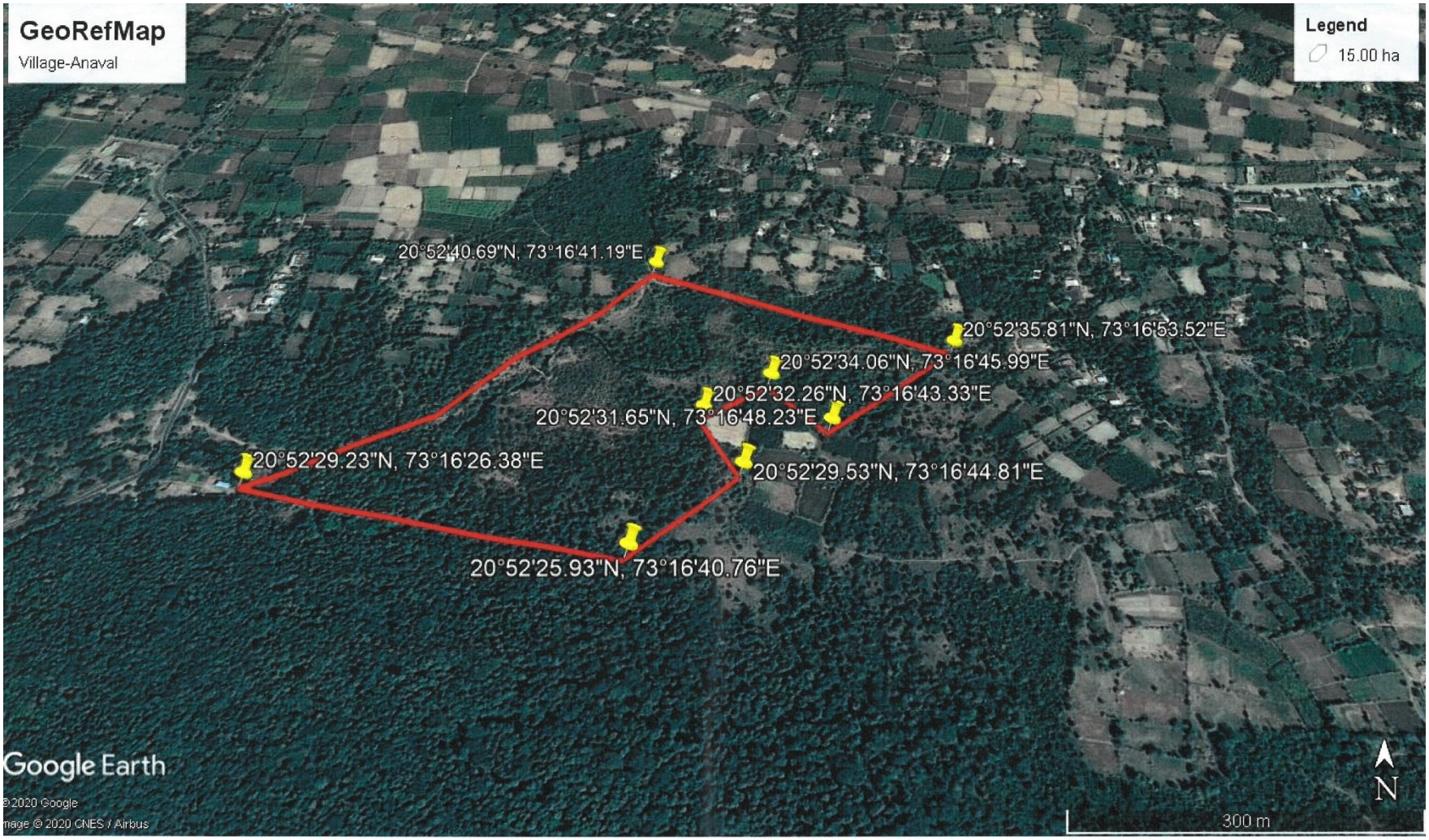
GMRCL has requested Forest Department for allotment of Land Parcel for Compensatory Afforestation. Forest Department has identified 15 ha of Forest land near village Anaval for the same. Location map of identified Forest Land is shown in Figure 7.1.

Transplantation

The trees recommendation for transplantation will be transplanted in the nearby open area. The management plan for transplantation of trees is summarized below:

- Preliminary root investigation should be carried out,
- Health diagnosis of the tree should be carried out for treating infected trees,
- Soil condition where the tree has to be transplanted is thoroughly checked & necessary treatments are applied to the soil after digging a pit,
- The pit size has to be kept in accordance with the root ball of the tree,
- Packing material should be strong enough to hold the soil around the root zone,
- Crane should be used to lift the packed tree and a trolley or truck should be used to transport the tree,
- Timely feeding of the plant should be done with soluble fertilizers and watering,
- JCB should be used for digging pits,
- There should be regular monitoring for fertilizer schedules and the chemicals like insecticides and pesticides.
- Scaffolding should be used wherever required to support the trees,
- Any broken stems during transplantation should be removed cautiously.
- After transplantation, there are chances of external infections to the tree which needs maintenance for at least 2-3 months.

Figure 7.1: Location of Land Parcel for Compensatory Afforestation



Dy. Conservator of Forests
Surat.

The plant species recommended for afforestation with their botanical names to be planted are indicated in the Table 7.3.

Table 7.3: Scientific Names Of Tree For Plantation

S. No	Local Name	Botanical Name
1.	Gulmohar	*Delonix regia*
2.	Siris	*Albizia lebbeck*
3.	Peltoforam	*Peltoforam pterocarpum*
4.	Neem	*Azadirachta indica*
5.	Ashoka	*Polyalthia longifolia*
6.	Jamun	*Syzygium cumini*
7.	Desi Badam	*Terminalia catappa*
8.	Pongam	*Millettia Pinnata*
9.	Karanj	*Pongamia pinnata*
10.	Amalatash	*Cassia fistula*
11.	Pipal	*Ficus religiosa*
12.	Umbar/Gular	*Ficus resimosa*

By implementing these afforestation measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.2.2. Right of Way, Alignment, Track design and Architecture

Alignment has been kept elevated where adequate width of right of way on roads is available. Geometric design of the alignment will be such as to optimize curvature. Track design will incorporate welded rails and elastic fittings. Viaduct and elevated stations shall be shaped to minimize visual intrusion.

7.2.3. Spatial Planning of Stations and Inter-Modal Integration

Adequate and well-laid out space shall be designed for concourses and platforms, escalators, elevators and staircases, lighting, turnstiles for normal and abnormal operating conditions; optimal height / depth of the stations, forced ventilation shall be provided. Physical and operational integration of metro with other modes shall be planned. Adequate design of stations and multimodal integration prevents and mitigates congestion at stations and safety is improved. Consumption of energy for climate control, lighting and other facilities at stations is significantly reduced by proper design of passenger flow inside stations, space & facilities inside stations. Use of evaporative cooling process in air conditioning can reduce water usage (Manual on norms and standards for environment clearance of large construction projects, MoEFCC, 2007).

7.2.4. Robust Design

The project area lies in Zone III as per revised Seismic Zoning Map of India corresponding to moderate seismic hazard. Engineering construction shall be done so as to meet codal provisions.

7.2.5. Green Buildings

Green building (also known as sustainable building) refers to both a structure and the using of processes that are environmentally responsible and resource-efficient throughout a building's life-cycle: from siting to design, construction, operation, maintenance, renovation, and demolition. Green buildings help in better preservation of environment as in such structures there are provisions for better saving of energy, water and CO_2. Such buildings also have better waste management arrangements. All stations and Depot buildings can be designed as green buildings.

The Indian Green Building Council (IGBC), part of the Confederation of Indian Industry (CII) was formed to enable a sustainable built environment for all and facilitate India to be one of the global leaders in the sustainable built environment by 2025. IGBC has launched green Mass Rapid Transit System (MRTS) rating system for all stations and green Factory Building rating system for Depots. These rating systems are tools to enable new rail based MRTS to apply green concepts during design & construction, so as to further reduce environmental impacts that are measurable.

The proposed metro project should address the green features under the following categories to get the IGBC green MRTS and green factory building rating system:

- Site Selection and Planning
- Water Efficiency
- Energy Efficiency
- Material Conservation
- Indoor Environment and Comfort
- Innovation in Design & Construction

7.2.6. Solar Power

For the utilization of renewable energy, wherever feasible, installations for solar power can be implemented on the roof of elevated stations and in Depots. It is assumed that 75% of roof area of each station is considered for solar power generation. Solar energy generation per year is estimated to be 3.45 Giga-watt-hr for Sarthana to Dream City corridor and 4.44 Giga-watt-hr for Bhesan to Saroli Corridor. Details of solar power generation for each corridor are given in Table 7.4. Energy potential and cost for installations in Depots has not been estimated in this report because amount of land/structure surface area available for this purpose can be estimated upon evaluation of the Depot layouts. The installation cost for solar system is about Rs 48.68 Lakh for Sarthana to Dream City corridor and Rs 62.59 Lakh for Bhesan to Saroli Corridor. However this cost is not included in estimated cost of EMP since installation and maintenance of solar power infrastructure is proposed to be awarded to developer along with power purchase agreement.

Table 7.4: Solar Power Generation Potential for each Corridor

S. No	Corridor	Stations		Power Generation in GWH per Year	Total Cost in Rs. Lakh
		No	Total Area in Sq.m		
1.	Sarthana to Dream City	14	43120	3.45	48.68
2.	Bhesan to Saroli	18	55440	4.44	62.59

7.2.7. Use of Energy and Water

Requirement of electrical energy for climate control, lighting and other facilities at stations shall be optimized by proper use of natural day/night light and design of passenger flow inside stations and on streets outside stations. Installations for solar power will be implemented at stations and Depot where feasible.

Water supply in stations for air conditioning, cleaning and use of staff and passengers will be procured from municipal supply – this will be supplemented by re-use of treated waste water generated by staff and passengers. Water for depots will be sourced from municipal supply: this will be supplemented by re-use of used water from coach wash. Construction purpose water will be from treated water of nearby STP.

7.2.8. Utility Plan

The proposed Metro alignment run along major roads of the city and is required to negotiate sub-surface, surface and overhead utility services. Prior to the execution of work at site, detailed investigation of all utilities will be undertaken and plans for their retention in situ with precautions or temporary/permanent diversions prepared and got approved by respective agencies. As such, these may affect construction and project implementation time schedule/costs, for which necessary planning/action needs to be initiated in advance. In case of underground utility services running across the alignment, the spanning arrangement of the viaduct may be suitably adjusted. The Organizations / Departments responsible for concerned utility services are reported in Table 7.5. The stakeholder engagement and information disclosure for shifting of utilities should be planned in line ESS 10 of World Bank to maintain the constructive relationship throughout the project construction stage. By implementing these measures, the project is expected to comply with the WB ESF guidelines.

Table 7.5: Organizations Responsible for Utilities and Services

S. No.	Organization/ Department	Utility/Services
1.	PWD / NHAI	Road
2.	Surat Municipal Corporation, SUDA	Sewerage and drainage lines. Water mains and their service lines, including hydrants and fountains etc, water treatment plants, pumping stations, Roads, surface water drains, nallahs, sewer lines,

S. No.	Organization/ Department	Utility/Services
		street lights, high mast lights etc.
3.	Telephone Operators	Tele cables, junction boxes, telephone posts, O.H lines
4.	Power Grid Corporation of India Ltd.	HT towers, cables
5.	Irrigation Dept.	Canal
6.	Oil Corporations	Gas pipe lines
7.	Gujrat State Electricity Board	HT/other overhead Power lines

7.2.9. Site Selection Criteria for Construction Yards

It is a storage yard or assembly yard for building materials and equipment directly related to a specific construction project and subject to removal at completion of construction. Site selection criteria for a construction yard are as under:

- Away from the right-of-way of a highway,
- Away from habitation and settlements,
- Away from the CRZ Notified area,
- Away from the flood plains, water bodies,
- Away from the Notified Areas- Reserved Forests, Nature Protection, Sanctuary, Wild life Sanctuary, Eco-sensitive zones, Historical Monuments, places of tourist interest, etc.

7.4. EMP During Construction

Measures to mitigate impacts observed during construction shall be implemented by Contractor and duly monitored by Owner in accordance with approved method statements. Their cost forms part of engineering and track cost.

7.3.1. Supply of Construction Material

The procurement source of the construction materials will be decided by the Contractor, but it will be from the licensed supplier.

7.3.2. Pre-casting yards and Material Stockpiling

Sites for casting of structural concrete elements and material stockpiling will be decided before start of construction. The sites will be identified by GMRCL, such that displacement of persons is not involved to the extent possible.

7.3.3. Construction Material Management and Housekeeping

Procedures for storage, handling and transport of construction material and its reuse shall be prescribed in EHS guidelines. The EHS guidelines should be prepared in line with World Bank ESS guidelines and National regulations.

Housekeeping is to keep the working environment cleared of all unnecessary waste, thereby providing a first-line of defense against accidents and injuries. It is the responsibility of Contractor and all site personnel. Some of the measures are listed below:

- Full height fence, barriers, barricades etc. shall be erected around the site in order to prevent the surrounding area from excavated soil, rubbish etc, which may cause inconvenience to and endanger the public.
- All stairways, passageways and gangways shall be maintained without any blockages or obstructions. All emergency exits passageways, exits fire doors, break-glass alarm points, fire-fighting equipment, first aid stations, and other emergency stations shall be kept clean, unobstructed and in good working order.
- All surplus earth and debris shall be removed/disposed off from the working areas to officially designated dumpsites. Trucks carrying sand, earth and any pulverized materials etc. shall be covered while moving.
- Unused/surplus cables, steel items and steel scrap within the working areas shall be removed to pre-identified locations.
- All wooden scrap, empty wooden cable drums and other combustible packing materials, shall be removed from work place to identified locations.
- Empty cement bags and other packaging material shall be properly stacked and removed.
- Proper and safe stacking of material is of paramount importance at yards, stores and such locations for future use. The storage area shall be well laid out with easy access and material stored / stacked in an orderly and safe manner.
- Flammable chemicals/compressed gas cylinders shall be safely stored.

By implementing these EHS guidelines, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.4. Hazardous Waste Management

The Hazardous waste management should be handled by authorised/licensed agent as per guidelines of Hazardous and Other Wastes (Management and Trans boundary Movement) Rules, 2016 & its amendments and ESS 3 of World Bank. It shall be the responsibility of the contractor to ensure that hazardous wastes are labeled, recorded, stored in impermeable containment and for periods not exceeding mandated periods and in a manner suitable for handling storage and transport. The contractor shall maintain a record of sale, transfer, storage of such waste and make these records available for inspection. The contractor shall

approach only Authorized Recyclers for treatment and disposal of Hazardous Waste, under intimation to the Project Authority. The treatment and disposal sites will be identified by GMRCL in consultation with SMC such that pollution of water bodies and green areas are not impacted and displacement of persons is not involved. By implementing these measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.5. Construction and Demolition Waste Management

The C&D waste generated will be reused and remaining will be disposed in a manner that will not affect the human health and environment in line with C&D Waste Management Rules, 2016 & its amendments and ESS 3 of World Bank. Construction and Demolition (C&D) waste is part of solid waste that results from land clearing, excavation, construction, demolition, remodeling and repair of structures, roads and utilities. C&D waste has the potential to save natural resources (stone, river sand, soil etc.) and energy, reduce transportation over long distances for dumping, and reduce space occupied at landfill sites. C&D waste generated from metro construction has potential use after processing, grading solid waste and recycling.

- Segregation and temporary storage of reusable and recyclable materials at identified locations. Transport recyclable materials to construction sites.
- Sale of metal scrap and other saleable waste
- Identification of intended transport means and route.
- Obtaining permission, where required, for treatment of the hazardous component and its disposal.
- Concrete material shall be brake into coarse size and reutilised in filling.
- The treatment and disposal sites will be identified by GMRCL in consultation with SMC such that pollution of water bodies and green areas are not impacted and displacement of persons is not involved. Before dumping, recyclable material will be removed. The disposal sites will be cleaned and then treated so that leached water does not contaminate the ground water.

By implementing these waste management measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.6. Muck Disposal

The excavated material shall be graded such that part can be re-used in construction; balance will be disposed. Before excavation, the Contractor will be required to test the soil quality including heavy metals and the results will be compared with standards. If the soil is contaminated, disposal will be done with due treatment or isolation of such muck.

Disposal sites will be identified by GMRCL in consultation with SMC such that pollution of water bodies and green areas are not impacted and displacement of persons is not involved. The following activities are involved:

- Disposal sites shall be cleaned and then treated so that leached water does not contaminate the Ground Water.
- Material will be stock-piled with suitable slopes
- Material will be stabilised each day by watering or other accepted dust suppression techniques. The muck shall be filled in the dumping site in layers and compacted mechanically.
- Once the filling is complete, the entire muck disposal area shall be provided with a layer of good earth on the top and covered with vegetation.

Dumping site selection criteria, precautions to be adopted during disposal are given in Annexure 7.1 to avoid the human health and environment as specified in ESS 3 of World Bank. By implementing these site selection criteria, the project is expected to comply with the WB ESF guidelines.

7.3.7. Energy Management

The contractor shall use and maintain lighting, tools and equipment of appropriate specifications so as to conserve energy as per EHS guidelines.

7.3.8. Labour Camp

The Contractor during the progress of work will provide, erect and maintain necessary (temporary) living accommodation for construction workers at locations away from construction sites in line with National regulations and World Bank ESS guidelines.

Water supply, waste water and sewage treatment: Uncontaminated water for drinking, cooking and washing, health care, latrines and urinals, system for conveyance, treatment and disposal of sewage and solid waste; adequate and clean washing and bathing places shall be provided. Wastewater shall be discharged to the existing sewage network or will be disposed off in septic tank and soak pit.

Solid Waste Management: Municipal solid waste generated will be collected and transported to local municipal bins for onward disposal to disposal site by municipality. Solid waste management facilities will be arranged by the construction contractors.

Health care awareness and clinics: Construction workers are more prone to Infectious diseases such as HIV/AIDS. It should be prevented by following actions: Counselling, community events, clinic, and coordination with local health authorities.

By implementing these measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.9. Welfare of Labour on Construction Site

Shelter at Workplace: At every workplace, shelter shall be provided free of cost, separately for use of men and women labourers. The height of shelter shall not be less than 3m from floor level to lowest part of the roof. Sheds shall be kept clean and the space provided shall be on the basis of at least 0.5m^2 per head.

Canteen Facilities: A cooked food canteen on a moderate scale shall be provided for the benefit of workers wherever it is considered necessary. The contractor shall conform generally to sanitary requirements of local medical, health and municipal authorities and at all times adopt such precautions as may be necessary to prevent soil pollution of the site.

First aid facilities: At every workplace, a readily available first-aid unit including an adequate supply of sterilized dressing materials and appliances will be provided. Suitable transport will be provided to facilitate taking injured and ill persons to the nearest hospital.

By implementing these measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.10. Safety of Labour

Construction works shall be executed as laid down in the Environment, Health and Safety (EHS) manual. The manual should be prepared in line with World Bank ESS guidelines and National regulations. The EHS manual should address the issues like labour camp, safety and welfare of Labour. The EHS manual

- describes the EHS interfaces between Employer and the Contractor.
- details the processes by which the contractor shall manage EHS issues while carrying out the work under the contract.
- describes by reference, the practices and procedures

The construction works shall be undertaken in accordance with all applicable legislation and Indian statutory requirements and guidelines-OHSAS 18001-2007: Occupational Health and Safety Management System and ISO 14001-2015: Environmental Management Systems. The key elements of the EHS manual are as follows:

1. Identification of the unit responsible for co-ordinating and monitoring the Contractor's EHS performance;

2. Procedures for identifying and estimating hazards, and the measures for addressing the same; a list of EHS hazards anticipated
3. EHS training courses and emergency drills
4. EHS inspections to identify any variation in construction activities and operations, machineries, plant and equipment and processes against the EHS Plan and its supplementary procedures and programs: Planned General Inspection, Routine Inspection, Specific Inspection and Other Inspection
5. Safety Audit: EHS Audit to assess potential risk, liabilities and the degree of compliance of construction Safety, Health & Environmental plan and its supplementary procedures and programs against applicable and current EHS legalisation regulations and requirements of the employer; Electrical Safety Audit; External EHS Audit
6. EHS Communication to communicate the Safety, Occupational health and Environment management measures through posters campaigns / billboards / banners / glow signs being displayed around the work site
7. EHS Reporting – reports, minutes, inspection reports, audit reports
8. Accident reporting and investigation
    - Reports of all accidents (fatal/injury) and dangerous occurrences to the Employer
    - Reporting to Govt. organisations
9. Investigations of Accidents and Dangerous Occurrences, Near misses and minor accidents
10. Prepare an Emergency Response Plan for all work sites including injury, sickness, evacuation, fire, chemical spillage, severe weather and rescue.

Workplace safety and occupational health shall be ensured with special focus on following areas:

a) Housekeeping
b) Working at Height and Falling objects and Danger areas
c) Lifting Appliances
d) Launching Operation
e) Construction machinery, tools equipment - Safe worthiness
f) employ qualified electrical personnel on site and requirements of electrical equipment, distribution etc
g) Lighting
h) Exposure of worker to use of exhaust or harmful gases in confined locations
i) Fire prevention, protection and fighting system
j) Corrosive substances
k) Demolition
l) Excavation and Tunnelling
m) Traffic Management
n) Personal Protective Equipment (PPE)
o) Reporting which will contain results of monitoring and inspection programs
p) Process of response to Inquiries, complaints and requests for information from private and government entities

q) Physical fitness of workmen
r) Medical Facilities on site : Occupational Health Centre, Ambulance van and room HIV/ AIDS prevention and control
s) Exposure to Noise – prevention measures
t) Ventilation and illumination

By implementing these measures proposed in the EHS manual, the project is expected to comply with the Indian regulations and WB ESF guidelines.

7.3.11. Air Pollution Control Measures

During the construction period, the impact on air quality will be mainly due to increase in Particulate Matter (PM) along haul roads and emission from vehicles and construction machinery. The mitigation measures proposed for air pollution reduction are in line with Air (Prevention and Control of Pollution) Act, 1981 & its amendments and ESS 3 of World Bank. Mitigation measures which shall be adopted to reduce the air pollution are presented below:

- Contractor's transport vehicles and other equipment shall conform to emission standards fixed by Statutory Agencies of Government of India or the State Government from time to time.
- The Contractor shall carry out periodical checks and undertake remedial measures including replacement, if required, so as to operate within permissible norms.
- Dust control (*Dust Control Manual for Industrial Minerals Mining and Processing, CDC NIOS, RI 9689/2012):*
- Water should be sprayed during construction phase, wherever it is required to avoid dust.
- Vehicles delivering materials should be covered to reduce spills and dust blowing off the load.
- Vehicles and machinery are to be regularly maintained so that emissions conform to National and State AAQ Standards.
- Processes such as pneumatic filling of silos, transportation by road, drilling and blasting, crushing, screening, bulk/bag unloading generate dust.

Local exhaust ventilation systems (LEVS) as shown in Figure 7.2, capture dust generated and then transport this dust via ductwork to a dust collection filtering device. By capturing the dust at the source, it is prevented from becoming liberated into the processing plant and contaminating the breathing atmosphere of the workers. LEVS possess a number of advantages:

- The ability to capture and eliminate very fine particles that are difficult to control using wet suppression techniques;

- The option of reintroducing the material captured back into the production process or discarding the material so that it is not a detriment later in the process; and
- Consistent performance in cold weather conditions because of not being greatly impacted by low temperatures.

Figure 7.2: Silo Top Aero Filter

For outdoor areas where potential dust sources are uncontrolled, such as haul roads, stockpiles, and miscellaneous unpaved areas dust suppression by water spraying is required. It is a combination of direct spraying of the material to prevent dust from becoming airborne (prevention) and knocking down dust already airborne by spraying the dust cloud and causing the particles to collide, agglomerate, and fallout from the air (suppression). Water spraying has only a limited residual effect due to evaporation, and will need to be reapplied at various points throughout the process to remain effective. The keys to effective wet spray dust control are proper application of moisture, careful nozzle location, controlling droplet size, choosing the best spray pattern and spray nozzle type, and proper maintenance of equipment.

Good quality haul roads with added petroleum emulsions and adhesives, water spraying with surfactants, speed control, traffic control, load covers are other required dust control measures during transportation. Compaction of topsoil stockpile shall be followed with chemical binder coating and vegetation.

The drilling process is used on surface and underground for blasting operations which are conducted to fragment the rock. Drilling operations are notorious sources of respirable dust, which can lead to high exposure levels for the drill operator, drill helper, and other personnel in the local vicinity during operation. Therefore, dust controls on drills are necessary and involve both wet and dry methods. Operator cabs are increasingly becoming an acceptable method for protecting the drill operator from respirable dust generated by the drilling operation. The most common method of dust control for underground drilling is

using wet drilling techniques. Dry collectors are not commonly used due to the bulkiness of the collectors and their associated maintenance issues.

Blasting occurs infrequently enough that it is not considered to be a significant contributor to PM10. The most common method of dust control is to allow the dust and gases from blasting to be dispersed and removed through the ventilation system in the case of underground operations or through atmospheric dispersion in the case of surface operations. Underground operations generally schedule the blasting during off-shift times to allow sufficient time for the area to ventilate, to disperse, and to remove the dust and gases from the blasting. If off-shift blasting is not feasible, then areas affected by the blasting should be cleared and work should not commence until the dust and gases are removed.

Aggregate crushing and screening operations can be major sources of airborne dust. Worker exposure may be managed through engineering controls to suppress or enclose the dust sources or by isolating the worker from the dust source- Operator Booths, Control Rooms, and Enclosed Cabs and personal protective equipment. The performance of installed dust control systems should be periodically evaluated, maintained, and, when necessary, modified to maximize performance. Wet technique is suitable. During bulk loading, loading spouts with dust collectors and physical barriers can be used.

By implementing these air pollution control measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.12. Noise Control Measures

Noise control measures during construction will be required to minimize noise levels on existing noise-sensitive land uses. All construction activities will have to comply with Noise Pollution (Regulation and Control) Rules, 2000 & amendments and ESS 3 of World Bank. Night time work could require a variance for local noise regulations.

The noise control measures discussed in the following sections are examples of those that will be incorporated into the Project's construction phase. The contractor will be required to comply with any noise regulations permits acquired for the Project. Construction hours could be set, and construction activity noise level emission criteria could be determined and compliance required during construction.

(I) Design Considerations

During the early stages of construction plan development, natural and artificial barriers could be considered for use as shielding against construction noise. Strategic placement of stationary equipment, such as compressors and generators, could reduce effects at sensitive receivers.

(II) Alternate Construction Methods

Certain phases of rail construction work, such as pile driving, may produce noise levels in excess of acceptable limits, even when feasible noise reduction methods are used. Using alternate methods of construction may reduce these effects. For pile driving, vibratory or hydraulic insertion could be used, depending on a variety of factors (i.e., vibratory pile driving is not always quieter). Drilling holes for cast-in-place piles is an alternative construction method that would produce significantly lower levels of noise.

(III) Source Control

The contractor will comply with standard specifications and all local sound control and noise level rules, regulations, and ordinances that apply to any work performed pursuant to the contract. Each internal combustion engine used for any purpose on the project or related to the project will be equipped with a muffler of a type recommended by the manufacturer. No internal combustion engine will be operated without a muffler.

(IV) Time and Activity Constraints

Noisier activities involving large machinery will be limited to daytime hours as practical, when most people normally affected are either not present or engaged in less noise-sensitive activities. Night time construction will require a variance. Compliance with local noise ordinances would mitigate effects associated with construction noise. To comply with these ordinances, all construction activities adjacent to residential uses will be limited to daytime hours (7:00 AM to 6:00 PM) Monday through Saturday.

In addition to these some of the measures are given below to minimise the high level of noise;

- Use of electric equipment instead of diesel powered equipment,
- Use of hydraulic tools instead of pneumatic tools,
- Acoustic enclosures should be provided for individual noise generating construction equipment like DG sets,
- Scheduling work to avoid simultaneous activities that generates high noise levels,
- Job rotation where feasible
- Sound proof control rooms etc.

Automation of equipment and machineries, wherever possible, should be done to avoid continuous exposure of workers to noise. The workers employed in high noise level area should be provided with protective devices.

By implementing these noise pollution control measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.13. Vibration Management

Underground Section: In these sections, tunnels will be constructed by State of Art Technology i.e. Tunnel Boring Machine (TBM) and stations will be constructed by Cut and Cover method as per DPR. The vibration levels generated due TBM are within the permissible limits.

Elevated Section: Pile-driving is one of the greatest sources of vibration associated with equipment used during construction of a project. Continuous operation of Vibratory piles at a fixed frequency may be more noticeable to nearby residents, even at lower vibration levels. The steady-state excitation of the ground may induce a growth in the resonant response of building components. The mitigation measures to reduce the construction vibration are:

- Route heavily-loaded trucks away from residential and sensitive areas,
- Operate earth-moving equipment on the construction site as far away from vibration-sensitive sites as possible,
- Phase demolition, earth-moving and ground-impacting operations so as not to occur in the same time period. The total vibration level produced could be significantly less when each vibration source operates separately,
- Avoid night time construction activities near residential and sensitive areas,
- Avoid impact pile-driving where possible in vibration-sensitive areas, and
- Avoid vibratory rollers near sensitive areas.

The information available during the preliminary engineering phase will not be sufficient to define specific construction vibration mitigation measures. The contractor should prepare a mitigation plan and implement the same during the final design and construction phases of the project. This scheme shall include:

- Monitoring requirements for vibrations at regular intervals throughout the construction period.
- Pre-construction structural integrity inspections of protected monuments, heritage assets or other sensitive structures.
- Information dissemination about the construction method, probable effects, quality control measures and precautions to be used.
- vibration monitoring plan during final design and the implementation of a compliance monitoring program during construction

Construction activities shall be scheduled such that demolition, earthmoving and ground-impacting operations do not occur in the same time period. Unlike noise, the total vibration produced could be significantly less when vibration sources operate separately.

Construction activities shall be avoided during night hours when people are more aware of vibration.

7.3.14. Water pollution Control

Wastewater generated from the site during the construction contains suspended materials, spillage and washings which can pollute surface and ground water; such washings shall be led through separate drains into precipitation chambers before discharge into the sewage drain to the standards prescribed for disposal. These mitigation measures should be in accordance with Water (Prevention and Control of Pollution) Act, 1974 & its amendments and ESS 3 of World Bank. By implementing these water pollution control measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.15. Traffic Diversion/Management

In order to retain satisfactory levels of traffic flow during the construction period; traffic management and engineering measures need to be taken. They can be road widening, traffic segregation, one-way movements, traffic diversions, acquisition of service lanes, etc.

- All construction workers should be provided with high visibility jackets
- Warn the road user clearly and sufficiently in advance.
- Provide safe and clearly marked lanes, buffer and work zones for guiding road users.

Various construction technologies like cut and cover can be employed to ensure that traffic impedance is minimized. Capital and operating cost are included in engineering cost and therefore is not included in EMP.

By implementing these control measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.16. Soil Erosion and land subsidence control

The surface area of erodible earth material exposed by clearing and grubbing, excavation shall be limited to the extent practicable. Works such as construction of temporary berms, temporary mulches, seeding or other methods as necessary to control erosion shall be implemented. Mitigation measures include careful planning, timing of cut and fill operations and re-vegetation in accordance with Water (Prevention and Control of Pollution) Act, 1974 & its amendments and ESS 3 of World Bank. In general, construction works are stopped during monsoon season.

To manage land subsidence measures including maintaining adequate distance of the trench from existing structures adjacent to the trench, measures to support the walls of the trench as well strengthen soil underneath adjacent structures will be required.

By implementing these control measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.3.17. Dewatering of underground works

Water from underground works shall be led by construction drains into sumps and then to trunk sewers or used to recharge groundwater or re-use for construction. Capital and operating cost is included in engineering cost and therefore is not included in EMP.

7.3.18. Management Plan for Flamingos

The Management Plans proposed for Flamingo Conservation in the project area are as given below:

- During construction period there may be generation of dust and emission from working machinery which would be restricted and contained by proper maintenance of machinery and providing higher curtain/ barriers to minimize the effect of dust and emissions
- No Bentonite shall be allowed to be spilled or stacked near flamingo feeding areas of Tapi River.
- Monitoring of River water quality (Phytoplankton, Zooplankton & Benthos) and comparing the diversity index with Shannon Weinner Diversity Index (SWDI) and Palmar Pollution Index (PPI) in order to know the food availability for Flamingos.
- No C&D waste shall be allowed to be stacked or stored along the river bed and near the flamingo feeding areas or any other water body.
- There shall not be any spillage of construction material in the area close to the Tapi River.
- Care shall be taken to avoid disposal of any waste material in the area close to Tapi river flamingo areas.
- Construction activities related to project activities close to flamingo areas (within 500m) may be restricted during night hours, particularly during winter months to avoid adverse impacts on migratory as well as native birds.
- Night working near the flamingo areas in Tapi River shall be minimized. Only covered lights with directional shielding shall be used, if the night working is not avoidable.
- The following are the measures to protect flamingo:
  - A conservation scheme for the feeding colonies in downstream of Tapi River involves protection from human disturbance, prevention of erosion, habitat maintenance.
  - Remove the algal bloom (Cyanophaceae) from both the banks
  - Liming of river bed to enhance the water quality near construction sites.
  - Training shall be imparted to GMRCL officers about flamingo conservation.
  - Publishing of pamphlets, brochures, hoardings to increase Public Awareness

A Lump sum provision of Rs 10.00 lakh is provided in the cost estimate for flamingo conservation under the supervision of Deputy Conservator of Forest, Surat. By implementing these measures, the project is expected to comply with the WB ESF guidelines.

7.5. EMP During Operation

7.4.1. Noise and Vibration Management

Noise: Use of ballast-less track with elastic and absorbent fittings is a standard provision for noise control. At depots use of green belt with vegetation of thick foliage helps reduce noise; where site layout permits barrier blocks of less-vulnerable buildings can be used; alternatively freestanding barrier walls can be built (*Manual on norms and standards for environment clearance of large construction projects, MoEFCC, 2007*). Screening of noise shall be ensured by providing parabolic noise barriers on each side of the track along the curved portion of the viaduct and at sensitive receptors during operation. Polycarbonate noise barriers 15 mm to 25 mm thick are known to reduce noise level by between 30 dB to 33 dB.

Even though the predicted noise levels due to metro operations are less than the existing ambient noise levels, noise barriers are proposed at sharp curved portions, structures falling within 20 m distance from the alignment and at sensitive receptors like educational institutes and hospitals where noise impact criteria is severe. Details of noise barriers proposed for each corridor is given in Table 7.6. Project proponent need to monitor the noise environment at these locations before erecting the noise barriers during operation.

Table 7.6: Details of Noise Barriers

Chainage in m		Length of Barrier in m	Total Cost of the Barrier in Rs. Lakh	Remarks
From	To			
Corridor I: Dream City - Sarthana				
-240	-50	190	22.8	LHS
85	400	315	37.8	LHS
675	825	150	18	LHS
875	1000	125	15	LHS
1325	1500	350	42	LHS
1500	1560	60	7.2	LHS
1650	1800	150	18	Both
1825	2040	430	51.6	Both
2040	2180	280	33.6	both
2180	3125	1890	226.8	Both
11128	11650	1044	125.28	both
11825	12240	830	99.6	Both
12650	12750	100	12	RHS
12850	12930	80	9.6	RHS
12930	13000	70	8.4	LHS

Chainage in m		Length of Barrier in m	Total Cost of the Barrier in Rs. Lakh	Remarks
From	To			
12975	13025	50	6	RHS
13050	13300	500	60	BOTH
13400	13550	300	36	Both
13690	13730	80	9.6	Both
13780	13830	100	12	Both
13980	14550	1140	136.8	Both
14915	15050	135	16.2	LHS
15000	15100	100	12	RHS
15100	15325	225	27	LHS
15575	15780	410	49.2	Both
15975	16025	50	6	LHS
16100	16160	60	7.2	RHS
16250	16300	50	6	RHS
19150	19225	75	9	RHS
Sub-Total		**9339**	**1120.68**	
Corridor II: Bhesan - Saroli				
-200	-135	65	7.8	LHS
1225	1450	450	54	Both
1825	1900	75	9	LHS
1950	2050	200	24	Both
3600	3725	125	15	LHS
5450	5700	250	30	LHS
5700	5840	280	33.6	Both
5840	6820	980	117.6	LHS
6150	6820	670	80.4	RHS
6820	7050	230	27.6	LHS
7575	7750	350	42	Both
8075	8150	150	18	Both
8450	8550	100	12	LHS
9120	10100	980	117.6	LHS
10240	10400	160	19.2	RHS
10500	10720	220	26.4	RHS
10830	10865	35	4.2	RHS
11520	11650	130	15.6	RHS
11600	12050	450	54	LHS
11825	11900	75	9	RHS
12350	12400	50	6	RHS
12400	12425	25	3	LHS
12675	12775	100	12	LHS
13475	13565	90	10.8	LHS
13725	13875	150	18	LHS
13875	13950	75	9	RHS

Chainage in m		Length of Barrier in m	Total Cost of the Barrier in Rs. Lakh	Remarks
From	To			
14125	14350	225	27	LHS
Sub-Total		**6690**	**802.80**	
Total		**16029**	**1923.48**	

Vibration: Vibration can be reduced by proper design and maintenance of track and rolling stock. In case of ballast less track, the following are some measures for vibration damping: resilient soft base plates between rail and track slab; resilient rubber between the base plate and track slab; soft elastic fastening system[11]. Deep and narrow trenches in the ground shall be tested at vibration-sensitive structures.

By implementing these measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.4.2. Water Supply and Sanitation at Stations

The public health facilities such as water supply, sanitation and toilets are much needed at each station. Water supply for air conditioning, cleaning and use of staff and passengers will be procured from municipal supply – this will be supplemented by re-use of treated waste water generated by staff and passengers. Total quantity of waste water generation at all stations of Sarthana to Dream city alignment will be 246 KLD; and at all stations of Bhesan to Saroli alignment will be 252 KLD. For the cost estimation of biodigester waste water generated for the year 2021 has been considered. Total cost of bio digester is estimated as **Rs 19.64 Lakh** and **Rs 20.14 Lkah** for Sarthana to Dream city corridor and Bhesan to Saroli corridor respectively. The waste water will be treated by installing bio digesters at each stations. Number of Bio Digesters and cost at each station is given in Table 7.7.

Table 7.7: Capacity of Bio Digester at Each Station

S. No.	Station Name	Waste Water in KLD	Capacity of Bio Digester System in KLD	Cost of Bio Digester in Rs Lakh
Sarthana to Dream City				
1	Dream City	0.78	1	0.06
2	Convention Center	0.36	0	0.03
3	Bhimrad	0.85	1	0.07
4	Woman ITI	0.34	1	0.03
5	VIP Road	21.22	21	1.70
6	Althan Gam	7.34	7	0.59
7	AlthanTenament	6.09	6	0.49
8	Rupali Canal	5.10	5	0.41
9	Majura Gate	23.29	23	1.86

[11]Current state of practice in railway track vibration isolation: an Australian overview, SakdiratKaewunruen & Alex M. Remennikov,Jan 2016, Australian Journal of Civil Engineering

S. No.	Station Name	Waste Water in KLD	Capacity of Bio Digester System in KLD	Cost of Bio Digester in Rs Lakh
10	Kadarsha Ni Nal	3.97	4	0.32
11	Gandhi Baug	30.29	30	2.42
12	Maskati Hospital	8.84	9	0.71
13	Surat Railway Station	23.60	24	1.89
14	Central Warehouse	9.60	10	0.77
15	Labheshwar Chowk	13.76	14	1.10
16	Kapodra	50.49	50	4.04
17	Shri Swaminarayan Mandir Kalakunj	12.45	12	1.00
18	VarchaChopati Garden	5.49	5	0.44
19	Nature Park	12.68	13	1.01
20	Sarthana	9.00	9	0.72
	Total	**246**	**246**	**19.64**
Bhesan to Saroli				
1	Bheshan	1.63	2	0.13
2	Botanical Garden	3.56	4	0.28
3	UgatVaarigruh	8.18	8	0.65
4	Palanpur Road	7.93	8	0.63
5	L.P. Savani School	9.19	9	0.74
6	Performing Art Centre	4.52	5	0.36
7	Adajan Gam	5.21	5	0.42
8	Aquerium	6.99	7	0.56
9	Badri Narayan Temple	6.83	7	0.55
10	AthwaChopati	3.53	4	0.28
11	Majura Gate	26.79	27	2.14
12	Udhna Darwaja	22.84	23	1.83
13	Kamela Darwaja	16.18	16	1.29
14	Anjana Farm	9.02	9	0.72
15	Model Town	15.02	15	1.20
16	Magob	41.81	42	3.34
17	Bharat Cancer Hospital	1.28	1	0.10
18	Saroli	61.22	61	4.90
	Total	**252**	**252**	**20.14**

Bio Digesters: GMRCL should install eco-friendly bio-digesters at all stations for onsite disposal of human waste. Defence Research and Development Organisation (DRDO) had developed this technology. The bio-digester based eco-friendly sanitation technology will convert human faecal waste into water and gas. This technology was developed by Defense

Research and Development Organization (DRDO) to solve the sanitation problem faced by soldiers in high altitude. This technology has 2 components

(i) The anaerobic microbial consortium which is formulated from group of bacteria that are very efficient to perform the biodegradation of the human waste in wide range of temperature and in presence of toxic chemicals.

(ii) The vessel in which fermentation is carried out with the help of bacteria is called bio-digester.

The process of bio-degradation of human waste used in the present technology is carried out in strict anaerobic environment (devoid of oxygen) by anaerobic microbial consortium. It involves multiple biochemical steps and is of complex nature. These steps are sequential in nature and interdependent.

Bio-digester technology treats human waste at the source. A collection of anaerobic bacteria that has been adapted to work at temperatures as low as -5°C and as high as 50°C act as inocula (seed material) to the bio-digesters and convert the organic human waste into water, methane, and carbon-dioxide. The anaerobic process inactivates the pathogens responsible for water-borne diseases and treats the fecal matter without the use of an external energy source. The only by-products of the waste treatment process are pathogen-free water, which is good for gardening, and bio-gas, which can be used for cooking. Bio-toilets do not require sewage connectivity and because the process is self-contained, bio-toilets are also maintenance-free. Anticipated influent & effluent wastewater quality parameters from the bio-digesters are given in Table 7.8 and typical cross section of a Bio Digester is shown in Figure 7.3. Organic waste will be segregated and treated by in-site bio-composter technique.

Table 7.8: Wastewater Quality Parameters (Influent & Effluent)

S. No	Parameter	Unit	Characteristics	
			Influent	Effluent
1.	pH	-	7.0 - 7.5	7.0 - 7.5
2.	Turbidity	NTU	70 - 90	2 - 5
3.	Total Suspended Solids	mg/l	90 - 120	50 - 80
4.	Total Dissolved Solids	mg/l	350 - 450	100 - 300
5.	Bio Chemical Oxygen Demand at 5 days and 20^{O}C	mg/l	70 - 120	2-4
6.	Chemical Oxygen Demand	mg/l	250 - 300	15 -25
7.	Coliforms	MPN/100 ml	300 - 350	0 - 12

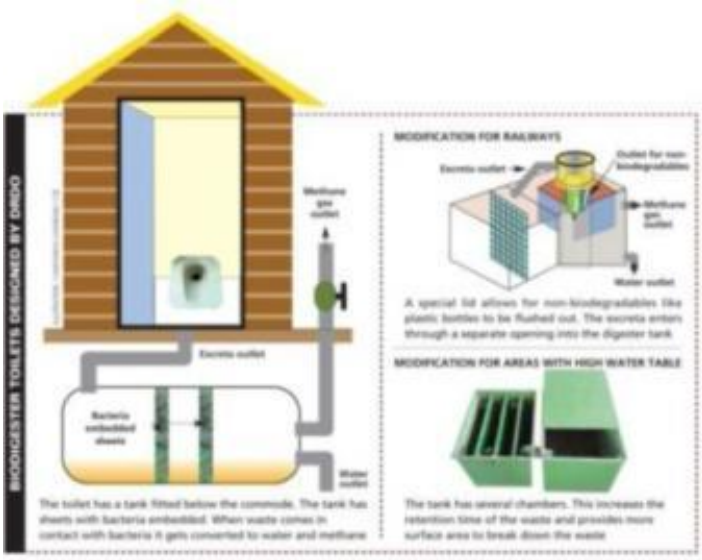
Source: DRDO Website

By implementing these measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.

7.4.3. Rain Water Harvesting

To augment the storage of groundwater, it is proposed to construct rainwater harvesting structure of suitable capacity at the elevated stations and in the elevated alignment. Each pillar can have inbuilt downpipes to collect the rainwater from the viaduct and led into underground tanks; water collected will percolate down to the subsoil through layers of sand and gravel.

Figure 7.3: Typical Cross Section of Bio Digester



Average annual rainfall of Surat is 1192 mm. The annual rainwater harvesting potential of elevated stations and viaduct is estimated at 4.12 lakh cubic meters per year. Estimated cost for rainwater harvesting for viaduct and elevated stations is **Rs 324.54 Lakh** for Sarthana to Dream city and **Rs 401.71 Lakh** for Bhesan to Saroli corridor.

7.4.4. Management Plan for Depots

- Water supply
- Rain water harvesting
- Waste Water Treatment
- Oil Pollution
- Surface Drainage
- Green Belt Development
- Solid Waste Disposal
- Top Soil Preservation, reuse and dust control

a) Water Supply

Water will be sourced from municipal supply. This will be supplemented by re-use of treated water of effluent treatment plant. Water required for the domestic use will be 46 KLD and metro train washing purpose will be 259 KLD for both the dopts.

b) Rain Water Harvesting

To augment storage of groundwater, it is proposed to construct roof top rainwater harvesting structure of suitable capacity in the depots. Rainwater harvesting potential of depots will be 48.92 Thousand cubic meter for for both the depots. The estimated cost of rainwater harvesting will be **Rs. 17 lakh** for each depot.

c) Waste water Treatment

Bio Digester with Reed bed (DRDO, India) system which can handle human waste as well as wastewater is proposed at each Depot. The details of waste water generated and estimated cost of each depot is given in Table 7.9. The treated waste water could be reused for horticulture, washing and flushing purpose in the depot area. Total estimated cost for waste water treatment is **Rs 41.32 Lakh** and **Rs 29.50 Lakh** for Dream city and Bhesan depot respectively.

Table 7.9: Capacity Estimated Cost of Bio Digester & ETP at Depots

S. No.	Depot Name	Domestic Waste Water in KLD	Waste Water from washing KLD	Capacity of Bio Digester System in KLD	Estimated Cost of Bio Digester in Rs Lakh	Estimated cost of ETP Rs Lakh
1	Dream city Depot	25.51	146	26	2.08	39.24
2	Bhesan Depot	15.75	88	16	1.28	28.22

d) Oil Pollution

Oil spillage from during change of lubricants, cleaning and repair processes in the maintenance Depot cum workshop for maintenance of rolling stock should be trapped in oil and grease traps and disposed off to authorised collectors, so as to avoid any underground/ surface water contamination. These traps need to be installed before effluent treatment plant.

e) Surface Drainage

The Storm water of the depot will be collected through the drain. Rain water harvesting pits are provided at different locations in the drains and for surplus storm water, the drainage system is connected to a nearby disposal site.

f) Green Belt Development

The greenbelt development / plantation in the depot area harmonizes the depot with surrounding environment and acts as pollution sink / noise barrier. It will check soil erosion. Compensatory afforestation cost at depot area is included in the compensatory afforestation cost of the project as given in Section 7.3.1. Treated waste water can be used for watering the green belt.

g) Solid Waste Disposal

Solid waste generated from the Depots which includes hazardous waste will be taken away by the cleaning contractor weekly and recycled/treated and disposed of at designated waste disposal sites.

7.4.5. Management Plan for Flamingos

The following measures need to be considered during construction at feeding & roosting area:

- Maintaining, and restoring favourable hydrological conditions and water quality.
- Minimizing human disturbance from boating, fishing, and tourists.

7.6. EMP Reporting Arrangement and Institutional Strengthening

Supervision involves periodic checking to ascertain whether activities are going according to the plans. It provides necessary feedback for project management team to keep the program on schedule. The supervision and reporting process with respect to implementation status of mitigation measures during construction will initiate from the contractor at the lowest rung who will report to the Project Implementation Agency (PIA) through the Project Management Consultant (PMC).

During construction phase of the project, the EMP implementation comprises of the following key activities:

- Implementing various mitigation and enhancement measures within the time frame recommended
- Overseeing the implementing various mitigation and enhancement measures and fine tuning/advocating more measures, if needed, depending on site conditions;
- Project level monitoring of key performance indicators to evaluate the implementation of EMP measures at the recommended intervals.
- Periodical reporting of status of EMP implementation and monitoring results and key performance indicators and
- Constant evaluation of EMP measures implemented based on the data available from project level monitoring and status reports and providing directions accordingly.

These activities to be carried out by various agencies that will be involved in the implementation of metro project. It is also to be noted that all these activities will be carried out concurrently or at regular intervals and at different duration and location. This makes it pertinent that all agencies involved work within a predefine setup. The coordination model proposed during construction and operation phases is presented in Figure 7.4 and Figure 7.5 respectively. The identified agencies and their sphere of work are presented in following section.

Figure 7.4: Institutional Mechanism for EMP Implementation (Construction Phase)

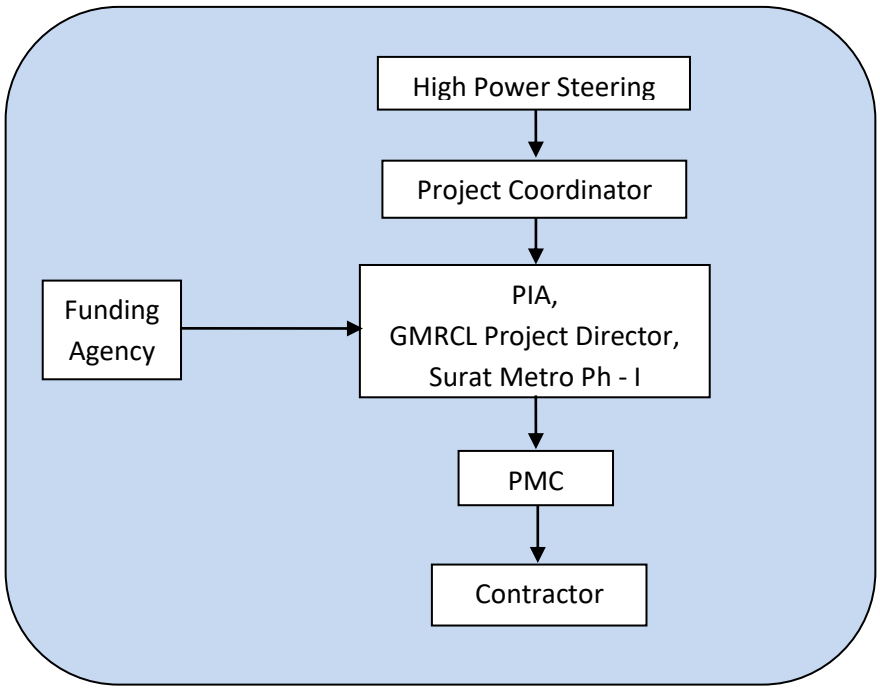
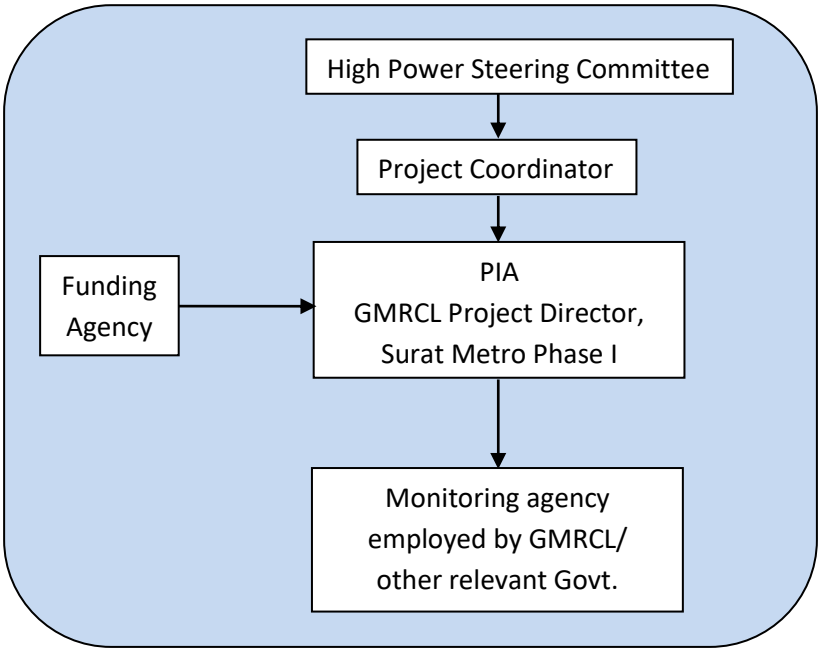


Figure 7.5: Institutional Mechanism for EMP Implementation (Operation Phase)



Project Implementation Agency (PIA): The responsibility of implementing environmental mitigation measures lies with the PIA. PIA in this project will be Gujarat Metro Rail Corporation Limited (GMRCL). The responsibility also includes various tasks such as notifying various affected parties such as the resident and commercial establishment, facilitate the relocation of people, notify other utility departments such as telephone, water supply, sewerage etc. which used the road for providing public utility services.

Project Management Consultant (PMC): The PIA will get the EMP implanted through the Project Management Consultant (PMC) appointed for managing engineering and construction related activity. The PIA will deliver the responsibility of the implementation as per the contract agreement. In order to effectively discharge the duties PMC will have an environmental officer/expert in the project management unit. The environmental officer will work for a full time basis at the site office. The officer must possess experience in the environmental management of metro projects.

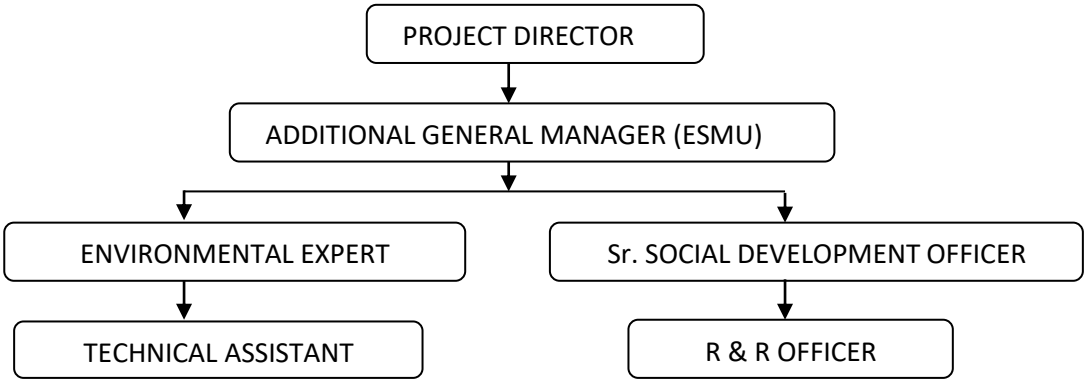
Project Contractor: Project contractor will implement the EMP measures, enhancement measures and measures as directed by PIA and PMC. The responsibility to implement the EMP measures will be built in to the contractual agreement. The contractor shall submit a report on compliance of environmental mitigation measures periodically to the PMC. The PMC will review and approve the Environmental Compliance Report (ECR) submitted by the contractor and forward the ECR to PIA after approval. The PIA will then submit the ECR to Joint Project Director (JPD), environment which after review and monitoring will submit to Independent Monitoring Panels through the Project Director, GMRCL. The Project Director accordingly submits report to the JICA.

GMRCL: GMRCL as an apex organization shall initiate to coordinate process among the concern organization for EMP implementation. GMRCL shall take lead in

- Reviewing the progress of the project for the subsequent year- institution wise
- Reviewing and discussing the salient features of the report in the year on environmental aspects and their violations
- Organizing and coordinating training programs for all member organization

Environmental Monitoring Division needs to be set up by the GMRCL for implementation of EMP and Environmental Monitoring Plan (EMoP). The Environmental Monitoring Division comprises staff and structure is shown in Figure 7.6.

Figure 7.6: EMP Monitoring Structure



7.7. Disaster Risk Management

Hazard is a threat or event which can cause damage; disaster is a major hazard event. Disaster risk is expressed as the likelihood of loss of life, injury or destruction and damage from a disaster.

The recommended approach (UNISDR) is to manage disaster risk rather than managing disasters. Disaster risk is the combination of the severity and frequency of a hazard, the numbers of people and assets exposed to the hazard, and their vulnerability to damage. The main opportunity in reducing risk lies in reducing exposure and vulnerability. Disaster Risk Management includes the following actions:

i. **Reduction and Prevention:** Measures to reduce existing and avoid new disaster risks, for instance relocating exposed people and assets away from a hazard area. In case of mass transit like Metro such measures are not actionable.
ii. **Mitigation:** The lessening of the adverse impacts of hazards and related disasters. For instance implementing strict land use and building construction codes. This aspect is accounted for in design and construction of the project.
iii. **Transfer:** The process of formally or informally shifting the financial consequences of particular risks from one party to another, for instance by insurance. This is not yet available.
iv. **Preparedness:** The knowledge and capacities of governments, professional response and recovery organisations, communities and individuals to effectively anticipate, respond to, and recover from the impacts of hazard events or conditions, for instance installing early warning systems, identifying evacuation routes and preparing emergency supplies.

Risk Management process[12] comprises the following stages:

a) Description of the system that is at risk

b) Identify the potential hazards or sources of risk (the list of initiating events or scenarios of events leading to the undesired outcome – technological and human)

c) Risk analysis to estimate the likelihood of the scenarios or events occurring and each scenario's consequence

d) Compare and rank the various risk drivers

e) Action plan in response to the identified major risks

f) Regular monitoring, review and updation of the process.

[12]A. Berrado, Em El-Koursi, A. Cherkaoui, M. Khaddour. A Framework for Risk Management in Railway Sector: Application to Road-Rail Level Crossings. Open transportation Journal, Bentham Open, 2010, 19p. HAL Id: hal-00542424 https://hal.archives-ouvertes.fr/hal-00542424 Submitted on 2 Dec 2010

g) For example, the system at risk needs to be defined as to include inter-modal integration.

h) Examples of potential hazards are fire risk or security alarms or failure of train control or motive power or passenger doors / escalators / platform screen doors on trains or in stations; staff training and work environment; inadequate maintenance.

Action plan shall include the following;

Procedures and Records: Evaluation of progress and effectiveness of EMP and EMoP,response to inquiries, complaints and requests for informationsurveillance, incident reporting, corrective and preventive actions, emergencies, training and emergency exercises, response to emergencies.

Identification of resources: Sources of repair equipment, personnel, transport and medical aid for use during emergency will be identified.

Emergency systems: Back-up systems for ventilation, communication and train control, lighting etc shall be established.

Evacuation procedures: Evacuation procedures will be prepared in consultation with local administration and notified. To ensure coordinated action, an Emergency Action Committee shall be constituted.

Communication System: Primary and back-up system shall be put in place

Review and Updation: Drawing inputs from the incident reporting system the Action Plan shall be reviewed at pre-decided intervals and upon occurrence of defined ``trigger events`` and suitably updated.

7.8. Provisions at Metro Stations/Other Installations

Preventive Action

Once the likelihood of a disaster is suspected, action has to be initiated to prevent a failure. Engineers responsible for preventive action should be aware of availability of repair equipments, materials, labour and expertise for use during emergency.

Reporting Procedures

The level at which a situation will be termed a disaster shall be specified. This shall include the stage at which the surveillance requirements should be increased both in frequency and details. The Engineer-in-Chief should notify the officer for the following information:

- Exit points for the public and

- Nearest medical facilities.

Communication System

An efficient communication system is absolutely essential for the success of any disaster management plan. This has to be worked out in consultation with local authorities. More often, the entire communication system gets disrupted when a disaster occurs. The damage areas need to be clearly identified and provided with temporary and full proof communication system.

Emergency Action Committee

To ensure coordinated action, an Emergency Action Committee should be constituted. Chairman cum Director, GMRCL will be the Chairman of this Committee. The committee may comprise of:

- Head of operations,
- Head of technical services,
- Head of security,
- Fire brigade,
- Police representatives, and
- NGO

Emergency Action Committee will prepare the evacuation plan and procedures for implementation based on local needs and facilities available. The plan should include :

- Demarcation of the areas to be evacuated with priorities,
- Safe route to be used, adequacy of transport for evacuation, and traffic control,
- Safe area and shelters,
- Security of property left behind in the evacuated areas,
- Functions and responsibilities of various members of evacuation teams,
- Setting up of Joint Control Room.

All personnel involved in the Emergency Action Plan should be thoroughly familiar with all the elements of the plan and their responsibilities. They should be trained through mock drills for the Emergency Action Plan. The staff at the site should be trained for problem detection, evaluation and emergency remedial measures. Individual responsibility to handle the segments in emergency plan must be allotted.

Success of an emergency plan depends on public participation, their response to warning notifications and timely action. Public has to be educated on the hazards and key role in disaster mitigation by helping in the planned evacuation and rescue operations.

It is essential to communicate by whom and how a declared emergency will be terminated. There should be proper notification to the public on de-alert signals regarding termination of the emergency. The notification should be clear so that the evacuees know precisely what to do when re-entering or approaching the affected areas.

7.9. Emergency Measures

The emergency measures are adopted to avoid any failure in the system such as lights, fire, means of escape, ventilation shafts etc. The aim of Emergency Action Plan is to identify areas, population and structures likely to be affected due to a catastrophic event of accident. The action plan should also include preventive action, notification, warning procedures and co-ordination among various relief authorities. These are discussed in following sections.

Emergency Lighting

The emergency lights operated on battery power should be provided at each station. The battery system should supply power to at least 25% of the lights at the station, platforms, and viaducts for a period of 2 hours.

Fire Protection

The building materials should be of appropriate fire resistance standard. For underground structures the fire resistance period should be at least 4 hours, and 2 hours for surface or over head structures. Wood shall not be used for any purpose, excluding artificial wood products, which are flame resistant. The materials which have zero surface burning characteristics need to be used. The electrical systems shall be provided with automatic circuit breakers activated by the rise of current as well as activated by over current. The design of a station will include provision for the following:

- Fire prevention measures,
- Fire control measures,
- Fire detection systems,
- Means of escape,
- Access for fireman, and
- Means of fire fighting.

Accumulations of refuse of inflammable material like paper, plastic cartons constitute a major fire hazards and should not be permitted. Smoking should be strictly prohibited at all locations of MRTS.

All aspects of fire prevention and control will be dealt in close collaboration with the city fire fighting authority. Smoke control will be achieved by the following means:

- Down stand bulkheads of a minimum depth of 600 mm to provide smoke containment. These will be provided around openings for escalators, lifts and stairs in underground stations, and
- In underground stations the ventilation system will be designed to extract smoke in the event of fire
- A minimum of 30 minutes supply of water is to be assured in the case of fire. The pumps/overhead tanks shall have the capacity to discharge the water at the rate of 1100 litres per minute at a head of 21 m at nozzle mouth.
- The storage capacity in an underground or overhead tank may be divided into two parts i.e. dead storage and running storage. Fire fighting pumps shall be provided with a diesel pump as a standby arrangement, in case of power failure.
- Fire of electrical origin, water cannot be used until the electric system has been made dead and earthen. For electrical fires, non-aqueous extinguishers like chemical dry powder or CO2 gas are utilized for fire fighting. Fire extinguishers with these agents shall be liberally provided at static installations and on the rolling stock.
- Generally there are often more casualties from smoke inhalation than from burning. Smoke needs to be transported away from the site of the fire. In order to achieve this, fresh air has to be introduced and exhaust gases should be sucked out.
- Openings, including ducts and passages, between MRTS property and any adjoining structures which allow free access into the MRTS property will be protected by fire doors, fire shutters, fire dampers etc. as appropriate. Fire detection and alarm systems will be provided as per the prevailing state of art technology.

A. Fire Prevention and Safety Measures

Fire prevention measures will be designed and implemented to minimize the risk of outbreak of fire by appropriate choice, location and installation of various materials and equipment. In stations planning, potential sources of fire can be reduced by:

i. Fire Prevention

- Use of non-combustible or smoke retardant materials where possible,
- Rolling stock is provided with fire retarding materials, low smoke zero halogen type electric cable is also provide,
- Provision of layout which permits ease of maintenance for equipment and cleaning of the station premises,
- Provision of special storage spaces for combustible materials such as paint and oil,
- Prohibition of smoking in fire prone areas,
- Good housekeeping.

ii. Safety

Following provisions will be required from fire safety point of view:

- Automatic sprinkler/detection system to be provided if floor area exceeds 750 sq.m
- One wet riser-cum-down comer per 1000 sqm floor area with static underground storage tank, overhead tanks and pumps of suitable capacity with hydrants, first-aid reel, etc.
- Portable fire non-aqueous extinguishers of Carbon di Oxide, chemical dry powder etc. at suitable places.
- Automatic smokes venting facilities.
- Two separate means of exit shall be provided, if more than 10 persons are working and the area exceeds 1400 sq.m.
- Fire resisting doors shall be provided at appropriate places along the escape routes to prevent spread of fire and smoke.
- The travel distance for fire escape shall not exceed 20 m where escape is available in more than one direction; the distance could be upto 40 m.

B. Fire Alarm and Detection System

A complete fire detection system with equipment complying with the requirements of Surat Fire Services shall be provided through out each station and ancillary buildings including entrance passageways, subways etc. to give visual and audible indication of alarm conditions actuated by the operation of break glass contact or fire sensors e.g. detector heads, linear heat detecting cables etc. The system shall be operated from 24 V DC Power sources.

Manually operated call points shall be provided at every hydrant and nose reel points, station head wall, tail wall and other locations. Alarm bells shall be installed in each plant room complex at both platform and concourse level and shall be clearly audible at all points in the room/area. Heat detector shall be installed at roof level, ceiling and floor cavity.

Smoke probe units shall be installed in rooms/compartments. When an alarm point is operated, the fire pump shall start to operate automatically. A station fire control and indicating panel shall be provided/ installed in the station controllers room, for the control, indication and monitoring of the whole detection and fire fighting systems. While designing the fire fighting system, the zone of Surat Fire Services shall be taken into account for linking with the same.

C. Fire Control Measures

Control of the spread of fire and smoke will be achieved by partition of fire risk areas, planning for smoke extraction, and arrangement for smoke containment. Partition is aimed at limiting the extent of a fire. The openings must be capable of being sealed in the event of fire. With the exception of station public areas, a fire compartment will not exceed 1500 m^2.

Partition of the public areas in stations is not practicable for operational reasons. The fire resistance period of this separated area should be about 3 hours.

D. Access for Fireman

A secondary access to the station, not used by passengers for evacuation, shall be available to fireman when the need arises. The entry point shall be easily accessible from the road. Access shall be available to all levels of the station. The minimum width of the stairs is 1.0 m and maximum height should not exceed 60 cm.

E. Emergency Door

The rolling stock is provided with emergency doors at both ends of the cab to ensure directed evacuation of passengers in case of any emergency including fire in the train.

7.10. Training

The training for engineers, managers and workers will be imparted by GMRCL on regular basis to a) monitor implementation of approved EMS by Contractor b) monitor environmental status during operation and c) monitor disaster management during operation. The cost is estimated to be Rs 7.10 Lakh for each corridor as given in Table 7.10. The curriculum for all the corridors will be same. One extension officer for each corridor (is considered and cost is distributed equally among the each corridor.

Table 7.10: Cost for Training Program

Sl. No.	ITEM	Cost (Rs)
1	Curriculum Development and course preparation 1 month Rs.50000 LS	50,000
2	Extension Officer (1 year) Rs.30,000/month	360,000
3	Instructor 6 sessions of 3 days each (@ 30000/session)	180,000
4	Demonstration/Presentation Aids LS	70,000
5	Miscellaneous (Material etc) LS	50,000
	Total	**7,10,000**

7.11. Environment Management System (EMS)

Environment Management System is intended to facilitate implementation, tracking and reporting on Environment Management Plan and Environment Monitoring Plan proposed for the project. Roles and responsibilities for preparation and Implementation of Environmental Management Plan (EMP) and Environmental Monitoring Plan (EMoP) are summarized in Table 7.11.

Table 7.11: Roles and Responsibilities – Preparation and Implementation EMP

S. No	Environmental Impact	Mitigation Measure		Implementing Entity	Responsible Entity
		Corridor 1: Sarthana - Dream City	Corridor 2: Bhesan - Saroli		
EMP During Location and Design Phase					
1.	Displacement and private property acquisition, impact of environmentally sensitive areas.	Alignment design to avoid or minimize impact.		DPR and design consultant	PIU
2.	Loss of trees and water bodies	Alignment designs to avoid or minimize impact are in line with ESS1 and ESS 6 of World Bank and Forest Conservation Act 1980.		DPR and design consultant	PIU
3.	Visual intrusion	To design aesthetic structures of viaduct and stations on elevated sections.		DPR and design consultant	PIU
4.	Archaeological Monuments	No Impact on legally protected Archaeological Monuments due to proposed alignments; accordingly ESS 8 of World Bank and The Ancient Monuments and Archaeological sites and Remains Act, 1958 amended in 2010 are not triggered.		DPR and design consultant	PIU
EMP during Pre-construction Phase					
1.	Disclosure	Disclose to stakeholders the EMP/EMoP measures proposed to be implemented; upon feedback, revise the measures if necessary. These measures are to be implemented are in compliance with ESS 10.		PIU	PIU
2.	Displacement and private property acquisition.	Implement R&R Plan in accordance with ESS 5, RFCTLARR Act 2013, RFCTLARR (Gujarat Amendment) Act 2016 and Government of Gujarat Revenue Department resolution number LOA 22-2014/54/5-declaration of Consult Award under section 23A.		PIU	PIU
3.	Loss of trees	The compensatory Afforestation plan are in line with Saurashtra Felling of trees (infliction of Punishment) Act, 1951 & amendment Rule 2008, Forest Conservation Act 1980 and ESS 6 of World Bank.		Contractor	PIU
		Implement compensatory afforestation for 836 trees along the corridor and 176 trees at Depot.	Implement compensatory afforestation for 762 trees along the corridor and 319 trees at Depot		

S. No	Environmental Impact	Mitigation Measure		Implementing Entity	Responsible Entity
		Corridor 1: Sarthana - Dream City	Corridor 2: Bhesan - Saroli		
4.	Site measures	Prepare Environment, Health and Safety (EHS) Manual in line with World Bank ESS guidelines and National regulations.		Contractor	PIU
5.	Environmental Management and Monitoring	Implement institutional requirements for implementation of EMP and EMoP.		Contractor	PIU
EMP during Construction Phase					
1.	Soil erosion, fugitive dust generation, muck disposal and C&D waste management	Implement suitable construction methods and as per EHS Manual • Careful planning, timing of cut and fill operations and re-vegetation shall reduce the Soil Erosion and dust generation • Contractor shall dispose the muck/dry soil generated at construction sites at a mutually agreed location by GMRCL and Contractor. • Contractor shall carry out the reconciliation for the disposed soil and quantities shall submit to GMRCL on quarterly basis. • Dry wheel wash facilities shall be provided at exit gate, from where soil disposal shall be carried. • Sufficient staff shall be made available at site to control the disposal of muck/soil from site such as a supervisor, labors for wheel cleaning; brooms for wheel cleaning and concrete pad where wheels will be cleaned. • The dumpers carrying the muck/dry soil has to be covered while plying on the roads on the way to disposal location. • Contractor shall take due care that muck generated during piling/tunnelling works does not get contaminated with any contaminant. • The onsite muck shall be monitored quarterly at random location during piling works in progress. In case any polluted muck is produced; the muck shall be handled and disposed as per provisions of Hazardous and Other Wastes (Management and Trans boundary Movement) Rules, 2016 and its amendments • Construction and Demolition Waste shall be disposed in accordance with the provisions of C & D waste Handling Rules 2016 and its amendments.		Contractor	PIU
2.	Air and Noise Pollution	• Vehicles and machinery are to be maintained to emission standards; periodic check of machinery and vehicles; dust collectors and physical barriers at bulk		Contractor	PIU

S. No	Environmental Impact	Mitigation Measure		Implementing Entity	Responsible Entity
		Corridor 1: Sarthana - Dream City	Corridor 2: Bhesan - Saroli		
		loading and un-loading areas; machinery noise muffles etc and personal protective gear to workers. • By implementing the proposed air and Noise mitigation measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.			
		The monitoring cost for Air and Noise monitoring will be Rs. 34.06 Lakh	The monitoring cost for Air and Noise monitoring will be Rs. 34.06 Lakh		
3.	Vibration	Implement vibration monitoring and building condition surveys at sensitive structures.		Contractor	PIU
		Monitoring at Seventeen locations and cost will be Rs. 61.20 Lakh	Monitoring at Eight locations and cost will be Rs. 28.80 Lakh		
4.	Water pollution	• Implement measures such as sedimentation tanks on site for batching plants • Contractor shall try and reduce the water consumption through use of energy efficient water fixtures at sites and project offices • Leakage of water should not be allowed through pipes and valves • Reuse of water used for curing and for other uses to be planned • By implementing the proposed measures, the project is expected to comply with Water (Prevention and Control of Pollution) Act, 1974 & its amendments and ESS 3 of World Bank.		Contractor	PIU
		Monitoring will be at 6 locations and cost will be Rs. 8.40 Lakh	Monitoring will be at 7 locations and cost will be Rs. 9.80 Lakh		
5.	Soil pollution	• Implement measures to prevent ingress of toxic / heavy metals. Suitable storage area for such materials shall be prepared and equipment shall be made available for handling of these materials. • Contractor shall take all necessary precautions such that construction material, diesel, grease, waste oil, chemicals etc. does not spill on ground. • Regular monitoring of groundwater and soil leachate shall be conducted at muck disposal areas where possibility of ground water contamination is anticipated • By implementing the proposed measures, the project is expected to comply with		Contractor	PIU

S. No	Environmental Impact	Mitigation Measure		Implementing Entity	Responsible Entity
		Corridor 1: Sarthana - Dream City	**Corridor 2: Bhesan - Saroli**		
		Water (Prevention and Control of Pollution) Act, 1974 & its amendments and ESS 3 of World Bank.			
		Monitoring will be at 5 locations & 3 muck samples per km of UG section and cost will be Rs. 5.97 Lakh	Monitoring will be at 5 locations and cost will be Rs. 5.00 Lakh		
6.	Water supply; waste water and solid waste disposal from construction activities	• Arrange for water supply; Implement measures as per EHS Manual. • The contractor shall prepare the waste management plan and submit to the Project Authority for concurrence. • Contractor shall dispose-of hazardous wastes as per the provisions of Hazardous and other wastes (management and trans boundary movement) rules 2016 and its amendments • Contractor shall dispose-off nonhazardous solid wastes, nonhazardous liquid wastes, and biomedical wastes as per the provisions of EHS manual.		Contractor	PIU
7.	Labour camp: water supply; sewage and solid waste disposal; healthcare	• Implement measures as per EHS Manual. • By implementing these measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.		Contractor	PIU
8.	Welfare and safety of Labour	• Implement measures as per EHS Manual • By implementing these measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.		Contractor	PIU
9.	Incident Management	Prepare Incident Management Plan with reporting formats.		Contractor	PIU
10.	Environmental Monitoring	Prepare Environmental Monitoring Plan for Air, Noise, Soil, Water, Vibration and Ecological		Contractor	PIU
		The environmental monitoring cost will be Rs. 114.64 Lakh	The environmental monitoring cost will be Rs. 82.67 Lakh		
11.	Availability of institutional capacity	Implement training and establish environment unit as per EHS manual		Contractor	PIU
12.	Tree protection/ Cutting and	• The compensatory Afforestation is planned in line with Saurashtra Felling of trees (infliction of Punishment) Act, 1951 & amendment Rule 2008, Forest		Contractor	PIU

S. No	Environmental Impact	Mitigation Measure		Implementing Entity	Responsible Entity
		Corridor 1: Sarthana - Dream City	Corridor 2: Bhesan - Saroli		
	Disposal	Conservation Act 1980 and ESS 6 of World Bank. • Contractor shall prepare an action plan for 2093 trees to be affected (Cut: 932 trees and Transplant: 1161) and trees proposed to be planted (about 3957 saplings) as per compensatory afforestation norms • The trees proposed for plantation in the project area are Gulmohar, Pipal, Neem, Karanj, Siris etc. • Other than Contractor, no one is allowed to cut the identified trees which are falling in the ROW. • Biomass shall not be stored at site for more than 15 days. • By implementing the proposed Afforestation measures, the project is expected to comply with World Bank ESF guidelines.			
		• Number of Trees along the alignment and depot are 836 and 176 respectively. • Number of Trees to be transplanted are 524 and samplings to be planted are 1988. • Compensatory Afforestation and Transplantion cost is Rs. 76.25 Lakh	• Number of Trees along the alignment and depot are 762 and 319 respectively. • Number of Trees to be transplanted are 637 and samplings to be planted are 1969. • Compensatory Afforestation and Transplantion cost is Rs. 87.32 Lakh		
13.	Energy Management /Conservation	The contractor shall use and maintain lighting, tools and equipment of appropriate specifications so as to conserve energy as per EHS guidelines.		Contractor	PIU
14.	Traffic Diversion/Management	• Traffic Management and Engineering measures like road widening, traffic segregation, one-way movements, traffic diversions, acquisition of service lanes, etc shall be implemented. • By implementing these control measures, the project is expected to comply with the Indian Standards and WB ESF guidelines.		Contractor	PIU
15.	Management Plan for Flamingos	• No spillage or disposal of construction waste close to Tapi River; • Restriction of Construction activities near Tapi River during winters;		Contractor	PIU

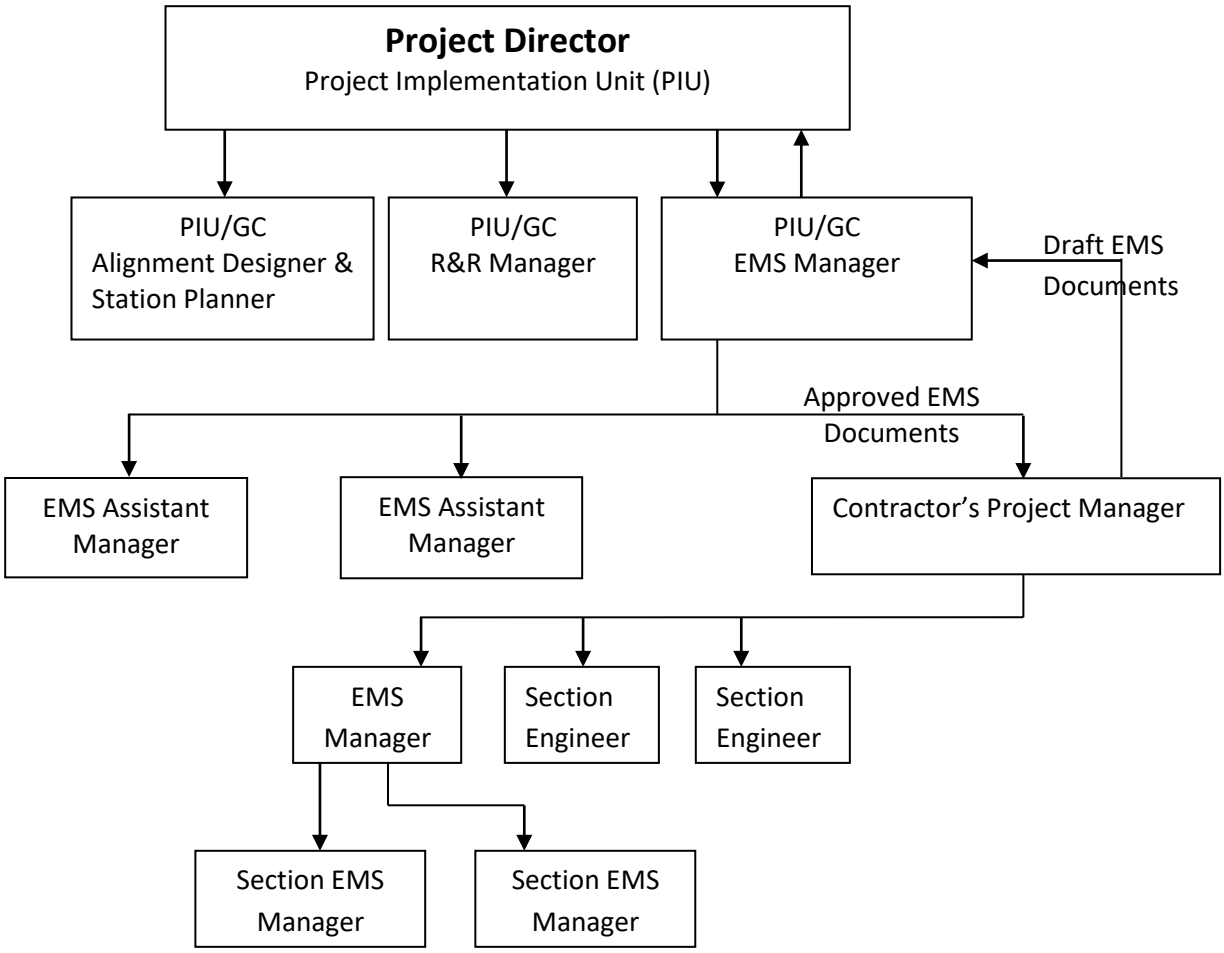
S. No	Environmental Impact	Mitigation Measure		Implementing Entity	Responsible Entity
		Corridor 1: Sarthana - Dream City	Corridor 2: Bhesan - Saroli		
		• Regular monitoring of River Water Quality; • An amount of Rs. 10.00 Lakh provided for Flamingo Conservation; • The project is expected to comply with the WB ESF guidelines by Implementing the management plan for flamingos			
16.	Construction Material Management and House Keeling	• Full height fence, barriers, barricades etc. shall be erected around the site • All stairways, passageways and gangways shall be maintained without any blockages or obstructions • All surplus earth and debris shall be removed/disposed off to officially designated dumpsites • By implementing the proposed mitigation measures, the project is expected to comply with Indian Standards and WB ESF guidelines		Contractor	PIU
EMP during Operation Phase					
1.	Air Pollution	• Air Quality monitoring need to be carried out at all stations and depots locations.		PIU	PIU
		Monitoring will be at all metro stations and Depot and cost will be Rs. 22.68 Lakh.	Monitoring will be at all metro stations and Depot and cost will be Rs. 20.52 Lakh.		
2.	Noise Pollution	• Noise monitoring need to be carried out at all stations and depots locations. • Wherever required erect Noise barriers and maintain noise barriers on viaduct. • These mitigations measures proposed for Noise Management are in accordance with Noise Pollution (Regulation and Control) Rules, 2000 and its amendment and ESS 3 of World Bank.		PIU	PIU

S. No	Environmental Impact	Mitigation Measure		Implementing Entity	Responsible Entity
		Corridor 1: Sarthana - Dream City	Corridor 2: Bhesan - Saroli		
		Monitoring will be at all metro stations and Depot and cost will be Rs. 3.78 Lakh. Noise Barrier proposed for a length of 9339 m and the cost will be Rs. 1120.68 Lakh.	Monitoring will be at all metro stations and Depot and cost will be Rs. 3.42 Lakh. Noise Barrier proposed for a length of 6690 m and the cost will be Rs. 802.80 Lakh.		
3.	Vibration	Implement vibration monitoring and building condition surveys at sensitive structures, if required		PIU	PIU
		Monitoring will be at Seven locations and cost will be Rs. 25.20 Lakh.	Monitoring will be at two locations and cost will be Rs. 7.20 Lakh.		
4.	Soil Monitoring	Monitoring will be at 5 locations and cost will be Rs. 3.00 Lakh	Monitoring will be at 5 locations and cost will be Rs. 3.00 Lakh	PIU	PIU
5.	Water supply, sewage and solid waste disposal at stations and depots	• Implement measures including treatment and reuse of waste water, rain water harvesting to augment ground water. • Organic waste shall be segregated and treated by in-site bio composter technique. • By implementing the proposed mitigation measures, the project is expected to comply with Indian Standards and ESS 3 of World Bank		PIU	PIU
		Monitoring cost will be Rs. 23.88 Lakh for water, waste water and solid waste.	Monitoring cost will be Rs. 23.04 Lakh for Water, waste water and solid waste.		
6.	Incident Management	Implement Incident Management Plan in accordance with EHS Manual		PIU	PIU

S. No	Environmental Impact	Mitigation Measure		Implementing Entity	Responsible Entity
		Corridor 1: Sarthana - Dream City	Corridor 2: Bhesan - Saroli		
7.	Management Plan for Flamingos	• Maintaining, and restoring favourable hydrological conditions and water quality at Tapi River. • Minimizing human disturbance from boating, fishing, and tourists etc near the alignment crossing at Tapi River • The project is expected to comply with the WB ESF guidelines by Implementing the management plan		PIU	PIU
8.	Environmental Monitoring	Implement Environmental Monitoring Plan as mentioned in Section 9.1 of EIA report.		PIU	PIU
		Environmental Monitoring cost during operation phase will be Rs. 81.54 Lakh	Environmental Monitoring cost during operation phase will be Rs. 60.18 Lakh		
9.	Monitoring and Grievances	• Implement mechanism to monitor progress of implementation of the EMP/EMoP measures and results achieved. • Implement mechanism for project-level grievance redressal.		PIU	PIU

A typical EMS organization is depicted in Figure 7.7. One indicative activity i.e, approval of EMS documents is shown in this organisation chart.

Figure 7.7: EMS Organization



* GC: General Consultant as Project Management Consultant

A. REPORTING SYSTEM

The monitoring report of environmental parameters (Air, Noise, Water and soil) will be prepared by the environmental engineer and submitted to the Project Management Consultant.

- The contractor will report to Construction Supervision Consultant (CSC) and CSC will report to GMRCL on compliance. GMRCL may disseminate the information to all interested parties.
- Non compliance of the monitoring will be seen by the GMRCL.
- Photographic monitoring record will be maintained by the contractor. All material source points, disposal locations, plant locations, camp locations, etc should be photographed.
- A full record of construction activities will be kept as a part of normal contract monitoring system under the various stages of construction.

- The reporting format for various activities during construction is given at Annexure 7.2 to Annexure 7.12.

B. RECORD KEEPING

Monitoring forms will need to be devised for use, focusing attention on environmental issues and providing feedback for future improvement. Mitigation and enhancement measures adopted in the final design will be explicitly under the bill of quantities (BOQ), so that performance and completion is readily documented. Project diaries would record environmental problems (soil erosion, air quality, water quality, noise level etc), as well as safety incidents and will be retained as part of the accepted environmental management.

Chapter 8 : Public Consultations

8.1 Introduction

"Public Consultation" refers to the process by which the concerns of local affected persons and others who have plausible stake in the environmental impacts of the project or activity are ascertained with a view to taking into account all the material concerns in the project or activity design as appropriate. Consultation is used as a tool to inform stakeholders about the proposed action both before and after the development decisions are made. It assists in identification of the problems associated with the project. Initial Public consultation has been carried out in the project areas with the objectives of minimizing probable adverse impacts of the project and to achieve speedy implementation of the project through bringing in awareness among the community about the benefits of the project.

8.2 Methods & Approach for Consultation

Public consultation was accomplished to collect the opinion/views of the stakeholders for the construction of the project. Open discussion was held by conducting one to one meeting at project level discussing the issue on the desk. People in meeting were explained about the project activities and their consequences in brief. Queries of the stakeholder were replied by the environmental expert and at the same time their suggestions were also endorsed. It was held at the doorstep of the people or at a common place of people's gathering. In one to one meeting, people expressed open view and asked openly to understand the project while in gathering combined opinion of the people was collected.

8.3 Public Consultation at Project Level

The consultations were conducted from November 8 to November 9, 2019 during the field visit and based on informal unstructured interviews and stakeholder consultation. The objective of the consultation was to disseminate the project information and ascertain stakeholder's views on probable environmental impacts that may arise due to the implementation of the proposed project. The public consultation helps to gather the variety of views about proposed development activity as well as valuable suggestions of public is received which is very important to finalize management plan to mitigate the impacts. The valued mitigation plan in planning phase will save the cost to mitigate the stratified impacts.

EIA Consultant officials explained about need of the project describing environmental issues like anticipated positive & negative impacts and use of techniques during construction and time frame of construction during public consultations. People from different community, sex, age participated in project level public consultations at 10 places for the Bhesan to Saroli

Corridor and Sarthana to Deam City Corridor which is summarized in **Table 8.1.** Issues raised with their valuable suggestions were noted down as depicted at **Annexure 8.1**. The issues discussed were overall need of the project, project location and environmental concerns.

Table 8.1: Minutes of Public Consultations

S. No.	Venue of the Public Consultation	Date	Number of Participants	Issues	Suggestions/Opinions
1	Bhesan (near proposed Bhesan Depot site)	08/11/2019	10	Air and Noise pollution	At present no air and noise pollution and need to be sustained during construction phase also.
				Tree cutting	10 lakh trees given by SMC and we have planted in our places. We love environment. No issues if some trees are going to cut in proposed site of Bhesan depot.
				Job	Local people expecting jobs during construction and operation phases of metro.
2	Near L P Savni School, Khodiyar Chawk, Adajan Gaam	08/11/2019	12	Air and Noise pollution	It's a clean city, no air and noise pollution at all. Not anticipated during operation as well.
				Tree	Two giant African Baobab trees are here at this location. We don't think it will be problematic to these trees if metro corridor is passing over them. These trees must be protected during construction and operation as well.
				Parking	There should be enough parking near Khodiyar Temple. Lot of people coming to the temple in festival season and they parked their vehicle beside the road.
				Business	Business may increase due to proposed metro station here.
3	Majura Gate	08/11/2019	10	Traffic	People will reach timely to their places due to metro. Road traffic will reduce specially autorikshaw.
				Trees	Trees to be transplanted, less trees to be cut.
				Construction Time	Construction period should not be delayed. It should be complete fast.
				Fare	Fare should be the minimum so everybody can afford.
4	Model Town Chawk	08/11/2019	8	Aesthetic view	City will look smart due to metro.
				Traffic	Road traffic will reduce so reduction in accident and safety will be increased.

S. No.	Venue of the Public Consultation	Date	Number of Participants	Issues	Suggestions/Opinions
				Air pollution	No air pollution is anticipated because metro running on electricity.
				Fare	Metro fare should be affordable as compare to BRT.
				Trees	More trees to be planted for trees to be cut.
				Train frequency	Metro train frequency should be more during peak time.
5	Kumbhariya Saroli	08/11/2019	6	Business	Business will increase for those who will have shops nearby stations.
				Travelling	Fast travelling within city
					Trains should be run out of city as well as to connect nearby city.
6	Bhatar Char Rasta Chawk	08/11/2019	7	Job	Local labour should get the job during construction and operation phases.
				Business	Shops located nearby proposed metro station will have good business.
				Traffic, Air Pollution	Traffic congestion and air pollution is contributed more by autorikshws so after metro it may reduce.
				Tree	For the infrastructure development we have to loss some trees.
7	Bhimrad Village	09/11/2019	16	Travel time	Travelling time will be save compare to present situations
				Easy travelling	Due to proposed metro station at Bhimarad village we can easily get in and travel through this station
				Safety	Due to metro travelling will be more safety.
8	Bhimrad Village	09/11/2019	7	Land and Structure	Gandhi Ground should not be affected due to construction of metro. Structures of the village should not be affected. Metro line should be 200m away from the village cricket ground. It should not pass through village agriculture land.
				Job	Preference to be given to the villagers during construction and operation.
9	Sarthana Jakat Naka (near Honda Service centre)	09/11/2019	6	Travelling time and Safety	Travelling time will be less and it will be more safety to travel in metro.
				Aesthetic view	City will look beautiful after construction of metro.

S. No.	Venue of the Public Consultation	Date	Number of Participants	Issues	Suggestions/Opinions
				Trees	More trees should be planted in the cost of cutting trees.
				Business	Business will grow who will nearby proposed metro stations
10	Ghanshyam Nagar, L S Road	09/11/2019	5	Convenience	People having at least two vehicles at each households so they feel more convenient with their own vehicle rather than travelling in metro. Autorikshaw is more convenient other than own vehicles.
				Business	We don't think we will get more benefit due station location
				Tree	Tree samplings is good provision at the cost of tree cutting
				Aesthetic view	City will look beautiful after construction of metro
				Passenger	City buses on this route are running empty so how will metro get passengers.

Major Findings of Preliminary Public Consultations:

The major findings of public consultation are as follows:

1. Travel time reduces as metro caters the people from all over the city
2. People believe that metro rail will enhance the aesthetic looks of the city
3. Metro will be the good transport mode as the city population increases rapidly.
4. People do not have any problem if the metro passes above / adjacent to the two giant African Baobab trees. However, one tree out of these two Giant Biobab tree collapsed due to weather disturbance on 02-05-2020.
5. People near Khodiyar Temple at Adajan Gaam suggested to have more parking space near the temple. Large people visit the temple in the festival season, especially Navaratri.
6. adequate compensation needs to be provided for Project Affected Families and no agricultural land to be affected due to the proposed metro.
7. Few people have the opinion that commuting by own vehicle/three wheelers is better than travelling by metro. They also expressed how the metro will get passengers when the present public transport system like buses is running empty.
8. Construction period of the metro shall not be delayed as it creates problems for the people commuting in the city.

Figure 8.1: Photographs of Public Consultations

Sarthana Jakat Naka

Ghanshyam Nagar, L S Road

Near Bhesan Depot Site

Bhatar Char Rasta Chawk

Model Town Chawk

Bhimrad Village

Majura Gate

Bhimrad Village

Chapter 9 : Environmental Monitoring Plan

9.1. Environment Monitoring Plan

9.1.1. Pre-Construction Phase

The environmental monitoring programme helps in signalling the potential problems resulting from the proposed project activities and will allow for prompt implementation of corrective measures. Pre-construction phase monitoring has been done for the proposed project for air, noise, vibration, water and soil quality as part of this report. The results so obtained are documented in Chapter 4. This will have to be followed by monitoring of afforestation/ transplantation.

9.1.2. Construction Phase

Monitoring schedule for the entire period of construction (5 Years) is summarized in Table 9.1. The number of locations could be modified based on need when the construction commences. Monitoring should be carried out by NABL Accredited/MoEFCC recognized private or Government agency. The contractor will be responsible for carrying out monitoring during construction under the supervision of PIU. The results of air quality, water quality, waste water, vibration monitoring will be submitted to management quarterly during construction phase.

Table 9.1: Construction Stage Monitoring Schedule

Parameter	Locations and Frequency	Reference/Standard	Implementation by / Approval by
Air	One day monitoring for Twice in Month at four locations for each corridor for entire construction period.	• Guidelines for Ambient Air Quality Monitoring , CPCB, 2003 • National Ambient Air Quality Standards 2009	Contractor/GMRCL
Noise	One day monitoring for Twice in Month at four locations for each corridor for entire construction period.	• Protocol for Ambient Level Noise Monitoring, CPCB, May 2015 • ISO/ TC 108 (vibration)	Contractor/ GMRCL
Vibration	Once in a month for a Year at 17 locations for Corridor 1 and 8 locations for Corridor 2		
Water (surface and ground water) and waste water	Once in a season, four seasons in a year at baseline monitoring locations	• Guide Manual – Water and waste water analysis, CPCB • Drinking water – Specifications IS 10500: 2012 and CPHEEO Manual	Contractor/ GMRCL
Soil	Once in a season, four seasons in a year at baseline	US EPA test protocols	Contractor/ GMRCL

Parameter	Locations and Frequency	Reference/Standard	Implementation by / Approval by
	monitoring locations		
Ecology	Terraistrial: Four times per year along each corridor, at depots and afforestation sites. Aquatic: 3 samples & 3 times in a year at Tapi River for Phytoplankton, Zooplankton and Benthos	As per Forest authorities & SMC	Contractor/ GMRCL, SMC & Forest Department
Worker safety	As per EHS/EMS		

Ecological Monitoring: The plantation/afforestation of trees will be reviewed four times a year during construction phase by Contractor or PIU.

Workers Health and Safety: Epidemiological studies at construction sites will be performed to monitor the potential spread of diseases. Regular inspection and medical checkups shall be carried out to workers health and safety monitoring. Any recurrence of health incidents shall be recorded and appropriate mitigation measures shall be taken. Contractor will be responsible to take care of health and safety of workers during construction and project proponent is responsible to review/audit the health and safety measures/plans.

The environmental monitoring cost during construction phase is estimated as **Rs 114.64 Lakh** and **Rs 84.92 Lakh** for Sarthana to Dream city and Bhesan to Saroli corridor respectively. The estimated cost towards environmental monitoring during construction will be part of civil contract and details are given in Table 9.2.

Table 9.2: Environmental Monitoring Cost during Construction

S No	ITEM	QUANTITY	TOTAL COST (Rs in Lakh)
Corridor I: Sarthana to Dream City			
i.	Air Quality Monitoring	4 Samples X 24 times in a year X 5 Years	29.20
ii.	Noise Quality Monitoring	4 Samples X 24 times in a year X 5 Years	4.86
iii.	Soil Quality Monitoring	5 Samples X 4 times in a year X 5 Years	5.00
iv.	Soil Quality Monitoring for Muck	3 muck samples per km for the UG section	0.97
v.	Water Quality Monitoring	6 Samples X 4 times in a year X 5 Years	8.40
vi.	Vibration Monitoring	17 Sample X 12 times in a year X 1 Year	61.20
vii.	Ecological Monitoring	4 times in a year X 5 Years	5.00
Env. Monitoring Cost for Corridor I: Sarthana to Dream City			**114.64**
Corridor II: Bhesan - Saroli			
i.	Air Quality Monitoring	4 Samples X 24 times in a year X 5 Years	29.20

S No	ITEM	QUANTITY	TOTAL COST (Rs in Lakh)
ii.	Noise Quality Monitoring	4 Samples X 24 times in a year X 5 Years	4.86
iii.	Soil Quality Monitoring	5 Samples X 4 times in a year X 5 Years	5.00
iv.	Water Quality Monitoring	7 Samples X 4 times in a year X 5 Years	9.80
v.	Vibration Monitoring	8 Sample X 12 times in a year X 1 Year	28.80
vi.	Ecological Monitoring	4 times in a year X 5 Years	5.00
vii.	Ecological Monitoring: Aquatic	3 times in a year X 5 Years	2.25
Env. Monitoring Cost for Corridor II: Bhesan - Saroli			**84.92**

9.1.3. Operation Phase

The environmental monitoring schedule during operation phase for 3 years is presented in Table 9.3. The results of air quality, water quality, waste water, vibration will be submitted to management bi-annually during operation phase.

Table 9.3: Operation Stage Monitoring Schedule

Parameter	Frequency	Reference/Standard	Implementation by / Approval by	Period (years)
Air	2x24 hours in a week for each season, Three seasons in a year at all metro stations and Depot	• Guidelines for Ambient Air Quality Monitoring , CPCB, 2003 • National Ambient Air Quality Standards 2009	GMRCL	3
Noise	2x24 hours in a week for each season, Three seasons in a year at all metro stations and Depot	Metro Rail Transit System, Guidelines for Noise and Vibrations, RDSO, Ministry of Railways, September 2015	GMRCL	3
Vibration	24 hours, once in two months at Sensitive locations at 7 locations for Corridor 1 and 2 locations for Corridor 2			
Water (surface and ground water)	Once in a season, four seasons in a year at baseline monitoring locations	• Guide Manual – Water and waste water analysis, CPCB • Drinking water – Specifications IS 10500: 2012 and CPHEEO Manual 2012	GMRCL	3
Waste	Once in a season, Four			

Parameter	Frequency	Reference/Standard	Implementation by / Approval by	Period (years)
Water	seasons in a year at all stations and Depot			
Solid Waste	Once in a season, Four seasons in a year at Depot	• Solid Waste Management Rules 2016	GMRCL	3
Soil	Once in a season, four seasons in a year at baseline monitoring locations	US EPA test protocols	GMRCL	3
Ecology	Once in a season, four season in a year at Afforestation Sites	As per Forest authorities & SMC	Contractor/ GMRCL, SMC & Forest Dept.	3

The environmental monitoring cost during operation phase is **Rs 81.54 Lakh** & **Rs 61.53 Lakh** for Sarthana to Dream city and Bhesan to Saroli corridor respectively. The estimated cost towards environmental monitoring during operation will be the responsibility of PIU and details are given in Table 9.4.

Table 9.4: Environmental Monitoring Cost during Operation

S No	ITEM	QUANTITY	TOTAL COST (Rs in Lakh)
Corridor I: Sarthana to Dream City			
i.	Air Quality Monitoring	21 Samples X 2 days X 3 times in a year for 3 Years	22.68
ii.	Noise Quality Monitoring	21 Samples X 2 days X 3 times in a year for 3 Years	3.78
iii.	Soil Quality Monitoring	5 Samples X 4 times in a year X 3 Years	3.00
iv.	Water Quality Monitoring	6 Samples X 4 times in a year X 3 Years	5.04
v.	Vibration Monitoring	7 Samples X 4 times in a year X 3 Year	25.20
vi.	Quality of Waste Water	21 Samples X 4 times in a year for 3 Years	17.64
vii.	Ecological Monitoring	4 times in a year X 3 Years	3.00
viii.	Solid Waste Characteristics at Depot	1 Sample X 4 times in a year X 3 Year	1.20
Env. Monitoring Cost for Corridor I: Sarthana to Dream City			**81.54**
Corridor I: Bhesan - Saroli			
i.	Air Quality Monitoring	19 Samples X 2 days X 3 times in a year for 3 Years	20.52
ii.	Noise Quality Monitoring	19 Samples X 2 days X 3 times in a year for 3 Years	3.42

S No	ITEM	QUANTITY	TOTAL COST (Rs in Lakh)
iii.	Soil Quality Monitoring	5 Samples X 4 times in a year X 3 Years	3.00
iv.	Water Quality Monitoring	7 Samples X 4 times in a year X 3 Years	5.88
v.	Vibration Monitoring	2 Samples X 4 times in a year X 3 Year	7.20
vi.	Quality of Waste Water	19 Samples X 4 times in a year X 3 Years	15.96
vii.	Ecological Monitoring	4 times in a year X 3 Years	3.00
viii.	Ecological Monitoring: Aquatic	3 times in a year X 3 Years	1.35
ix.	Solid Waste Characteristics at Depot	1 Sample X 4 times in a year X 3 Year	1.20
Env. Monitoring Cost for Corridor II: Bhesan - Saroli			**61.53**

9.2. Establishment of Environmental Division

It is recommended that Project Authority establishes an Environment Division at the initial stage of the project itself. This division should have an Environmental Officer and an Environment Engineer. The task of the division would be to monitor implementation of environmental mitigation measures and environmental monitoring and it should report directly to Chief Engineer of the Project Authority. Progress of the division should be reviewed by an Environmental Advisor once in a year. The Environmental Advisor should be an experienced expert familiar with environmental management in similar projects. Cost for the first ten years (including 10% annual increase) is given in Table 9.3. The estimated cost for 10 years is **Rs 166.31 Lakh** for each corridor. The division will be setu up at one location for both the corridor and capital & recurring costs are distributed equally among two corridor.

Table 9.5: Environmental Division Cost

S No	ITEM	COST (Rs)	
		Corridor I	Corridor II
A	**Capital Cost**		
	Office Furnishings (Computer, furniture etc) LS	250,000	
B	**Recurring Cost**		
	Man Power Cost (For 12 months)		
	Environmental Engineer/Officer @ Rs. 40,000/month	480,000	
	Environmental Assistant @30000/month	360,000	360,000
	Office Maintenance and consumables @ Rs. 10,000/month	120,000	
C	**Sub Total (A+B)**	**785,000**	**785,000**
	Miscellaneous and unforeseen expenses, LS (10 % of C)	78,500	78,500
Total cost for establishment of cell for 1 Year		**863,500**	**863,500**
Total cost for 10 years with 10% annual increse		**16,631,450**	**16,631,450**

Chapter 10 : Cost Estimate

10.1. Summary of Costs

The total estimated environmental management and monitoring cost for the proposed project is **Rs 3656.44 Lakh** as indicated corridor wise in Table 10.1. The details of each component given in respective chapters. The cost towards environmental monitoring and rainwater harvesting during construction phase will be the part of Civil Contract and remaining cost will be the part of Project Implementation Unit (PIU).

Table 10.1: Cost of Environmental Management Plan

S. No	Item	Amount in Rs. Lakh		Responsibility	Remarks
		Sarthana to Dream City	Bhesan to Saroli		
1.	Compensatory Afforestation & Transplantation			PIU/GMRCL	Section 7.3.1 of Chapter 7
	Compensatory Afforestation @ Rs. 1200 per Tree	17.57	15.98		
	Transplantation @ 9999 per Tree	52.39	63.69		
	Add. tree for transplantation @ Rs. 1200 per Tree	7.64	6.29		
2.	Noise Barriers @ 0.12 Lakh/meter	1120.68	802.80	PIU/GMRCL	Section 7.5.1
3.	Rainwater Harvesting @ 22 lakh/km	341.10	418.27	Contractor	Section 7.5.3
4.	Flamingo Conservation	-	10.00	Contractor	Section 7.4.18
5.	Environmental Monitoring				
	A. During Construction				
	Air Quality Monitoring	29.20	29.20	Contractor	Section 9.1.2
	Noise Quality Monitoring	4.86	4.86	Contractor	Section 9.1.2
	Soil Quality Monitoring	5.97	5.00	Contractor	Section 9.1.2
	Water Quality Monitoring	8.40	9.80	Contractor	Section 9.1.2
	Vibration Monitoring	61.20	28.80	Contractor	Section 9.1.2
	Ecological Monitoring	5.00	5.00	Contractor	Section 9.1.2
	Ecological Monitoring - Aquatic	-	2.25	Contractor	Section 9.1.2

S. No	Item	Amount in Rs. Lakh		Responsibility	Remarks
		Sarthana to Dream City	Bhesan to Saroli		
	B. During Operation				
	Air Quality Monitoring	22.68	20.52	PIU/GMRCL	Section 9.1.3
	Noise Quality Monitoring	3.78	3.42	PIU/GMRCL	Section 9.1.3
	Soil Quality Monitoring	3.00	3.00	PIU/GMRCL	Section 9.1.3
	Water Quality Monitoring	5.04	5.88	PIU/GMRCL	Section 9.1.3
	Vibration Monitoring	25.20	7.20	PIU/GMRCL	Section 9.1.3
	Quality of Waste Water	17.64	15.96	PIU/GMRCL	Section 9.1.3
	Ecological Monitoring	3.00	3.00	PIU/GMRCL	Section 9.1.3
	Ecological Monitoring - Aqiatic	1.35	-	PIU/GMRCL	Section 9.1.3
	Solid Waste Characteristics at Depot	1.20	1.20	PIU/GMRCL	Section 9.1.3
6.	Training and Extension	7.10	7.10	PIU/GMRCL	Section 7.10
7.	Environment Division (Cell)	166.31	166.31	PIU/GMRCL	Section 9.2
8.	Waste Water Treatment through Bio Digesters	21.72	21.42	PIU/GMRCL	Section 7.5.2
9.	Effluent Treatment Plant (ETP)	39.24	28.22	PIU/GMRCL	Section 7.5.4
	Total in Rs. Lakh	**1971.27**	**1685.17**		
	Total Cost in Rs. Lakh for both the Corridors	**3656.44**			

ANNEXURE 2.1

DRINKING WATER QUALITY STANDARDS (IS 10500:2012)

S. No.	Characteristic	Requirement (Acceptable Limit)	Permissible limit in the absence of alternate source	Remarks
Essential Characteristics				
1	Colour, Hazen units, Max	5	15	Extended to 15 only, if toxic substances are not suspected in absence of alternate source
2	Odour	Agreeable	Agreeable	a) Test cold and when heated b) Test at several dilutions
3	pH Value	6.5 to 8.5	No relaxation	-
4	Taste	Agreeable	Agreeable	Test to be conducted only after safety has been established
5	Turbidity NTU, max	1	5	-
6	Total dissolved solids, mg/l, Max	500	2000	-
7	Aluminium (as Al), mg/l Max	0.03	0.2	-
8	Ammonia (as total ammonia-N), mg/l Max	0.5	No relaxation	-
9	Anionic detergents (as MBAS), mg/l, Max	0.2	1.0	-
10	Barium (as Ba), mg/l, max	0.7	No relaxation	-
11	Boron (as B), mg/l Max	0.5	1.0	-
12	Calcium (as Ca) mg/l, Max	75	200	-
13	Chloramines (as Cl_2), mg/l, Max	4.0	No relaxation	-
14	Chloride (as Cl) mg/l, Max	250	1000	-
15	Copper (as Cu) mg/l, Max	0.05	1.5	-
16	Fluoride (as F) mg/l, Max	1.0	1.5	-
17	Free residual Chlorine, mg/l, Min	0.2	1	To be applicable only when water is chlorinated. Tested at consumer end. When protection against viral infection is required, it should be minimum 0.5 mg/l
18	Iron (as Fe) mg/l, max	0.3	No relaxation	Total concentration of manganese (as Mn) and iron (as Fe) shall not exceed 0.3mg/l
19	Magnesium (as Mg) mg/l, Max	30	100	-
20	Manganese (as Mn) mg/l, Max	0.1	0.3	-

S. No.	Characteristic	Requirement (Acceptable Limit)	Permissible limit in the absence of alternate source	Remarks
21	Mineral oil, mg/l Max	0.5	No relaxation	-
22	Nitrate (as NO_3) mg/l, Max	45	No relaxation	-
23	Phenolic compounds (as C_6H_5OH) mg/l, Max	0.001	0.002	-
24	Selenium (as Se), mg/l, Max	0.01	No relaxation	-
25	Silver (as Ag), mg/l, Max	0.1	No relaxation	-
26	Sulphate (as SO_4) mg/l, Max	200	400	May be extended to 400 provided that Magnesium does not exceed 30
27	Sulphide (as H_2S) mg/l, max	0.05	No relaxation	-
28	Total alkalinity as calcium carbonate, mg/l Max	200	600	-
29	Total Hardness (as $CaCO_3$) mg/l, Max	200	600	-
30	Zinc (as zn), mg/l, Max	5	15	-
31	Cadmium (as Cd), mg/l, Max	0.003	No relaxation	-
32	Cyanide (as CN), mg/l, Max	0.05	No relaxation	-
33	Lead (as Pb), mg/l, Max	0.01	No relaxation	-
34	Mercury (as Hg) mg/l, Max	0.001	No relaxation	-
35	Molybdenum (as Mo) mg/l, max	0.07	No relaxation	-
36	Nickle (as Ni), mg/l, max	0.02	No relaxation	-
37	Polychlorinated biphenyls, mg/l, max	0.0005	No relaxation	-
38	Polynuclear aromatic hydrocarbons (as PAH) mg/l, Max	0.0001	No relaxation	-
39	Total Arsenic (as As), mg/l, Max	0.01	0.05	-
40	Total Chromium (as Cr) mg/l, Max	0.05	No relaxation	-
41	Trihalomethanes Bromoform, mg/l, max Dibromochloromethane, mg/l, max Bromodichloromethane, mg/l, max Chloroform, mg/l, max	 0.1 0.1 0.06 0.2	 No relaxation No relaxation No relaxation No relaxation	-
42	Radioactive materials a) Alpha emitters Bq/l max b) Beta emitters pci/l, Max	 0.1 1.0	 No relaxation No relaxation	-

ANNEXURE 2.2

EFFLUENT DISCHARGE STANDARDS

S.No.	Parameter	Unit	Effluent Discharge Standards (Inland Surface Water), GOI	Treated Sanitary Sewage Discharges by IFC
1	Colour &Odor	--	All efforts should be made to remove colour and unpleasant odor as far as practicable.	-
2	Suspended Solids Max.	mg/l	100	50
3	Particle size of Suspended Solids	--	Shall pass 850 micron IS Sieve	-
4	pH value	--	5.5 to 9.0	6.0-9.0
5	Temperature, Max.	°C	Shall not exceed 5°C above the receiving water temperature	-
6	Oil and grease, Max.	mg/l	10.0	10.0
7	Total residual Chlorine, Max.	mg/l	1.0	-
8	Ammonical Nitrogen (as N), Max.	mg/l	50	-
9	Total Kjeldah Nitrogen (as N), Max.	mg/l	100	-
10	Free Ammonia (as NH_3), Max.	mg/l	5	-
11	Biochemical Oxygen Demand (5 days at 20°C), Max.	mg/l	30	30
12	Chemical Oxygen Demand Max.	mg/l	250	125
13	Arsenic (as As), Max.	mg/l	0.2	-
14	Mercury (as Hg), Max.	mg/l	0.01	-
15	Lead (as Pb), Max.	mg/l	0.1	-
16	Cadmium (as Cd), Max.	mg/l	2.0	-
17	Hexavalent Chromium (as Cr^{+6}), Max.	mg/l	0.1	-
18	Total Chromium (as Cr) Max.	mg/l	2.0	-
19	Copper (as Cu), Max.	mg/l	3.0	-
20	Zinc (as Zn), Max.	mg/l	5.0	-
21	Selenium (as Se), Max.	mg/l	0.05	-
22	Nickel (as Ni), Max.	mg/l	3.0	-
23	Cyanide (as CN), Max.	mg/l	0.2	-
24	Fluorides (as F), Max.	mg/l	2.0	-
25	Dissolved phosphates (as P), Max.	mg/l	5.0	-
26	Sulphides (as S), Max.	mg/l	2.0	-
27	Phenolic compounds (as C_6H_5OH), Max.	mg/l	1.0	-
28	Radioactive Materials α Emitters, µcurie/ml, Max. β Emitters, µcurie/ml, Max.	mg/l	 10-7 10-6	-
29	Bio-assay test	mg/l	90% survival of fish after 96 hours in 100% effluent	-
30	Manganese (as Mn)	mg/l	2.0	-
31	Iron (as Fe)	mg/l	3.0	-
32	Vanadium (as V)	mg/l	0.2	-
33	Nitrate Nitrogen	mg/l	10.0	-
34	Total Nitrogen	mg/l	-	10.0
35	Total phosphorus	mg/l	-	2.0

Source: Central Pollution Control Board Notification, GOI and
General EHS Guidelines: Environmental Wastewater and ambient Water Quality, IFC, 2007

ANNEXURE 2.3

TOLERANCE LIMITS FOR INLAND SURFACE WATER QUALITY

Characteristic	Designated Use Class of Inland Waters				
	A	B	C	D	E
pH value	6.5 to 8.5	6.5 to 8.5	6.5 to 8.5	6.5 to 8.5	6.0 to 8.5
Dissolved Oxygen, mg/l, Min.	6	5	4	4	-
Biochemical Oxygen Demand (5 days at 20^0C), mg/l	2	3	3	-	-
Total coliform organisms, MPN/100 ml. Max.	50	500	5000	-	-
Colour Hazen units	10	300	300	-	-
Chlorides (as Cl), mg/l Max.	250	-	600	-	600
Sodium Adsorption ratio Max.	-	-	-	-	26
Boron (as B), mg/l. Max.	-	-	-	-	2
Sulphates (as SO_4), mg/ l	400	-	400	-	1000
Nitrates (as NO), mg/l Max.	20	-	50	-	-
Free Ammonia (as NH_3), mg/l	-	-	-	1.2	-
Conductivity at 25° C microhm / cm Max.	-	-	-	1000	2250
Arsenic (as As), mg/l. Max.	0.05	0.2	0.2	-	-
Iron (as Fe), mg/l	0.3	-	50	-	-
Fluorides (as F), mg/l	1.5	1.5	1.5	-	-
Lead (as Pb), mg/l. Max.	0.1	-	0.1	-	-
Copper (as Cu), mg/l	1.5	-	1.5	-	-
Zinc (as Zn) mg/l/ Max.	1.5	-	1.5	-	-
Manganese (as Mn), mg/l	0.5	-	-	-	-
Total Dissolved Solids, mg/l	500	-	1500	-	2100
Total Hardness ($CaCO_3$), mg/l	300	-	-	-	-
Magnesium (as Mg), mg/l	100	-	-	-	-
Chlorides (as Cl), mg/l	250	600	-	-	600
Cyanides (as CN), mg/l	0.05	0.05	0.05	-	-

A: Drinking Water Source without conventional treatment but after disinfections;
B: Outdoor bathing organized;
C: drinking water source with conventional treatment followed by disinfections;
D: propagation of wildlife and fisheries;
E: irrigation, industrial cooling, controlled waste disposal.
Source: Central Pollution Control Board

ANNEXURE 2.4

AMBIENT AIR QUALITY STANDARDS

Pollutant	National Ambient Air Quality Standards, GOI			Air Quality Standards by IFC	
	Time Weighted Average	Industrial, Residential, Rural & Other Area	Ecologically Sensitive Area (notified by Central Government)	Averaging Period	Guideline Value in µg/m3
Sulphur Dioxide (SO_2), µg/m^3	Annual	50	20	10 minute	500
	24 Hours**	80	80	24-Hour	20
Nitrogen Dioxide as NO_2, µg/m^3	Annual	40	30	1-Year	40
	24 Hours**	80	80	1-Hour	200
Particulate Matter (size less than 10µm) or PM_{10} µg/m^3	Annual	60	60	1-Year	20
	24 Hours**	100	100	24-Hour	50
Particulate Matter (size less than 2.5µm) or PM2.5 µg/m^3	Annual *	40	40	1-Year	10
	24 Hours**	60	60	24-Hour	25
Ozone (O3) µg/m^3	8 hours**	100	100	8-Hour daily maximum	100
	24 Hours**	180	180	-	-
Lead (Pb µg/m^3	Annual *	0.50	0.50	-	-
	24 Hours**	1.0	1.0		
Carbon Monoxide (CO) mg/m^3	8 Hours**	02	02	-	-
	1 Hour**	04	04		
Ammonia (NH_3) µg/m^3	Annual *	100	100	-	-
	24 Hours**	400	400		
Benzene (C_6H_6) µg/m^3	Annual *	05	05	-	-
Benzo (a) pyrene (BaP)particulate phase only ng/m^3	Annual *	01	01	-	-
Arsenic (AS) ng/m^3	Annual *	06	06	-	-
Nickle (Ni) ng/m^3	Annual *	20	20	-	-

Source: *Central Pollution Control Board Notification dated* 18th November 2009, GOI and General EHS Guidelines: Environmental Air Emissions and Ambient Air Quality, IFC, 2007

** Annual arithmetic mean of minimum 104 measurements in a year at a particular site taken twice a week hourly at uniform intervals*

*** 24 hourly or 08 hourly or 01 hourly monitored values, as applicable, shall be complied with 98% of the time in a year. 2% of the time, they may exceed the limits but not on two consecutive days of monitoring.*

ANNEXURE 2.5

NATIONAL AMBIENT NOISE STANDARDS

Category of Zones	National Ambient Noise Standards, GOI		Noise Standards by IFC	
	Leq in dB (A)		One Hour LAeq (dBA)	
	Day Time 6.00 AM to 10.00 PM	Night Time 10.00 PM to 6.00 PM	Day Time 7.00 AM to 10.00 PM	Night Time 10.00 PM to 7.00 PM
Industrial	75	70	70	70
Commercial	65	55		
Residential	55	45	55	45
Silence Zone: Institutional, Educational	50	40		

Source: Central Pollution Control Board, GOI and
General EHS Guidelines: Environmental Noise Management, IFC, 2007

1. Silence zone is an area comprising not less than 100 metres around hospitals, educational institutions, courts, religious places or any other area which is declared as such by the competent authority
2. Mixed categories of areas may be declared as one of the four above mentioned categories by the competent authority.

* dB(A) Leq denotes the time weighted average of the level of sound in decibels on scale A which is relatable to human hearing.
A "decibel" is a unit in which noise is measured.
"A", in dB(A) Leq, denotes the frequency weighting in the measurement of noise and corresponds to frequency response characteristics of the human ear.
Leq: It is an energy mean of the noise level over a specified period

ANNEXURE 5.1

DETAILS OF TREE INUMERATION & INVENTORY

FOR SARTHANA TO DREAM CITY

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
1.	Start to Sarthana	2T	Kadam	*2Anthocephalus kadamba*	30	4	T
2.		1T	Kadam	*Anthocephalus kadamba*	40	4	T
3.		1T	Kadam	*Anthocephalus kadamba*	30	4	T
4.		1T	Kadam	*Anthocephalus kadamba*	40	6	T
5.		1C	Subabool	*Leucaena leucocephala*	*40*	*6*	C
6.		2T	Kadam	*2Anthocephalus kadamba*	40	6	T
7.		1T	Devils tree	*Alstonia scholaris*	*60*	*6*	T
8.		1T	Kadam	*Anthocephalus kadamba*	40	8	T
9.		(9T+1C) 10					
10.	Sarthana st	1T	Kadam	*Anthocephalus kadamba*	40	4	T
11.		1T	kadam	*Anthocephalus kadamba*	30	3	T
12.		(2T) 2					
13.	Sarthana to nature park station	1C	*Gulmohar*	*Delonix regia*	120	10	C
14.		5T	*Gulmohar*	*5Delonix regia*	100	10	T
15.		(1C+5T)6					
16.	Nature park st	1C	*Pipal*	*Ficus religiosa*	160	12	C
17.		(1C)1					
18.	Nature park station to Varacha chopati garden	1T	*Neem*	*Azadirachta indica*	40	3	T
19.		2T	*Devils tree*	*2Alstonia scholaris*	*40*	*3*	T
20.		1C	*Pili kaner*	*Nerium indicum*	40	3	C
21.		3T	*Devils tree*	*3Alstonia scholaris*	*30*	*3*	T
22.		1T	*Devils tree*	*Alstonia scholaris*	*40*	*4*	T
23.		1T	*Devils tree*	*Alstonia scholaris*	*50*	*8*	T
24.		1T	*Neem*	*Azadirachta indica*	50	8	T
25.		5T	*Neem*	*5Azadirachta indica*	30	3	T

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
26.		1T	*Neem*	*Azadirachta indica*	60	4	T
27.		4T	*Devils tree*	*4Alstonia scholaris*	*50*	*6*	T
28.		3T	*Devils tree*	*3Alstonia scholaris*	*40*	*5*	T
29.		2T	*Devils tree*	*2Alstonia scholaris*	*30*	*3*	T
30.		1T	*Neem*	*Azadirachta indica*	80	8	T
31.		1C	*pipal*	*Ficus religiosa*	180	11	C
32.		1T	*Devils tree*	*Alstonia scholaris*	*40*	*4*	T
33.		(26T+1C+1C)28					
34.	Varacha chopati garden st	1T	*Semal*	*Ceiba pentandra*	80	10	T
35.		3T	*Neem*	*3Azadirachta indica*	30	3	T
36.		3T	*Devils tree*	*3Alstonia scholaris*	*40*	*3*	T
37.		1T	*Devils tree*	*Alstonia scholaris*	*60*	*8*	T
38.		3T	*Devils tree*	*3Delonix regia*	*40*	*6*	T
39.		7T	*Asoka*	*7Polyalthia longifolia*	*40*	*8*	T
40.		(18T)18					
41.	Vracha chopati garden st to sri swamirnarayan mandir kalakunj	1T	*Devils tree*	*2Alstonia scholaris*	*40*	*4*	T
42.		1C	*Karanj*	*Pongamia pinnata*	60	6	C
43.		1C	*Karanj*	*Pongamia pinnata*	40	3	C
44.		1T	*Devils tree*	*Alstonia scholaris*	*60*	*4*	T
45.		4C	*Karanj*	*4Pongamia pinnata*	60	6	C
46.		1T	*Bargad*	*Ficus benghalensis*	40	3	T
47.		1T	*Bargad*	*Ficus benghalensis*	80	10	T
48.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	60	8	C
49.		1T	*Yellow flame tree*	*Ficus benghalensis*	180	12	C
50.		2C	*Karanj*	*2Pongamia pinnata*	60	6	C
51.		1C	*Karanj*	*Pongamia pinnata*	60	4	C
52.		1C	*Karanj*	*Pongamia pinnata*	60	8	C
53.		2C	*Yellow flame tree*	*2Peltoforam pterocarpum*	30	3	C

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
54.		1T	*Devils tree*	*Alstonia scholaris*	*60*	*6*	T
55.		2T	*Devils tree*	*2Alstonia scholaris*	*50*	*8*	T
56.		1C	*Bargad*	*Ficus benghalensis*	110	10	C
57.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	110	6	C
58.		1C	*Coconut*	*Coccos nucifera*	40	12	C
59.		2T	*Neem*	*2Azadirachta indica*	30	3	T
60.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	110	6	C
61.		1T	*Devils tree*	*Alstonia scholaris*	*80*	*10*	T
62.		1T	*Neem*	*Azadirachta indica*	60	10	T
63.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	45	6	C
64.		2T	*Neem*	*2Azadirachta indica*	60	6	T
65.		1T	*Devils tree*	*Alstonia scholaris*	*40*	*6*	T
66.		1T	*Neem*	*Azadirachta indica*	60	10	T
67.		1T	*Neem*	*Azadirachta indica*	40	10	T
68.		1T	*Neem*	*Azadirachta indica*	60	8	T
69.		1C	*Neem*	*Azadirachta indica*	180	12	C
70.		(15C+5C+17T) 37					
71.	Sri swaminarayan mandir kalakunj st	0		-			
72.		0					
73.	Sri swaminarayan mandir kalakunj st to start of the ramp	1T	*Bargad*	*Ficus benghalensis*	80	10	T
74.		1T	*Pipal*	*Ficus religiosa*	90	8	T
75.		1C	*Karanj*	*Pongamia pinnata*	30	3	T
76.		(2T+1C)3					
77.	Ramp and end of ramp	1T	*Bargad*	*Ficus benghalensis*	80	10	T
78.		1T	*Pipal*	*Ficus religiosa*	60	8	T
79.		3C	*Gulmohar*	*3Delonix regia*	50	8	C
80.		(2T+3C) 5					

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
81.	Kapodra station	4T	Neem	4Azadirachta indica	80	10	T
82.		2T	Neem	2Azadirachta indica	60	10	T
83.		(6T)6					
84.	Labheshwar chowk station	2T	Neem	2Azadirachta indica	40	8	T
85.		1T	Kadam	Anthocephalus kadamba	60	10	T
86.		1T	Neem	Azadirachta indica	40	4	T
87.		1T	Badam	Terminalia catapa	40	6	T
88.		1T	Bargad	Ficus benghalensis	40	4	T
89.		1C	Karanj	Pongamia pinnata	40	6	C
90.		1C	Karanj	Pongamia pinnata	70	8	C
91.		1C	Karanj	Pongamia pinnata	60	8	C
92.		(6T+3C)9					
93.	Central warehouse station	1T	Jamun	Syzygium cumini	80	12	T
94.		1T	Pipal	Ficus religiosa	80	12	T
95.		1C	Yellow flame tree	Peltoforam pterocarpum	40	6	C
96.		1T	Pipal	Ficus religiosa	80	10	T
97.		1T	Pipal	Ficus religiosa	80	10	T
98.		1T	Kachnar	Bahunia varigata	80	10	T
99.		1T	Pipal	Ficus religiosa	80	8	T
100.		1T	Pipal	Ficus religiosa	60	8	T
101.		1T	Bargad	Ficus benghalensis	60	10	T
102.		1C	Yellow flame tree	Peltoforam pterocarpum	60	10	C
103.		1T	Pipal	Ficus religiosa	60	10	T
104.		1C	Devils tree	Alstonia scholaris	120	4	C
105.		1T	Pipal	Ficus religiosa	80	10	T
106.		1T	Pipal	Ficus religiosa	60	8	T
107.		3T	Devils tree	3Alstonia scholaris	30	3	T
108.		1C	Pipal	Ficus religiosa	110	12	C
109.		1C	Sadafuli	3Taberna –e-montana	30	3	C

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
110.		1C	Safed champa	Plumerea rubra	40	3	C
111.		1T	Semal	Bombax ceiba	90	12	T
112.		16C	Yellow flame tree	16Peltoforam pterocarpum	110	12	C
113.		1T	Asoka	Polyalthia longifolia	80	12	T
114.		7C	Asoka	7Polyalthia longifolia	60	12	C
115.		1C	Pipal	Ficus religiosa	110	12	C
116.		1C	Karanj	Pongamia pinnata	40	4	C
117.		2T	Asoka	2Polyalthia longifolia	80	12	T
118.		1T	Asoka	Polyalthia longifolia	80	4	T
119.		1T	Asoka	Polyalthia longifolia	60	12	T
120.		1T	Butterfly tree	Bahunia purpurea	80	10	T
121.		1T	Pipal	Ficus religiosa	80	10	T
122.		1T	Pipal	Ficus religiosa	60	10	T
123.		(12C+19C+25T)56					
124.	Surat railway station	2T	Bargad	2Ficus benghalensis	60	4	T
125.		1T	Asoka	Polyalthia longifolia	40	10	T
126.		1T	Jamun	Syzygium cumini	60	10	T
127.		1T	Neem	2Azadirachta indica	60	10	T
128.		4C	Gulmohar	4Delonix regia	40	6	C
129.		1T	Pipal	Ficus religiosa	40	4	T
130.		1T	Jamun	Syzygium cumini	110	12	C
131.		2T	Asoka	2Polyalthia longifolia	30	10	T
132.		2T	Asoka	2Peltoforam pterocarpum	80	12	T
133.		3T	Neem	3Azadirachta indica	50	6	T
134.		(4C+15T)19					
135.	Maskati hospital	2T	Asoka	2Polyalthia longifolia	30	2	T
136.		1T	Jamun	Syzygium cumini	150	12	C
137.		1C	Kaner	Nerium indicum	40	6	C

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
138.		**1T**	*Asoka*	*Polyalthia longifolia*	30	10	**T**
139.		**1C**	*Jungle jalebi*	*Pithecellobium dulce*	30	3	**C**
140.		**(2C+4T)6**					
141.	**Chowk bazaar**	**1T**	*Pipal*	*Ficus religiosa*	30	8	**T**
142.		**3T**	*Asoka*	*3Polyalthia longifolia*	40	8	**T**
143.		**1C**	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	**C**
144.		**1C**	*Semal*	*Bombax ceiba*	110	12	**C**
145.		**2T**	*Semal*	*2Bombax ceiba*	60	10	**T**
146.		**1C**	*Neem*	*Azadirachta indica*	110	10	**C**
147.		**1C**	*Gular*	*Ficus racemosa*	180	12	**C**
148.		**(1C+3C+6T) 10**					
149.	Ramp	4T	*Asoka*	*4Polyalthia longifolia*	80	12	T
150.		1T	*Arjuna*	*Terminalia arjuna*	120	12	T
151.		1T	*Pipal*	*Ficus religiosa*	30	10	T
152.		1C	*Nilgiri*	*Eucalyptus globulus*	80	12	C
153.		1C	*Neem*	*Azadirachta indica*	120	12	C
154.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	120	12	C
155.		1C	*Pipal*	*Ficus religiosa*	125	10	C
156.		10T	*Asoka*	*10Polyalthia longifolia*	80	12	T
157.		4C	*Tad*	*4 Borassus flaberifer*	50	12	C
158.		1C	*Coconut*	*Coccos nucifera*	120	12	C
159.		1C	*Spanish cherry*	*Mimusops elengi*	110	12	C
160.		2C	*Suru*	*2Casuarina equisetifolia*	60	10	C
161.		18T	*Asoka*	*18Polyalthia longifolia*	80	12	T
162.		4T	*Asoka*	*4Polyalthia longifolia*	60	12	T
163.		(5C+7C+38T) 50					
164.	Ramp end to kadarsha ni naal	1T	*Semal*	*Bombax ceiba*	60	12	T
165.		2T	*Shisam*	*2Dlabergia sisoo*	60	6	T
166.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	110	12	C

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
167.		1T	*Pipal*	*Ficus religiosa*	60	10	T
168.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	4	T
169.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	40	6	T
170.		(1C+7T) 8					
171.	kadarshahi mall st	1T	*Gular*	*Ficus recemosa*	60	6	T
172.		1T	*Asoka*	*Polyalthia longifolia*	40	10	T
173.		4T	*Yellow flame tree*	*4Polyalthia longifolia*	30	30	T
174.		1C	*Karanj*	*Pongamia pinnata*	60	10	C
175.		1C	*Subabool*	*Leucaena leucocephala*	60	10	C
176.		10T	*Asoka*	*10Polyalthia longifolia*	40	10	T
177.		5T	*Asoka*	*5Polyalthia longifolia*	60	5	T
178.		1C	*Karanj*	*Pongamia pinnata*	60	10	C
179.		1T	*Gular*	*Ficus recemosa*	60	6	T
180.		1T	*Asoka*	*Polyalthia longifolia*	40	10	T
181.		1T	*Asoka*	*Polyalthia longifolia*	40	4	T
182.		(3C+24T) 27					
183.	Kadarshahi mall to majura gate	3T	*Devils tree*	*3Alstonia scholaris*	30	4	T
184.		1T	*Devils tree*	*Alstonia scholaris*	60	10	T
185.		3T	*Devils tree*	*3Alstonia scholaris*	50	6	T
186.		1T	*Devils tree*	*Alstonia scholaris*	40	6	T
187.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	80	10	T
188.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	110	12	C
189.		1T	*Devils tree*	*Alstonia scholaris*	30	3	T
190.		3T	*Yellow flame tree*	*3Peltoforam pterocarpum*	30	3	T
191.		1T	*Pipal*	*Ficus religiosa*	60	10	T
192.		1T	*Pipal*	*Ficus religiosa*	80	12	T
193.		1T	*Bargad*	*Ficus benghalensis*	60	10	T
194.		1T	*Devils tree*	*Alstonia scholaris*	60	10	T
195.		1T	*Semal*	*Bombax ceiba*	60	10	T

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
196.		1T	*Devils tree*	*Alstonia scholaris*	40	10	T
197.		1T	*Asoka*	*Polyalthia longifolia*	40	10	T
198.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	12	T
199.		1T		*Alstonia scholaris*	40	6	T
200.		(1C+22T)23					
201.	Majura gate station	1C	*Yellow flame tree*	*Peltoforam pterocarpum*	120	12	T
202.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	80	12	T
203.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	12	T
204.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	50	12	T
205.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
206.		1T	*Devils tree*	*Alstonia scholaris*	50	6	T
207.		1T	*Pipal*	*Ficus religiosa*	50	10	T
208.		1C	*Pipal*	*Ficus religiosa*	260	12	C
209.		1T	*Devils tree*	*Alstonia scholaris*	60	10	T
210.		1T	*Pipal*	*Ficus religiosa*	50	6	T
211.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	50	10	T
212.		1T		*Alstonia scholaris*	60	4	T
213.		(2C+11T)13					
214.	Majura gate station to roopali canal	1T	*Pipal*	*Ficus religiosa*	60	12	T
215.		2T	*Devils tree*	*2Alstonia scholaris*	60	12	T
216.		1T	*Devils tree*	*Alstonia scholaris*	80	12	T
217.		10C	*Yellow flame tree*	*10Peltoforam pterocarpum*	120	12	C
218.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	230	12	C
219.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	180	10	C
220.		1T	*Semal*	*Bombax ceiba*	80	10	T
221.		1C	*Semal*	*Bombax ceiba*	120	12	C
222.		2C	*Yellow flame tree*	*2Peltoforam pterocarpum*	120	12	C
223.		2C	*Yellow flame tree*	*2Peltoforam pterocarpum*	100	12	T

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
224.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	180	12	C
225.		1T	*Jamun*	*Syzygium cumini*	40	4	T
226.		1T	*Bargad*	*Ficus benghalensis*	60	10	T
227.		4T	*Kadam*	*4Anthocephalus kadamba*	60	6	T
228.		1T	*Shisam*	*Dalbergia sisoo*	60	6	T
229.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	60	6	T
230.		1T	*Pipal*	*Ficus religiosa*	80	12	T
231.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	12	T
232.		4T	*Devils tree*	*4Alstonia scholaris*	50	12	T
233.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	120	12	C
234.		3T	*Shisam*	*3Dalbergia sisoo*	60	6	T
235.		1T	*Shisam*	*Dalbergia sisoo*	90	12	T
236.		23C	*Karanj*	*23 Pongamia pinnata*	60	10	C
237.		1T	*Semal*	*Bombax ceiba*	60	10	T
238.		1T	*Devils tree*	*Alstonia scholaris*	60	10	T
239.		1T	*Shisam*	*Dalbergia sisoo*	60	10	T
240.		1T	*Shisam*	*Dalbergia sisoo*	60	6	T
241.		(19C+23C+28T)70					
242.	Roopali canal station	3T	*Shisam*	*3Dalbergia sisoo*	60	6	T
243.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	12	T
244.		4T	*Yellow flame tree*	*4Peltoforam pterocarpum*	60	10	T
245.		(8T) 8					
246.	Rooplai canal station to althan tenanmnet	3T	*Devils tree*	*3Alstonia scholaris*	60	12	T
247.		1T	*Kadam*	*Anthocephalus kadamba*	60	10	T
248.		1C	*Pipal*	*Ficus religiosa*	120	12	C
249.		3T	*Yellow flame tree*	*3Peltoforam pterocarpum*	80	12	T
250.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	12	T

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
251.		2T	*Shisam*	*2Dalbergia sisoo*	60	6	T
252.		1C	*Bargad*	*Ficus benghalensis*	110	10	C
253.		1T	*Shisam*	*Dalbergia sisoo*	60	6	T
254.		3C	*Yellow flame tree*	*3Peltoforam pterocarpum*	160	12	C
255.		1T	*Kadam*	*Anthocephalus kadamba*	60	6	T
256.		1T	*Email*	*Tamarindus indicus*	80	4	T
257.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
258.		1C	*Kavith*	*Limonia acidissima*	90	12	C
259.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	6	T
260.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
261.		1C	*Kavith*	*Limonia acidissima*	60	10	C
262.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	60	10	T
263.		1T	*Butterfly tree*	*Bahuni a purpurea*	60	12	T
264.		2T	*Bargad*	*2Ficus benghalensis*	60	6	T
265.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	6	T
266.		1T	*Neem*	*Azadirachta indica*	60	6	T
267.		1T	*Asoka*	*Polyalthia longifolia*	60	10	T
268.		1T	*Neem*	*Azadirachta indica*	80	10	T
269.		1T	*Pipal*	*Ficus religiosa*	60	10	T
270.		2C	*Bargad*	*2Ficus benghalensis*	310	12	C
271.		1T	*Pipal*	*Ficus religiosa*	90	12	T
272.		1T	*Neem*	*Azadirachta indica*	60	6	T
273.		3T	*Kadam*	*3Anthocephalus kadamba*	60	10	T
274.		1T	*Semal*	*Bombax ceiba*	80	12	T
275.		1C	*Coconut*	*Coccos nicifera*	40	12	T
276.		1T	*Asoka*	*Polyalthia longifolia*	30	10	T
277.		1T	*Badam*	*Terminalia catapa*	60	10	T
278.		(7C+2C+35T) 44					
279.	Althan tenanment st	2C	*Karanj*	*2Pongamia pinnata*	60	10	C

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
280.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	80	12	T
281.		1T	*Pipal*	*Ficus religiosa*	60	10	T
282.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
283.		2C	*Kadam*	*2Anthocephalus kadamba*	100	6	C
284.		1T	*Pipal*	*Ficus religiosa*	40	6	T
285.		1T	*Pipal*	*Ficus religiosa*	110	12	T
286.		13T	*Devils tree*	*13Alstonia scholaris*	60	6	T
287.		1T	*Pipal*	*Ficus religiosa*	180	12	T
288.		1C	*Suru*	*Eucalyptus species*	60	12	C
289.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	12	T
290.		1T	*Arjuna*	*Terminalia arjuna*	60	10	T
291.		(3C+2C+22T)27					
292.	Althan tenanment to Althan gaam st	1T	*Neem*	*Azadirachta indica*	60	10	T
293.		3T	*Yellow flame tree*	*3Peltoforam pterocarpum*	60	10	T
294.		1C	*Pipal*	*Ficus religiosa*	120	12	C
295.		1T	*Pipal*	*Ficus religiosa*	60	6	T
296.		1C	*Talwar babool*	*Acacia auriculoformis*	60	6	C
297.		1T	*Neem*	*Azadirachta indica*	60	6	T
298.		1T	*Asoka*	*Polyalthia longifolia*	40	10	T
299.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	80	6	T
300.		2T	*Neem*	*2Azadirachta indica*	60	10	T
301.		1C	*Suru*	*Eucalyptus globulus*	60	12	C
302.		5T	*Gulmohar*	*5Delonix regia*	60	10	T
303.		1T	*Pipal*	*Ficus religiosa*	40	4	T
304.		1T	*Kala siris*	*Albizia odoratissima*	80	10	T
305.		1T	*Kala siris*	*Albizia odoratissima*	60	10	T
306.		1T	*Kala siris*	*Albizia odoratissima*	50	10	T
307.		1T	*Neem*	*Azadirachta indica*	60	6	T

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
308.		(1C+2C+21T) 24					
309.	Althan gaam st	1T	*Badam*	*Terminalia catapa*	60	6	T
310.		2T	*Gulmohar*	*2Delonix regia*	80	10	T
311.		1C	*Jungle jalebi*	*Pithecellobium dulce*	60	10	C
312.		8T	*Asoka*	*8Polyalthia longifolia*	60	8	T
313.		1T	*Badam*	*Terminalia catapa*	50	6	T
314.		3T	*Jamun*	*3 Syzygium cumini*	60	6	T
315.		5T	*Asoka*	*5Polyalthia longifolia*	40	4	T
316.		1T	*Pipal*	*Ficus religiosa*	60	10	T
317.		1T	*Neem*	*Azadirachta indica*	60	10	T
318.		1T	*Devils tree*	*Alstonia schollaris*	60	10	T
319.		(1C+23T)24					
320.	Althan gaamst to VIP road	1C	*Jamun*	*Syzygium cumini*	120	12	C
321.		3T	*Devils tree*	*3Alstonia schollaris*	60	8	T
322.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	40	6	T
323.		4T	*Devils tree*	*4Alstonia schollaris*	30	3	T
324.		37T	*Kadam*	*37Anthocephalus kadamba*	60	10	T
325.		10	*Yellow flame tree*	*10Peltoforam pterocarpum*	60	6	T
326.		8T	*Devils tree*	*8Alstonia schollaris*	50	5	T
327.		2T	*Pipal*	*2Ficus religiosa*	80	10	T
328.		4T	*Bargad*	*4Ficus bengalensis*	60	10	T
329.		1C	*Karanj*	*Pongamia pinnnta*	60	10	C
330.		2C	*Talwar babool*	*2Acacia auriculoformis*	60	10	C
331.		(3C+70T) 73					
332.	VIP road station	1T	*Devils tree*	*Alstonia schollaris*	60	6	T
333.		7T	*Yellow flame tree*	*7Peltoforam pterocarpum*	60	6	T
334.		7C	*Karanj*	*7Pongamia pinnata*	80	6	C
335.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	50	6	T
336.		2T	*Devils tree*	*2Alstonia schollaris*	40	4	T

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
337.		1C	*Khejari*	*Prosopis julifera*	60	6	C
338.		2T	*Kadam*	*2Anthocephalus kadamba*	60	6	T
339.		1T	*Pipal*	*Ficus religiosa*	60	6	T
340.		(8C+14T)22					
341.	VIP road to surat women IIT	20T	*Devils tree*	*20Alstonia schollaris*	60	6	T
342.		5T	*Kadam*	*5Anthocephalus kadamba*	60	10	T
343.		21C	*Tad*	*21Borassus flaberiformis*	50	12	C
344.		11T	*Yellow flame tree*	*11Peltoforam pterocarpum*	60	8	T
345.		6C	*Karank*	*6Pongamia pinnata*	50	8	C
346.		5C	*Tad*	*5Borassus flaberiformis*	50	12	C
347.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	100	10	T
348.		(33C+36T)69					
349.	Surat women IIT station	2C	Jungle jalebi	*2Pithecellobium dulce*	60	10	C
350.		4T	*Jamun*	*4Syzygium cumini*	60	4	T
351.		1T	*Pipal*	*Ficus religiosa*	60	10	T
352.		(2C+5T)7					
353.	Surat women IIT station to Bhimrad	1C	*Karanj*	*Pongamia pinnata*	40	4	C
354.		1C	*Karanj*	*Pongamia pinnata*	80	12	C
355.		1C	*Tad*	*Borassus flaberiformis*	80	12	C
356.		1C	*Jungle jalebi*	*Pithecellobium dulce*	60	10	C
357.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
358.		1C	*Karanj*	*Pongamia pinnata*	80	10	C
359.		1T	*Jamun*	*Syzygium cumini*	50	10	T
360.		1T	*Pipal*	*Ficus religiosa*	60	10	T
361.		1C	*Kachnar*	*Bahunia varigarta*	60	10	C
362.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	80	10	T
363.		(6C+4T)10					
364.	Bhimrad st	2C	*Khejari*	*2Prosopis julifera*	40	3	C

Sr. No	Place	Number	Common Name	Botanical name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/ Save (S)
365.		14C	*Talwar babool*	*14 Acacia auriculoformis*	60	10	C
366.		8T	*Yellow flame tree*	*8Peltoforam pterocarpum*	80	10	T
367.‘		3T	*Gulmohar*	*3Delonix regia*	80	10	T
368.		1T	*Talwar babool*	*Acacia auriculoformis*	60	10	T
369.		(16C+12T)28					
370.	Bhimrad st to Convention centre	28C	*Khejari*	*28 Prosopis julifera*	40	4	C
371.		(28C) 28					
372.	Convention centre st	1C	*Khejari*	*Prosopis julifera*	30	3	C
373.		1T	*Bargad*	*Ficus bengalensis*	40	4	T
374.		1C	*Tad*	*Borassus flaberiformis*	80	12	C
375.		1C	*Karanj*	*Pongamia pinnata*	60	6	C
376.		(3C+1T)4					
377.	Convention centre st to Dream city	17C	*Khejari*	*17Prosopis julifera*	40	10	C
378.		(17C)17					
379.	Dream city station	20C	*Khejari*	*20 Prosopis julifera*	60	8	C
380.		(20C)20					
381.	Dream city to end	24C	*Khejari*	*24 Prosopis julifera*	60	10	C
382.		(24C)24					
383.	Connecting line to depot	0					
384.	***Dream city depot (25.66 ha)***	(176C) 176					
385.		176 C	*Trees*	*Trees*	30	6	C
386.	**Total**	**1012**					

Note: Red marked trees – Tree to be cut for girth more than 100cm, Blue marked trees – Trees to be cut species wise

DETAILS OF TREE INUMERATION & INVENTORY
FOR BHESAN TO SAROLI

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
1.	***Bhesan depot (16.95 ha)***	42C	*Tad*	*42 Borassus flaberiformis*	80	12	C
2.		25C	*Suru*	*25 Eucalyptus globules*	60	12	C
3.		160C	*Subabool*	*160 Leucaena leucocephala*	60	6	C
4.		10C	*Khejari*	*10 Prosophis julifera*	40	4	C
5.		35T	*Mango*	*35 Mangifera indica*	60	5	T
6.		7T	*Neem*	*7 Azadirachta indica*	50	4	T
7.		40C	*Babool*	*40 Acacia nilotica*	60	4	C
8.		(277C+42T)319					
9.	Start to Bhesan	1C	*Neem*	*Azadirachta indica*	150	12	C
10.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	100	12	T
11.		1C	*Neem*	*Azadirachta indica*	110	12	C
12.		3C	*Yellow flame tree*	*3Peltoforam pterocarpum*	110	10	C
13.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	108	10	C
14.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	90	10	T
15.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	80	10	T
16.		2C	*Yellow flame tree*	*2Peltoforam pterocarpum*	130	10	C
17.		(8C+3T)11					
18.	Bhesan st	1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	6	T
19.		1C	*Neem*	*Azadirachta indica*	120	10	C
20.		(1C+1T)2					
21.	Bhesan st to botanical garden	1T	*Jamun*	*Syzygium cumini*	80	10	T
22.		1C	*Karanj*	*Pongamia pinnata*	60	10	C
23.		1T	*Email*	*Tamarindus indicus*	80	10	T
24.		1T	*Pipal*	*Ficus religiosa*	100	12	T
25.		(1C+3T)4					
26.	Botanical garden st	0					
27.		0					
28.	Botanical garden st to Ugat varigruha	3T	*Devils tree*	*3 Alstonia scholaris*	40	4	T

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
29.		1T	*Devils tree*	*Alstonia scholaris*	60	6	T
30.		1T	*Devils tree*	*Alstonia scholaris*	50	4	T
31.		(5T)5					
32.	Ugat varigruha st	12T	*Devils tree*	*12 Alstonia scholaris*	50	12	T
33.		(12T)12					
34.	Ugat varigruha st to palampur road station	10C	*Karanj*	*10Pongamia pinnata*	60	8	T
35.		7T	*Devils tree*	*7 Alstonia scholaris*	50	10	T
36.		3T	*Gulmohar*	*3Delonix regia*	60	8	T
37.		6T	*Yellow flame tree*	*6Peltoforam pterocarpum*	60	6	T
38.		1T	*Pipal*	*Ficus religiosa*	100	12	T
39.		1C	*Bottle-brush*	*Callistemon citrinus*	40	3	C
40.		39C	*Karanj*	*39Pongamia pinnata*	40	6	C
41.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	120	10	C
42.		(40C+1C+27T)68					
43.	Palampur road station	1T	*Gular*	*Ficus racemosa*	60	6	T
44.		3C	*Karanj*	*3Pongamia pinnata*	60	10	C
45.		1T	*Neem*	*Azadirachta indica*	80	10	T
46.		1C	*Booganvel*	*Bougainvillea glabra*	*60*	*3*	C
47.		5C	*Karanj*	*5Pongamia pinnata*	60	6	C
48.		2T	*Devils tree*	*2 Alstonia scholaris*	60	10	T
49.		1T	*Amrud*	*Psidium guajava*	*30*	*3*	T
50.		2T	*Neem*	*2Azadirachta indica*	30	3	T
51.		1T	*Kadam*	*Anthocephalus kadamba*	60	12	T
52.		1T	*Asoka*	*Polyalthia longiflia*	45	6	T
53.		2T	*Devils tree*	*2 Alstonia scholaris*	60	12	T
54.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	6	T
55.		(9C+12T)21					
56.	Palampur road station to L P Savani school	1T	*Pipal*	*Ficus religiosa*	60	6	T

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
57.		24T	*Devils tree*	*24 Alstonia scholaris*	40	6	T
58.		6C	*Karanj*	*6Pongamia pinnata*	60	10	C
59.		3T	*Gulmohar*	*3Delonix regia*	60	10	T
60.		4C	*Babool*	*4Acacia nilotica*	60	10	C
61.		13T	*Devils tree*	*13 Alstonia scholaris*	40	4	T
62.		1T	*Neem*	*Azadirachta indica*	60	10	T
63.		1T	*Gulmohar*	*Delonix regia*	60	6	T
64.		1C	*Karanj*	*Pongamia pinnata*	60	6	C
65.		1T	*Neem*	*Azadirachta indica*	60	6	T
66.		36T	*Devils tree*	*36 Alstonia scholaris*	60	5	T
67.		(11C+80T)91					
68.	L P Savani school st	1T	*Shisam*	*Dalbergia sissoo*	60	6	T
69.		1C	*Karanj*	*Pongamia pinnata*	80	12	C
70.		4T	*Yellow flame tree*	*4Peltoforam pterocarpum*	60	12	T
71.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	80	12	T
72.		1T	*Pipal*	*Ficus religiosa*	60	6	T
73.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
74.		2T	*Devils tree*	*2 Alstonia scholaris*	40	6	T
75.		1T	*Neem*	*Azadirachta indica*	30	4	T
76.		(1C+11T)12					
77.	L P Savani school to Performing art centre	4C	*Pili Kaner*	*4Cascabela thevetia*	*40*	*3*	*T*
78.		1T	*Neem*	*Azadirachta indica*	60	8	T
79.		2T	*Neem*	*2Azadirachta indica*	40	6	T
80.		10T	*Devils tree*	*10Alstonia scholaris*	40	4	T
81.		1T	*Semal*	*Bombax ceiba*	80	10	T
82.		1T	*Neem*	*Azadirachta indica*	60	6	T
83.		1T	*Kala siris*	*Albizia lebbeck*	*80*	*14*	T
84.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	110	14	C
85.		3T	*Gulmohar*	*3Delonix regia*	60	10	T

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
86.		1T	*Kadam*	*Anthocephalus kadamba*	60	6	T
87.		1T	*Devils tree*	*Alstonia scholaris*	60	6	T
88.		(1C+25T)26					
89.	Performing art centre st	1T	*Neem*	*Azadirachta indica*	60	10	T
90.		1T	*Taravad*	*Cassia auriculata*	60	10	T
91.		1T	*Devils tree*	*Alstonia scholaris*	60	10	T
92.		2T	*Devils tree*	*2Alstonia scholaris*	80	10	T
93.		1C	*Coconut*	*Coccos nucifera*	50	10	C
94.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
95.		1T	*Pipal*	*Ficus religiosa*	60	10	T
96.		1C	*Kala siris*	*Albizia lebbeck*	150	12	C
97.		2T	*Gulmohar*	*2Delonix regia*	80	10	T
98.		3C	*Kala siris*	*3Albizia lebbeck*	110	12	C
99.		(1C+4C+9T)14					
100.	Performing art centre st to Adajaan gam	53T	*Devils tree*	*53 Alstonia scholaris*	60	10	T
101.		19T	*Devils tree*	*19Alstonia scholaris*	80	10	T
102.		**1S**	***The great African baobab***	***Adansonia digitata***	**1120**	**14**	**SAVE**
103.		**1S**	***The great African baobab***	***Adansonia digitata***	**1020**	**14**	**Collapsed due to rain on 2-05-20**
104.		1T	*Pipal*	*Ficus religiosa*	60	10	T
105.		1T	*Pipal*	*Ficus religiosa*	50	10	T
106.		1T	*Asoka*	*Polyalthia longifolia*	60	10	T
107.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	80	10	T
108.		24T	*Devils tree*	*24 Alstonia scholaris*	40	4	T
109.		(2S+100T)102					
110.	Adajaan gam st	12T	*Devils tree*	*12 Alstonia scholaris*	60	6	T
111.		1T	*Neem*	*Azadirachta indica*	60	10	T
112.		2T	*Kadam*	*2Anthocephalus kadamba*	60	6	T

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
113.		1T	Pipal	Ficus religiosa	60	6	T
114.		(16T)16					
115.	Adajaan gam to Aquarium	5T	Yellow flame tree	5Peltoforam pterocarpum	60	10	T
116.		1C	Karanj	Pongamia pinnata	60	10	C
117.		2T	Gulmohar	2Delonix regia	60	10	T
118.		4T	Neem	4Azadirachta indica	40	6	T
119.		1T	Kadam	Anthocephalus kadamba	50	4	T
120.		2T	Kadam	2Anthocephalus kadamba	30	4	T
121.		(1C+14T)15					
122.	Aquarium st	1T	Kadam	Anthocephalus kadamba	30	3	T
123.		1T	Kadam	Anthocephalus kadamba	40	6	T
124.		1T	Gulmohar	Delonix regia	80	10	T
125.		1C	Bargad	Ficus benghalensis	180	12	C
126.		1T	Yellow flame tree	Peltoforam pterocarpum	80	8	T
127.		1T	Pipal	Ficus religiosa	80	12	T
128.		1C	Shevaga	Moringa oleifera	70	6	C
129.		1T	Pipal	Ficus religiosa	80	10	T
130.		1T	Neem	Azadirachta indica	60	10	T
131.		2T	Gulmohar	2Delonix regia	60	10	T
132.		3T	Asoka	3Polyalthia longiflolia	60	10	T
133.		1T	Gular	Ficus recemosa	60	12	T
134.		(2C+13T)15					
135.	Aquarium to Bhadri narayan temple	2T	Neem	2Azadirachta indica	90	12	T
136.		1C	Karanj	Pongamia poinnata	80	10	C
137.		1T	Neem	Azadirachta indica	60	12	T
138.		1T	Pipal	Ficus religiosa	80	12	T
139.		1T	Neem	Azadirachta indica	40	12	T
140.		1C	Gular	Ficus recemosa	110	12	C
141.		1T	Yellow flame tree	Peltoforam pterocarpum	80	12	T
142.		1C	Karanj	Pongamia pinnata	90	12	T

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
143.		1T	*Safed champa*	*Plumeria rubra*	60	6	T
144.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	120	14	C
145.		1C	*Karanj*	*Pongamia pinnata*	30	8	C
146.		1T	*kassod tree*	*Cassia siamea*	60	6	T
147.		1T	*Neem*	*Azadirachta indica*	40	10	T
148.		2C	*Karanj*	*2Pongamia poinnata*	60	10	C
149.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
150.		1T	*Pipal*	*Ficus religiosa*	60	10	T
151.		1C	*Babool*	*Acacia nilotica*	60	8	C
152.		1C	*Karanj*	*Pongamia pinnata*	50	10	C
153.		1C	*Beri*	*Ziziphus jujube*	60	10	C
154.		1T	*Neem*	*Azadirachta indica*	80	12	T
155.		1C	*Bargad*	*Ficus benghalensis*	180	12	C
156.		1T	*Devils tree*	*Alstonia scholaris*	60	12	T
157.		1T	*Neem*	*Azadirachta indica*	100	12	T
158.		1T	*Neem*	*Azadirachta indica*	90	12	T
159.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	110	12	C
160.		2T	*Asoka*	*2Polyalthia longifolia*	40	12	T
161.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	110	12	C
162.		(5C+3C+22T)30					
163.	Bhadri narayan temple st	1C	*Pipal*	*Ficus religiosa*	280	12	C
164.		1T	*Neem*	*Azadirachta indica*	80	10	T
165.		1C	*Gular*	*Ficus recemosa*	160	12	C
166.		1C	*Coconut*	*Coccos nucifera*	60	6	C
167.		1C	*Karanj*	*Pongamia pinnata*	60	7	C
168.		1C	*Coconut*	*Coccos nucifera*	60	10	C
169.		(2C+3C+1T)6					
170.	Bhadri narayan temple st to Ahathwa chopati	1T	*Asoka*	*Polyalthia longifolia*	40	12	T
171.		1C	*Karanj*	*Pongamia pinnata*	60	10	C

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
172.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
173.		2T	*Neem*	*2Azadirachta indica*	60	10	T
174.		1T	*Asoka*	*Polyalthia longifolia*	50	10	T
175.		1T	*Asoka*	*Polyalthia longifolia*	60	10	T
176.		10C	*Khejari*	*10Prosopis julifera*	30	6	C
177.		**4T**	***Mangrove***	***4Avicenia marina* (mangrove)**	**30**	**3**	T
178.		5C	*Khejari*	*5Prosopis julifera*	40	4	C
179.		1C	*Karanj*	*Pongamia pinnata*	60	6	C
180.		8T	*Asoka*	*8Polyalthia longifolia*	50	12	T
181.		17C	*Asoka*	*17Polyalthia longifolia*	60	12	C
182.		2T	*Neem*	*2Azadirachta indica*	80	12	T
183.		1T	*Neem*	*Azadirachta indica*	60	8	T
184.		5T	*Asoka*	*5Polyalthia longifolia*	40	10	T
185.		1C	*Suru*	*Casuarina equisetifolia*	60	10	C
186.		***1C***	***Bamboo***	***Dendrocalamus strictus (Grass)***	***120***	***12***	C
187.		2C	*Tad*	*2Borassus flaberiformis*	60	10	T
188.		2C	*Yellow flame tree*	*2Peltoforam pterocarpum*	110	10	C
189.		(36C+3C+28T) 67					
190.	Ahathwa chopati st	1T	*Neem*	*Azadirachta indica*	60	10	T
191.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	60	4	T
192.		4C	*Yellow flame tree*	*4Peltoforam pterocarpum*	110	10	C
193.		1T	*Badam*	*Terminalia catapa*	60	10	T
194.		4C	*Tad*	*4Borassus flaberiformis*	60	12	C
195.		5T	*Gulmohar*	*5Delonix regia*	60	10	T
196.		5C	*Tad*	*5Borassus flaberiformis*	90	12	T
197.		1C	*Karanj*	*Pongamia pinnata*	110	18	C
198.		5T	*Asoka*	*5Polyalthia longifolia*	60	12	T
199.		5T	*Neem*	*Azadirachta indica*	60	10	T
200.		2C	*Yellow flame tree*	*2Peltoforam pterocarpum*	110	12	C

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
201.		(7C+4C+20T)31					
202.	Ahathwa chopati st to Majura gate	1C	*Neem*	*Azadirachta indica*	190	12	C
203.		7T	*Yellow flame tree*	*7Peltoforam pterocarpum*	60	10	T
204.		2C	*Karanj*	*2Pongamia pinnata*	60	10	C
205.		5T	*Devils tree*	*5Alstonia scholaris*	50	10	T
206.		3T	*Badam*	*3Terminalia catapa*	60	8	T
207.		1C	*Karanj*	*Pongamia pinnata*	40	6	C
208.		(2C+2C+15T)19					
209.	Majura gate st	4T	*Yellow flame tree*	*4Peltoforam pterocarpum*	60	4	T
210.		1C	*Subabool*	*Leucaena leucocephala*	40	4	C
211.		1T	*Devils tree*	*Alstonia scholaris*	40	4	T
212.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	40	4	
213.		(1C+6T)7					
214.	Majura gate st to Udhana darwaja st	1T	*Gulmohar*	*Delonix regia*	60	6	T
215.		1T	*Badam*	*Terminalia catapa*	60	12	T
216.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	80	12	T
217.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	90	12	T
218.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	60	10	T
219.		1T	*kuda*	*Holarrhena pubescens*	60	6	T
220.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	80	10	T
221.		6T	*Gulmohar*	*6Delonix regia*	70	10	T
222.		2C	*Karanj*	*2Pongamia pinnata*	60	12	C
223.		1T	*Gulmohar*	*Delonix regia*	60	10	T
224.		1C	*Neem*	*Azadirachta indica*	120	12	C
225.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	110	12	C
226.		1T	*Semal*	*Bombax ceiba*	90	12	T
227.		2T	*Yellow flame tree*	*2Peltoforam pterocarpum*	60	12	T
228.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	4	T
229.		1T	*Pipal*	*Ficus religiosa*	60	12	T
230.		2T	*Pipal*	*2Ficus religiosa*	60	10	T

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
231.		1T	*Semal*	*Bombax ceiba*	80	10	T
232.		1T	*Neem*	*Azadirachta indica*	80	10	T
233.		1C	*Karanj*	*Pongamia pinnata*	60	10	C
234.		1C	*Bargad*	*Ficus benghalensis*	110	10	C
235.		2T	*Pipal*	*2Ficus religiosa*	80	12	T
236.		1T	*Devils tree*	*Alstonia scholaris*	60	10	T
237.		(5C+1C+27T) 33					
238.	Udhana darwaja st	1C	*Karanj*	*Pongamia pinnata*	60	10	C
239.		1T	*Pipal*	*Ficus religiosa*	60	10	T
240.		1T	*Pipal*	*Ficus religiosa*	80	10	T
241.		1T	*Pipal*	*Ficus religiosa*	80	12	T
242.		1T	*Neem*	*Azadirachta indica*	60	10	T
243.		2T	*Devils tree*	*2Alstonia scholaris*	60	10	T
244.		1T	*Neem*	*Azadirachta indica*	80	10	T
245.		1C	*Karanj*	*Pongamia pinnata*	60	10	C
246.		1T	*Pipal*	*Ficus religiosa*	60	12	T
247.		(2C+8T)10					
248.	Udhana darwaja st to Kamela Darwaja	6T	*Devils tree*	*6Alstonia scholaris*	80	10	T
249.		(6T)6					
250.	Kamela Darwaja st	1T	*Semal*	*Bombax ceiba*	100	8	T
251.		1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	12	T
252.		1C	*Babool*	*Acacia nilotica*	60	10	C
253.		1T	*Pipal*	*Ficus religiosa*	60	12	T
254.		1T	*Pipal*	*Ficus religiosa*	50	10	T
255.		(1C+4T)5					
256.	Kamela Darwaja st to Anjana farm	1T	*Yellow flame tree*	*Peltoforam pterocarpum*	60	10	T
257.		1T	*Semal*	*Bombax ceiba*	80	12	T
258.		1T	*Devils tree*	*Alstonia scholaris*	60	6	T
259.		1T	*Semal*	*Bombax ceiba*	40	10	T
260.		10T	*Devils tree*	*10Alstonia scholaris*	30	3	T

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
261.		(14T)14					
262.	Anjana farm ST	1T	*Devils tree*	*Alstonia scholaris*	30	6	T
263.		11T	*Devils tree*	*11Alstonia scholaris*	30	3	T
264.		1T	*Devils tree*	*Alstonia scholaris*	30	8	T
265.		(13T)13					
266.	Anjana farm st to Model town st	5T	*Devils tree*	*5Alstonia scholaris*	30	6	T
267.		1T	*Devils tree*	*Alstonia scholaris*	60	10	T
268.		45T	*Kadam*	*45Anthocephalus kadamba*	60	10	T
269.		(51T)51					
270.	Model town st	1T	*Pipal*	*Ficus religiosa*	60	10	T
271.		1C	*Karanj*	*Pongamia pinnata*	80	10	C
		6T	*Asoka*	*6 Polyalthia longifolia*	40	6	T
272.		1C	*Pipal*	*Ficus religiosa*	120	12	C
273.		1C	*Neem*	*Azadirachta indica*	120	12	C
274.		1C	*Karanj*	*Pongamia pinnata*	40	6	C
275.		1T	*Gulmohar*	*Delonix regia*	60	6	T
276.		1T	*Pipal*	*Ficus religiosa*	40	4	T
277.		1C	*Karanj*	*Pongamia pinnata*	40	3	T
278.		1T	*Pipal*	*Ficus religiosa*	60	4	T
279.		1T	*Pipal*	*Ficus recemosa*	50	8	T
280.		1C	*Gulmohar*	*Delonix regia*	110	12	C
281.		22T	*Kadam*	*22Anthocephalus kadamba*	60	10	T
282.		4T	*Kadam*	*4Anthocephalus kadamba*	60	6	T
283.		(3C+2C+ 38T) 43					
284.	Model town st to Magob	1T	*Kadam*	*Anthocephalus kadamba*	60	6	T
285.		1T	*Kadam*	*Peltoforam pterocarpum*	60	10	T
286.		1T	*Kadam*	*Anthocephalus kadamba*	60	10	T
287.		1C	*Safed champa*	*Plumeria rubra*	40	8	C
288.		1T	*Sagwan*	*Tectona grandis*	60	10	T
289.		1C	*Tad*	*Borassus flaberiformis*	50	6	C

Sr No	Place	Number	Common Name	Botanical Name	Girth (cm)	Height (m)	Cut (C)/ Transplant (T)/Save(S)
290.		1C	*Tad*	*Borassus flaberiformis*	50	4	C
291.		1T	*Pipal*	*Ficus religiosa*	60	10	T
292.		1T	*Pipal*	*Ficus religiosa*	60	6	T
293.		1C	*Yellow flame tree*	*Peltoforam pterocarpum*	110	12	C
294.		(3C+1C+6T)10					
295.	Magob st	0					
296.		0					
297.	Magob st to Bharat cancer hospital	0					
298.		0					
299.	Bharat cancer hospital st	1T	*Badam*	*Terminalia catapa*	80	10	T
300.		1T	*Neem*	*Azadirachta indica*	80	10	T
301.		(2T)2					
302.	Bharat cancer hospital st to saroli	1T	*Neem*	*Azadirachta indica*	40	6	T
303.		(1T)1					
304.	Saroli st	1	*Safed champa*	*Plumeria rubra*	30	3	T
305.		(1T)1					
306.	Saroli st to end	0					
307.		0					
308.	**Total**	**1081**					

Note: Red marked trees – Tree to be cut for girth more than 100cm, Blue marked trees – Trees to be cut species wise

ANNEXURE 5.2

VIBRATION LEVELS AT SENSITIVE RECEPTORS

S. No	Receptors	Distance from C/L	Elevated/ UG	Vibration in mm/sec				Vibration Threshold Criteria in mm/sec		
				Vibratory Pile Driver	Impact Pile Driver	TBM	Metro Operations	Older Residential Structures	Historic and Old Buildings	Fragile Buildings
Corridor I: Dream City - Sarthana										
	Educational									
1.	SGE College	16	Elevated	8.50	22.70	-	0.04	7.62	-	-
2.	Sardar Patel Vidhya Bhawan	26	Elevated	5.56	15.24	-	0.03	7.62	-	-
3.	Tapovan Science And Commerce Academy	13	Elevated	6.96	22.73	-	0.05	7.62	-	-
4.	Suman School	53	Elevated	1.01	6.00	-	0.02	7.62	-	-
5.	Little Flower High School	66	UG	-	-	0.75	0.02	7.62	-	-
6.	Anglo Urdu High School	82	UG	-	-	0.57	0.02	7.62	-	-
7.	IP Mission School	93	UG	-	-	0.49	0.02	7.62	-	-
8.	J J Training College	47	UG	-	-	1.13	0.02	-	6.35	
9.	Andrew Library	14	UG	-	-	3.54	0.04	7.62	-	-
10.	Ramkrishna Vidhya Bhavan	21	UG	-	-	2.63	0.04	7.62	-	-
11.	Gandhi College	98	Elevated	0.20	1.50	-	0.02	7.62	-	-
12.	ICFAI University	52	Elevated	1.08	6.00	-	0.02	7.62	-	-
13.	Institute Of Design And Center	25	Elevated	2.09	12.08	-	0.03	7.62	-	-
14.	Kinder Dream Pre School	33	Elevated	1.82	9.44	-	0.03	7.62	-	-
15.	Ishwarlal Gulabbhai Desai Vidyalaya	51	Elevated	1.16	6.10	-	0.02	7.62	-	-
16.	School	13	Elevated	3.45	19.22	-	0.05	7.62	-	-
17.	Symga High School	100	Elevated	0.20	1.40	-	0.02	7.62	-	-
	Hospital									
1.	Shree Physioteraphy And Fitness Clinic	18	Elevated	8.28	21.43	-	0.04	-	-	2.54
2.	Shree Krishna Dental Hospital	16	Elevated	8.50	22.70	-	0.04	-	-	2.54
3.	Patel Hospital And Health Center	12	Elevated	9.35	25.66	-	0.05	-	-	2.54

S. No	Receptors	Distance from C/L	Elevated/ UG	Vibration in mm/sec				Vibration Threshold Criteria in mm/sec		
				Vibratory Pile Driver	Impact Pile Driver	TBM	Metro Operations	Older Residential Structures	Historic and Old Buildings	Fragile Buildings
4.	New Life Hospital	32	Elevated	5.36	13.23	-	0.03	-	-	2.54
5.	Vedanta Hospital & ICU	45	Elevated	-	-	-	0.02	-	-	2.54
6.	Vatsalya Hospital & Apple Skin Care	8	UG	-	-	4.53	0.06	-	-	2.54
7.	Vatsalya Hospital	75	UG	-	-	0.64	0.02	-	-	2.54
8.	Santosh Hospital	72	UG	-	-	0.67	0.02	-	-	2.54
9.	Women Hospital And Baby Tube Center	74	UG	-	-	0.65	0.02	-	-	2.54
10.	Radhey Hospital And Maternity Home	40	UG	-	-	1.37	0.03	-	-	2.54
11.	Akshar Hospital	92	UG	-	-	0.50	0.02	-	-	2.54
12.	Rainbow Children Hospital	32	UG	-	-	1.75	0.03	-	-	2.54
13.	Burhani Hospital	42	UG	-	-	1.29	0.03	-	-	2.54
14.	Dumaswala Hospital	100	UG	-	-	0.45	0.02	-	-	2.54
15.	Maskati Dharmarth Hospital	30	UG	-	-	1.87	0.03	-	-	2.54
16.	Eye Hospital	73	UG	-	-	0.66	0.02	-	-	2.54
17.	Kapse Children Hospital	25	UG	-	-	2.24	0.03	-	-	2.54
18.	Paavan Orthopedic Hospital	27	UG	-	-	2.08	0.03	-	-	2.54
19.	District TB Center	23	Elevated	2.21	13.05	-	0.03	-	-	2.54
20.	Girish Group Of Hospital	25	Elevated	2.09	12.08	-	0.03	-	-	2.54
21.	Rahul Hospital And Clinic	26	Elevated	2.05	11.72	-	0.03	-	-	2.54
22.	Swagat Hospital	27	Elevated	2.01	11.36	-	0.03	-	-	2.54
23.	Ketan Children Hospital	100	Elevated	0.20	1.40	-	0.02	-	-	2.54
24.	Leela Health Care	18	Elevated	2.62	15.77	-	0.04	-	-	2.54
25.	Sharda Hospital	13	Elevated	3.45	19.22	-	0.05	-	-	2.54
26.	Vasan Eye Care Hospital	26	Elevated	2.05	11.72	-	0.03	-	-	2.54
27.	Baby Beat Heart Center	34	Elevated	1.80	9.16	-	0.03	-	-	2.54
28.	Harihar ENT Hospital	12	Elevated	3.69	20.00	-	0.05	-	-	2.54

S. No	Receptors	Distance from C/L	Elevated/ UG	Vibration in mm/sec				Vibration Threshold Criteria in mm/sec		
				Vibratory Pile Driver	Impact Pile Driver	TBM	Metro Operations	Older Residential Structures	Historic and Old Buildings	Fragile Buildings
29.	Smt. Deepika & Dr. Hasmukhlal C Vankawala Children Hospital	42	Elevated	1.64	7.47	-	0.03	-	-	2.54
30.	Arora Surgical Hospital	28	Elevated	1.98	10.99	-	0.03	-	-	2.54
31.	Tulsi Hospital	46	Elevated	1.53	6.55	-	0.02	-	-	2.54
32.	Pooja Hospital	11	Elevated	3.94	20.77	-	0.05	-	-	2.54
33.	Shree Guru Nanak Dharmrath Hospital	75	Elevated	0.40	3.40	-	0.02	-	-	2.54
34.	Park Nursing Home	68	Elevated	0.50	4.00	-	0.02	-	-	2.54
35.	Rainbow Hospital	32	Elevated	1.85	9.71	-	0.03	-	-	2.54
36.	Pal Hospital	13	Elevated	3.45	19.22	-	0.05	-	-	2.54
37.	Sanjivani Hospital & ICCU Center	78	Elevated	0.40	3.00	-	0.02	-	-	2.54
38.	Amruta Hospital	54	Elevated	0.93	5.50	-	0.02	-	-	2.54
39.	Sanjivani Hospital	16	Elevated	2.84	17.04	-	0.04	-	-	2.54
40.	Jugal Hospital	9	Elevated	4.37	22.26	-	0.05	-	-	2.54
41.	Sonal Hospital and Diabetes Clinic	95	UG	-	-	0.48	0.02	-	-	2.54
42.	Bombay Multi Speciality Hospital	32	Elevated	1.85	9.71		0.03	-	-	2.54
	Religious Places									
1.	Shyam Dham Mandir	54	Elevated	6.60	10.20	-	0.02	7.62	-	-
2.	Shyam Temple	47	Elevated	7.11	11.80	-	0.02	7.62	-	-
3.	Kalakunj Swami Naryan Mandir	41	Elevated	1.66	7.64		0.03	-	6.35	-
4.	Shri Sambanath Jain Temple	54	UG	-	-	0.96	0.02	7.62	-	-
5.	Ramkrishna Vidhya Bhavan	21	Elevated	2.34	14.02		0.04	7.62	-	-
6.	Tarani Devi Temple	30	UG	-	-	1.87	0.03	7.62	-	-
7.	Lambey Hanuman Temple	33	UG	-	-	1.69	0.03	-	6.35	
8.	Shri Goverdhan Nathji Bethakji	27	UG	-	-	2.08	0.03	-	6.35	
9.	Momin Masjid	5	UG	-	-	4.97	0.07	7.62	-	-
10.	Jagnat Mahadev Mandir	91	UG	-	-	0.50	0.02	7.62	-	-

S. No	Receptors	Distance from C/L	Elevated/ UG	Vibration in mm/sec				Vibration Threshold Criteria in mm/sec		
				Vibratory Pile Driver	Impact Pile Driver	TBM	Metro Operations	Older Residential Structures	Historic and Old Buildings	Fragile Buildings
11.	Shree Santinath Bhagwan Jain Derasar	100	Elevated	0.20	0.20	-	0.02	7.62	-	-
12.	Aebi Gebi Hanuman Temple	76	UG	-	-	0.63	0.02	7.62	-	-
13.	Patalia Hanuman	31	UG	-	-	1.81	0.03	7.62	-	-
14.	Minara masjid	15	UG	-	-	3.39	0.04	-	6.35	-
15.	Bhagal masjid	1	UG	-	-	5.31	0.16	-	6.35	-
16.	Chand sahid masjid	60	UG	-	-	0.84	0.02	-	6.35	-
17.	Hussain masjid	50	UG	-	-	1.05	0.02	-	6.35	-
18.	Markaz Masjid	51	UG	-	-	1.03	0.02	-	6.35	-
19.	Badi Masjid	67	UG	-	-	0.74	0.02	-	6.35	-
20.	Masjid e Kadim	6	UG	-	-	4.84	0.07	-	6.35	-
21.	Sabiri Masjid	53	UG	-	-	0.98	0.02	-	6.35	-
22.	Chruch Of North India	73	UG	-	-	0.66	0.02	-	6.35	-
23.	Bethany House of Worship	70	UG	-	-	0.70	0.02	7.62		-
24.	Masjid e Hamayu Sarai	77	UG	-	-	0.62	0.02	-	6.35	-
25.	Shri Adinath Jain Derasar	55	UG	-	-	0.94	0.02	-	6.35	-
26.	Raj Rajeshwar Mahadev	5	UG	-	-	4.97	0.07	7.62	-	-
27.	Masjid e Nagina	13	UG	-	-	3.70	0.05	7.62	-	-
28.	CNI Christ Chruch	1	UG	-	-	5.31	0.16	-	6.35	-
29.	Bakriwali Masjid	23	UG	-	-	2.43	0.03	-	6.35	-
30.	Jaghnath Mahadev	57	UG	-	-	0.90	0.02	7.62	-	-
31.	Mubarak Mosque	10	UG	-	-	4.19	0.05	-	6.35	-
32.	Botwala Masjid	32	UG	-	-	1.75	0.03	-	6.35	-
33.	Chandraprabhu Swamis Jain Temple	62	Elevated	0.80	4.90	-	0.02	7.62		-
34.	Mulla Hashim Masjid	41	Elevated	1.66	7.64	-	0.03		6.35	-
35.	Hatimi Masjid	62	Elevated	0.80	4.90	-	0.02	7.62		-

S. No	Receptors	Distance from C/L	Elevated/ UG	Vibration in mm/sec				Vibration Threshold Criteria in mm/sec		
				Vibratory Pile Driver	Impact Pile Driver	TBM	Metro Operations	Older Residential Structures	Historic and Old Buildings	Fragile Buildings
36.	Aksa Masjid	43	Elevated	1.63	7.31	-	0.03	-	6.35	-
37.	Tulja Maa Mandir	40	Elevated	1.67	7.81	-	0.03	-	6.35	-
38.	Sayed Jafer Sahid Bawa Dargah	78	Elevated	0.40	3.00	-	0.02	-	6.35	-
39.	Khankah E Chishtiya Nizamiya Rehmaniya	50	Elevated	1.23	6.30	-	0.02	-	6.35	-
40.	Ahle Sunnat Wal Jamaat Chhoti Masjid	35	Elevated	1.77	8.88	-	0.03	-	6.35	-
41.	Hazrat Murad Shah Bava Rahmatullahi Alaihi	32	Elevated	1.85	9.71	-	0.03	-	6.35	-
42.	Ahle Sunnat Wal Jamat Hazrat Jamalshah Masjid	21	Elevated	2.34	14.02	-	0.04	-	6.35	-
43.	Sai Mandir	37	Elevated	1.73	8.45	-	0.03	7.62	-	-
44.	Hazrat Peer Syed Hasan Shahid Rahmatullahi Alaihi	21	Elevated	2.34	14.02	-	0.04	7.62	-	-
45.	Shree Ananddayak Vasu Pujay Swami Jain Sang	21	Elevated	2.34	14.02	-	0.04	7.62	-	-
46.	Shri Ram Temple	10	Elevated	4.19	21.55	-	0.05	7.62	-	-
47.	Shankeshwar Da jain Temple	62	Elevated	0.80	4.90	-	0.02	7.62	-	-
48.	Baba Amarnath Panchmukhi Temple	39	Elevated	1.69	8.02	-	0.03	7.62	-	-
49.	Khodiyar Mata Ji Temple	24	Elevated	2.15	12.56	-	0.03	-	6.35	-
50.	SAI Baba Temple	43	Elevated	1.63	7.31	-	0.03	-	6.35	-
51.	Dade Elahi Masjid	27	UG	0.00	0.00	2.08	0.03	-	6.35	-
52.	Temple	26	Elevated	2.05	11.72	-	0.03	7.62	-	-
53.	Temple	17	Elevated	2.73	16.40	-	0.04	7.62	-	-
54.	Masjid	73	Elevated	0.50	3.50	-	0.02	7.62	-	-
55.	Temple	32	Elevated	1.85	9.71	-	0.03	7.62	-	-
56.	Kahar Machi Samaj Temple	68	Elevated	0.60	4.00	-	0.02		6.35	-
57.	Mosque	12	Elevated	3.69	20.00	-	0.05	7.62	-	-
58.	Mastana Shah Dargah	28	Elevated	1.98	10.99	-	0.03	-	6.35	-
59.	Masjid	14	Elevated	3.20	18.45	-	0.04	7.62	-	-

S. No	Receptors	Distance from C/L	Elevated/ UG	Vibration in mm/sec				Vibration Threshold Criteria in mm/sec		
				Vibratory Pile Driver	Impact Pile Driver	TBM	Metro Operations	Older Residential Structures	Historic and Old Buildings	Fragile Buildings
Corridor II: Bhesan - Saroli										
	Educational									
1.	South Gujarat Medical College	62	Elevated	1.90	5.40	-	0.02	7.62	-	-
2.	Perfact Brilliant Classess	17	Elevated	3.29	16.95	-	0.04	7.62	-	-
3.	Lord Krishna School	93	Elevated	1.10	2.20	-	0.02	7.62	-	-
4.	Vidhya Kunj School	54	Elevated	1.95	6.50	-	0.02	7.62	-	-
5.	L P Savani School	37	Elevated	2.28	9.01	-	0.03	7.62	-	-
6.	Friends Classess	37	Elevated	2.28	9.01	-	0.03	7.62	-	-
7.	Dakshina	11	Elevated	4.49	21.32	-	0.05	7.62	-	-
8.	Nutan Vidhyalay Madhyamik School	15	Elevated	4.74	22.10	-	0.04	7.62	-	-
9.	KP College Of Commerce	77	Elevated	0.22	2.80	-	0.02	7.62	-	-
10.	INFID	89	Elevated	0.10	2.20	-	0.02	7.62	-	-
11.	Dr S.S. Gandhi Engineering College	100	Elevated	0.20	1.40	-	0.02	7.62	-	-
12.	G & G. V. Kadilwal	92	Elevated	0.20	2.00	-	0.02	7.62	-	-
13.	RMG Maheswari Public School	10	Elevated	4.19	21.55	-	0.05	7.62	-	-
14.	Rose Buds Nursery School	26	Elevated	2.05	11.72	-	0.03	7.62	-	-
15.	Lord Krishna International School	100	Elevated	0.20	1.40	-	0.02	7.62	-	-
16.	Kidzee School	64	Elevated	0.70	4.30	-	0.02	7.62	-	-
	Religious Places									
1.	Sati Mata Temple	42	Elevated	2.19	7.94	-	0.03	-	6.35	-
2.	Shree Sidhi Vinayak Temple	37	Elevated	2.28	9.01	-	0.03	7.62	-	-
3.	Shanidev Temple	35	Elevated	2.32	9.44	-	0.03	7.62	-	-
4.	Shree Ram Temple	80	Elevated	1.30	2.00	-	0.02	7.62	-	-
5.	Shigontar Mata Mandir	47	Elevated	2.10	6.87	-	0.02	7.62	-	-
6.	Shri Rameshwar Neelkanth Temple	97	Elevated	1.10	1.10	-	0.02	7.62	-	-

S. No	Receptors	Distance from C/L	Elevated/ UG	Vibration in mm/sec				Vibration Threshold Criteria in mm/sec		
				Vibratory Pile Driver	Impact Pile Driver	TBM	Metro Operations	Older Residential Structures	Historic and Old Buildings	Fragile Buildings
7.	Amba Mata Temple	17	Elevated	3.29	16.95	-	0.04	-	6.35	-
8.	Jalaram Temple	32	Elevated	2.40	10.27	-	0.03	7.62	-	-
9.	Bhrugudeep Jalram Temple	16	Elevated	3.40	17.59	-	0.04	7.62	-	-
10.	Aaj Shree Khodiyar Mata Ji Mandir	15	Elevated	3.51	18.22	-	0.04	7.62	-	-
11.	Badrinarayan Temple	10	Elevated	4.74	22.10	-	0.05	-	6.35	-
12.	Bholenath Temple	20	Elevated	2.95	15.05	-	0.04	7.62		-
13.	Surat Yatim Khana Masjid	65	Elevated	1.00	4.50	-	0.02	-	6.35	-
14.	Gajjarwadi Jain Temple	90	Elevated	1.20	0.90	-	0.02	7.62		-
15.	Udhwana Darwaja Masjid	41	Elevated	1.66	7.64	-	0.03	-	6.35	-
16.	Bileshwar Mahadev Mandir	50	Elevated	1.23	6.30	-	0.02	-	6.35	-
17.	Masjide Aaisha	11	Elevated	3.94	20.77	-	0.05	7.62	-	-
18.	Shanti Nath Swetambari Mandir	67	Elevated	0.50	4.30	-	0.02	7.62	-	-
19.	Sai Baba Temple	17	Elevated	2.73	16.40	-	0.04	7.62	-	-
20.	Masjid/Temple	5	Elevated	5.43	25.43	-	0.07	7.62	-	-
	Hospital									
1.	Palanpor Urban Health Center	99	Elevated	1.00	1.80	-	0.02	-	-	2.54
2.	Sai Medical Hospital & Maternity Home	30	Elevated	2.45	10.82	-	0.03	-	-	2.54
3.	Arpan Children Hospital	23	Elevated	2.77	13.60	-	0.03	-	-	2.54
4.	Shubham Hospital	37	Elevated	2.28	9.01	-	0.03	-	-	2.54
5.	Divyesh Eye Hospital	25	Elevated	2.65	12.64	-	0.03	-	-	2.54
6.	Dhameliya Kidney Hospital & Lithotripsy Centre	15	Elevated	3.51	18.22	-	0.04	-	-	2.54
7.	BAPS Pramukh Swami Hospital	43	Elevated	1.63	7.31	-	0.03	-	-	2.54
8.	Adventist Work Hardt Hospital	50	Elevated	1.19	5.70	-	0.02	-	-	2.54
9.	Care Hospital	27	Elevated	2.01	11.36	-	0.03	-	-	2.54
10.	Bridge Cross Hospital	62	Elevated	0.70	4.90	-	0.02	-	-	2.54

S. No	Receptors	Distance from C/L	Elevated/ UG	Vibration in mm/sec				Vibration Threshold Criteria in mm/sec		
				Vibratory Pile Driver	Impact Pile Driver	TBM	Metro Operations	Older Residential Structures	Historic and Old Buildings	Fragile Buildings
11.	Kalpa Hospital	20	Elevated	2.40	14.50	-	0.04	-	-	2.54
12.	Tri Star Hospital	46	Elevated	1.24	1.24	-	0.02	-	-	2.54
13.	Tender Touch Children Hospital	55	Elevated	0.86	5.60	-	0.02	-	-	2.54
14.	G D Mehata Mahavir Heart Hospital	38	Elevated	1.71	8.24	-	0.03	-	-	2.54
15.	Shreyas Ano-Rectal Hospital & Research Centre	10	Elevated	4.19	21.55	-	0.05	-	-	2.54
16.	Arora Surgical Hospital	47	Elevated	1.57	6.63	-	0.02	-	-	2.54
17.	Nirmal Hospital	11	Elevated	3.94	20.77	-	0.05	-	-	2.54
18.	Navsarjan Hospital	80	Elevated	0.30	2.90	-	0.02	-	-	2.54
19.	Apple Hospital	48	Elevated	1.56	6.47	-	0.02	-	-	2.54
20.	Vatsalya Children Hospital	72	Elevated	0.60	4.00	-	0.02	-	-	2.54
21.	Chirayu Hospital	60	Elevated	0.80	5.00	-	0.02	-	-	2.54
22.	Bharat Cancer Hospital	26	Elevated	2.05	11.72	-	0.03	-	-	2.54

ANNEXURE 5.3

Cumulative Noise Levels at Receptors due to Metro Train Operations

S. No	Receptor	Distance from Metro in m	Noise Baseline Data in dB(A) of nearest monitoring location			Cumulative Noise for Year 2021		Noise Criteria
			Location	Lday	Lnight	Lday	Lnight	
Corridor I: Dream City - Sarthana								
	Educational							
1.	SGE College	16	N19	68.5	59.6	69.2	60.8	No Impact
2.	Sardar Patel Vidhya Bhawan	26	N10	69.2	57.3	69.4	58.1	No Impact
3.	Tapovan Science and Commerce Academy	13	N10	69.2	57.3	70.1	59.9	No Impact
4.	Suman School	53	N12	68.6	57.1	68.7	57.3	No Impact
5.	Gandhi College	98	N6	69.8	50.4	69.8	50.7	No Impact
6.	ICFAI University	52	N6	69.8	50.4	69.9	51.4	No Impact
7.	Institute Of Design And Center	25	N15	69.0	49.4	69.0	50.3	No Impact
8.	Kinder Dream Pre School	33	N15	69.0	49.4	69.2	52.0	No Impact
9.	Ishwarlal Gulabbhai Desai Vidyalaya	51	N16	65.3	54.9	65.5	55.3	No Impact
10.	School	13	N16	65.3	54.9	67.3	58.8	No Impact
	Hospital							
1.	Shree Physioteraphy And Fitness Clinic	18	N19	68.5	59.6	69.1	60.6	No Impact
2.	Shree Krishna Dental Hospital	16	N19	68.5	59.6	69.2	60.8	No Impact
3.	Patel Hospital And Health Center	12	N19	68.5	59.6	69.7	61.6	No Impact
4.	New Life Hospital	32	N19	68.5	59.6	68.7	59.9	No Impact
5.	Vedanta Hospital & ICU	45	N10	69.2	57.3	69.3	57.6	No Impact
6.	Paavan Orthopedic Hospital	27	N14	67.1	57.2	67.5	58.0	No Impact
7.	District TB Center	23	N6	69.8	50.4	70.1	54.0	No Impact
8.	Girish Group Of Hospital	25	N6	69.8	50.4	70.0	53.6	No Impact
9.	Rahul Hospital And Clinic	26	N6	69.8	50.4	70.0	53.5	No Impact
10.	Swagat Hospital	27	N6	69.8	50.4	70.0	53.3	No Impact
11.	Ketan Children Hospital	100	N6	69.8	50.4	69.8	50.7	No Impact
12.	Leela Health Care	18	N6	69.8	50.4	70.2	55.4	No Impact
13.	Sharda Hospital	13	N6	69.8	50.4	70.6	57.5	No Impact
14.	Vasan Eye Care Hospital	26	N6	69.8	50.4	70.0	53.5	No Impact
15.	Baby Beat Heart Center	34	N6	69.8	50.4	69.9	52.4	No Impact
16.	Harihar ENT Hospital	12	N6	69.8	50.4	70.0	52.9	No Impact
17.	Smt. Deepika & Dr. Hasmukhlal C Vankawala Children Hospital	42	N6	69.8	50.4	69.8	50.7	No Impact
18.	Arora Surgical Hospital	28	N6	69.8	50.4	69.8	51.0	No Impact
19.	Tulsi Hospital	46	N6	69.8	50.4	69.9	51.6	No Impact
20.	Pooja Hospital	11	N15	69.0	49.4	70.3	58.6	Severe Impact -

S. No	Receptor	Distance from Metro in m	Noise Baseline Data in dB(A) of nearest monitoring location			Cumulative Noise for Year 2021		Noise Criteria
			Location	Lday	Lnight	Lday	Lnight	
								Night
21.	Shree Guru Nanak Dharmrath Hospital	75	N15	69.0	49.4	69.0	50.0	No Impact
22.	Park Nursing Home	68	N15	69.0	49.4	69.0	50.2	No Impact
23.	Rainbow Hospital	32	N15	69.0	49.4	69.2	52.1	No Impact
24.	Pal Hospital	13	N15	69.0	49.4	69.9	57.3	Severe Impact - Night
25.	Sanjivani Hospital & ICCU Center	78	N15	69.0	49.4	69.0	50.0	No Impact
26.	Amruta Hospital	54	N15	69.0	49.4	69.1	50.5	No Impact
27.	Sanjivani Hospital	16	N15	69.0	49.4	69.6	55.9	Severe Impact - Night
28.	Jugal Hospital	9	N15	69.0	49.4	70.8	60.1	Severe Impact - Night
29.	Bombay Multi Speciality Hospital	32	N16	65.3	54.9	65.7	55.8	No Impact
	Religious Places							
1.	Shyam Dham Mandir	54	N19	68.5	59.6	68.6	59.7	No Impact
2.	Shyam Temple	47	N19	68.5	59.6	68.5	59.6	No Impact
3.	Kalakunj Swami Naryan Mandir	41	N11	65.9	54	66.1	54.7	No Impact
4.	Shri Sambanath Jain Temple	54	N12	68.6	57.1	68.7	57.3	No Impact
5.	Ramkrishna Vidhya Bhavan	21	N12	68.6	57.1	69.0	58.4	No Impact
6.	Tarani Devi Temple	30	N13	65.7	53.8	66.1	55.1	No Impact
7.	Lambey Hanuman Temple	33	N13	65.7	53.8	66.0	54.9	No Impact
8.	Shri Goverdhan Nathji Bethakji	27	N13	65.7	53.8	66.2	55.4	No Impact
9.	Jagnat Mahadev Mandir	91	N13	65.7	53.8	65.7	54.0	No Impact
10.	Shree Santinath Bhagwan Jain Derasar	100	N13	65.7	53.8	65.7	53.9	No Impact
11.	Aebi Gebi Hanuman Temple	76	N13	65.7	53.8	65.8	54.0	No Impact
12.	Patalia Hanuman	31	N13	65.7	53.8	66.1	55.0	No Impact
13.	Minara masjid	15	N13	65.7	53.8	67.1	57.6	Severe Impact - Night
14.	Chand sahid masjid	60	N13	65.7	53.8	65.8	54.2	No Impact
15.	Hussain masjid	50	N13	65.7	53.8	65.8	54.3	No Impact
16.	Markaz Masjid	51	N13	65.7	53.8	65.8	54.3	No Impact
17.	Badi Masjid	67	N13	65.7	53.8	65.8	54.1	No Impact
18.	Sabiri Masjid	53	N13	65.7	53.8	65.8	54.3	No Impact
19.	Chruch Of North India	73	N14	67.10	57.2	67.2	57.3	No Impact
20.	Bethany House of Worship	70	N14	67.10	57.2	67.2	57.3	No Impact
21.	Masjid e Hamayu Sarai	77	N14	67.10	57.2	67.1	57.3	No Impact
22.	Shri Adinath Jain Derasar	55	N14	67.10	57.2	67.2	57.4	No Impact
23.	Masjid e Nagina	13	N14	67.10	57.2	68.5	59.9	No Impact

S. No	Receptor	Distance from Metro in m	Noise Baseline Data in dB(A) of nearest monitoring location			Cumulative Noise for Year 2021		Noise Criteria
			Location	Lday	Lnight	Lday	Lnight	
24.	Bakriwali Masjid	23	N14	67.10	57.2	67.6	58.3	No Impact
25.	Jaghnath Mahadev	57	N14	67.10	57.2	67.2	57.4	No Impact
26.	Mubarak Mosque	10	N14	67.10	57.2	69.2	61.1	Severe Impact - Night
27.	Botwala Masjid	32	N14	67.10	57.2	67.4	57.8	No Impact
28.	Chandraprabhu Swamis Jain Temple	62	N14	67.10	57.2	67.2	57.4	No Impact
29.	Mulla Hashim Masjid	41	N14	67.10	57.2	67.3	57.6	No Impact
30.	Hatimi Masjid	62	N14	67.10	57.2	67.2	57.4	No Impact
31.	Aksa Masjid	43	N14	67.10	57.2	67.2	57.5	No Impact
32.	Tulja Maa Mandir	40	N14	67.10	57.2	67.3	57.6	No Impact
33.	Sayed Jafer Sahid Bawa Dargah	78	N14	67.10	57.2	67.1	57.3	No Impact
34.	Khankah E Chishtiya Nizamiya Rehmaniya	50	N14	67.10	57.2	67.2	57.4	No Impact
35.	Ahle Sunnat Wal Jamaat Chhoti Masjid	35	N14	67.10	57.2	67.3	57.7	No Impact
36.	Hazrat Murad Shah Bava Rahmatullahi Alaihi	32	N14	67.10	57.2	67.4	57.8	No Impact
37.	Ahle Sunnat Wal Jamat Hazrat Jamalshah Masjid	21	N14	67.10	57.2	67.7	58.4	No Impact
38.	Sai Mandir	37	N6	69.8	50.4	69.9	52.2	No Impact
39.	Hazrat Peer Syed Hasan Shahid Rahmatullahi Alaihi	21	N6	69.8	50.4	70.1	54.5	Severe Impact - Night
40.	Shree Ananddayak Vasu Pujay Swami Jain Sang	21	N15	69.0	49.4	69.4	54.1	Severe Impact - Night
41.	Shri Ram Temple	10	N15	69.0	49.4	70.5	59.3	Severe Impact - Night
42.	Shankeshwar Da jain Temple	62	N6	69.8	50.4	69.8	51.1	No Impact
43.	Baba Amarnath Panchmukhi Temple	39	N15	69.0	49.4	69.0	49.8	No Impact
44.	Khodiyar Mata Ji Temple	24	N16	65.3	54.9	65.4	55.2	No Impact
45.	SAI Baba Temple	43	N15	69.0	49.4	69.1	51.1	No Impact
46.	Dade Elahi Masjid	27	N13	65.7	53.8	66.2	55.4	No Impact
47.	Kahar Machi Samaj Temple	68	N14	67.10	57.20	67.2	57.3	No Impact
48.	Mastana Shah Dargah	28	A5	64.5	51.9	65.1	54.0	No Impact
Corridor II: Bhesan - Saroli								
	Educational							
1.	South Gujarat Medical College	62	N1	65.3	48.9	65.5	51.1	No Impact
2.	Perfact Brilliant Classess	17	N2	68.7	57.5	69.8	60.9	Severe Impact - Night
3.	Lord Krishna School	93	N2	68.7	57.5	68.7	57.7	No Impact
4.	Vidhya Kunj School	54	N2	68.7	57.5	68.8	58.0	No Impact

S. No	Receptor	Distance from Metro in m	Noise Baseline Data in dB(A) of nearest monitoring location			Cumulative Noise for Year 2021		Noise Criteria
			Location	Lday	Lnight	Lday	Lnight	
5.	L P Savani School	37	N3	69.6	58.5	69.8	59.3	No Impact
6.	Friends Classess	37	N3	69.6	58.5	69.8	59.3	No Impact
7.	Dakshina	11	N4	70.6	57.1	72.2	63.3	Severe Impact - Night
8.	Nutan Vidhyalay Madhyamik School	15	N5	64.5	51.9	67.5	60.1	Severe Impact - Night
9.	KP College Of Commerce	77	N5	64.5	51.9	64.7	52.7	No Impact
10.	INFID	89	N6	69.8	50.4	69.8	51.3	No Impact
11.	Dr S.S. Gandhi Engineering College	100	N6	69.8	50.4	69.8	51.1	No Impact
12.	G & G. V. Kadilwal	92	N6	69.8	50.4	69.8	51.2	No Impact
13.	RMG Maheswari Public School	10	N8	64.2	51.9	69.6	63.2	Severe Impact
14.	Rose Buds Nursery School	26	N6	69.8	50.4	70.2	56.0	Severe Impact - Night
15.	Lord Krishna International School	100	N8	64.2	51.9	64.3	52.4	No Impact
16.	Kidzee School	64	N8	64.2	51.9	64.5	53.1	No Impact
	Religious Places							
1.	Sati Mata Temple	42	N1	65.3	48.9	65.7	52.7	Severe Impact - Night
2.	Shree Sidhi Vinayak Temple	37	N1	65.3	48.9	65.9	53.4	Severe Impact - Night
3.	Shanidev Temple	35	N2	68.7	57.5	69.0	58.6	No Impact
4.	Shree Ram Temple	80	N2	68.7	57.5	68.8	57.7	No Impact
5.	Shigontar Mata Mandir	47	N2	68.7	57.5	68.9	58.1	No Impact
6.	Shri Rameshwar Neelkanth Temple	97	N2	68.7	57.5	68.7	57.7	No Impact
7.	Amba Mata Temple	17	N2	68.7	57.5	69.8	60.9	Severe Impact - Night
8.	Jalaram Temple	32	N3	69.6	58.5	69.9	59.5	No Impact
9.	Bhrugudeep Jalram Temple	16	N3	69.6	58.5	70.7	61.7	Severe Impact - Night
10.	Aaj Shree Khodiyar Mata Ji Mandir	15	N4	70.6	57.1	71.6	61.4	Severe Impact - Night
11.	Badrinarayan Temple	10	N5	64.5	51.9	69.6	63.2	Severe Impact
12.	Bholenath Temple	20	N5	64.5	51.9	64.7	52.7	No Impact
13.	Surat Yatim Khana Masjid	65	N5	64.5	51.9	64.7	53.0	No Impact
14.	Gajjarwadi Jain Temple	90	N5	64.5	51.9	64.6	52.5	No Impact
15.	Udhwana Darwaja Masjid	41	N7	72.5	60.9	72.6	61.3	No Impact
16.	Bileshwar Mahadev Mandir	50	N7	72.5	60.9	72.6	61.2	No Impact
17.	Masjide Aaisha	11	N7	72.5	60.9	73.6	64.5	Severe Impact - Night
18.	Shanti Nath Swetambari Mandir	67	N8	64.2	51.9	64.2	52.0	No Impact

S. No	Receptor	Distance from Metro in m	Noise Baseline Data in dB(A) of nearest monitoring location			Cumulative Noise for Year 2021		Noise Criteria
			Location	Lday	Lnight	Lday	Lnight	
19.	Sai Baba Temple	17	N8	64.2	51.9	66.9	59.2	Severe Impact - Night
20.	Masjid/Temple	5						
	Hospital							
1.	Palanpor Urban Health Center	99	N2	68.7	57.5	68.7	57.7	No Impact
2.	Sai Medical Hospital & Maternity Home	30	N2	68.7	57.5	69.1	58.9	No Impact
3.	Arpan Children Hospital	23	N2	68.7	57.5	69.4	59.7	No Impact
4.	Shubham Hospital	37	N2	68.7	57.5	69.0	58.5	No Impact
5.	Divyesh Eye Hospital	25	N2	68.7	57.5	69.3	59.4	No Impact
6.	Dhameliya Kidney Hospital & Lithotripsy Centre	15	N4	70.6	57.1	70.7	57.5	No Impact
7.	BAPS Pramukh Swami Hospital	43	N4	70.6	57.1	70.7	57.9	No Impact
8.	Adventist Work Hardt Hospital	50	N6	69.8	50.4	69.9	52.7	No Impact
9.	Care Hospital	27	N5	64.5	51.9	65.7	56.3	Severe Impact - Night
10.	Bridge Cross Hospital	62	N8	64.2	51.9	64.5	53.1	No Impact
11.	Kalpa Hospital	20	N8	64.2	51.9	66.3	58.1	Severe Impact - Night
12.	Tri Star Hospital	46	N6	69.8	50.4	69.9	53.1	No Impact
13.	Tender Touch Children Hospital	55	N6	69.8	50.4	69.9	52.4	No Impact
14.	G D Mehata Mahavir Heart Hospital	38	N6	69.8	50.4	70.0	53.9	Severe Impact - Night
15.	Shreyas Ano-Rectal Hospital & Research Centre	10	N6	69.8	50.4	72.0	63.1	Severe Impact - Night
16.	Arora Surgical Hospital	47	N6	69.8	50.4	69.9	53.0	No Impact
17.	Nirmal Hospital	11	N6	69.8	50.4	71.7	62.4	Severe Impact - Night
18.	Navsarjan Hospital	80	N6	69.8	50.4	69.8	51.5	No Impact
19.	Apple Hospital	48	N6	69.8	50.4	69.9	52.9	No Impact
20.	Vatsalya Children Hospital	72	N7	72.5	60.9	72.5	61.0	No Impact
21.	Chirayu Hospital	60	N9	69.4	59.4	69.5	59.7	No Impact
22.	Bharat Cancer Hospital	26	N9	69.4	59.4	69.8	60.6	No Impact

ANNEXURE 7.1

MUCK DUMPING SITE

The locations of Disposal sites have to be selected such that:

- No residential areas are located downwind side of these locations,
- Dumping sites are to be located at least 1000 m away from sensitive locations like Settlements, Water body, notified forest areas, Sanctuaries or any other sensitive locations.
- Public perception about the location of debris disposal site has to be obtained before finalizing the location.
- Permission from the Villager/local community is to be obtained for the Disposal site selected
- The Plan must be approved by GMRCL and SMC

A. PRECAUTIONS TO BE ADOPTED DURING DISPOSAL OF MUCK

While disposing debris / muck, the contractor will take into account the wind direction and location of settlements to ensure against any dust problems. The contractor shall take the following precautions while disposing off the muck:

- During the site clearance and disposal of muck, the contractor will take full care to ensure that public or private properties are not damaged/ affected.
- Contractor will dispose off muck/debris at the identified places or at other places only with prior permission of Engineer-in-Charge of works.
- In the event of any muck or debris from the sites being deposited on any adjacent land, the contractor will immediately remove all such spoil debris and restore the affected area to its original state to the satisfaction of the Engineer-in-Charge of works.
- At all times, the contractor will ensure that the canals and drains within or adjacent to the site are kept safe and free from any debris.
- Contractor will utilize effective water sprays during the delivery and handling of materials when dust is likely to be created and to dampen stored materials during dry and windy weather.
- Materials having the potential to produce dust will not the loaded to a level higher than the side and tail boards and will be covered with a tarpaulin in good condition.
- Any diversion required for traffic during disposal of debris shall be provided with traffic control signals and barriers after the discussion with local people and with the permission of Engineer-in-Charge of works.
- During the debris disposal, contractor will take care of surrounding features and avoid any damage to it.
- Some of the dumpsites could be used either for plantation or for growing agricultural produce
- Care should always be taken to maintain the hydrological flow in the area.
- Display Boards: The capacity of the disposal locations, name of the location etc shall be written in an Information board at each identified disposal locations.

B. PROPOSED DESIGN

Contractors need to get approvals for specific design for each identified disposal area. Contractor needs to plan the disposal in the following way.

- Identify the disposal area
- Need to record the present land use and condition of the area
- Consult with all stake holders
- Get written agreements from all concerned
- Prepare a suitable design for the safe disposal
- Construct all required structures (e.g. retaining wall)
- Planting of fast growing poplar trees on the outer portion of the retaining wall in the form of a linear wall parallel to the retaining wall
- Estimate the quantities
- Compact of the materials after disposal
- Prepare a Contractors debris disposal plan with design drawings for each identified area

ANNEXURE 7.2

DETAILS OF BATCHING PLANT
(To be filled by the Contractor)

Name of Location _____
Report for Batching Plant
Reporting Month……..
Date of Submission……..

1. Environment Features of the surrounding area

1.1	Name and location of Batching Plant	
1.2	Wind direction	
1.3	Name (s), distance population and type of settlements in a 1.5 km radius of site.	

2. Details of Batching Plant and Mitigation Measures taken

2.1	Installed Capacity	
2.2	Average Utilization	
2.5	Last maintenance date	

3. Explain Air Pollution Control Measures taken at the Batching Plant site

4. Explain Noise Pollution Control Measures taken at the Batching Plant site

Remark

Submitted	Checked	Approved
Signature …….	Signature ……	Signature ……
Name ……	Name ……	Name ……
Designation …….		
Contractor	Environmental Engineer of Construction Supervision Consultant	In-charge Officer (PIU)

ANNEXURE 7.3

DETAILS OF MUCK DUMPING OPERATIONS

(To be filled by the Contractor)

Dumping site location _____
Reporting Month……..
Date of Submission …….

1. Environment Features of the surrounding area

1.1	Location of Dumping site	
1.2	Capacity of Dumping site	
1.3	Safety measure taken at Dumping site (s)	
1. 2. 3. 4. 5.		

Remark

Submitted	**Checked**	**Approved**
Signature …….	Signature ……	Signature ……
Name ……	Name ……	Name ……
Designation …….		
Contractor	Environmental Engineer of Construction Supervision Consultant	In-charge Officer (PIU)

ANNEXURE 7.4

DETAILS OF MACHINERY DURING CONSTRUCTION

(To be filled Monthly by the Contractor)

Location Name _____
Reporting Month……..
Date of Submission ……..

1. Details of Machinery Operation

1.1	Total machinery in operation (Nos.)	
1.2	Number of pavers	
1.3	Number of rollers	
1.4	Number of excavators	
1.5	Number of graders	
1.6	Number of dumpers	
1.7	Number of Cranes	
1.8	No. of workshops with repairs facility (furnish location and type of facility provided)	Workshop on Facility Location Provided
1.9	Number of vehicles in repair at each location	
1.10	Details of waste disposal	
1.11	Others	

Remark

Submitted	**Checked**	**Approved**
Signature …….	Signature ……	Signature ……
Name ……	Name ……	Name ……
Designation …….		
Contractor	Environmental Engineer of Construction Supervision Consultant	In-charge Officer (PIU)

ANNEXURE 7.5

SAFETY CHECK LIST
(To be filled by the Contractor)

1	Contract No.	
2	Name of Contractor	
3	Representation	
4	Name of Safety Officer	
5	Date of Inspection	

Location 1 …… Location 2 …… Location 3 ……..

Adequate at time of Inspection Needs Improvement Needs Immediate Attention	Location 1			Location 2			Location 3			Remark
	A	B	C	A	B	C	A	B	C	
General										
House keeping										
Stacking of Material										
Passageway										
Lighting										
Ventilation										
Others										
Electrical										
Switches										
Wirings										
Fixed Installation										
Portable Lighting										
Portable Tool										
Welding Machine										
Others										
Fire Prevention										
Fire Fighting Appliance										
Dangerous Goods Store										
Gas Welding Cylinders										
Others										
Others										
Dust Control										
Noise Control										
First Aid Equipment										
Washing Facility										
Latrine										
Canteen										
Provision of Personal Protective										
Helmet										
Eye Protector										
Ear Protector										
Respirator										
Safety Shoes										
Safety Belts										
Others										

Remark

Submitted	**Checked**	**Approved**
Signature …….	Signature ……	Signature ……
Name ……	Name ……	Name ……
Designation …….		
Contractor	Environmental Engineer of Construction Supervision Consultant	In-charge Officer (PIU)

ANNEXURE 7.6

ACCIDENT REPORT
(To be completed on Occurrence of Injury by the Safety Officer)

Type of Accident

	Fall of person from a height		Explosion
	Slip, trip or fall on same level		Fire
	Struck against fixed objects		Contact with hot or corrosive substances
	Struck by flying or falling objects		Contact with poisonous gas or toxic substances
	Struck by moving objects		Contact with electric current
	Struck / caught by cable		Hand tool accident
	Stepping on hail etc.		Vehicle / Mobile plant accident
	Handling without machinery		Machinery operation accident
	Crushing / burying		Other (please specify)
	Drowning or asphyxiation		

Agent Involved in Accident

	Machinery		Excavation / underground working
	Portable power appliance		Floor, ground, stairs or any working, surface
	Vehicle or associated equipment / machinery		Ladder
	Material being handled, used or stored		Scaffolding/gondola
	Gas, vpour, dust, fume or oxygen		Construction formwork, shuttering and falsework
	Hand tools		Electricity supply cable, wiring switchboard and associated equipment
	Floor edge		Nail, sllnter or chipping
	Floor opening		Other (Please specify)
	Left shaft		
	Stair edge		

Unsafe Action Relevant to the Accident

	Operating without authority		Failure to use proper footwear
	Failure to secure objects		Failure to use eye protector
	Making safety devices inoperative		Failure to use respirator
	Working on moving or dangerous equipment		Failure to use proper clothing
	Using un-safety equipment		Failure to use warn others or given proper signals
	Adopting unsafe position or posture		Horseplay
	Operating or working at unsafe speed		No unsafe action
	Unsafe loading, Placing, mixing etc.		Others (please specify)

	Failure to use helmet		

	No Protective gear		Unsafe layout of job, traffic etc.
	Defective protective gear		Unsafe process of job methods
	Improper dress / footwear		Poor housekeeping
	Improper guarding		Lack of warming system
	Improper ventilation		Defective tool, machinery or materials
	Improper illumination		No unsafe condition
	Improper procedure		Others (please specify)

Personal Factor Relevant to the Accident

	Incorrect attitude / motive		Unsafe act by another person
	Lack of knowledge or skill		No unsafe personal factor
	Physical defects		Other (please specify)

Remark

Submitted
Signature ……
Name …….
Designation ……..
Contractor

Checked
Signature ……..
Name ……
Designation ……..
Environmental engineer.
Construction Supervision Consultant

Approved
Signature ……
Name ……..
Designation ……
In-charge Officer (PIU)

ANNEXURE 7.7

POLLUTION MONITORING

Construction site location _____

Construction Stage: Report – Date: _____Month _____Year_____

Mitigation measures suggested in last report complied or Not……

If not reasons thereof……

(Location at which monitoring to be conducted as per EMP)

Sl. No.	Chainage (km)	Details of locations	Duration of monitoring	Instruments used	Completion			Monitoring Parameters	Standards	Results	Reasons for exceeding standards	Mitigation Measures suggested	Type of area (Residential / Industrial / Commercial)	Remark
1. Air Monitoring (As per National Ambient Air Quality Standards, CPCB (2009)														
		As per decision of Engineer in Charge	As per Section 8.1					$PM_{2.5}$	60 µg/m3					
								PM_{10}	100 µg/m3					
								SO_2	80 µg/m3					
								CO	02 mg/m3					
								NOx	80 µg/m3					
2. Water Monitoring (As per Drinking Water Quality Standards, IS 10500, 2012)														
		As per decision of Engineer in Charge	As per Section 8.1					pH	6.5-8.5					
								BOD	Nil					
								COD	Nil					
								TDS	500 mg/l					
								Chlorides	250 mg/l					
								Nitrates	45 mg/l					
								Sulphates	200 mg/l					
								Iron	0.3 mg/l					
								Calcium	75 mg/l					
								Lead	0.01 mg/l					
3. Soil Monitoring														
		As per decision of Engineer in Charge	As per Section 8.1					pH	<7.0 Acid 6.5–7.5 Neutral >7.5 Alkaline					
								Organic Matter	0.5 -0.75 %					
								Sodium	0-1 %					
								Potassium	2-6 %					

Sl. No.	Chainage (km)	Details of locations	Duration of monitoring	Instruments used	Completion			Monitoring Parameters	Standards	Results	Reasons for exceeding standards	Mitigation Measures suggested	Type of area (Residential / Industrial / Commercial)	Remark
								Chloride	0-1 %					
								Available Nitrogen	280-560 kg/hac					
								Phosphorous	11.5 – 24.5 kg/hac					
								Arsenic	< 20 mg/kg					
								Cadmium	< 1 mg/kg					
								Mercury	< 1 mg/kg					
								Lead	< 35 mg/kg					
								Electric Conductivity	0.0-2.0 Non Saline 4.1-8.0 Saline 16.0 Strongly Saline					
4. Noise Monitoring (As per National Ambient Noise Standards, CPCB														
		As per decision of Engineer in Charge	As per Section 8.1					L_{day}	Residential-55 dB(A) Commercial-65 dB(A)					
								L_{night}	Residential-45 dB(A) Commercial-55 dB(A)					
Remark														

Submitted
Signature ……
Name …….
Designation ……..
Contractor

Checked
Signature ……..
Name ……
Environmental Engineer of Construction Supervision Consultant

Approved
Signature ……
Name ……..
In-charge Officer (PIU)

ANNEXURE 7.8

Format for Vibration monitoring

Construction site location _____
Construction Stage: Report – Date: _____Month _____Year_____
Mitigation measures suggested in last report complied or Not……
If not reasons thereof……
(Location at which monitoring to be conducted as per EMP)

	Format for Vibration monitoring								
Sampling code	**Location**	**Date**	**Start Time**	**Stop Time**	**Vibration level (PPV in mm/s)**	**Latitude**	**Standard**	**Longitude**	**Remarks**
1							DGMS (Directorate General of Mines and Safety)		
2									
3									

Submitted
Signature ……
Name ……..
Designation ……..
Contractor

Checked
Signature ……..
Name ……
Environmental Engineer of
Construction Supervision Consultant

Approved
Signature ……
Name ……..
In-charge Officer (PIU)

ANNEXURE 7.9

RESTORATION OF CONSTRUCTION SITES
(To be filled by the Contractor)

Construction site location _____
(Reporting by Contractor to PIU)
Construction stage: Monthly Report – Date ……Month ……Year.

Sl. No.	Contract Package	Labor Camp		Construction Camp		Plant Site		Disposal Locations		Top Soil	
		O	R	O	R	O	R	O	R	Preserved	Restored

Remark

Submitted
Signature ……
Name …….
Designation ……..
Contractor

Checked
Signature ……..
Name ……
Designation ……..
Environmental engineer of Construction Supervision Consultant

Approved
Signature ……
Name ……..
Designation ……
In-charge Officer (PIU)

ANNEXURE 7.10

FORMAT FOR KEEPING RECORDS OF CONSENT OBTAINED BY CONTRACTOR

Construction site location _____
Construction Stage: Report – Date: _____Month_____ Year_____

Sl. No.	Contractor's Name	Clearance	Applicable Acts	Agencies	Obtained on	Valid up to	Remarks
	Construction site location						

Remark

Submitted
Signature ……
Name …….
Designation ……..
Contractor

Checked
Signature ……..
Name ……
Designation ……..
Environmental engineer of Construction Supervision Consultant

Approved
Signature ……
Name ……..
Designation ……
In-charge Officer (PIU)

ANNEXURE 7.11

CHECKLIST FOR ENVIRONMENT INSPECTION
(Points / Issues to be covered)

Construction site location _____

Date of Inspection_____

S. No.	ESMP Measures
1	Provision of a personnel accountable for implementation of ESMP / Safety Measures with Contractor
2	Consent of PCB to Establish Batching Plant
3	Consent of PCB to operate Batching Plant
4	Compliance of PCB Conditions for Batching Plant installation and operation
5	Whether compliance reported through monthly Progress report to In-Charge (PIU)
6	PUC taken for all Construction vehicles
7	Concrete platform with trap under bitumen boiler, Fuel Tank for Batching Plant and generator set provided or not
8	Precautions to prevent contamination of soil by emulsion, oil and lubricant taken while storing
9	Providing cover to fine construction material & bituminous mix during transportation
10	Muck /debris disposal:
	a) Present status of land
	b) Closure and completion plan
11	Site specific traffic Safety management Plan:
	a) Contractor installed the warning / regulatory Traffic signs at the construction site
	b) The arrangement adequate
12	Safety equipment i.e helmet, gloves, gumboot, mask, earplugs etc. provided to workers
13	Health Facility at camp and work site i.e. First Aid kit & suitable vehicle for conveyance in case of emergency / accident
14	Permit for Procuring River sand
15	Licence from Department of mines for quarrying
16	Consent to establish / operation of crusher
17	Provision of labour camp with sanitation & potable water
18	Fire precautions at Plant and site Office
19	Air and noise monitoring done in camp site
20	Whether any cultural property is being impacted
21	Status of drainage provision in camp area
22	General House Keeping

Remarks:……..

Submitted	**Checked**	**Approved**
Signature ……	Signature ……..	Signature ……
Name …….	Name ……	Name ……
Designation ……..	Designation ……..	Designation ……
Contractor	Environmental engineer of Construction Supervision Consultant	In-charge Officer (PIU)

ANNEXURE 7.12

SUMMARY SHEET

(To be filled monthly by supervisory staff and Submitted to GMRCL)

Construction site location _____
Month_____Date_____

S No.	Description	Remarks
1	***No Objection Certificate***	
A	Cement Batching Plant	
	Location 1	
	Location 2	
	Location 3	
2	***Pollution Under Certificate***	
	Vehicles	
	Machineries	
3	**No Objection Certificate for Diesel Gen set**	
	Location 1	
	Location 2	
4	**Labour Camps**	
	No. of sites Identified	
	Approved	
	Opened	
	Conforms to conditions imposed at the time of opening of sites	
	Closed	
5	***Workers***	
	No of workers employed	
	No of male workers	
	No of female workers	
	No of day workers	
6	**Borrow Area**	
	No. of sites identified	
	Approved	
	Opened	
	Quantity of available material	
	Quantity of material Utilized	
	Quantity of Topsoil preserved	
	Quantity of top soil used	
	No of sites closed	
	No. of sites Rehabilitated	
7	**Quarry**	
	No. of sites identified	
	Approved	
	Opened	
	Material available	
	Material obtained	
	No. of sites Rehabilitated	

S No.	Description	Remarks
8	**Disposal Locations**	
	No. of sites identified	
	Approved	
	Opened	
	Amount of Waste disposed	
	Type of waste disposed	
	No. of sites Rehabilitated	
9	**Road Safety**	
	Road Safety norms and approved Traffic plan	
10	**Cleaning of Culvert/ drains**	
	No. of culverts/ drains	
	Nos Cleaned	
11	**Trees**	
	No of trees marked for cutting in field	
	No of trees cut	
	No of trees to be Planted	
	Trees Planted	
12	**Haul Roads**	
	Adequacy of maintenance of Haul Road Network	

Remarks:…….

Submitted
Signature ……
Name ……
Designation …….
Contractor

Checked
Signature ……..
Name ……
Designation ……..
Environmental engineer of Construction Supervision Consultant

Approved
Signature ……
Name ……..
Designation ……
In-charge Officer (PIU)

ANNEXURE 8.1

Public Consultations

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 8/11/19
Time: 10:10
No of Participants: 10
Perception about the project:

1. Do you think that metro is necessary for Surat City? ~~1-Yes~~ 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

Positive Impacts:

Save travel time
Easy traveling for person who don't have vehicles.
Asthetic view (looks good)

Negative Impacts:

who have the shops near alignment they may have problem of obstruction due to project.

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
Bhesan Near STP (Near proposed Bhesan Depot)	Air pollution	Not anticipated.	
	Trees	10L trees are given by smc so we planted in our places. we love env. No issues of cutting trees in Bhesan Depot	
	Noise	Agriculture land is nearby so no issues of noise. Don't feel noise	

	Job.	pollution during operation People will get job due to metro. Happy for the project, it will give very good looks to the city	

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Hitesh Patel	35	M	service		9898587930
2	Nehal Patel	33	M	"		9825417766
3	Kantibhai Patel	59	M.	Shop.		9825649895
4.	Josnaben Patel	53	F.	"		
5	Pravinbhai	48	M	Service.		
6	Dharmendra Patel	28	M	"		9913725528
7	Rameshbhai Patel	50	M	Farmer		9879405851
8	Bhavnaben Patel	45	F	"		
9	Thakurbhai Pasma	52	M	Tea stall		[illegible] 9879 68787
10.	Heraben Thaku	45	F	"		

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 08/11/19
Time: 11:25 AM
No of Participants: 11
Perception about the project:

1. Do you think that metro is necessary for Surat City? ~~1-~~Yes 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

Positive Impacts:

- time saving
- safety
- fast traveling

Negative Impacts:

No any structures may be broken
Roads width may be reduced.

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
Near L.P. Savani school, Kholdiyar Chowk.	Air/Noise	Its a clean city no air pollution & noise pollution at all. Not anticipating any running of metro as well.	
Adajan Gam.	Trees	Two giant tree are here so people are worksup it sometime only. Needs to be saved/protected.	
		We don't see any problem if metro going elevated near trees.	

	parking (Khodiyar Mata Temple)	There should be enough parking near temple. People coming more in festival so they need space to park vehicle at temple.	
	—	Business will increase due to station location	
	—	Trees are evergeen since many year we don't think if there is no any vehicle exaust or smoke than it will not hazard to trees.	

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Omprakash Rawat	35	M	service	O.P.R.	9913462625
2	Tasun Patel	32	M	Bsg. Gala.	Tasun	9687455991
3	Ankit Chauhan	32	M	service		9969229089
4	Rahul Bhai	31	M	snack stall		7487871630
5	Atul Bhai	32	M	"	Atul	"
6	Ravi Dholare	33	M	Rikshaw Auto	रवि	8401158061
7	Rameshwar Gaikwad	37	M	sugarcane juice stall		7[illegible]35896
8	Swapnil Dalan	29	M	"		
9	Govindbhai Patel	74	M	Retaired.	V. N. Patel	—
10	Jeety Mali	40	M	Flower shop at toyle	J. P. Mali	9726395670
11	Jayesh Kumar Kundan das.	48	M	service at twp.	[illegible]	8849681735
12	Indra Devi		F	Retaired service.	[illegible]	8128550855

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 8/11/19
Time: 12:05
No of Participants:
Perception about the project:

1. Do you think that metro is necessary for Surat City? ~~1-Yes~~ 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

Positive Impacts:

- Happy to have metro.
- Traveling time reduced.

Negative Impacts:

Auto Riksha Drivers may loose their jobs.

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
majura Gate Chawk	-	Line should be cater the people from all city Traveling fast so people can reach timely. very helpul to the people.	
	Traffic	Reduced pollution reduced due reduction in traffic specially auto.	

	Trees construction	Trees to be transplanted less trees to be cut. construction time should be less, should be complete fast. Fuel save. Fare should be minimum so people can affordable.	

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Chetan Gadhiya	34	M	Job.	C.V.G.	9714260775
2	Pankaj Sojitra	30	M	service	..	9913339089
3	Pradip Jain	42	M	Business		9173978289
4	Abhishek Singh	22	M	"		9558702585
5	Pradeep Gupta	27	M	Parking		9726112708
6	Sanjay Misra	26	M	"		—
8	Golu Pandey	29	M	"		—
9	Nayeem Bhai	38	M	service		9926527890
10	Vilas Bhai	46	M	"		—

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 8/11/19
Time: 1:00
No of Participants: 8
Perception about the project:

1. Do you think that metro is necessary for Surat City? (1-Yes) 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

Positive Impacts:

- Less travel time.
- Safety of people
- less air pollution.

Negative Impacts:

Structure may be demolish which is coming in alignment.

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
Model Town Chowk	Asthetic	Traveling time will be less. Population are increasing fastly so metro will be good option. safety for public city looks smart due to metro. Traffic will reduce. Accident reduce No Air pollution because train running on electricity.	

Model Town Chowk		construction & work should be safe & long life.
	Tree -	Fare should be affordable to people as compare to BRT.
	Frequency.	Trees to be planted for cutted trees.
		Frequency & trains should be online during peak time

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Dinesh Bhojiriya	25	M	salesman		6353613990
2	Raghav Bathya	25	M	"		8871019950
3	Prakash Patidar	32	M	Teastall		9687455831
4	Vagh Devram	35	M	Traffic Police		9723797808
5	Bakare Mukesh	37	M	"	Mukesh	7778333732
6	Krishna	46	M	service		9106517998
7	Yogesh	44	M	"		9664395721
8	Aman	33	M	shop		9998777212

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 8/11/19
Time: 1:50
No of Participants: 6
Perception about the project:

1. Do you think that metro is necessary for Surat City? 1-Yes 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

Positive Impacts:

- Fast service within city
- Time save

Negative Impacts:

- structures may be demolished.
- Business may affect.

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
Kumbhariya Sarali	Fasility	Lines are proposed through good area where tragic congetion is more. It will facilitate the people.	
	Business	Business may be inorese which are close to the station	
	Fast	Fast traveling within city.	

		frashs should be run out of city as well as nearby places.	

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Arvind Pal	32	M	Shop.	अरविन्द	9904594352
2	Rohit Kumar	19	M	Daily weges	रोहित कुमार	-
3	Arjun	19	M	"	अर्जुन	-
4	Shekh Jamaluddin	42	M	Shop		6353691132
5	Rusat Banu shekh	35	F	"		"
6	Pramod Londhe	29	M	service		-

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 03/11/19
Time:
No of Participants: 7
Perception about the project:

1. Do you think that metro is necessary for Surat City? ~~1~~-Yes 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

Positive Impacts:

- less air pollution
- less traffic congestion
- fast moving traffic.

Negative Impacts:

- Tree will cut
- Traffic congestion due to autorikshaw
- air pollution due to autorikshaw

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
Bhatar char Rasta chowk.	Job	Local labours are plenty in the city so they may get job during construction of metro.	
	Business	shopes nearby station will have more good business.	
	Traffic Air pollution	Autorikshq are very much which contributed traffic congestion & also pollution. so after metro these thing will stop.	
	Tree	for the development some trees needs to be cut.	

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Pravinbhai	35	M	Business		9825170839
2	Jadhavji bhai	45	M	"		9879564913
3	Raju Damor	25	M	service		8000375555
4	Manish Sowan	20	M	worker		—
5	Sagar	22	M	"	S.G.	8469579190
6	Rushi Patil	39	M	service		9898467668
7	Ramesh Pandey	43	M	Panshop		9974180049

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 9/11/19
Time:
No of Participants: 16
Perception about the project:

1. Do you think that metro is necessary for Surat City? ~~1-Yes~~ 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

Positive Impacts:

- Time will save
- Easy for us to travel
- Fast travelling

Negative Impacts:

- Tree cutting

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
Bhimrad village	Travel Time	Traveling time will be less as compare to present situation.	
	Easy traveling	Due to proposed station at Bhimrad village we can commute easily.	

	Safety	Traveling will be more safe & fast for us.	
	Trees	More trees to be planted for cutted trees.	

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Indu ben	52	F	House wife	K. B. Patel	
2	Alpa ben	49	F	"	A. R. Patel	
3	Hemu ben	51	F	"	H. P. Patel	
4	Kalpna ben	48	F	"	K. Patel	
5	Damu bhai	49	F	"	D. Patel	
6	Sushila ben	55	F	"	S. Patel	
7	Bejihumu ben	40	F	"	B. Patel	
8	Anuradha	45	F	"	A. Patel	
9	Savita Behan	50	F	"	S. Patel	
10	Raksha ben	47	F	"	R. Patel	
11	Kusum ben	53	F	"	K. Patel	
12	Bharati ben	43	F	"	B. Patel	
13	Sapna ben	40	F	"	S. Patel	
14	Kamu ben	41	F	"	K. Patel	
15	Chandrika ben	39	F	"		
16	Setna Behan	44	F	"		
17.						

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 9/11/19
Time:
No of Participants: 7
Perception about the project:

1. Do you think that metro is necessary for Surat City? ✓Yes 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

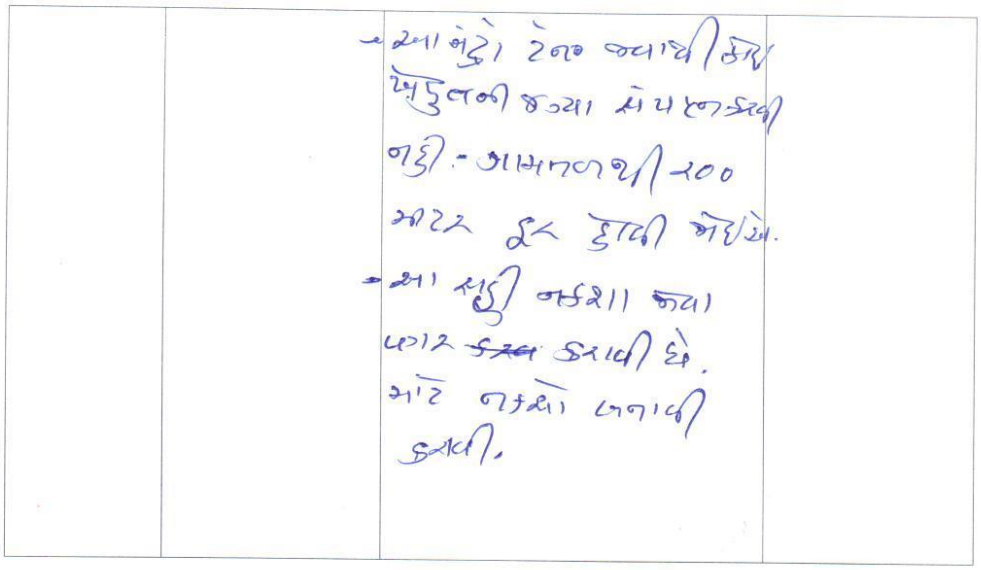
Positive Impacts:

- Fast & safe Traveling

Negative Impacts:

- No structures & agriculture land affected.
- ~~Job~~

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
Bhimrad village	Land & structures	- metro should not affect the gandhi ground. - structures should be protected of this village	
	Job	- Preference should be given to the villagers during construction. - Fast & safe traveling - આ ગામ ને રેલ્વે સ્ટેશન ગામ માંથી દૂર ના જોઈએ - આ ગામ ના ગામતળ વિસ્તાર ને ગામ કરિયાણાં બિસ્તાર ફરવું ના જોઈ	



- આ મેટ્રો રેલના વ્યાપારી તેમજ
મુસાફરોની સગવડ માટે ખૂબજ
સારી. - ગામમાંથી ૨૦૦
મીટર દૂર હોવાથી અગવડ.
- આ મેટ્રો બનવાથી કયા
પ્રકારે ~~ફરક~~ ફરકી છે.
માટે બનવું જરૂરી
છે.

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Bankim Patel	54	M	Business	બકીમ	9723693163
2	Navneet Patel	51	M	"	નવનીત પટેલ	7043292999
3	~~Hitesh~~ Navin Patel	42	M	Service	નવીન.એ.પટેલ	~~9825[illegible]~~
4	~~Nilesh~~ Patel	39	M	Business	[illegible]	98798-62263
5	Mukesh	41	M	"	મુકેશભાઈ	
6	વરજાંગ ટી પટેલ	38	M	"	વરજાંગ ટી પટેલ	9974074448
7	Anil Patel.	39	M.	"	Anil.	9825334810

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 09/11/19
Time: 1:50
No of Participants: 6
Perception about the project:

1. Do you think that metro is necessary for Surat City? 1-Yes 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

Positive Impacts:

Fast traveling
safety
Asthetically looks good.

Negative Impacts:

- trees to be cut
- inconvinent during construction

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
Santhanam Jakat Naka Hazira	Traveling time safety	less travel time and safety for travellers.	
	Asthetic view	It will looks beautiful after construction	
	Trees	Trees should be planted properly.	

	Business	Business will grow which are located near proposed station locations.	

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Arbaj Khan	19	M	service		7990874849
2	Hardik chahsoliya	23	M	Design		7984399481
3	Nikhil Parmar	24	M	service		7984581524
4	Tejas Mistry	33	M	"		9737550165
5.	Maitri Patel	40	M	Service		8758550345.
6.	Ankit Laheri	32	M	"		9824379116
7.	Ratnesh Bhai	41	M	"		6355856213
8	Nitin Bhai	45	M	"		9979978841

FORMAT FOR PUBLIC CONSULTATION MEETING

Name of the Project: Surat Metro Rail Project, Phase-I
Date: 09/11/19
Time: 2:15
No of Participants: 5
Perception about the project:

1. Do you think that metro is necessary for Surat City? 1-Yes 2-No
2. What impacts, both positive and negative of the metro project do you foresee?

Positive Impacts:

- City look beautiful

Negative Impacts:

- Tree cutting

Location/ Venue	Issues Discussed	Suggestions by Stakeholders	Remarks
Ghanshyam Nagar Road L.S.	Convinienly	people having two vehicles at each households so we dont think people will travel in metro in this place.	
	Business	we don't think we will get more benifit in business due to station location.	
	Tree	Tree ~~cutting~~ sampling/ afforestation Provision D good at the cost of 1 cutting tree.	

	Aesthetic view	Auto rikshaw is more convinient. City look beautiful after metro operation. Buses are running empty now, so now you will get passengers in metro.	

S.No	Name	Age	Sex	Occupation	Signature	Mob No
1	Deepak Zala	37	M	Business	D.K.Zala	9998769272
2	Gopal Mandaliya	33	M	Service		9624604204
3	Kamlesh Hingu	29	M	Diamond Job		9825009177
4	M.R.Rathod	32	M	"		8980164764
5	Sunil Rathod	34	M	Business		8141238623